Mio-Pliocene aridity in the south-central Andes associated with Southern Hemisphere cold periods

William H. Amidon[a,1], G. Burch Fisher[b,c], Douglas W. Burbank[c], Patricia L. Ciccioli[d], Ricardo N. Alonso[e], Andrew L. Gorin[a], Perri H. Silverhart[a], Andrew R. C. Kylander-Clark[c], and Michael S. Christoffersen[b]

[a]Geology Department, Middlebury College, Middlebury, VT 05753; [b]Jackson School of Geosciences, University of Texas at Austin, Austin, TX 78712; [c]Earth Research Institute, University of California, Santa Barbara, CA 93106; [d]Departamento de Ciencias Geológicas, Instituto de Geociencias Básicas, Aplicadas y Ambientales de Buenos Aires, Facultad de Ciencias Exactas y Naturales, Universidad de Buenos Aires-Consejo Nacional de Investigaciones Científicas y Técnicas, Buenos Aires C1428EHA, Argentina; and [e]Instituto Superior de Correlación Geológica, Universidad Nacional de Salta-Consejo Nacional de Investigaciones Científicas y Técnicas, 4400 Salta, Argentina

Edited by Thure E. Cerling, University of Utah, Salt Lake City, UT, and approved May 12, 2017 (received for review January 9, 2017)

Although Earth's climate history is best known through marine records, the corresponding continental climatic conditions drive the evolution of terrestrial life. Continental conditions during the latest Miocene are of particular interest because global faunal turnover is roughly synchronous with a period of global glaciation from ∼6.2–5.5 Ma and with the Messinian Salinity Crisis from ∼6.0–5.3 Ma. Despite the climatic and ecological significance of this period, the continental climatic conditions associated with it remain unclear. We address this question using erosion rates of ancient watersheds to constrain Mio-Pliocene climatic conditions in the south-central Andes near 30° S. Our results show two slowdowns in erosion rate, one from ∼6.1–5.2 Ma and another from 3.6 to 3.3 Ma, which we attribute to periods of continental aridity. This view is supported by synchrony with other regional proxies for aridity and with the timing of glacial "cold" periods as recorded by marine proxies, such as the M2 isotope excursion. We thus conclude that aridity in the south-central Andes is associated with cold periods at high southern latitudes, perhaps due to a northward migration of the Southern Hemisphere westerlies, which disrupted the South American Low Level Jet that delivers moisture to southeastern South America. Colder glacial periods, and possibly associated reductions in atmospheric CO_2, thus seem to be an important driver of Mio-Pliocene ecological transitions in the central Andes. Finally, this study demonstrates that paleo-erosion rates can be a powerful proxy for ancient continental climates that lie beyond the reach of most lacustrine and glacial archives.

Messinian | Miocene | Pliocene | M2 | precipitation

Earth's climate has played a fundamental role in the evolution of terrestrial life (1, 2). Whereas marine-based climate records offer broad insights into ancient climates with high temporal resolution (3), it is the associated changes in continental climate that govern the evolution of terrestrial life. Past changes in continental climate, however, are typically difficult to measure and the effect of global climate events on terrestrial life is commonly unclear. For example, how do changes in global temperature translate to changes in the hydrologic cycle at specific locations on Earth? To what extent does averaging of signals in the marine climatic record mask the severity of continental climatic shifts and their impact on terrestrial fauna? Answers to these questions remain particularly elusive for pre-Quaternary climatic events, which are commonly beyond the reach of continental lacustrine or glacial climate proxies. Additional continental climate proxies for the Mio-Pliocene time can thus help reveal the role of global climate excursions in driving well-known faunal transitions.

Continental conditions during latest Miocene cooling are of particular interest, because major faunal turnovers during this time influenced the evolution of many species (4, 5). For example, the late Miocene (∼7.0–5.5 Ma) saw expansion of plant species that use a C4 photosynthetic pathway, which began to outcompete existing C3 plants across subtropical regions of South America, Africa, and Asia (6–8). However, the exact timing and continental climatic conditions that drove this expansion have been the subject of much discussion. C4 plants tend to thrive in warmer conditions, as long as ample moisture is present during the growing season (9). However, because they concentrate CO_2 during photosynthesis, they likely outcompete C3 plants under low-CO_2 and low-temperature conditions, such as during cold glacial periods. For example, at an atmospheric CO_2 concentration of 250 ppm C4 plants would be favored at all growing season temperatures above ∼18 °C (8). It is thus logical to speculate that the late Miocene spread of C4 plants may have resulted from low-CO_2 conditions. Although several studies have argued against reduced global CO_2 in the late Miocene (10, 11), recent work shows a significant decrease beginning around 7 Ma (12, 13).

Although the timing of onset of C4 expansion varies by location, the major shift in $\delta^{13}C$ is commonly observed in the very latest Miocene between ∼6–5.5 Ma (7, 14, 15). This timing is remarkably similar to the timing of the Messinian Salinity Crisis (MSC), in which the Mediterranean Sea shrank and ultimately dried up from 5.96 to 5.33 Ma (16). The cause of the MSC has also been the subject of much debate (17, 18), but recent work argues it was triggered by glacio-eustatic sea-level fall in response to expansion of Antarctic ice sheets at ∼6 Ma (19) and that the period of maximum aridity (the "Messinian gap") corresponds to one or two glacial periods near 5.53 ± 0.06 Ma (16, 20). These glacial periods are part of a long-recognized, but somewhat enigmatic, sequence of latest Miocene glacials between ∼6.2 and 5.5 Ma, which included up to 18 glacial-interglacial cycles on a

Significance

This paper identifies two periods of enhanced aridity that are synchronous with faunal turnovers in southern South America. Close temporal coincidence with marine climate proxies suggests a period of latest Miocene aridity was associated with a global glacial period and the expansion of C4 vegetation. We thus argue that continental aridity in the south-central Andes is associated with cold periods at high southern latitudes and propose a model to link global and regional (continental) climate via shifting of the Southern Hemisphere westerlies. This paper also provides an example of how ^{10}Be paleo-erosion rates can be used as a climate proxy and demonstrates how ^{10}Be and ^{36}Cl can be combined to reduce uncertainties associated with this method.

Author contributions: W.H.A., G.B.F., and D.W.B. designed research; W.H.A., G.B.F., D.W.B., P.L.C., R.N.A., A.L.G., P.H.S., A.R.C.K.-C., and M.S.C. performed research; W.H.A. and G.B.F. analyzed data; and W.H.A. wrote the paper.

The authors declare no conflict of interest.

This article is a PNAS Direct Submission.

[1]To whom correspondence should be addressed. Email: wamidon@middlebury.edu.

This article contains supporting information online at www.pnas.org/lookup/suppl/doi:10.1073/pnas.1700327114/-/DCSupplemental.

41,000-y obliquity-paced timescale (21, 22). Evidence of latest Miocene ice accumulation in both the Northern (23) and Southern Hemispheres (24) suggest these glacial periods were global in nature and could have been associated with CO_2 levels low enough to favor C4 vegetation on a global scale.

Testing whether late Miocene faunal shifts were driven by CO_2 forcing or temperature/ precipitation forcing is difficult given the relative paucity of proxies for continental paleoclimate. Much of our knowledge of latest Miocene terrestrial climate is derived from plant fossils, hydrocarbon biomarkers, and the isotopic composition of paleosoils and fossil teeth, features that are rare or poorly preserved in most sedimentary sections (25, 26). Likewise, most terrestrial sediments are deposited in association with active mountain belts, creating the additional challenge of deconvolving regional climatic signals from local tectonic–orographic signals. These issues are especially challenging in the south-central Andes where a regionally extensive climate transition in the latest Miocene (14, 27, 28) appears synchronous with significant tectonic uplift at this time (29–32). It thus remains unclear whether the timing and extent of latest Miocene landscape ecologic changes in the south-central Andes were primarily controlled by tectonic–orographic effects or by global-scale changes in climate (e.g., refs. 33–35).

This study addresses these questions by using erosion rates of ancient watersheds as a proxy for Mio-Pliocene (~7–2.8 Ma) precipitation in the south-central Andes (~30° S, 68° W). Because erosion rate in semiarid watersheds of the central Andes is primarily correlated with precipitation (36), transient excursions in erosion rate over <400,000-y timescales should be a robust proxy for paleoprecipitation. Erosion rates are estimated by measuring cosmogenic ^{10}Be in ancient fluvial sediments deposited in foreland basin sequences, which record the erosion rate in their source watersheds (37–39). This approach yields a powerful proxy because it integrates landscape processes over a large region (10^4–10^5 km^2) and provides a continuous record that can be precisely dated using magnetostratigraphy and tephrachronology. Our results show two decreases in erosion rate that are synchronous with the latest Miocene glacial period described above and with a second glacial period from roughly 3.6–3.3 Ma. We attribute these slowdowns to enhanced continental aridity during cold periods at high southern latitudes. Finally, we suggest that both latest Miocene aridity and the coeval expansion of C4 vegetation may have been driven by a latest Miocene glacial period and an associated drop in global CO_2.

Study Area and Methods

This section presents a summary of the study area and methods, with full details provided in *SI Appendix*. We developed paleo-erosion-rate records on the Huaco and Toro Negro sections, which were deposited in the central and northern regions of the Bermejo Foreland Basin, respectively (Fig. 1). The two sections were chosen because they are well exposed in deep road cuts, a prerequisite for applying the paleocosmogenic technique. The sections are composed of coarse fluvial sandstones and conglomerates shed from the eastern slopes of the Andes over the last 15 My. A paleomagnetic chronology was developed along the Huaco section, constraining the age of our samples to ~6.4 to ~2.6 Ma (*SI Appendix*, Fig. S2). The Toro Negro section lacks suitable deposits for paleomagnetic dating and has been previously dated to ~7–2.3 Ma by U-Pb zircon dating of tephra (40).

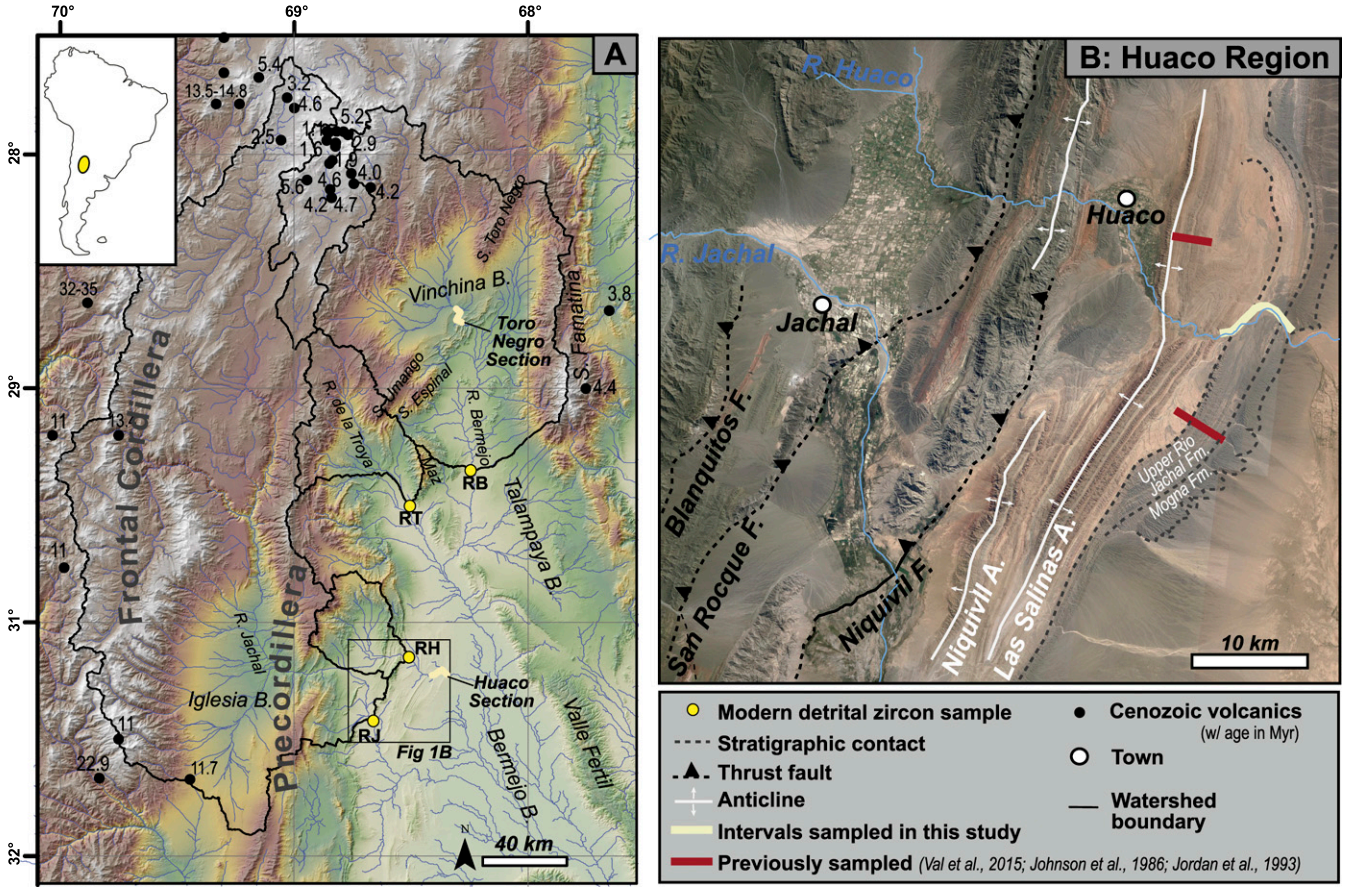


Fig. 1. (*A*) Shaded relief map of the study area derived from a 90-m digital elevation model showing the extent of modern watersheds and a compilation of young volcanic ages (73, 74). (*B*) Satellite image of the Huaco study area showing the location of the sampling transect used in this and previous studies, as well as key tectonic structures (75).

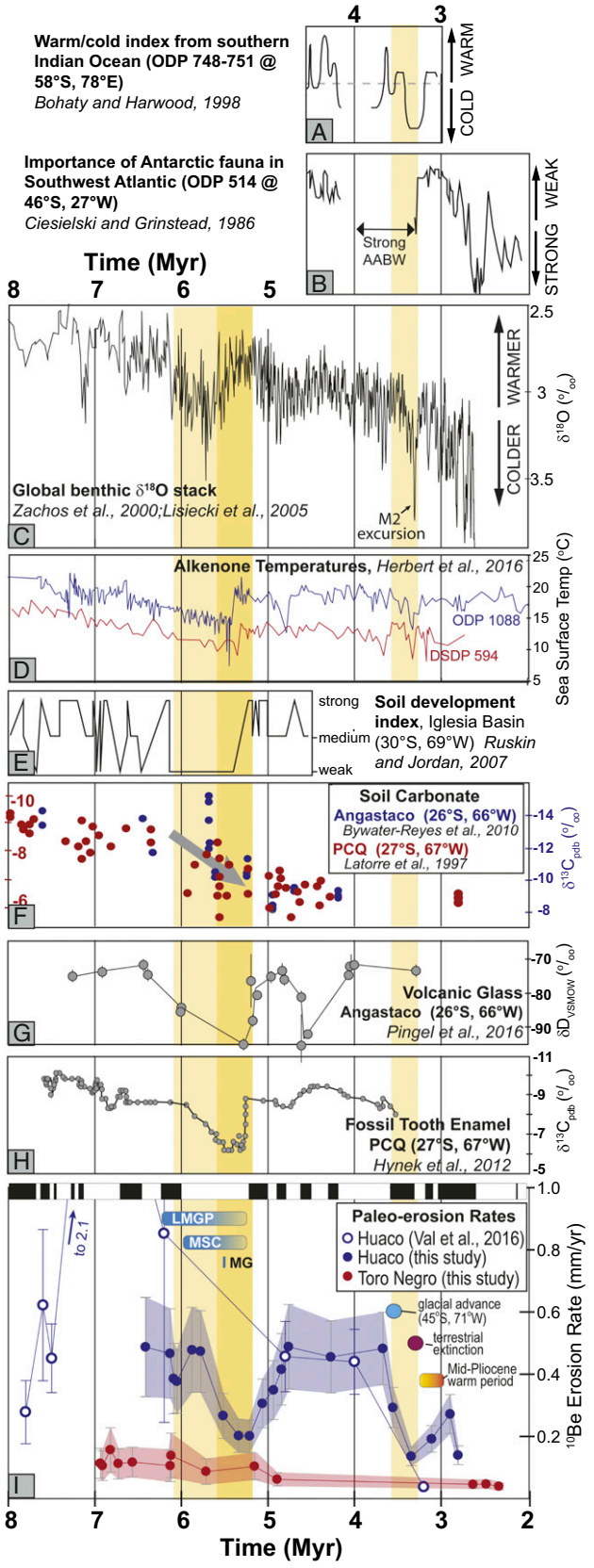


Fig. 2. (*A*–*H*) Compilation of regional and global climate proxies. (*I*) Paleo-erosion rate records compared with major global and regional climatic events. LMGP denotes the latest Miocene glacial period (20). MSC denotes the Messinian Salinity Crisis (16) MG denotes the Messinian Gap when aridity peaked in the Mediterranean (16). The timing of glaciation, extinction, and the mid-Pliocene warm period are depicted (49, 54, 76). The paleo-erosion rates of Val et al. (38) were also developed on the Huaco section, but along a different sampling transect (see Fig. 1). Larger uncertainties on their dataset result from using surface samples that were not shielded from modern cosmic-ray exposure. Such contrasting sampling strategies may explain disagreement between the two datasets for older samples in which small amounts of modern exposure ($C_{exhumation}$) could swamp the ancient (and significantly decayed) ^{10}Be signal (C_{paleo}).

Samples for measurement of cosmogenic ^{10}Be were collected from sandstones and conglomerates exposed at the base of 3.5- to 6-m-high road cuts to minimize modern cosmogenic nuclide production that could skew paleo-erosion rate estimates. Our approach to estimating paleo-erosion rates in foreland sediments is similar to that originally described by Charreau et al. (37) and also applied in more recent studies (38, 39). Measured ^{10}Be represents the sum of at least three components:

$$C_{paleo} = C_{measured} - C_{burial} - C_{exhumation}. \quad [1]$$

C_{paleo} is the concentration of ^{10}Be accumulated during exposure in the paleo watershed and is the quantity used to estimate paleo-erosion rates. C_{burial} is the amount of ^{10}Be acquired during deposition of the sample and is estimated using the sedimentation rate. $C_{exhumation}$ is the amount of ^{10}Be in the sample acquired during recent exhumation. At Huaco it is estimated using long-term exhumation rates derived from apatite U-Th/He dating (29). At Toro Negro it is estimated from exhumation rates computed from the concentration of muogenic ^{36}Cl in K-rich feldspars. The rationale for this approach is that because sandstone samples have depositional ages >5.15 Ma and ^{36}Cl has a half-life of ∼0.3 Ma, all cosmogenic ^{36}Cl originally created during paleoexposure or burial has decayed away. Likewise, because the samples were collected from shielded depths, modern spallation production is negligible and any measured ^{36}Cl is derived from either radiogenic or muogenic production.

A major assumption of our paleo-erosion approach is that the extent of the watershed and the distribution of quartz-bearing lithologies have not changed significantly over time. This assumption is tested by using U-Pb dating of detrital zircons to establish sediment provenance. Detrital zircon provenance of the Toro Negro section was previously summarized in Amidon et al. (40), who inferred no major changes in provenance. At Huaco, 10 detrital zircon samples were analyzed from the Huaco section and compared with four samples from modern rivers (*SI Appendix*, Figs. S3 and S4). Results show that sedimentary provenance did not change significantly during the period of interest and that sediments in the Huaco section were sourced primarily from Precordillera units immediately to the east (Fig. 1).

Results and Discussion

Evidence for Latest Miocene Aridity. The Huaco record shows generally stable erosion rates of ∼0.5 mm/y marked by two ∼70% slowdowns from ∼6.1–5.2 Ma and ∼3.6–3.3 Ma (Fig. 2*I*). These decreases occur rapidly over timescales of 200,000–400,000 y and are followed by returns to the apparent "background" erosion rate. The slowdowns seem to be robust (e.g., not artifacts of the data), given that successive samples show monotonic increases and decreases, and the data show generally good agreement with the coarser resolution results of Val et al. (38). Likewise, because the slowdowns show rapid recovery of erosion rates on 100,000-y timescales, it seems unlikely they can be explained by changes in the rate or pattern of tectonism, as previously suggested (38). More detailed discussion of tectonic scenarios is presented in *SI Appendix*, section S4. Given that erosion rates in semiarid catchments of the central Andes have been shown to depend primarily on precipitation (36), we equate transient slowdowns in erosion rate to periods of enhanced aridity.

Observations from neighboring regions support the interpretation of enhanced aridity. For example, a period of weak soil development and few preserved paleosols in the Iglesia basin (∼31.5° S) records enhanced aridity between 6.2–5.3 Ma (Fig. 2*E*) (41). Likewise, δD isotopic measurements in volcanic glass suggest a period of aridity in the Angastaco basin (∼25.5° S) from ∼6.3–5.1 Ma (Fig. 2*G*) (28, 35). Inferred aridity is broadly synchronous with carbon isotope excursions thought to record the

expansion of C4 vegetation. For example, $\delta^{13}C$ in paleosol carbonate from the Puerta Corral Quemado section (~28° S) and the Angastaco basin become significantly heavier beginning at ~6 Ma (Fig. 2*F*) (14, 27, 28). In the Puerta Corral Quemado section, $\delta^{13}C$ in fossil tooth enamel shows a well-defined, 3‰ excursion from ~6–5.3 Ma, which is synchronous with our observed slowdown in erosion rates, and with the δD excursion. The fossil tooth record is less noisy than the paleosol record and provides strong evidence of a transient shift toward C4 plants from ~6–5.3 Ma (Fig. 2*H*).

The timing of latest Miocene aridity and $\delta^{13}C$ shifts is broadly synchronous with a global glacial period between 6.2 and 5.5 Ma, inferred from a rapid sea-level drawdown (22, 42, 43), $\delta^{18}O$ in the marine sedimentary record (20, 21), and glacial advances in Greenland (23) and South America (24). Recently, evidence from marine records in the Southern Ocean has provided evidence for major advances of the Antarctic ice sheet beginning about 6 Ma (19). This latest Miocene glacial period created particularly cold conditions at high southern latitudes. Alkenone temperature records from marine cores at 40–50° S show ~5 °C drops in sea-surface temperature between about 7 and 5 Ma, with minima at ~5.5 Ma (Fig. 2*D*). Likewise, a Mio-Pliocene core from Deep Sea Drilling Project (DSDP) site 284 (41° S, 167° E) shows major cooling of bottom waters between ~6.2 and 5.2 Ma, as recorded by the relative proportions of *Neogloboquadrina pachyderma* and *Globigerina falconensis*, which prefer colder and warmer conditions, respectively (44, 45). Finally, the northward limit of ice-rafted debris in Ross Sea sediments shows a major advance in the latest Miocene, interpreted as a northward advance of cold conditions associated with a strengthening Antarctic Polar Front (APF) (46).

The timing of the latest Miocene erosion-rate slowdown at Huaco agrees well with nearby continental proxies (Fig. 2 *E–H*), but the continental proxies are not perfectly synchronous with the global benthic $\delta^{18}O$ stack (Fig. 2*C*). Most notably, the recovery of erosion rate, δD, and $\delta^{13}C$ beginning ~5.3 Ma seems to lag recovery of the $\delta^{18}O$ record by ~0.4 Ma. Taken at face value this offset suggests that aridity and/or low CO_2 in the south-central Andes continued for about 0.4 My after global ice volumes had diminished. However, the minima of Southern Hemisphere sea-surface temperature also lags recovery of the global benthic record by ~0.3 Ma, suggesting that the Southern Hemisphere may have stayed cold for longer or reached peak cold ~0.3 Ma after the peak in Northern Hemisphere ice volume.

Evidence for Mid-Pliocene Aridity. The second slowdown in erosion rates from 3.6–3.3 Ma also corresponds to a period of faunal turnover and glaciation. The slowdown corresponds tightly with the M2 isotope excursion, a 0.75‰ dip in the global benthic $\delta^{18}O$ stack from ~3.5–3.2 Ma (Fig. 2*C*), which presumably records a glacial period (47). Evidence for Southern Hemisphere glaciation comes from glacial till in the southern Andes (46° S) dated to ~3.6 Ma (48). Evidence for aridity comes from a major extinction event observed in the Chapadmalal Cliffs of southeastern Argentina (~38° S, 58° W), in which 53% of species go extinct at ~3.3 Ma (49). This extinction has been attributed to the onset of aridity associated with abrupt changes in the global ocean–atmosphere system (49) and alternatively to an impact event at 3.3 Ma (50). Given sedimentary evidence for aridity in the Chapadmalal Cliffs and the observation of roughly synchronous extinctions in the Uquia Formation of northwestern Argentina (51), we favor a climatic explanation.

Again, aridity and glaciation correspond to cold temperatures at high southern latitudes. A cooling event at ~3.5 Ma is recorded by relative variations of diatoms, silicoflagellates, and radiolarian species in the Southern Ocean, which have varying tolerances for warm vs. cold water (45, 52, 53). For example, changes in faunal abundance reveal a major northward advance of the APF between ~3.7 and 3.2 Ma, suggesting colder conditions over Antarctica during this time. Because the APF exists where the cold waters of the Antarctic displace warmer waters northward, its location is a proxy for Antarctic surface temperature: When Antarctica is colder the APF shifts northward then recedes southward as Antarctica warms (Fig. 3*C*). Ocean Drilling Program (ODP) holes 748–751 in the southernmost Indian Ocean (~58° S, 78° E) record a major northward shift of the APF from 3.6 to 3.2 with a maximum of cold conditions from 3.35 to 3.2 Ma (Fig. 2*A*) (54). Nearby DSDP site 266 (~56° S, 110° E) shows peak "Antarctic" cool conditions between 3.7–3.2 Ma (55). In the southwest Atlantic, ODP hole 514 (~46° S, 27° W) records stratigraphic evidence for intensification of Antarctic bottom water flow (e.g., cooling) from ~3.8–3.2 Ma (Fig. 2*B*) (56).

The correlation between high-latitude temperature and erosion rate continues into the mid-Pliocene warm period, when erosion rates rebounded between ~3.3–2.9 Ma as global temperatures warmed (Fig. 2*I*). The APF migrated ~6° southward during this warm period, and sea-surface temperatures were roughly 3–4° warmer than present (57). In the southwest Atlantic, DSDP hole 514 shows a pronounced shift toward warmer species from roughly 3.15–2.95 Ma (56). In the southern Indian Ocean, ODP holes 748–751 also show a swing back to warmer temperatures from roughly 3.15–3.0 Ma (54). It thus seems that as high southern latitudes temporarily warmed during the mid-Pliocene warm period erosion rates at Huaco rebounded, adding further support for a close link between high-latitude colder temperature and aridity in east-central Argentina.

Global-to-Regional Climate Linkages. When combined with continental and marine proxies, our data suggest that when high southern latitudes were colder conditions in the south-central

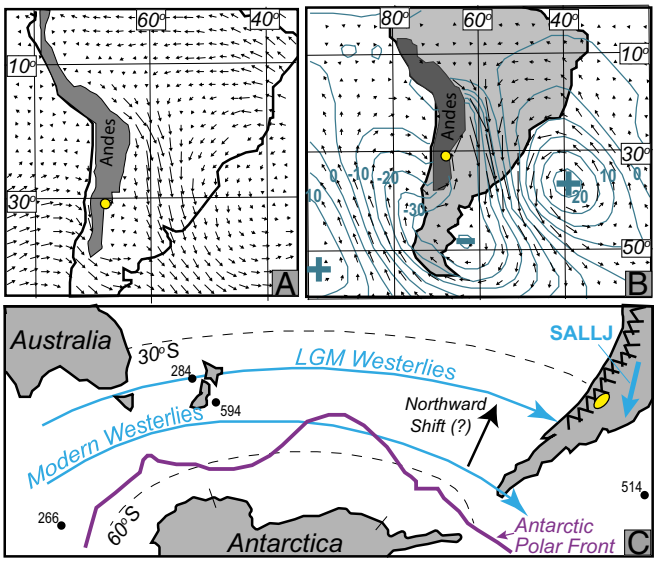


Fig. 3. (*A*) Patterns of low-level winter wind (850 hPa) from 2001 to 2005 based on National Centers for Environmental Prediction reanalysis data, as published in ref. 60. Reproduced from ref. 60. Note the westerlies impinging on the Andes south of 30° S and the easterly trade winds deflected south along the flank of the Andes. Yellow dots denote the Huaco study area in all plots. (*B*) Difference between the mean circulation at 850 hPa and the circulation during a mean Chaco Jet event from 2001 to 2005 as published in ref. 60. Reproduced from ref. 60. Contours of geopotential height anomalies (blue) show the baroclinic wave commonly associated with the Chaco Jet. Note that winds arriving at the study area are primarily southerly and are associated with clockwise circulation around the low pressure system. (*C*) Cartoon showing the hypothetical northward shift of the main westerly flow between modern times and the Last Glacial Maximum.

Andes (25–31° S) were drier. We propose that this aridity represents a breakdown of the atmospheric circulation patterns that bring moisture to the region today. The modern Huaco region receives ∼90% of its moisture in the Austral summer (November–March), primarily from wind patterns associated with the South American Low Level Jet (SALLJ) and associated convective recycling (58, 59). The SALLJ is a southward flow of high-velocity winds originating in the Amazon basin (Fig. 3), following trajectories ranging from due south along the flank of the Andes to southeasterly toward Buenos Aires and eastern Argentina (60). Moisture delivered by the SALLJ is subsequently recycled during intense convective storms as warm air rises upward through colder westerly air masses flowing down off high Andean topography (59, 61). Here we use the term "Chaco Jet" interchangeably with SALLJ, considering it to be a particularly strong episode of the SALLJ with deeper southward penetration (62).

The SALLJ is primarily caused by a trough of low pressure that forms on the eastern side of the Andes in response to the setup of baroclinic wave trains over high topography (63–65). Baroclinic waves are changes in upper atmosphere air pressure (<500 hPa) that emanate from the south Pacific and are magnified as the westerlies interact with the orographic barrier of the Andes (64, 66, 67). The intensity and orientation of the SALLJ are thus sensitive to the strength and latitude of the westerly flow and the height and width of Andean topography where the westerlies cross the Andes (Fig. 3*C*). For example, Wang and Fu (66) showed that the intensity of the SALLJ is predicted by the intensity of westerly flow across the Andes at 700 hPa. Likewise, models suggest that as westerlies shift southward over the lower topography of the southernmost Andes the trajectory of the SALLJ becomes more southerly (64).

Given that SALLJ circulation is responsible for most modern precipitation near Huaco and that the SALLJ is itself dependent on the location and strength of the westerlies, we speculate that the ultimate cause of aridity at Huaco was a change in the westerly circulation and/or associated baroclinic waves during periods of high-latitude cold. Geologic evidence shows that the westerlies shifted equatorward during the Last Glacial Maximum (Fig. 3*C*) and would likely have done the same during periods of late Miocene and Pliocene cold (68, 69). If so, the baroclinic pressure trough should shift northward, changing the geometry and/or intensity of the SALLJ and potentially increasing aridity over the eastern Andes from 25 to 31° S.

Colder temperatures may also have directly diminished some of the processes that give rise to the SALLJ and convective recycling of precipitation. For example, colder temperatures in the troposphere during glacial periods are interpreted to have reduced baroclinicity, which would have made it harder to achieve ideal conditions to set up the SALLJ (70, 71). Likewise, colder temperatures in the austral summer may have reduced the vigor of mesoscale convective processes near the downstream end of the SALLJ, which currently strengthens low-level flow of the SALLJ and enhances moisture delivery (72).

Conclusions

Based on ^{10}Be concentrations in foreland basin sediment, we generate a record of paleo-erosion rates in the south-central Andes of Argentina near 30° S. The Huaco record shows two slowdowns in erosion rates: the latest Miocene (6.1–5.3 Ma) and the middle Pliocene (3.6–3.3 Ma). Both slowdowns are followed by rapid returns to a "background" erosion rate, suggesting they are not driven by tectonic reorganization. Given that erosion rates in semiarid watersheds of the central Andes are governed by precipitation (36), we attribute the periods of slower erosion to periods of enhanced aridity.

Both slowdowns are synchronous with cold temperatures at high southern latitudes, as evidenced from glacial deposits, sea-surface temperatures, faunal assemblages, and benthic δ^{18}O records. We thus suggest that periods of aridity are associated with global glacial periods. The arid glacial periods proposed here are also synchronous with major faunal changes in South America: a shift from C3 to C4 vegetation in the latest Miocene (27) and a small-mammal extinction event in the middle Pliocene (49). Given that C4 plants can only outcompete C3 plants if aridity is also associated with lower atmospheric CO_2 levels, we suggest that the expansion of C4 vegetation may have been driven by low CO_2 rather than a shift toward warmer and wetter growing seasons. Regardless of the exact cause, it is clear that global climate is a first-order control on aridity and ecology in the south-central Andes and that large parts of the region experienced synchronous changes over time.

We speculate that aridity during glacial periods was caused by circulation changes in the SALLJ, which is responsible for delivering much of the modern moisture to the study area. Because the SALLJ depends on a baroclinic trough generated by interaction of westerly air masses with Andean topography, we hypothesize that a northward shift of the westerlies during glacial periods changed the location of this trough, thereby altering the geometry and/or intensity of the SALLJ.

Finally, our data show that paleo-erosion rates can be a powerful proxy for ancient continental climate, especially when ^{10}Be and ^{36}Cl are combined to reduce uncertainties associated with modern exposure. The inherent spatial and temporal averaging of sediment in a watershed yields a robust spatially averaged signal that can place tight temporal constraints on periods of continental aridity. When combined with appropriate global and regional climate proxies paleo-erosion rates may thus provide more precise information about the timing and amplitude of extreme climatic events that might otherwise be averaged out by marine-based proxies. In this case, our data pinpoint the timing of peak aridity to ∼5.4 and 3.3 Ma, events that likely had a profound impact on the faunal evolution of the south-central Andes and neighboring regions.

ACKNOWLEDGMENTS. We thank Louise McCarren for field assistance and Ryan McElroy, C. C. Connard, and Andrew Hollyday for laboratory assistance. This work was supported by the NSF Geomorphology and Land Use Dynamics program through NSF Grant EAR-1148233.

1. Hewitt GM (2004) Genetic consequences of climatic oscillations in the Quaternary. *Philos Trans R Soc Lond B Biol Sci* 359:183–195, discussion 195.
2. deMenocal PB (2004) African climate change and faunal evolution during the Pliocene-Pleistocene. *Earth Planet Sci Lett* 220:3–24.
3. Zachos J, Pagani M, Sloan L, Thomas E, Billups K (2001) Trends, rhythms, and aberrations in global climate 65 Ma to present. *Science* 292:686–693.
4. Arakaki M, et al. (2011) Contemporaneous and recent radiations of the world's major succulent plant lineages. *Proc Natl Acad Sci USA* 108:8379–8384.
5. Dupont LM, Rommerskirchen F, Mollenhauer G, Schefuß E (2013) Miocene to Pliocene changes in South African hydrology and vegetation in relation to the expansion of C4 plants. *Earth Planet Sci Lett* 375:408–417.
6. Cerling TE, Wang Y, Quade J (1993) Expansion of C4 ecosystems as an indicator of global ecological change in the late Miocene. *Nature* 361:344–345.
7. Quade J, Cerling TE (1995) Expansion of C4 grasses in the Late Miocene of Northern Pakistan: Evidence from stable isotopes in paleosols. *Palaeogeogr Palaeoclimatol Palaeoecol* 115:91–116.
8. Ehleringer JR, Cerling TE, Helliker BR (1997) C4 photosynthesis, atmospheric CO_2, and climate. *Oecologia* 112:285–299.
9. Still CJ, Berry JA, Collatz GJ, DeFries RS (2003) Global distribution of C3 and C4 vegetation: Carbon cycle implications. *Global Biogeochem Cycles* 17: 6-1–6-14.
10. Pagani M, Freeman KH, Arthur MA (1999) Late miocene atmospheric $CO_{(2)}$ concentrations and the expansion of C(4) grasses. *Science* 285:876–879.
11. Freeman KH, Colarusso LA (2001) Molecular and isotopic records of C4 grassland expansion in the late miocene. *Geochim Cosmochim Acta* 65:1439–1454.
12. Bolton CT, Stoll HM (2013) Late Miocene threshold response of marine algae to carbon dioxide limitation. *Nature* 500:558–562.
13. Bolton CT, et al. (2016) Decrease in coccolithophore calcification and CO_2 since the middle Miocene. *Nat Commun* 7:10284.
14. Latorre C, Quade J, McIntosh WC (1997) The expansion of C4 grasses and global change in the late Miocene; stable isotope evidence from the Americas. *Earth Planet Sci Lett* 146:83–96.

15. Quade J, Cater JML, Ojha TP, Adam J, Harrison TM (1995) Late Miocene environmental change in Nepal and the northern Indian subcontinent: Stable isotopic evidence from paleosols. *Geol Soc Am Bull* 107:1381–1397.
16. Cosentino D, et al. (2013) Refining the Mediterranean "Messinian gap" with high-precision U-Pb zircon geochronology, central and northern Italy. *Geology* 41:323–326.
17. Pérez-Asensio JN, Aguirre J, Jiménez-Moreno G, Schmiedl G, Civis J (2013) Glacioeustatic control on the origin and cessation of the Messinian salinity crisis. *Global Planet Change* 111:1–8.
18. Garcia-Castellanos D, Villaseñor A (2011) Messinian salinity crisis regulated by competing tectonics and erosion at the Gibraltar arc. *Nature* 480:359–363.
19. Ohneiser C, et al. (2015) Antarctic glacio-eustatic contributions to late Miocene Mediterranean desiccation and reflooding. *Nat Commun* 6:8765.
20. Hilgen F, Kuiper K, Krijgsman W, Snel E, van der Laan E (2007) Astronomical tuning as the basis for high resolution chronostratigraphy: The intricate history of the Messinian Salinity Crisis. *Stratigraphy* 4:231–238.
21. Hodell DA, Curtis JH, Sierro FJ, Raymo ME (2001) Correlation of late Miocene to early Pliocene sequences between the Mediterranean and North Atlantic. *Paleoceanography* 16:164–178.
22. Hodell DA, Elmstrom KM, Kennett JP (1986) Latest miocene benthic $\delta^{18}O$ changes, global ice volume, sea level and the 'Messinian salinity crisis'. *Nature* 320:411–414.
23. Larsen HC, et al.; ODP Leg 152 Scientific Party (1994) Seven million years of glaciation in greenland. *Science* 264:952–955.
24. Mercer JH (1983) Cenozoic glaciation in the Southern Hemisphere. *Annu Rev Earth Planet Sci* 11:99–132.
25. Passey BH, Cerling TE (2006) In situ stable isotope analysis ($\delta^{13}C$, $\delta^{18}O$) of very small teeth using laser ablation GC/IRMS. *Chem Geol* 235:238–249.
26. Uno KT, Polissar PJ, Jackson KE, deMenocal PB (2016) Neogene biomarker record of vegetation change in eastern Africa. *Proc Natl Acad Sci USA* 113:6355–6363.
27. Hynek SA, et al. (2012) Small mammal carbon isotope ecology across the Miocene-Pliocene boundary, northwestern Argentina. *Earth Planet Sci Lett* 321-322:177–188.
28. Bywater-Reyes S, Carrapa B, Clementz M, Schoenbohm L (2010) Effect of late Cenozoic aridification on sedimentation in the Eastern Cordillera of northwest Argentina (Angastaco Basin). *Geology (Boulder)* 38:235–238.
29. Fosdick JC, Carrapa B, Ortíz G (2015) Faulting and erosion in the Argentine Precordillera during changes in subduction regime: Reconciling bedrock cooling and detrital records. *Earth Planet Sci Lett* 432:73–83.
30. Jordan TE, et al. (2001) Unsteady and spatially variable evolution of the Neogene Andean Bermejo foreland basin, Argentina. *J S Am Earth Sci* 14:775–798.
31. Jordan TE, Allmendinger RW, Damanti JF, Drake RE (1993) Chronology of motion in a complete thrust belt: The Precordillera, 30-31 degrees S, Andes Mountains. *J Geol* 101: 135–156.
32. Coughlin TJ, O'Sullivan PB, Kohn BP, Holcombe RJ (1988) Apatite fission-track thermochronology of the Sierras Pampeanas, central western Argentina: Implications for the mechanism of plateau uplift in the Andes. *Geology* 26:999–1002.
33. Strecker MR, et al. (2007) Tectonics and climate of the southern central Andes. *Annu Rev Earth Planet Sci* 35:747–787.
34. Pingel H, et al. (2014) Pliocene orographic barrier uplift in the southern Central Andes. *Geology* 42:691–694.
35. Rohrmann A, et al. (2016) Miocene orographic uplift forces rapid hydrological change in the southern central Andes. *Sci Rep* 6:35678.
36. Bookhagen B, Strecker MR (2012) Spatiotemporal trends in erosion rates across a pronounced rainfall gradient: Examples from the southern Central Andes. *Earth Planet Sci Lett* 327-328:97–110.
37. Charreau J, et al. (2011) Paleo-erosion rates in Central Asia since 9Ma: A transient increase at the onset of Quaternary glaciations? *Earth Planet Sci Lett* 304:85–92.
38. Val P, Hoke GD, Fosdick JC, Wittmann H (2016) Reconciling tectonic shortening, sedimentation and spatial patterns of erosion from ^{10}Be paleo-erosion rates in the Argentine Precordillera. *Earth Planet Sci Lett* 450:173–185.
39. Puchol N, et al. (2016) Limited impact of Quaternary glaciations on denudation rates in Central Asia. *Geol Soc Am Bull* 129:479–499.
40. Amidon WH, et al. (2016) U-Pb ages of detrital and volcanic zircons of the Toro Negro Formation, northwestern Argentina: Age, provenance and sedimentation rates. *J S Am Earth Sci* 70:237–250.
41. Ruskin BG, Jordan TE (2007) Climate change across continental sequence boundaries: Paleopedology and lithofacies of Iglesia Basin, northwestern Argentina. *J Sediment Res* 77:661–679.
42. Peck DM, Missimer TM, Slater DH, Wise SW, Jr, O'Donnell TH (1979) Late Miocene glacial-eustatic lowering of sea level: Evidence from the Tamiami Formation of South Florida. *Geology (Boulder)* 7:285–288.
43. Loutit TS, Keigwin LD, Jr (1982) Stable isotopic evidence for latest Miocene sea-level fall in the Mediterranean region. *Nature* 300:163–166.
44. Loutit TS, Kennett JP (1979) Application of carbon isotope stratigraphy to late miocene shallow marine sediments, new zealand. *Science* 204:1196–1199.
45. Shackleton NJ, Kennett JP (1975) Late Cenozoic oxygen and carbon isotopic changes at DSDP Site 284: Implications for glacial history of the Northern Hemisphere and Antarctica. *Initial Rep Deep Sea Drill Proj* 29:801–807.
46. Hayes DE, Frakes LA (1975) General synthesis, Deep Sea Drilling Project Leg 28. *Initial Rep Deep Sea Drill Proj* 28:919–942.
47. Lisiecki LE, Raymo ME (2005) A Pliocene-Pleistocene stack of 57 globally disturbed benthic $\delta^{18}O$ records. *Paleoceanography* 20:PA1003.
48. Mercer JH (1976) Glacial history of southernmost South America. *Quat Res* 6:125–166.
49. Vizcaino SF, Farina RA, Zarate MA, Bargo MS, Schultz P (2004) Palaeoecological implications of the mid-Pliocene faunal turnover in the Pampean region (Argentina). *Palaeogeogr Palaeoclimatol Palaeoecol* 213:101–113.
50. Schultz PH, Zarate M, Hames W, Camilion C, King J (1998) A 3.3-Ma impact in argentina and possible consequences. *Science* 282:2061–2063.
51. Ortiz PE, et al. (2012) Exceptional late Pliocene microvertebrate diversity in northwestern Argentina reveals a marked small mammal turnover. *Palaeogeogr Palaeoclimatol Palaeoecol* 361-362:21–37.
52. Hodell DA, Venz K (1992) Toward a high-resolution stable isotopic record of the Southern Ocean during the Pliocene-Pleistocene (4.8 to 0.8 Ma). *The Antarctic Paleoenvironment: A Perspective on Global Change: Part One*, eds Kennett JP, Warkne DA (American Geophysical Union, Washington, DC), Vol 56, pp 265–310.
53. Kennett JP, Hodell DA (1993) Evidence for relative climatic stability of Antarctica during the early Pliocene: A marine perspective. *Geogr Ann, Ser A* 75:205–220.
54. Bohaty SM, Harwood DM (1998) Southern Ocean Pliocene paleotemperature variation from high-resolution silicoflagellate biostratigraphy. *Mar Micropaleontol* 33: 241–272.
55. Keany J (1978) Paleoclimatic trends in early and middle Pliocene deep-sea sediments of the Antarctic. *Mar Micropaleontol* 3:35–49.
56. Ciesielski PF, Grinstead GP (1986) Pliocene variations in the position of the Antarctic convergence in the Southwest Atlantic. *Paleoceanography* 1:197–232.
57. Barron JA (1996) Diatom constraints on the position of the Antarctic Polar Front in the middle part of the Pliocene. *Mar Micropaleontol* 27:195–213.
58. Garreaud RD, Vuille M, Compagnucci R, Marengo J (2009) Present-day South American climate. *Palaeogeogr Palaeoclimatol Palaeoecol* 281:180–195.
59. Romatschke U, Houze RA, Jr (2010) Extreme summer convection in South America. *J Clim* 23:3761–3791.
60. Castañeda ME, Ulke AG (2015) Analysis of atmospheric conditions associated to CHACO events of the Low Level Jet East of the Andes and their implications for regional transport. *Int J Climatol* 35:4126–4138.
61. Rohrmann A, et al. (2014) Can stable isotopes ride out the storms? The role of convection for water isotopes in models, records, and paleoaltimetry studies in the central Andes. *Earth Planet Sci Lett* 407:187–195.
62. Nicolini M, Saulo AC (2006) Modeled Chaco low-level jets and related precipitation patterns during the 1997-1998 warm season. *Meteorol Atmos Phys* 94:129–143.
63. Salio P, Nicolini M, Saulo AC (2002) Chaco low-level jet events characterization during the austral summer season. *J Geophys Res, D, Atmospheres* 107:ACL 32-1–ACL 32-17.
64. Campetella CM, Vera CS (2002) The influence of the Andes mountains on the South American low-level flow. *Geophys Res Lett* 29:7-1–7-4.
65. Marengo JA (2004) Interdecadal variability and trends of rainfall across the Amazon basin. *Theor Appl Climatol* 78:79–96.
66. Wang H, Fu R (2004) Influence of cross-Andes flow on the South American low-level jet. *J Clim* 17:1247–1262.
67. Marengo JA, Soares WR, Saulo C, Nicolini M (2004) Climatology of the low-level jet east of the Andes as derived from the NCEP-NCAR reanalyses: Characteristics and temporal variability. *J Clim* 17:2261–2280.
68. Moreno PI, Lowell TV, Jacobson GL, Jr, Denton GH (1999) Abrupt vegetation and climate changes during the Last Glacial Maximum and last termination in the Chilean Lake District: A case study from Canal de la Puntilla (41°S). *Geogr Ann, Ser A* 81: 285–311.
69. Stuut JBW, Lamy F (2004) Climate variability at the southern boundaries of the Namib (southwestern Africa) and Atacama (northern Chile) coastal deserts during the last 120,000 yr. *Quat Res* 62:301–309.
70. Toggweiler JR, Russell JL, Carson SR (2006) Midlatitude westerlies, atmospheric CO_2, and climate change during the ice ages. *Paleoceanography* 21, 10.1029/2005PA001154.
71. Thompson DWJ, Solomon S (2002) Interpretation of recent Southern Hemisphere climate change. *Science* 296:895–899.
72. Saulo C, Ruiz J, García Skabar Y (2007) Synergism between the low-level jet and organized convection at its exit region. *Mon Weather Rev* 135:1310–1326.
73. Goss AR, Kay SM, Mpodozis C (2013) Andean adakite-like high-Mg andesites on the northern margin of the Chilean-Pampean flat-slab (27–28.5° S) associated with frontal arc migration and fore-arc subduction erosion. *J Petrol* 54:2193–2234.
74. Kay SM, Mpodozis C, Gardeweg M (2013) Magma sources and tectonic setting of Central Andean andesites (25.5°–28° S) related to crustal thickening, forearc subduction erosion and delamination. Special Publication (Geological Society of London, London), Vol 385, pp 303–334.
75. Allmendinger RW, Judge PA (2014) The Argentine Precordillera: A foreland thrust belt proximal to the subducted plate. *Geosphere* 10:1203–1218.
76. Mercer JH, Sutter JF (1982) Late Miocene-earliest Pliocene glaciation in southern Argentina: Implications for global ice-sheet history. *Palaeogeogr Palaeoclimatol Palaeoecol* 38:185–206.

Supplementary Appendix: text, figures, and tables

Mio-Pliocene aridity in the south-central Andes associated with Southern Hemisphere cold periods

S1. Geologic Overview

S1.1 Regional Geologic Evolution

The Cenozoic history of the eastern Andes is dominated by non-collisional orogenesis and volcanism associated with subduction off the west coast of Chile. As deformation in the Andean orogeny intensified about 19 Ma, significant shortening was accommodated in the Frontal Cordillera and Precordillera, causing sediments to be shed eastward into the Bermejo Basin, which extended 100s of km from north to south. By 10 Ma, flat-slab subduction began affecting the region, extinguishing volcanism, and eventually triggering uplift of Pampean ranges, such as the Valle Fértil, Umango, Espinal, and Maz ranges (1, 2). Uplift of Pampean blocks fragmented the Bermejo foreland by blocking existing sediment routing patterns and creating isolated basins (3). Over the past 2.5 Myr, a renewed phase of shortening has affected the region, causing inversion of foreland-basin sequences, such as at Huaco and Toro Negro, whose youngest sediments are roughly 2.5 Myr old (e.g. 4).

S1.2 The Huaco section

The Huaco depocenter is located near the center of the Bermejo basin at ~31°S and is well exposed in the Las Salinas anticline just east of the village of Huaco (Fig. 1). The section spans ~ 14-2.4 Ma, and is interpreted to have been deposited in a desert environment due to abundant gypsum mineralization, ubiquitous oxidation, and the near complete lack of plant and animal fossils (5). Our work focuses on the Río Jáchal Formation, which was deposited by ephemeral braided streams in an alluvial fan setting with a coarsening upward trend. The lower member (~940 m thick) is characterized by medium to thickly bedded sandstones interbedded with blue-green shale and siltstone. The upper member (~1435 m thick) marks the onset of abundant pebble conglomerates interbedded with coarse sandstones.

The Huaco section contains several stratigraphic transitions that are interpreted to record regional tectonic events. The most significant shift occurs around 10 Ma, when the sedimentation rate increases from ~0.2 to 0.9 mm/yr, and by 9.6 Ma, limestone clasts and a new

suite of heavy minerals appear for the first time (5, 6). The limestone clasts are likely sourced from San Luis Formation exposed in the San Roque Thrust Sheet 25 km to the west (Fig. 1B), and suggest renewed thrust activity at or before ~10 Ma (7).

Another important transition occurs at ~5.95 Ma, when extensive conglomerates first appear at the base of the upper Río Jáchal Formation, and the sedimentation rate begins to increase again from ~ 0.35 to ~0.6 mm/yr (6). This transition has traditionally been interpreted as the onset of uplift along the Niquivil fault 10-15 km to the west (7) (Fig. 1B). Shortening on the Niquivil fault continued throughout Plio-Quaternary times, as evidenced by a modern fault scarp cutting a late Quaternary deposit (7).

Finally, the overlying Mogna gravels are interpreted to record the early stages of growth on the Huaco anticline along a west-vergent blind thrust sometime after 2.6 Ma (7, 8). The Mogna gravels are distinguished by their very coarse and rounded conglomerates with a lower percentage of limestone clasts and are interpreted as growth strata deposited on the backlimb of the Huaco anticline (8, 9). Apatite U-Th/He cooling ages of ~ 2.5 Myr along the crest of the Las Salinas anticline (Fig. 1B) support this model and suggest that anticline growth has proceeded rapidly since 2.6 Ma, with vertical exhumation rates of ~ 2.4 mm/yr (10).

S1.2 The Toro Negro Section

The Toro Negro Formation was deposited in the Vinchina Depocenter at the northern end of the Bermejo basin near 28.5° (Fig. 1). It is ~2500 m thick, deposited between ~6.9 and 2.3 Ma, and composed of yellow-green fluvial sandstones with sparse conglomerate and shale intervals (11, 12). The fluvial facies alternate between semi-arid anastomosed channels and wider braidplains, and comprise dominantly feldspathic litharenites and litharenites, typically cemented by zeolites, sulfates, and carbonate cement (12, 13).

Regional tectonic evolution during deposition of the Toro Negro Formation remains unclear. The formation is deposited atop an erosional surface with high relief (up to 1000 m) in the uppermost Vinchina Formation, which likely signals a period of regional tectonic compression sometime between 9.3 and 6.9 Ma (11, 14). Accumulation of the Toro Negro Formation appears to have been characterized by large variations in sedimentation rate, but only

minor differences in provenance. The sequence directly overlying the basal unconformity (~ 6.95 to 5.7 Ma) was characterized by very fast sedimentation rates (~1.2 mm/yr), which then slowed dramatically to ~0.3 mm/yr after ~6 Ma. Sandstone petrography and detrital zircon signatures throughout the Toro Negro section show relatively little variation, suggesting that no major provenance change occurred during deposition of the unit (11, 13). At some point after ~2.3 Ma, tectonic compression was initiated, causing uplift and erosion of the Toro Negro Formation and ultimately deposition of the overlying gravels of the El Corral Formation (13).

S2. Methods

S2.1 Magnetostratigraphy

Both sections were measured using a 1.5-m Jacob staff, such that cosmogenic nuclide , paleomagnetic, and detrital samples were precisely located in the section. Paleomagnetic samples were collected from mudstones and siltstones along the Huaco section, but were not collected along the Toro Negro section due to the paucity of fine grained sediments. Instead, chronology of the Toro Negro section is based on U-Pb dating of seven tephra layers (15). The motivation for repeating the existing magnetostratigraphy at Huaco was to better constrain the age of cosmogenic nuclide samples from fresh roadcuts approximately 8-12 km north along strike from the original sites of Johnson et al. (1986). Due to the high uncertainty in paleo-erosion rate studies commonly associated with sample ages and post-burial exposure, sampling fresh, deep roadcuts coupled with high-resolution chronologies greatly reduces uncertainty in the erosion-rate estimates.

Approximately 84 oriented block samples were excavated at a stratigraphic spacing of ~25 m, drilled and cut using air and nonmagnetic bits and blades, and then measured at the UT Austin Paleomagnetic Laboratory. Remanent magnetizations for all samples were measured using a 2G Enterprises DC SQuID, three-axis cryogenic magnetometer housed in a magnetically shielded room and equipped with computer-controlled alternating field (AF) demagnetization coils and automated sample changer system. Background noise for the magnetometer is <1 pAm^2, and all thermal demagnetization steps were performed within a magnetically shielded ASC^{TM} oven.

Of the 84 collected sites, 9 were destroyed during transport and/or preparation for a yield of 75 accepted sites. Duplicate samples were analyzed for 1/3 of the sites yielding a total of 100

specimens from 75 sites. All specimens were first analyzed for natural remanent magnetization (NRM) and then subjected to one AF demagnetization at 7.5 mT in order to remove low-coercivity magnetizations. Lastly, stepwise thermal demagnetization was performed on the specimens in 50 °C steps from 100 °C to 500 °C and then in 20-50 °C steps up to ~650 °C.

Results from stepwise thermal demagnetization indicate a very consistent progression toward the origin throughout the thermal steps, with this singular component interpreted to represent the characteristic remanent magnetization (ChRM) at the time of deposition (Fig. S1. Due to the consistent nature of the thermal progression up to 650 °C and the cusp in intensity at ~580°C, the dominant magnetic carrier in our samples appears to be a mixture of hematite and magnetite.

ChRM directions for each specimen were determined using PaleoMag 3.1.0 b2 software (16). For each specimen, between 4 and 13 points (including the origin) were used to delimit magnetic vectors using principle component analysis (17) implemented within the software. We note that, in all cases, the origin was used to help constrain ChRM direction, because the demagnetization paths chosen always headed towards the origin. Of the 100 specimens analyzed, none were discarded due to incoherent orthogonal demagnetization diagrams, and all maximum angular deviations (MAD) were <9°. Virtual geomagnetic poles (VGPs) were calculated from the resulting tilt-corrected specimens and their latitudes used to delineate periods of normal and reversed polarity (Fig. 3). Although a formal fold test was not performed, the robust antipodal behavior and tight clustering of the reversed and normal polarity samples following tilt-correction (Fig. S1) argues for a well-behaved system and is in good agreement with previous findings in the section (5).

The magnetic polarity stratigraphy records 8 normal and 7 reversed magnetozones, which are roughly anchored by the chronostratigraphy of Johnson et al., (1986) (Fig. 2; Table S1). The correlation shows that the four normal periods in the Gilbert chron are well defined, as are the normal periods at the beginning of the Gauss and end of Chron 5. It appears that our magnetostratigraphy misses either a short normal or reversed period near the middle of the Gauss chron. We interpret this mismatch as the absence of the younger reversed subchron in the Gauss chron (3.116-3.032 Ma), because this interpretation yields sedimentation rates that vary least over time and match most closely with those published by Johnson et al. (1986) which captured both reversals. Adopting this correlation and using the magnetic timescale of Lourens et al.,

(2004), our cosmogenic nuclide samples span a time period from roughly 6.4 to 2.8 Ma, excluding samples collected from the overlying Mogna gravel, which have an unknown age (Table 1).

S2.2 Cosmogenic ^{10}Be in detrital quartz

Samples for cosmogenic nuclide analysis were collected from sandstones and gravels exposed in road cuts at least 3.5-m high and commonly 5- to 6-m high. Bulk material was exhumed, hand crushed, and immediately passed through a 500-1000 um sieve. Quartz preparation was undertaken at Middlebury College where samples were wet sieved to 300-1000 um and passed through a Frantz magnetic separator prior to acid etching. An initial 1 M HCl etch was performed for 12-18 hours, followed by three etchings in 2% 2:1 HF:HNO_3. Purity checks were performed by ICP-OES analysis at Middlebury College using commercially available standard solutions. Samples that showed high Fe or Al, or exhibited correlation of Na or K with Al were passed through an additional HCL etch and 1-3 additional HF:HNO_3 etches until their bulk chemistry was consistent with other samples from the section.

Quartz dissolution and target preparation was performed at UC Santa Barbara following publicly available protocols (http://www.geog.ucsb.edu/~bodo/pdf/bookhagen_chemSeparation_UCSB.pdf). Quartz was digested using concentrated hydrofluoric acid with addition of a low-ratio ^{9}Be spike with a ^{10}Be/^{9}Be ratio value of 3 x 10^{-15}. ^{10}Be was then extracted by ion exchange chromatography, targets were packed at UCSB and measured via accelerator mass spectrometry at the Purdue Rare Isotope Measurement Laboratory (PRIME) using the 07KNSTD standardization (18).

Measured ^{10}Be ($C_{measured}$) in modern samples ranges from 2.3 to 180 kat/g with laboratory blank corrections in the range of 0.1-10% and 1σ uncertainties ranging from 1 to 8% (Table 1). As described in several recent studies (19, 20), measured ^{10}Be represents the sum of at least three ^{10}Be components:

$$C_{measured} = C_{paleo} + C_{burial} + C_{exhumation} \quad \text{(S1)}$$

Our goal is to estimate C_{paleo}, which is the amount of remaining ^{10}Be in the sample that was originally produced during exposure on the hillslope and in transit. As formulated in equation S1, C_{paleo} has not yet been corrected for decay since the time of deposition. C_{burial} is the amount of remaining ^{10}Be that was originally accumulated during the burial process. $C_{exhumation}$ is the amount of ^{10}Be recently accumulated during exhumation of the sample. Because our samples are collected from roadcuts below the depth (>3 m) of significant spallation production, $C_{exhumation}$ can be overwhelmingly attributed to muogenic production, although the spallation component is also estimated. Our shallowest sample was collected at 3 meters depth (TN-7). The spallation component acquired during recent exhumation is estimated to be 31.4 at/g, which is 8% of the estimated muogenic signal and 0.006 % of the measured signal. Three samples were collected at a depth of 3.5 m. At 3.5 m the estimated spallation component drops to ~6.6 at/g, which is roughly 2 % of the muogenic component and roughly 0.2% of the measured concentration, even for samples Hu-1.1 and Hu-1, which are both > 6 Ma in age. Once C_{paleo} is estimated it is then corrected for the decay of ^{10}Be since the time of deposition yielding, $C_{paleo\text{-}c}$, which is the concentration in the sample at the moment it was buried.

As formulated in Charreau et al. (2011), the exposure during burial can be estimated as:

$$C_{burial} = \sum_{i,j,k} \left[\frac{P_{burial_{i,j,k}}}{\lambda + (\rho_{burial} * A_r)/\Lambda_{i,j,k}} \right] \qquad \text{(S2)}$$

Where P_{burial} denotes the production rate at the site of burial for spallation, slow, and fast muons (i,j, and k respectively). λ denotes the decay constant of ^{10}Be, ρ_{burial} denotes the density of freshly deposited sediment, A_r denotes the accumulation rate of sediment, and Λ denotes the attenuation length for spallation, slow, and fast muons (i,j, and k respectively).

The nuclide concentration acquired during exhumation can be described as:

$$C_{exhumation} = \sum_{i,j,k} \left[\frac{P_{outcrop_{i,j,k}} * e^{\frac{-\rho_{ss} * z_{sample}}{\Lambda_{i,j,k}}}}{\lambda + (\rho_{ss} * E_{exhum})/\Lambda_{i,j,k}} \right] \qquad \text{(S3)}$$

Where $P_{outcrop}$ denotes the production rate at the outcrop for spallation, slow, and fast muons (i,j, and k respectively), ρ_{ss} denotes the density of the sandstone, Z_{sample} denotes the modern depth from which the sample was collected, and E_{exhum} denotes the exhumation rate of the sample towards the surface.

C_{paleo} is then estimated by solving equation S1 and C_{paleo} is corrected for decay over the burial period to estimate $C_{paleo\text{-}c}$, which is the concentration of ^{10}Be in the sample just before burial.

$$C_{paleo-c} = C_{paleo}/e^{-\lambda t} \quad \text{(S4)}$$

Where t is the depositional age of the sample inferred from maegnetostratigraphy or tephra chronology.

The resulting quantity $C_{paleo\text{-}c}$ is then combined with the production rate of ^{10}Be in the paleo-watershed (P_{wshed}) to estimate erosion rate in the paleo-watershed (E_{wshed}) by solving equation S5 (21):

$$C_{paleo-c} = \frac{P_{wshed}}{E_{wshed}*\rho*\Lambda^{-1}+\lambda} \quad \text{(S5)}$$

We estimate C_{burial}, $C_{exhumation}$, and P_{wshed} using parameters extracted from the Cronus Matlab Code, such as the time-integrated scaling factors and surficial production rates for spallation, slow muons, and fast muons over the last 25 kyr. The arbitrary 25-kyr timescale used in all calculations is intended to generate scaling factors which are reasonable for production/burial at times in the ancient past when no proxies for magnetic scaling are available. In all cases, we have used the LSD nuclide-specific scaling scheme ('sa') along with an air pressure based on the ERA40 model (22). We adopt attenuation lengths of 160, 1500, and 5300 g cm^{-2} for spallation, slow, and fast muons in all cases (23, 24). Resultant scaling factors are 1.25 and 1.21 for spallation and muons at the Huaco site and 3.13 and 1.75, respectively, at the Toro Negro site. These scaling factors yield ^{10}Be production rates of 5.09, 0.051, and 0.029 at g^{-1} yr^{-1} for spallation, slow muons, and fast muons at the Huaco site and 12.80, 0.074, and 0.035 at g^{-1} yr^{-1} , respectively, at the Toro Negro site.

To estimate C_{burial}, we use the undecompacted sediment accumulation rates computed from either paleomagenetic chronology (Huaco) or tephrachronology (Toro Negro), such that each sample is assigned the mean rate over the time interval into which it falls. Application of the parameters described above yields ^{10}Be concentrations of 0.6 to 30 kat/g due to cosmogenic nuclide production during burial, which would represent 2-19% of measured ^{10}Be in the modern sample (Table 1).

To estimate $C_{exhumation}$ at the Huaco site, we adopt a long-term exhumation rate of 3.1 mm/yr at the crest of the anticline based on the apatite U-Th/He ages of Fosdick et al, 2015. Following Val et al. (2016), we assume this exhumation rate tapers linearly to zero away from the crest of the anticline, giving site-specific rates from 2.4 to 0.35 mm/yr. At Toro Negro, we adopt the lowest exhumation rate (2.3 mm/yr) given by the ^{36}Cl calculations described below. These exhumation rates yield $C_{exhumation}$ values between 0.2 and 1.9 kat/g at Huaco and 0.2 to 0.4 kat/g at Toro Negro, or roughly 3-12% of the measured ^{10}Be in modern samples (Table 1).

To estimate paleo-erosion rates (E_{wshed}), we compute mean watershed elevations using a 90-m SRTM DEM: 3600 m for the Río Jáchal watershed (Huaco) and 3325 m for the Río Jaguel watershed (Toro Negro). These elevations yield scaling factors of 7.82 and 6.66, respectively. Assuming a single elevation value to represent cosmogenic nuclide growth in a watershed over 5 Myr make several implicit assumptions whose uncertainty is difficult to quantify. Recognizing this, we compute uncertainties on paleo-erosion rates using a Monte Carlo simulation with 5000 random iterations. 1σ uncertainties were assumed to be 20% on C_{burial}, $C_{exhumation}$, and the paleo-watershed scaling factor. 1-sigma uncertainties on depositional age were assumed to be 0.1 Ma for Huaco where paleomagentic age control is tight and 0.2 Ma for Toro Negro where interpolated tephra ages leave more uncertainty. These simulations yield 1-sigma uncertainties on paleo-erosion rate (E_{wshed}) of 20 -30% for Huaco and 25-50% for Toro Negro.

S2.3 Cosmogenic ^{36}Cl analysis and bulk geochemistry

^{36}Cl concentrations were measured on feldspar separates at four sites from the Toro Negro section in an effort to quantify the muon-produced component and use it to estimate exhumation rates (Table S2). The rationale for this approach is that because these samples have depositional ages >5.15 Ma and ^{36}Cl has a half-life of ~0.3 Ma, all cosmogenic ^{36}Cl created by paleo-exposure or burial has decayed away. Likewise, because the samples were collected from

below the depth of most modern spallation production (4-20 m depth), any measured ^{36}Cl ($^{36}Cl_{measured}$) is dominated by radiogenic ($^{36}Cl_{radiogenic}$) and muogenic ($^{36}Cl_{muogenic}$) production. Spallogenic ^{36}Cl can be neglected because it is minor relative to the ~20% uncertainty on the radiogenic component. For example, at a depth of 4 meters we estimate spallogenic production rates of 0.013 at g^{-1} yr^{-1} for sample V15, which is ~ 6% of the muogenic production rate at the same depth. Considering integrated production below 4 m, spallation is only 1.6% of muogenic production, and only 0.03% of radiogenic ^{36}Cl for an assumed erosion rate of 2 mm/yr.

If the muogenic component can be isolated, it can then be used to estimate recent exhumation rates over the depth of muon production, which is roughly 25 m before production drops to less than 10% of the surface production rate. This relationship can be summarized as:

$$^{36}Cl_{measured} = {}^{36}Cl_{radiogenic} + {}^{36}Cl_{muogenic} \quad \text{(S6)}$$

Potassium-rich feldspar was deliberately target because of the relatively high production rate of ^{36}Cl from ^{39}K (25). Potassium-rich feldspar separates were produced from the 200-800 um size fraction by first extracting non-magnetic grains and then using LST heavy liquids to isolate a density fraction of < 2.58 g/cm^3. These separates were then leached in 2% HF/HNO_3 for three 6-hour cycles to remove plagioclase feldspar and other lithics. Major element chemistry of the feldspar separates was analyzed via ICPMS at Middlebury College. The resultant separates contained 1.3 to 7 weight % K and roughly 0.2 weight % Ca (Table S6). Finally, samples were powdered to < 125 um and submitted to PRIME lab for chemical preparation and analysis. ^{36}Cl was extracted and analyzed under the supervision of Dr. Greg Chmiel at PRIME lab following the methodology described in Marrero et al. (25).

Interpretation of ^{36}Cl concentrations requires bulk chemical analyses of the rock and mineral separates. These analyses were performed by ALS minerals, Reno, NV (Tables S3-S6). Major, trace, and rare earth elements were measured on whole rock samples using a Li-metaborate fusion followed by ICPMS analysis (protocol ME-MS81). These analyses excluded Li, B, and Cl, which are important due to their high neutron absorption cross sections. Li was measured in bulk rock samples by ICPMS following a four-acid digestion using HF, HNO_3, HCL, and -$HClO_4$ (protocol ME-MS61). B was measured in bulk rock samples by ICP-AES

following an aquaregia digestion (protocol ME-ICP41). Cl was analyzed in bulk rock samples by KOH fusion and ion chromatography (protocol Cl-IC881).

Measured ^{36}Cl ratios ranged from 12 to 15 x 10^{-15} with a blank ratio of 3.62 x 10^{-15}, yielding blank corrections of 25-30% and 1σ analytical uncertainties of 7-9% (Table S2). Measured ^{36}Cl concentrations ranged from 14-38 kat/g, which can be viewed as a combination of radiogenic and muogenic ^{36}Cl.

Radiogenic ^{36}Cl is produced by capture of low-energy radiogenic neutrons on ^{35}Cl, a process that occurs throughout the lifetime of the sample, regardless of burial depth. Radiogenic ^{36}Cl is estimated by computing a radiogenic neutron flux and multiplying it by the atom density and neutron capture cross-section of ^{35}Cl (e.g. Marrero et al., 2016). Radiogenic neutron flux in the rock depends on the concentration of U and Th, which produce neutrons via (α,n) reactions, as well as the concentration of neutron moderators such as Gd, B, Li, and Cl that slow neutrons to thermal energies (26). Bulk rock U contents range from 1.7 to 2.9 ppm, Th from 7.5 to 10 ppm, Gd from 4.3-6 ppm, B less than 10 ppm, Li from 20-30 ppm, and Cl from 160 to 550 ppm (Table S3-S6). A water content of 1.5% was assumed, and carbon concentrations were estimated from loss on ignition values. Cl concentration in the feldspar was estimated by isotope dilution and ranged from 63-116 ppm suggesting significant production of radiogenic ^{36}Cl from neutron capture on ^{35}Cl.

Radiogenic neutron production rates were initially calculated with the Cronus Matlab Code version 2.2, which gave unrealistically high ^{36}Cl concentrations: 4-6 times higher than measured concentrations (24). These results were then compared with the code used by (27), which yielded more realistic radiogenic production rates between 0.021 and 0.037 at g^{-1} yr^{-1}. Assuming secular equilibrium between production and decay, these rates translate to radiogenic ^{36}Cl concentrations of 10-16 kat/g, or roughly 30 to 110% of the measured ^{36}Cl concentration. Uncertainties on the radiogenic ^{36}Cl estimates are computed using a Monte Carlo calculation with 5,000 iterations, yielding 1σ uncertainties between 17 and 22%. The validity of our estimates is further supported by agreement between $^{36}Cl_{radiogenic}$ and $^{36}Cl_{measured}$ to within 10% for sample TN9, implying minimal muogenic production in this sample. This is expected given the sample was collected from the base of a 20-m-high roadcut; a depth where muogenic production is predicted to be slow and $^{36}Cl_{muogenic}$ negligible.

For the remaining three samples collected from 4- to 6-m-deep roadcuts, $^{36}Cl_{muogenic}$ can be used to solve for an exhumation rate. Samples TN11, TN15, and TN18 yield $^{36}Cl_{muogenic}$ concentrations of 12-13 kat/g after subtraction of $^{36}Cl_{radiogenic}$. These concentrations were converted to exhumation rates using a 1-d numerical code that simulates steady state erosion at 10 cm increments for depths up to 1000 meters (e.g. 28). Depth profiles of muogenic production rate were obtained from the Cronus Matlab Code, and steady-state erosion was simulated at successively different erosion rates until a rate was found that matched the observed muogenic ^{36}Cl concentration at the appropriate sampling depth. This process yields exhumation rates of 2.3, 3.8, and 3.4 m/kyr for samples V11, V15, and V18 (Table S2). These rates are rapid, but plausible given geomorphic evidence for rapid uplift of the Sierra Colorados. They are also broadly consistent with the long-term exhumation rates of ~3.1 mm/yr obtained at the Huaco section from low-temperature thermochronology (10).

S2.4 U-Pb dating of detrital zircon

U-Pb dating of detrital zircons in sandstone was used to establish sediment provenance. Ten detrital zircon samples were analyzed from the Huaco section, with four additional samples analyzed from modern rivers: the Río Huaco, Río Jáchal, Río Bermejo, and Río de La Troya (Fig. S2 and S3; Table S8). Detrital zircons from the Toro Negro section were presented in (15) and show no significant changes in provenance.

In the field, bulk material was exhumed, hand crushed, and sieved to <500 μm. In the lab, all samples were then wet sieved to < 210 μm, rinsed to remove clay-sized particles, and zircons were separated following standard magnetic and heavy liquid procedures. Zircons were then mounted in epoxy and polished parallel to the c-axis for analysis by laser ablation – inductively coupled plasma mass spectrometry (LA-ICPMS).

LA-ICPMS analyses were performed at the UCSB LASS facility using a Photon Machines Analyte 193-nm excimer laser and Nu Instruments Plasma MC-ICPMS, following the methods described in (29). Analyses targeted the cores of zircons free of cracks or inclusions using a spot size of 15 μm and a repetition rate of 4Hz. Results were standardized using the 91500 zircon reference material (30). Standards GJ1 (601.7± 1.3 Ma (31)) and Plesovice (337.13

± 0.37 Ma (32)) were used for secondary quality control, yielding mean $^{206}Pb/^{238}U$ ages of 599.3 ± 4.5 (n=244; MSWD = 1.15) and 337.9 ± 3.6 Ma (n=116; MSWD = 0.96).

Given the increasing possibility of Pb loss with age, 207-corrected $^{238}U/^{206}Pb$ ages are used for < 1200 Ma grains, and $^{207}Pb/^{206}Pb$ ages are used for >1200 Ma. Grains that showed greater than 15% discordance between $^{238}U/^{206}Pb$ and $^{235}U/^{207}Pb$ ages were rejected, except for grains <100 Ma, which were all utilized. Detrital zircon results are displayed using kernel density estimate (KDE) plots with smoothing kernels of 0.5 Myr for ages < 50 Ma and of 4 Ma for ages > 50 Ma. 207-corrected $^{206}Pb/^{238}U$ ages were computed for all grains <1200 Ma using ISOPLOT (33), and assuming a common $^{207}Pb/^{206}Pb$ ratio of 0.84 (34).

Detrital zircon age spectra contain several peaks that can be matched with specific source units, which are summarized here with additional detail available in recent publications (10, 15, 35). First, Mesoproterozoic ages (~1400-900 Ma) are potentially derived from Grenville-aged basement rocks of the Western Sierras Pampeanas, sedimentary units of the Precordillera, or metamorphic rocks of the Frontal Cordillera. Second, Neoproterozoic-Cambrian zircons (~900-490 Ma) are important in the spectra of the Frontal Cordillera and the Famatina System, but more weakly expressed in the Precordillera and Western Sierras Pampeanas. Third, Ordovician-Silurian zircons (~490-415 Ma) are mostly present in the younger meta-sediments of the Famatina System and the Frontal Cordillera and can also be sourced from granitoids currently exposed in the Famatina System.

Because the younger age populations (<360 Ma) represent zircons that formed after the major Paleozoic tectonic episodes, they have experienced less recycling over time and thus provide more useful diagnostic information. Early Carboniferous zircons (~360-320 Ma) are sourced from Carboniferous-Permian strata widely exposed in Toro Negro region which includes numerous volcanigenic intervals from ~348-340 Ma (36). Permian -Triassic ages (~305-238 Ma) are a minor, yet persistent component that can only be sourced from younger batholiths in the Frontal Cordillera, such as the Colangüil Batholith or Montosa-El Potro complex (37-39). Likewise, Cretaceous-Paleocene ages (~75-55 Ma) can only be sourced from volcanic rocks exposed near the crest of the Frontal Cordillera and the western part of Precordillera (38). Finally, Late Cenozoic ages from ~38-1 Ma are sourced from the northwestern part of the study area, where they are widely exposed today (40, 41). We note that there is an especially strong

population of 10-15 and 20-23 Ma zircons in the Río Jáchal, presumably derived from volcanics of this age exposed in the western part of the Río Jáchal watershed (Fig. 1).

Given the complexity of the detrital age distributions, it is difficult to objectively assess their similarities and differences based on the presence or absence of specific age peaks. Instead we rely on statistical comparisons using the Kolmogorov-Smirnov (K-S) test. The K-S test is a non-parametric statistical test designed to test the hypothesis that two sample distributions are drawn from the same underlying population. In the case of U-Pb age distributions, it computes cumulative density functions (CDF's) for two age distributions and then determines the maximum vertical distance between the two functions. Samples with CDF's that closely track each other are more likely to be drawn from the same population. Results show that most samples in the upper Río Jáchal Formation are similar to one another and are collectively most similar to the modern Río Huaco (Table S7). We thus conclude that sediments in the Upper Río Jáchal Formation were derived from the ancestral Río Huaco, which primarily sourced Precordilleran units from ~6.2 to 2.6 Ma and did not experience a major change in provenance during that time.

In contrast, the two youngest samples from the overlying Mogna Formation (Hu-14 and Hu-15) are different from the older detrital samples, and are statistically similar to the modern Río Jáchal. Visually this can be explained by huge populations of 240-280 Ma and 10-25 Ma zircons in both the modern Río Jáchal and the Mogna samples. We thus conclude that a major change in provenance occurred sometime after 2.6 Ma such that the Mogna gravels were primarily derived from Río Jáchal sediments, either directly or via reworked foreland strata.

S4. The role of tectonics

Although the transient nature of the two erosion-rate slowdowns strongly suggests a climatic driver, the latest Miocene slowdown also overlaps with a tectonic reorganization in the eastern Andes at 31° S (2, 10), when faulting simultaneously stepped eastward to the front of the Precordillera and back westwards into the Frontal Cordillera (Fig. 1). Just west of Huaco, the onset of uplift along the Niquivil fault is inferred from a change in sedimentation rate in the Huaco section at ~5.95 Ma (7). Just east of Huaco, uplift along the Valle Fértil lineament and of the Pie de Palo block to the south also began sometime between ~6-4 Ma (2, 4, 42).

Despite the temporal correlation, we argue against a strong influence of local tectonics on the Huaco record for two reasons. First, our detrital zircon data show little evidence for a change in provenance associated with either drop in erosion rate (Fig. S3 and S4). For example, sample Hu-5 represents the maximum extent of the latest Miocene slowdown and is not statistically different from samples Hu-1.1, and Hu-8 which represent sediment deposited just before and after the erosional slowdown (Table S7). Second, because our results suggest that sediments are consistently sourced from the east in the vicinity of the Niquivil fault (e.g., the modern Río Huaco watershed), it seems unlikely that accelerated uplift in this region would cause a decrease in erosion rate.

Although tectonics cannot explain the erosion-rate slowdowns observed in the Huaco section, the existence of a tectonically induced orographic barrier may explain the damped erosional response to aridity observed in the Toro Negro section. The Toro Negro section shows a roughly 5 times lower erosion rate than Huaco, which decreases gradually from ~0.1 to 0.05 mm/yr over 5 million years and shows no major oscillations, although very few data are available for the intervals of slowdowns (Fig. 2i). One possible explanation is that the Toro Negro watershed was already experiencing arid conditions at 7 Ma, perhaps due to existing orographic barriers created by mid-Miocene uplift of mountain ranges to the north and east associated with the Famatinian and Sierras Pampeanas systems (43). In contrast, the Río Huaco watershed is exposed entirely along the western flank of the Precordillera without major topographic shielding from the north, east, or south. An alternative explanation for the damped climatic signal from Toro Negro is that erosion rates were not high enough to record changes in precipitation, perhaps due to lower relief or lower rates of tectonic uplift.

S3. References Cited

1. Kay SM & Abbruzzi JM (1996) Magmatic evidence for Neogene lithospheric evolution of the central Andean "flat-slab" between 30 degrees S and 32 degrees S. *Tectonophysics* 259(1-3):15-28.
2. Jordan TE*, et al.* (2001) Unsteady and spatially variable evolution of the Neogene Andean Bermejo foreland basin, Argentina. *Journal of South American Earth Sciences* 14(7):775-798.
3. Jordan TE*, et al.* (1985) Foreland basin evolution in the Central Andes, Bermejo Basin, San Juan Province, Argentina. *Abstracts with Programs - Geological Society of America* 17(7):621.
4. Siame LL*, et al.* (2015) Active basement uplift of Sierra Pie de Palo (Northwestern Argentina): Rates and inception from ^{10}Be cosmogenic nuclide concentrations. *Tectonics* 34(6):1129-1153.

5. Johnson NM, Jordan TE, Johnsson PA, & Naeser CW (1986) Magnetic polarity stratigraphy, age and tectonic setting of fluvial sediments in an eastern Andean foreland basin, San Juan Province, Argentina. *Special Publication of the International Association of Sedimentologists* 8:63-75.
6. Jordan TE, Flemings PB, & Beer JA (1988) *Dating thrust-fault activity by use of foreland-basin strata* (Springer-Verlag, New York, NY).
7. Jordan TE, Allmendinger RW, Damanti JF, Drake RE, & Anonymous (1993) Chronology of motion in a complete thrust belt; the Precordillera, 30-31 degrees S, Andes Mountains. *Journal of Geology* 101(2):135-156.
8. Zapata TR & Allmendinger RW (1996) Thrust-front zone of the Precordillera, Argentina: A thick-skinned triangle zone. *AAPG Bulletin* 80(3):359-381.
9. Furque G (1983) Descripcion geologica de la Hoja 19 c, Cienaga de Gualilan, provincia de San Juan. (Direccion Nacional de Mineria y Geologia, Buenos Aires), p 111.
10. Fosdick JC, Carrapa B, & Ortíz G (2015) Faulting and erosion in the Argentine Precordillera during changes in subduction regime: Reconciling bedrock cooling and detrital records. *Earth and Planetary Science Letters* 432:73-83.
11. Amidon W*, et al.* (2016) U-Pb ages of detrital and volcanic zircons of the Toro Negro Formation, northwestern Argentina: age, provenance and sedimentation rates. *Journal of South American Earth Sciences*:In Press.
12. Ciccioli P (2008) Evolución paleoambiental, estratigrafía y petrología sedimentaria de la Formación Toro Negro (Neógeno), Sierras Pampeanas Noroccidentales (Provincia de La Rioja) Ph.D. (Universidad de Buenos Aires, Buenos Aires, Argentina).
13. Ciccioli PL, Limarino CO, Marenssi SA, Tedesco AM, & Tripaldi A (2011) Tectosedimentary evolution of the La Troya-Vinchina depocenters (northern Bermejo Basin, Tertiary), La Rioja Province, Argentina. *Cenozoic Geology of the Central Andes of Argentina*, eds Salfity JA & Marquillas RA (SCS, Salta), pp 91-110.
14. Limarino CO, Ciccioli PL, & Marenssi SA (2010) Analisis del contacto entre las formaciones Vinchina y Toro Negro (Sierra de los Colorados, provincia de La Rioja, Argentina), sus implicancias tectonicas. *Latin American Journal of Sedimentology and Basin Analysis* 17(2):113-132.
15. Amidon WH*, et al.* (2016) U-Pb ages of detrital and volcanic zircons of the Toro Negro Formation, northwestern Argentina: Age, provenance and sedimentation rates. *Journal of South American Earth Sciences* 70:237-250.
16. Jones CH (2002) User-driven integrated software lives: “Paleomag” paleomagnetics analysis on the Macintosh. *Computers & Geosciences* 28(10):1145–1151.
17. Kirschvink JL (1980) The Least-Squares Line and Plane and the Analysis of Palaeomagnetic Data. *Geophysical Journal International* (62):699-718.
18. Niishizumi K*, et al.* (2007) Absolute calibration of ^{10}Be AMS standards. *Nuclear Instruments and Methods in Physics Research* B 258:403-413.
19. Charreau J*, et al.* (2011) Paleo-erosion rates in Central Asia since 9Ma: A transient increase at the onset of Quaternary glaciations? *Earth and Planetary Science Letters* 304(1-2):85-92.
20. Val P, Hoke GD, Fosdick JC, & Wittmann H (2016) Reconciling tectonic shortening, sedimentation and spatial patterns of erosion from ^{10}Be paleo-erosion rates in the Argentine Precordillera. *Earth and Planetary Science Letters* 450:173-185.
21. Lal D & Peters B (1967) Cosmic ray produced radioactivity in the earth. *Handbuch der Physik*, (Springer-Verlag), pp 551-612.
22. Lifton N, Sato T, & Dunai TJ (2014) Scaling in situ cosmogenic nuclide production rates using analytical approximations to atmospheric cosmic-ray fluxes. - 2014/01/15/ *Earth and Planetary Science Letters* 386:149 - 160.

23. Braucher R, Brown ET, Bourles DL, & Colin F (2003) In situ produced Be-10 measurements at great depths: implications for production rates by fast muons. *Earth and Planetary Science Letters* 211(3-4):251-258.
24. Marrero SM*, et al.* (2016) Cosmogenic nuclide systematics and the CRONUScalc program. *Quaternary Geochronology* 31:160-187.
25. Marrero SM, Phillips FM, Caffee MW, & Gosse JC (2016) CRONUS-Earth cosmogenic ^{36}Cl calibration. *Quaternary Geochronology* 31:199-219.
26. Gosse J & Phillips F (2001) Terrestrial in situ cosmogenic nuclides: theory and application. *Quaternary Science Reviews* 20:1475-1560.
27. Amidon W, Farley KA, Burbank DW, & Pratt-Sitaula B (2008) Anomalous cosmogenic ^{3}He production and elevation scaling in the high Himalaya. *Earth and Planetary Science Letters* 265:287-301.
28. Amidon WH, Rood DH, & Farley KA (2009) Cosmogenic ^{3}He and ^{21}Ne production rates calibrated against ^{10}Be in minerals from the Coso volcanic field. *Earth and Planetary Science Letters* 280:194-204.
29. Kylander-Clark ARC, Hacker BR, & Cottle JM (2013) Laser ablation split stream ICP petrochronology. *Chemical Geology* 345:99-112.
30. Wiedenbeck M*, et al.* (1995) Three natural zircon standards for U-Th-Pb, Lu-Hf, trace element and REE analyses. *Geostandards Newsletter* 19:1-23.
31. Jackson SE, Pearson NJ, Griffin WL, & Belousova EA (2004) The application of laser ablation-inductively coupled plasma-mass spectrometry to in situ U-Pb zircon geochronology. *Chemical Geology* 211(1-2):47-69.
32. Sláma J*, et al.* (2008) Plešovice zircon - A new natural reference material for U-Pb and Hf isotopic microanalysis. *Chemical Geology* 249(1-2):1-35.
33. Ludwig KR (1991) ISOPLOT; a plotting and regression program for radiogenic-isotope data; version 2.53. *Open-File Report - U. S. Geological Survey*:39.
34. Stacey JS & Kramers JD (1975) Approximation of Terrestrial Lead Isotope Evolution by a 2-Stage Model. *Earth and Planetary Science Letters* 26(2):207-221.
35. Levina M, Horton BK, Fuentes F, & Stockli DF (2014) Cenozoic sedimentation and exhumation of the foreland basin system preserved in the Precordillera thrust belt (31-32 degrees S), southern Central Andes, Argentina. *Tectonics* 33(9):1659-1680.
36. Gulbranson EL*, et al.* (2010) High-precision U-Pb calibration of Carboniferous glaciation and climate history, Paganzo Group, NW Argentina. *Bulletin of the Geological Society of America* 122(9-10):1480-1498.
37. Caminos R (1972) Upper Paleozoic sedimentation and magmatism in the Cordillera Frontal, Argentina. *Anais da Academia Brasileira de Ciencias* 44, Suppl.:77-86.
38. Caminos R*, et al.* (1979) *Cordillera Frontal* (Acad. Nac. Cien., Cordoba).
39. Coughlin TJ (2000) Linked orogen-oblique fault zones in the central Argentine Andes; the basis of a new model for Andean orogenesis and metallogenesis. PhD (The University of Queensland).
40. Goss AR, Kay SM, & Mpodozis C (2013) Andean adakite-like high-Mg andesites on the northern margin of the Chilean-Pampean flat-slab (27-28.5 degrees S) associated with frontal arc migration and fore-arc subduction erosion. *Journal of Petrology* 54(11):2193-2234.
41. Kay SM, Mpodozis C, & Gardeweg M (2013) Magma sources and tectonic setting of Central Andean andesites (25.5-28 degrees S) related to crustal thickening, forearc subduction erosion and delamination. *Special Publication - Geological Society of London* 385(1):303-334, 333.
42. Coughlin TJ, O'Sullivan PB, Kohn BP, & Holcombe RJ (1988) Apatite fission-track thermochronology of the Sierras Pampeanas, central western Argentina: Implications for the mechanism of plateau uplift in the Andes. *Geology* 26(11):999-1002.

43. Davila FM & Astini RA (2007) Cenozoic provenance history of synorogenic conglomerates in western Argentina (Famatina Belt); implications for central Andean foreland development. *Geological Society of America Bulletin* 119(5-6):609-622.

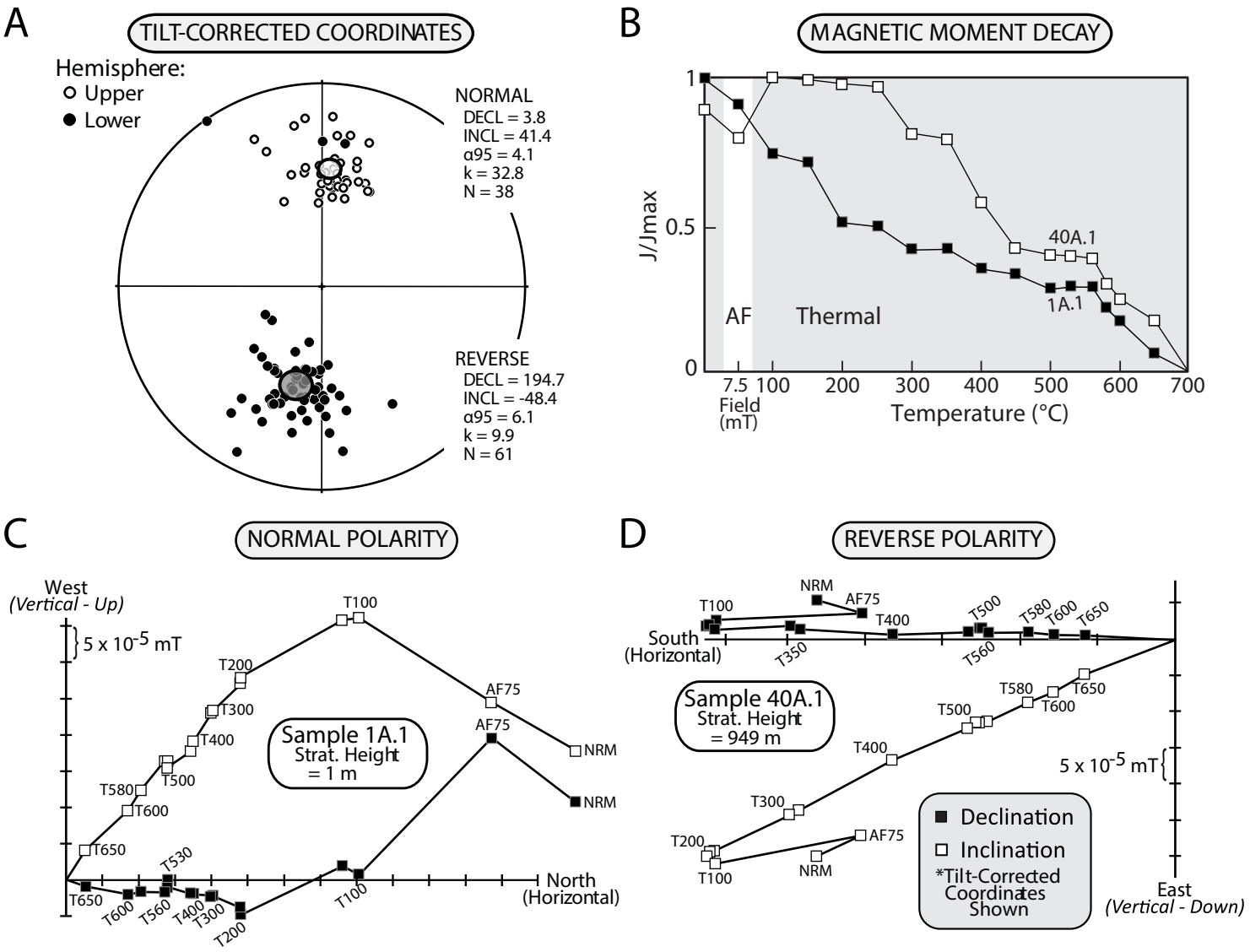


Figure S1. A) Stereographic equal-area plot of tilt-corrected remanent magnetizations directions and their statistics. Larger circles indicate the α95 error around the Fisher mean for the normal and reversed polarity samples. B) Magnetic moment decay during step-wise AF and thermal demagnetization steps for the representative normal and reversed polarity sample behavior shown in C and D, respectively.

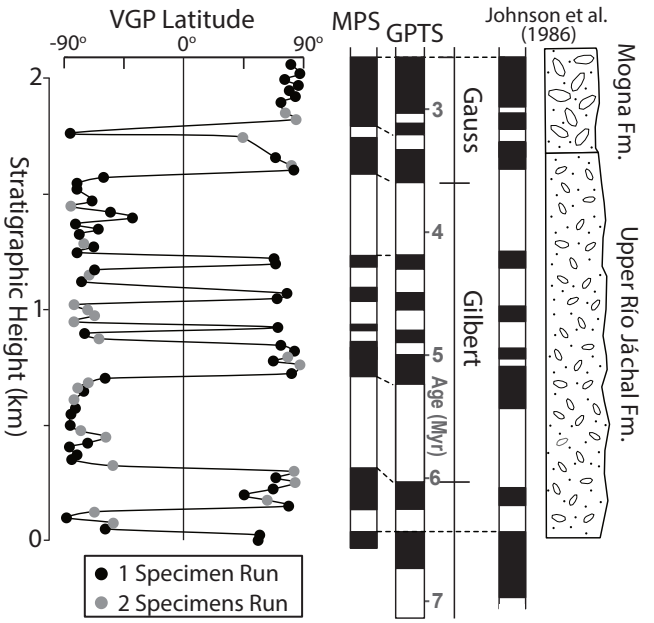


Figure S2. Paleomagnetic correlation of measured virtual geomagnetic pole (VGP) latitude (left) and a magnetic polarity stratigraphy (MPS) derived from it with the global polarity time scale (GPTS) (42). Simplified stratigraphy and original magnetic stratigraphy of Johnson et al. (5) are shown at right for comparison.

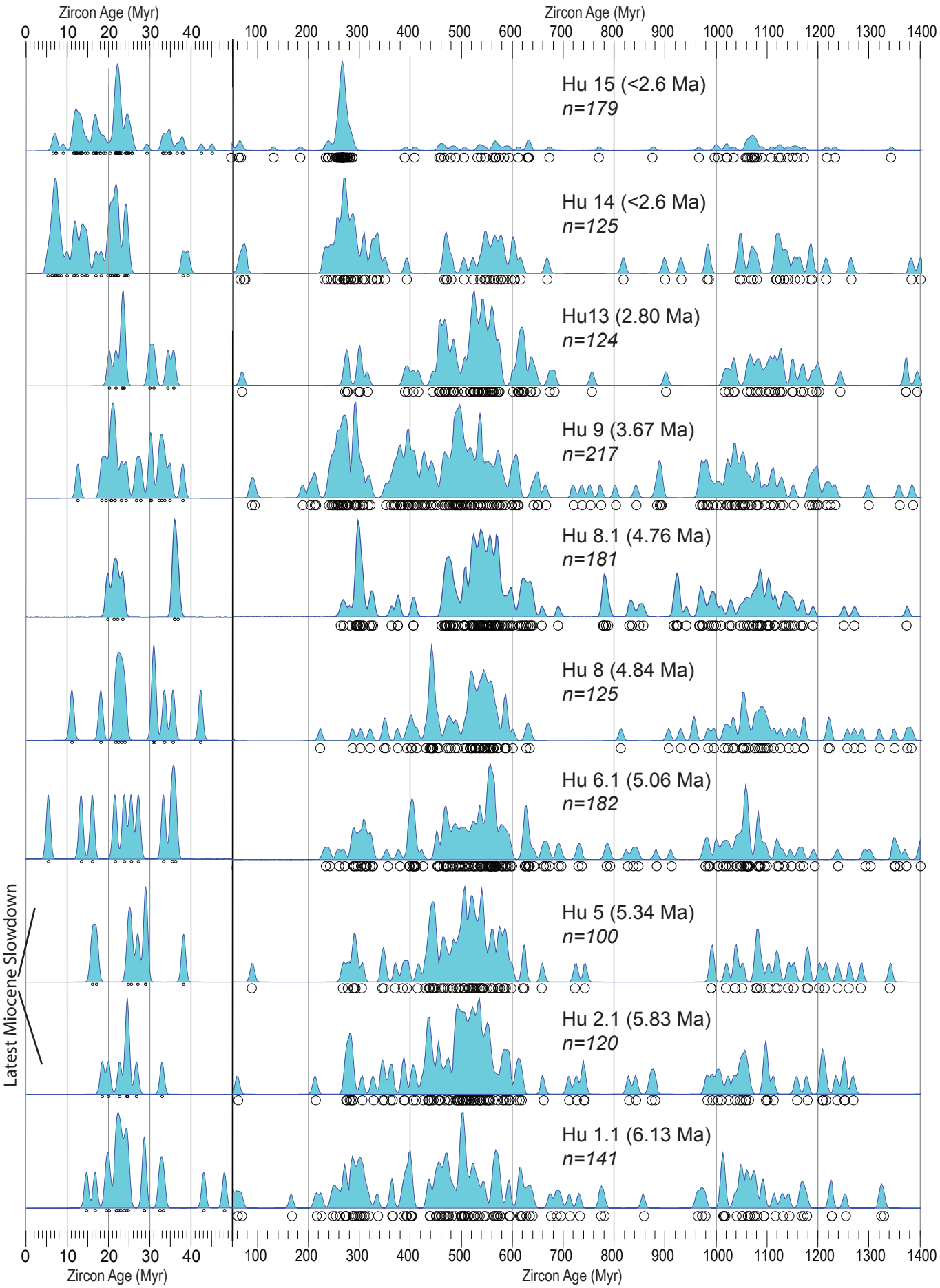


Figure S3. Kernel density plots showing the U-Pb age distributions of samples from the Río Huaco section. K-S tests (Table S8) show that samples in the Río Jáchal Formation (Hu-1.1 to Hu-13) are generally similar to each other, but samples from the overlying Mogna Formation (Hu-13 and Hu-14) are significantly different, suggesting a provenance shift sometime after 2.6 Ma. Note the change in scale at 50 Ma.

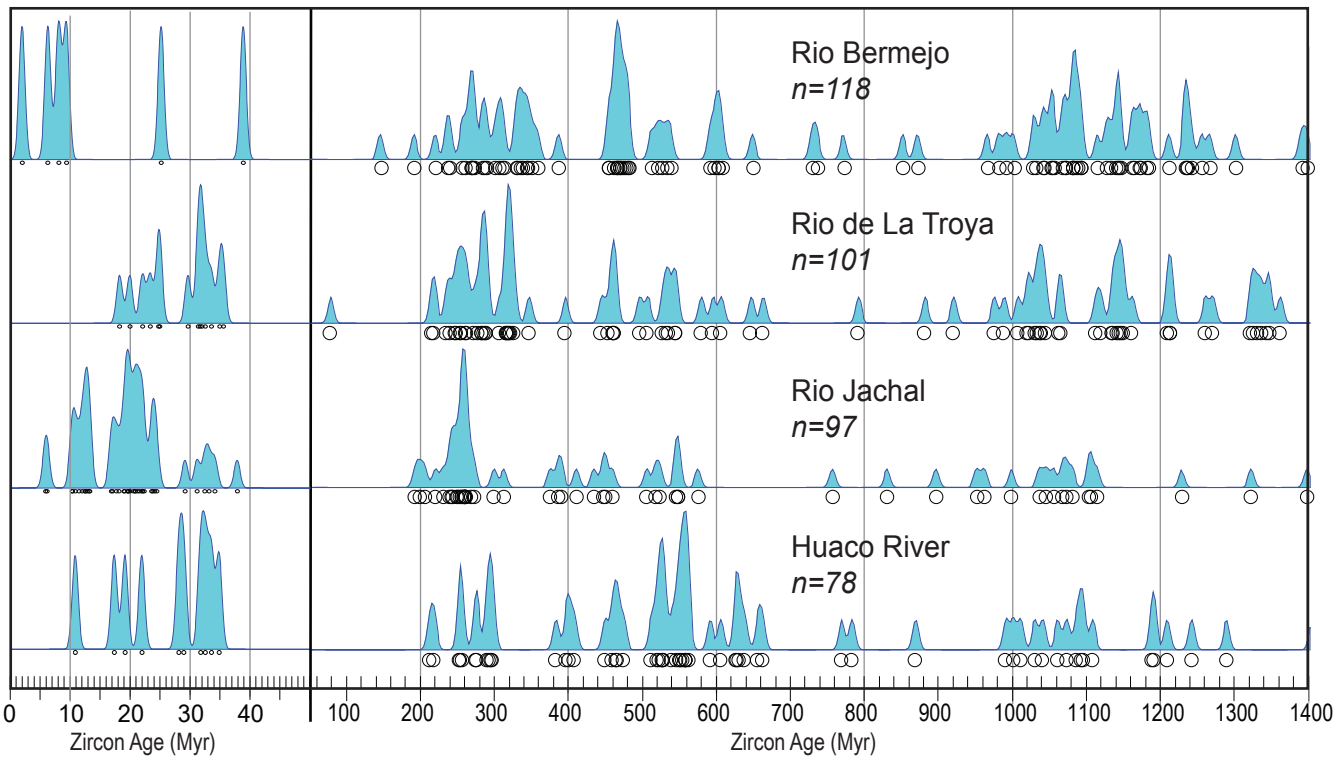


Figure S4. Kernel density plots showing the U-Pb age distributions of samples from modern watersheds in the study area, as shown in figure 1. K-S tests (Table S8) show that samples from the Río Jáchal Formation are most similar to the modern Río Huaco and samples from the Mogna Formation are most similar to the modern Río Jáchal. Note the change in scale at 50 Ma.

Table S1: Paleomagnetic Data

Sample	Lat WGS84	Lon WGS84	Polarity	MAD (deg)	VGP Lat (deg)
h-1A	-30.227309	-68.402823	N	2.2	55.5
h-2B	-30.227235	-68.402535	N	3.5	56.7
h-3B	-30.227163	-68.402182	S	7.4	-59.0
h-4A	-30.22709	-68.40184	S	3.3	-62.2
h-4B	-30.22709	-68.40184	S	3.6	-43.9
h-5A	-30.226722	-68.401448	S	3.6	-88.1
h-6B	-30.226595	-68.401112	S	4.7	-63.3
h-6B	-30.226595	-68.401112	S	5.1	-71.2
h-7B	-30.226434	-68.400736	N	3.4	78.6
h-8A	-30.226279	-68.40037	N	5	64.8
h-8B	-30.226279	-68.40037	N	5.2	60.2
h-9A	-30.226218	-68.40004	?	3.1	45.0
h-10A	-30.226226	-68.399714	N	2.5	66.8
h-11B	-30.226103	-68.399312	N	4	82.0
h-11B	-30.226103	-68.399312	N	5.3	84.9
h-12A	-30.225966	-68.398968	N	3.1	68.5
h-13A	-30.225719	-68.398466	N	3	84.8
h-13B	-30.225719	-68.398466	N	3.5	79.2
h-14A	-30.22535	-68.397975	S	6	-54.0
h-14B	-30.22535	-68.397975	S	3.4	-52.7
h-15A	-30.224298	-68.396939	S	2.7	-84.4
h-16A	-30.22176	-68.395073	S	1.4	-80.2
h-17C	-30.218435	-68.392707	S	1.9	-86.3
h-18A	-30.217581	-68.3918	S	4	-72.4
h-19A	-30.216975	-68.391149	S	3.4	-67.7
h-19B	-30.216975	-68.391149	S	3.3	-49.5
h-20A	-30.216126	-68.390226	S	1.4	-78.1
h-20B	-30.216126	-68.390226	S	2.2	-77.3
h-21B	-30.215809	-68.389679	S	3.2	-85.4
h-23B	-30.214949	-68.388549	S	7.7	-85.1
h-24B	-30.214444	-68.387895	S	2	-81.3
h-25A	-30.21419	-68.387186	S	6.1	-83.0
h-25B	-30.21419	-68.387186	S	3.3	-81.8
h-26A	-30.213132	-68.386169	S	6.6	-75.1
h-27A	-30.212832	-68.385871	S	3.7	-76.3
h-27B	-30.212832	-68.385871	S	4.2	-83.7
h-28A	-30.212401	-68.385416	S	5.4	-75.0
h-28A	-30.212401	-68.385416	S	6	-68.0
h-29A	-30.211993	-68.384938	S	2.7	-59.5
h-30A	-30.211426	-68.384387	N	2.8	80.5
h-32A	-30.211411	-68.38379	N	3.8	86.3
h-32B	-30.211411	-68.38379	N	1.9	87.4
h-33A	-30.211436	-68.383528	N	1.9	66.8

Table S1: Paleomagnetic Data

Sample	Lat WGS84	Lon WGS84	Polarity	MAD (deg)	VGP Lat (deg)
h-34A	-30.211503	-68.383296	N	2.5	73.4
h-34B	-30.211503	-68.383296	N	2.6	81.7
h-35A	-30.210647	-68.382028	N	1.3	82.6
h-36A	-30.208236	-68.379986	N	3.1	72.7
h-37A	-30.208138	-68.379299	S	2.3	-61.3
h-37B	-30.208138	-68.379299	S	3.1	-66.1
h-38A	-30.207732	-68.378601	S	2.8	-74.5
h-39A	-30.206417	-68.376919	N	9	70.6
h-40A	-30.206383	-68.376437	S	1.4	-84.4
h-40B	-30.206383	-68.376437	S	1.4	-81.2
h-41A	-30.206168	-68.375727	S	4.5	-55.4
h-41B	-30.206168	-68.375727	S	6.7	-79.5
h-42A	-30.206295	-68.375273	S	3	-73.9
h-42B	-30.206295	-68.375273	S	1.6	-70.5
h-43A	-30.206226	-68.374912	S	3.2	-86.9
h-43B	-30.206226	-68.374912	S	4.1	-78.2
h-44A	-30.205966	-68.374152	N	2.6	69.6
h-45A	-30.205865	-68.373629	N	3.7	77.3
h-47B	-30.205759	-68.372635	S	3.5	-76.8
h-48B	-30.205876	-68.372194	S	3.5	-76.2
h-48B	-30.205876	-68.372194	S	3.4	-66.5
h-49B	-30.206028	-68.371836	S	2.8	-67.4
h-50A	-30.205757	-68.371126	N	3.4	68.4
h-51B	-30.20548	-68.370579	N	1.7	67.6
h-52B	-30.205541	-68.37016	S	4.2	-80.5
h-53A	-30.205471	-68.369815	S	3.2	-68.0
h-54A	-30.20542	-68.369446	S	2.6	-75.9
h-54B	-30.20542	-68.369446	S	1.9	-74.5
h-55B	-30.205482	-68.36887	S	1.3	-78.7
h-56A	-30.205635	-68.368605	S	4.6	-64.6
h-57A	-30.205723	-68.36831	S	2.1	-81.4
h-58A	-30.205884	-68.367956	S	2.6	-38.6
h-59A	-30.205979	-68.367567	S	7.1	-55.3
h-60A	-30.206159	-68.367262	S	4.6	-83.0
h-60C	-30.206159	-68.367262	S	3.8	-86.5
h-61B	-30.2061	-68.366837	S	2.7	-69.0
h-63A			S	2.2	-80.5
h-64A	-30.20579	-68.365646	S	2.2	-80.1
h-65A	-30.205912	-68.365375	S	2	-60.4
h-66B	-30.206137	-68.365007	N	2.3	82.2
h-67A	-30.206254	-68.364804	N	3.5	82.7
h-67B	-30.206254	-68.364804	N	2.7	78.0
h-68A	-30.206439	-68.36443	N	6.4	68.6
h-72A	-30.207012	-68.363494	?	3.5	43.4

Table S1: Paleomagnetic Data

Sample	Lat WGS84	Lon WGS84	Polarity	MAD (deg)	VGP Lat (deg)
h-72B	-30.207012	-68.363494	?	4.8	44.4
h-73A	-30.207117	-68.363269	S	3	-85.4
h-75A	-30.207494	-68.362554	N	4.9	88.9
h-75B	-30.207494	-68.362554	N	7.3	79.5
h-76A			N	2.7	83.1
h-76B			N	5.2	69.3
h-78A	-30.207626	-68.362119	N	5.4	72.7
h-79A	-30.208182	-68.361227	N	4.3	83.3
h-80A	-30.208363	-68.360902	N	6.7	78.8
h-81A	-30.208534	-68.360656	N	5.7	85.5
h-82A	-30.20872	-68.360216	N	1.7	75.2
h-83A	-30.208913	-68.359863	N	3.1	86.6
h-84A	-30.209162	-68.359182	N	2.5	80.1

Table S2: ^{10}Be results from the Huaco (HU) and Toro Negro (TN) sections

Sample	Latitude (WGS 84)	Longitude (WGS 84)	Strat. Height (m)	Age (Myr)	2σ (Myr)	Outcrop Height (m)	Mass (g)	$^{10}Be/^{9}Be_{samp}$ ($x10^{-15}$)	1σ ($x10^{-15}$)	$^{10}Be/^{9}Be_{blk}$ ($x10^{-15}$)	$^{10}Be_{meas}$ (kat/g)	1σ (kat/g)	$^{10}Be_{burial}$ (kat/g)	$^{10}Be_{exhum}$ (kat/g)	$^{10}Be_{paleo\text{-}c}$ (kat/g)	1σ (at/g)	Erosion Rate (mm/yr)	1σ (mm/yr)
HU-1	-30.22714	-68.40218	50	6.41	0.05	3.8	235.1	43.0	2.6	2.91	2.30	0.14	0.4	0.28	42.1	9.7	0.49	0.16
HU-1.1A	-30.22620	-68.39945	235	6.13	0.05	3.5	326.7	72.1	4.0	8.13	2.64	0.15	0.3	0.32	43.9	9.2	0.47	0.14
HU-1.1	-30.22601	-68.39898	267	6.09	0.06	3.5	282.7	73.6	3.4	8.13	3.11	0.14	0.4	0.32	53.1	10.3	0.39	0.11
HU-2	-30.22572	-68.39847	297	6.05	0.01	6.0	416.1	106.2	4.2	8.13	3.17	0.13	0.4	0.26	54.3	10.6	0.38	0.10
HU-2.1	-30.22192	-68.39499	391	5.88	0.02	8.0	416.4	96.4	4.7	8.13	2.84	0.14	0.4	0.24	42.9	8.6	0.48	0.14
HU-3	-30.21757	-68.39164	441	5.78	0.03	5.0	342.4	86.1	3.4	8.13	3.06	0.12	0.4	0.31	43.3	8.2	0.47	0.13
HU-4	-30.21446	-68.38792	573	5.52	0.03	5.0	253.2	112.5	4.9	8.13	5.54	0.24	0.5	0.33	76.9	13.0	0.27	0.07
HU-5	-30.21285	-68.38584	663	5.34	0.02	4.0	275.7	164.5	6.8	8.13	7.72	0.32	0.5	0.39	101.2	15.9	0.20	0.05
HU-6	-30.21127	-68.38419	729	5.21	0.05	4.5	235.7	148.4	4.6	2.91	8.33	0.26	0.7	0.39	101.7	15.3	0.20	0.05
HU-6.1	-30.20465	-68.37821	821	5.06	0.03	4.0	212.9	102.9	4.2	2.91	6.31	0.26	0.7	0.43	67.0	11.0	0.31	0.08
HU-7	-30.20449	-68.37638	893	4.94	0.02	18.0	225.2	98.7	3.8	2.91	5.78	0.22	0.8	0.19	58.7	9.0	0.35	0.09
HU-8	-30.20393	-68.37521	926	4.84	0.02	10.0	274.2	119.6	5.7	8.13	5.48	0.26	0.9	0.31	49.3	8.2	0.42	0.11
HU-8.1	-30.20376	-68.37443	960	4.76	0.02	3.5	245.2	97.7	4.2	2.91	5.25	0.22	1.0	0.51	42.0	7.8	0.49	0.14
HU-8.2	-30.20485	-68.36858	1200	4.27	0.02	4.0	256.4	142.7	6.8	8.13	7.12	0.34	1.3	0.62	45.0	7.6	0.46	0.12
HU-9	-30.20586	-68.36574	1546	3.67	0.02	4.5	231.6	159.8	5.4	2.91	9.11	0.31	1.6	0.91	42.5	6.1	0.48	0.12
HU-9.1	-30.20625	-68.36494	1617	3.55	0.02	4.0	377.7	405.6	12.5	8.13	14.16	0.44	1.5	1.06	70.0	8.4	0.29	0.06
HU-10	-30.20698	-68.36349	1751	3.34	0.02	8.0	208.6	472.4	10.4	2.91	30.41	0.67	1.7	1.00	150.4	14.0	0.14	0.03
HU-11	-30.20759	-68.36240	1842	3.11	0.02	7.0	247.7	462.4	9.5	2.91	25.79	0.53	2.2	1.50	106.6	10.1	0.19	0.04
HU-12	-30.20829	-68.36109	1941	2.89	0.02	9.0	211.9	336.0	10.5	2.91	21.15	0.66	2.4	1.31	75.3	7.6	0.27	0.06
HU-13	-30.20867	-68.36056	1983	2.80	0.02	4.0	234.8	695.3	15.3	2.91	39.84	0.88	2.6	1.87	146.1	12.5	0.14	0.03
HU-14	-30.21895	-68.34872	NA	2.4*	NA	7.0	154.2	1506.3	25.2	2.91	132.02	2.21	1.9	1.50	433.1	7.4	0.00	0.00
HU-15	-30.21989	-68.34799	NA	2.4*	NA	7.0	143.7	1924.8	33.7	2.91	180.37	3.16	1.9	1.50	595.9	10.6	0.00	0.00
TN-7	-28.70683	-68.30521	60	6.94	0.35	7.0	350	141.5	9.5	8.0	5.09	0.34	0.24	0.28	153	44	0.11	0.05
TN-7.1	-28.70605	-68.30538	88	6.91	0.35	4.0	309	137.0	11.1	8.0	5.55	0.45	0.24	0.36	164	46	0.11	0.05
TN-8	-28.70381	-68.30588	199	6.82	0.34	8.0	369	120.1	8.1	8.0	4.04	0.27	0.25	0.26	111	32	0.16	0.07
TN-9	-28.70277	-68.30609	308	6.73	0.34	20.0	323	141.5	6.7	8.0	5.53	0.26	0.26	0.13	155	42	0.11	0.05
TN-11	-28.69692	-68.30657	500	6.57	0.33	6.0	352	164.4	4.8	8.0	5.93	0.17	0.29	0.30	148	38	0.12	0.05
TN-13	-28.68180	-68.30347	1025	6.13	0.31	7.0	306	215.3	8.0	8.0	9.02	0.34	1.36	0.28	164	47	0.11	0.05
TN-15	-28.68006	-68.30332	1038	6.12	0.31	4.0	254	148.5	5.3	8.0	7.39	0.27	1.36	0.36	125	37	0.14	0.07
TN-17	-28.67085	-68.29576	1132	5.72	0.29	12.0	165	168.1	14.8	8.0	12.91	1.14	1.67	0.20	199	53	0.09	0.04
TN-18	-28.66052	-68.28991	1348	5.15	0.26	4.0	244	280.2	8.9	8.0	14.90	0.47	2.22	0.36	167	39	0.10	0.04
TN-19	-28.65708	-68.28875	1415	4.89	0.24	10.0	191	367.5	15.1	8.0	25.02	1.03	1.29	0.23	279	53	0.06	0.02
TN-21	-28.65666	-68.30633	1864	2.63	0.13	4.5	192	1509.2	21.7	8.0	103.63	1.49	4.06	0.34	375	37	0.05	0.01
TN-23	-28.65640	-68.30811	1939	2.48	0.12	4.5	218	1812.0	60.3	8.0	109.86	3.65	4.38	0.34	368	37	0.05	0.01
TN-25	-28.65611	-68.31120	2080	2.33	0.12	3.0	191	2022.4	50.3	8.0	140.02	3.48	4.72	0.41	438	39	0.04	0.01

Age: *depositional age of sample ;* $^{10}Be_{meas}$ *: measured* ^{10}*Be in modern sample;* $^{10}Be_{burial}$ **:** *estimated* ^{10}*Be acquired during burial, corrected to modern concentration*

$^{10}Be_{exhum}$ **:** *^{10}Be attributed to recent exhumation of landscape* $^{10}Be_{paleo\text{-}c}$ **:** *amount of ^{10}Be in sediment sample prior to burial = ($^{10}Be_{meas}$ - $^{10}Be_{burial}$ - $^{10}Be_{exhum}$)*exp(- λ *Age)*

** denotes very rough estimate for two samples from the Mogna Gravels whose exact depositional age is unknown*

Table S3: ^{36}Cl results

	Mass (g)	^{36}Cl Ratio ($x10^{-15}$)	1σ ($x10^{-15}$)	$^{35}Cl/^{37}Cl$	1σ	Tot. Cl_{corr} (mg)	1σ (mg)	$^{36}Cl_{corr}$ (at/g)	1σ (at/g)	Cl Conc. (ppm)	1σ (ppm)	Roadcut height (m)	Rad. PR (at g^{-1} yr^{-1})	1σ (at g^{-1} yr^{-1})	Rad. Conc. (at/g)	1σ (at/g)	Slow Muon PR# (at g^{-1} yr^{-1})	Fast Muon PR# (at g^{-1} yr^{-1})	Exhum. rate (mm/yr)
Blk	-	3.62	0.58	18.348	0.669	3.619	20317	-	-	-	-	-	-		-		-	-	-
TN9	105.9	11.70	0.77	3.745	0.032	6.689	0.361	14,277	1,204	63.2	3.4	20	0.037	0.006	16,241	2,761	47.30	0.04	-
TN11	92.9	17.98	1.10	3.591	0.032	9.065	0.645	33,329	2,983	97.6	6.9	6	0.030	0.007	12,941	2,847	21.52	0.03	2.29
TN15	117.4	18.05	0.93	3.430	0.012	13.571	0.549	38,141	2,471	115.6	4.7	4	0.027	0.005	11,899	2,332	35.57	0.03	3.78
TN18	144.3	14.22	0.83	3.487	0.018	11.700	0.599	21,257	1,624	81.1	4.2	4	0.029	0.005	12,420	2,111	9.53	0.01	3.36

$^{36}Cl_{corr}$: *Corrected for chem blank and contamination*

Tot.Cl_{corr}: *Total chlorine mass corrected for contamination*

Rad. PR: denotes radiogenic production rate

denotes muon production rates at the surface, not at the sample (roadcut) depth

Rad Conc.: denotes estimated radiogenic concentration

Table S4: Bulk geochemistry of Toro Negro Samples by Li-Metaborate digestion and ICPMS (ALS minerals)

	Ba	Ce	Cr	Cs	Dy	Er	Eu	Ga	Gd	Hf	Ho	La	Lu	Nb	Nd	Pr	Rb	Sm	Sn	Sr	Ta	Tb	Th	Tm	U	V	W	Y	Yb	Zr
	ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm
TN9	456	68	70	2.61	5.83	3.65	1.21	13	6.08	11.8	1.22	32.6	0.56	13.3	32	8.18	65.3	6.71	2	442	1.1	0.97	10.1	0.54	2.87	80	1	33.2	3.43	422
TN11	491	54.6	50	1.89	3.61	2.16	1.15	14.3	4.36	5.1	0.75	27.3	0.28	11	24.8	6.43	62.1	4.92	2	417	0.8	0.63	7.37	0.33	1.95	76	1	19.3	2.13	182
TN15	507	55.7	50	2.2	4.2	2.55	0.99	12.3	4.59	5.9	0.81	26.8	0.36	11.2	25.3	6.57	77.6	5.39	2	277	0.8	0.7	8.21	0.36	1.79	67	1	22.6	2.4	206
TN18	488	58.4	70	3.01	3.66	2.13	1.31	16.9	4.45	6.5	0.67	28.6	0.31	11.8	26.6	6.87	65.3	5.5	2	497	0.7	0.62	7.53	0.33	1.66	97	2	19.8	2.07	228

Table S5: Bulk geochemistry of Toro Negro Samples by Li-Metaborate digestion and ICPMS (ALS minerals)

	SiO2	Al2O3	Fe2O3	CaO	MgO	Na2O	K2O	Cr2O3	TiO2	MnO	P2O5	SrO	BaO	LOI	Total
	%	%	%	%	%	%	%	%	%	%	%	%	%	%	%
TN9	71.1	10.9	4.63	3.03	1.62	2.15	1.88	0.01	0.71	0.08	0.15	0.05	0.05	3.87	100.2
TN11	66	12.25	4.63	3.45	2	2.73	2.12	0.01	0.63	0.06	0.15	0.05	0.05	6.03	100.2
TN15	73.1	10.55	4.23	2.37	1.27	2.15	2.22	0.01	0.57	0.06	0.12	0.03	0.05	3.38	100.1
TN18	65.7	13.6	4.99	3.6	2.58	2.62	1.82	0.01	0.76	0.07	0.17	0.06	0.05	4.34	100.4

Table S6: Bulk geochemistry of Toro Negro Samples by 4-acid digestion and ICPMS (ALS minerals)

	Ag	As	B**	Be	Bi	Cd	Ce	Cl*	Co	Cu	In	Li	Mo	Ni	Pb	Se	Sb	Sc	Zn
	ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm
TN9	0.07	12.7	<10	1.59	0.2	0.09	74.5	550	11.5	15	0.04	23.4	0.4	30.2	17.5	1	1.3	9.9	59
TN11	0.07	7.2	<10	1.68	0.16	0.08	62.4	340	10.7	10.6	0.04	18.7	0.5	21.3	14.6	1	0.6	9.6	58
TN15	0.05	5	<10	1.51	0.18	0.08	61.4	160	8.7	13	0.04	18.2	0.4	20.5	14.2	1	0.5	8.4	48
TN18	0.09	8	10	1.73	0.19	0.11	60.2	180	12.8	15.8	0.05	28.4	0.5	25.8	14.2	1	0.6	11	72

*Cl determined by KOH fusion and ion chromatography

**B determined by Aqua regia digestion and ICPMS

Table S7: Composition of Toro Negro feldpsar separates by Li-Metaborate digestion and ICPMS (Middlebury)

	Na	Mg	Al	K	Ca	Ti	Mn	Fe
	wt%	wt%	wt%	wt%	wt%	wt%	wt%	wt%
TN9	1.95	0.01	7.45	6.97	0.26	0.02	0.00	0.05
TN11	2.25	0.02	6.65	5.44	0.26	0.02	0.00	0.00
TN15	2.12	0.02	5.82	5.02	0.17	0.03	0.00	0.07
TN18	1.12	0.02	2.77	1.27	0.18	0.03	0.00	0.08

Table S8: P-values derived from stepwise K-S tests on all detrital zircon samples

Gray shading denotes samples are probably not drawn from the same distribution (p=0.05)

(e.g. they are likely to be different)

Hu-1.1	**Hu-1.1**												
Hu-2.1	0.08	**Hu-2.1**											
Hu-5	0.16	0.95	**Hu-5**										
Hu-6.1	0.00	0.07	0.03	**Hu-6.1**									
Hu-8	0.02	0.68	0.42	0.43	**Hu-8**								
Hu-8.1	0.00	0.04	0.01	0.84	0.39	**Hu-8.1**							
Hu-9	0.87	0.01	0.02	0.00	0.00	0.00	**Hu-9**						
Hu-13	0.01	0.36	0.22	0.58	0.61	0.60	0.00	**Hu-13**					
Hu-14	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**Hu-14**				
Hu-15	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**Hu-15**			
R. Huaco	0.29	0.32	0.49	0.08	0.16	0.03	0.09	0.20	0.00	0.00	**Huaco**		
R. Bermejo	0.01	0.07	0.01	0.14	0.16	0.02	0.00	0.08	0.00	0.00	0.14	**Bermejo**	
R. de la Troy	0.25	0.00	0.00	0.00	0.00	0.00	0.04	0.00	0.01	0.00	0.32	0.05	**Troya**
R. Jachal	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.09	0.00	0.00	0.00

Table S9: U-Pb zircon data from modern and paleo-detrital samples

Plot ages are calculated as follows:

Grains whose 207/235 age and 206/238 age differ by more than 15% are discarded, except for206/238 ages younger than 100 Ma, which are all are included

For ages <800 Ma the 206/238 age is used and for >1200, the 206/207 age is used

Latitude and longitude are given in WGS84 datum

Rio Huaco **Lat:** **-30.14824** **Lon:** **-68.50532**

Isotope Ratios							Apparent Ages								Concentrations		
$^{207}Pb/^{235}U$	± (2σ)	$^{206}Pb/^{238}U$	± (2σ)	Error Corr.	$^{207}Pb/^{206}Pb$	± (2σ)	$^{207}Pb/^{235}U$ (Ma)	± (Ma)	$^{206}Pb/^{238}U$ (Ma)	± (Ma)	$^{207}Pb/^{206}Pb$ (Ma)	± (Ma)	Plot Age (Ma)	± (Ma)	conc. (%)	U (ppm)	Th (ppm)
0.0030	0.0250	0.0025	0.0005	-0.17	0.3000	0.3300	-4	25	15.9	3.5	-	-	**10.8**	**3.5**	497.50	76	61
0.0134	0.0032	0.0027	0.0002	-0.21	0.0368	0.0097	13.4	3.2	17.1	1	-	-	**17.3**	**1.0**	27.61	817	973
0.0580	0.0360	0.0031	0.0006	-0.02	0.0770	0.0930	41	34	19.8	4.1	-	-	**19.1**	**4.1**	51.71	42	32
0.0157	0.0063	0.0034	0.0002	-0.13	0.0400	0.0220	15.3	6.3	21.8	1.4	-	-	**21.9**	**1.4**	42.48	706	187
0.0490	0.0170	0.0049	0.0005	0.07	0.1040	0.0690	44	17	31.2	3.1	-	-	**28.9**	**3.1**	29.09	86	58
0.0280	0.0130	0.0049	0.0005	0.04	0.0440	0.0260	25	13	31.6	2.9	-	-	**31.7**	**2.9**	26.40	153	69
0.0630	0.0450	0.0050	0.0008	0.29	0.1450	0.0810	54	42	32.1	5.3	-	-	**28.1**	**5.3**	40.56	49	27
0.0210	0.0350	0.0050	0.0010	0.08	0.0210	0.0630	16	35	32.3	6.2	-	-	**33.5**	**6.2**	101.88	30	23
0.0350	0.0031	0.0051	0.0002	0.07	0.0511	0.0047	34.8	3	32.7	1	-	-	**32.5**	**1.0**	6.03	1174	680
0.0250	0.0100	0.0055	0.0005	-0.12	0.0530	0.0200	24	10	35	3.2	-	-	**34.8**	**3.2**	45.83	188	112
0.2380	0.0230	0.0333	0.0011	0.05	0.0539	0.0055	218	19	211.7	6.8	300	190	**211.7**	**6.8**	2.89	178	391
0.2760	0.0200	0.0343	0.0010	-0.01	0.0572	0.0047	247	16	217.3	5.9	500	160	**217.3**	**5.9**	12.02	120	154
0.2940	0.0160	0.0398	0.0009	-0.04	0.0527	0.0032	264	13	251.8	5.3	340	120	**251.8**	**5.3**	4.62	265	204
0.2950	0.0190	0.0399	0.0010	0.00	0.0539	0.0037	264	15	252.3	5.9	350	130	**252.3**	**5.9**	4.43	114	69
0.2750	0.0230	0.0404	0.0013	0.06	0.0476	0.0042	246	18	255.2	8	110	150	**255.2**	**8.0**	3.74	56	41
0.3120	0.0120	0.0433	0.0007	0.20	0.0509	0.0020	273.8	9.5	273.5	4.5	232	78	**273.5**	**4.5**	0.11	289	111
0.3110	0.0170	0.0436	0.0008	0.06	0.0518	0.0029	273	13	274.9	5.2	280	110	**274.9**	**5.2**	0.70	209	208
0.3440	0.0180	0.0458	0.0009	0.15	0.0552	0.0031	298	14	288.8	5.8	370	110	**288.8**	**5.8**	3.09	329	221
0.3080	0.0280	0.0464	0.0016	0.12	0.0466	0.0041	273	22	292	10	70	150	**292**	**10.0**	6.96	131	83
0.3260	0.0190	0.0467	0.0011	0.01	0.0516	0.0034	283	15	294.2	6.6	250	120	**294.2**	**6.6**	3.96	249	366
0.3350	0.0150	0.0470	0.0010	0.23	0.0527	0.0025	293	11	296.4	6.2	290	96	**296.4**	**6.2**	1.16	489	186
0.5430	0.0250	0.0610	0.0014	0.36	0.0657	0.0029	438	16	381.9	8.3	784	88	**381.9**	**8.3**	12.81	824	828
0.4900	0.0200	0.0634	0.0013	-0.01	0.0558	0.0026	404	13	396.1	7.6	419	98	**396.1**	**7.6**	1.96	315	175
0.4910	0.0350	0.0641	0.0019	-0.15	0.0586	0.0047	406	23	400	12	480	150	**400**	**12.0**	1.48	119	62
0.5720	0.0120	0.0652	0.0008	0.34	0.0604	0.0012	458.7	7.9	407.1	5.1	613	44	**407.1**	**5.1**	11.25	342	377
0.5220	0.0370	0.0720	0.0018	0.18	0.0538	0.0038	452	23	448	11	300	140	**448**	**11.0**	0.88	252	337
0.5870	0.0130	0.0736	0.0011	0.10	0.0582	0.0016	467.5	8.6	457.9	6.3	513	59	**457.9**	**6.3**	2.05	248	174
0.5600	0.0240	0.0745	0.0016	0.13	0.0548	0.0025	452	15	463.2	9.7	381	96	**463.2**	**9.7**	2.48	267	412
0.5460	0.0320	0.0749	0.0017	0.03	0.0532	0.0034	449	21	465	10	340	120	**465**	**10.0**	3.56	168	93
0.6270	0.0150	0.0761	0.0010	0.32	0.0597	0.0013	493.8	9.2	473.2	6.1	576	47	**473.2**	**6.1**	4.17	732	139
0.6410	0.0370	0.0826	0.0021	0.08	0.0540	0.0036	501	24	511	12	420	120	**511**	**12.0**	2.00	186	97
0.6560	0.0340	0.0839	0.0021	-0.11	0.0558	0.0033	518	21	519	12	460	120	**519**	**12.0**	0.19	150	125
0.6620	0.0170	0.0842	0.0012	0.24	0.0573	0.0014	516	10	521.1	7.4	488	54	**521.1**	**7.4**	0.99	635	708
0.6770	0.0300	0.0850	0.0016	0.21	0.0578	0.0025	522	18	526	9.5	468	92	**526**	**9.5**	0.77	248	163
0.6730	0.0240	0.0851	0.0016	0.10	0.0572	0.0023	522	14	526.4	9.2	475	86	**526.4**	**9.2**	0.84	290	160
0.6680	0.0390	0.0852	0.0023	0.01	0.0590	0.0039	517	23	527	14	480	130	**527**	**14.0**	1.93	138	152
0.6840	0.0520	0.0872	0.0023	0.04	0.0579	0.0044	529	31	538	14	470	150	**538**	**14.0**	1.70	71	53
0.7440	0.0490	0.0883	0.0024	-0.01	0.0610	0.0042	559	29	545	14	570	140	**545**	**14.0**	2.50	78	62
0.8130	0.0430	0.0886	0.0025	0.07	0.0662	0.0037	597	25	549	15	760	120	**549**	**15.0**	8.04	59	9
0.7700	0.0310	0.0893	0.0017	-0.02	0.0613	0.0031	580	17	551	10	610	100	**551**	**10.0**	5.00	111	72
0.7280	0.0310	0.0896	0.0019	-0.01	0.0592	0.0028	556	18	553	11	525	96	**553**	**11.0**	0.54	179	112
0.8830	0.0720	0.0904	0.0023	0.52	0.0717	0.0054	625	36	557	14	850	130	**557**	**14.0**	10.88	117	219
0.7200	0.0200	0.0904	0.0010	0.18	0.0581	0.0017	549	12	557.7	6.2	529	62	**557.7**	**6.2**	1.58	490	215
0.7300	0.0210	0.0904	0.0012	0.13	0.0588	0.0017	555	13	557.9	6.8	524	66	**557.9**	**6.8**	0.52	555	796
0.7700	0.0310	0.0909	0.0017	0.09	0.0586	0.0026	580	18	561	10	552	91	**561**	**10.0**	3.28	130	133
0.8150	0.0270	0.0959	0.0017	0.67	0.0623	0.0018	602	15	590.1	9.9	685	62	**590.1**	**9.9**	1.98	959	204
0.8120	0.0310	0.0984	0.0016	0.09	0.0595	0.0023	604	18	604.8	9.6	563	83	**604.8**	**9.6**	0.13	177	227
0.8200	0.0220	0.1017	0.0017	0.23	0.0593	0.0015	606	13	625	10	566	57	**625**	**10.0**	3.14	557	203
0.8680	0.0500	0.1021	0.0024	-0.04	0.0607	0.0038	630	27	626	14	590	120	**626**	**14.0**	0.63	101	100
0.8460	0.0330	0.1027	0.0017	0.35	0.0594	0.0022	622	18	629.9	9.9	558	80	**629.9**	**9.9**	1.27	274	272
0.8310	0.0700	0.1041	0.0038	-0.08	0.0576	0.0054	602	41	637	22	470	180	**637**	**22.0**	5.81	61	30
0.8990	0.0290	0.1071	0.0018	0.25	0.0616	0.0022	651	15	655	11	636	76	**655**	**11.0**	0.61	299	247
0.8870	0.0330	0.1080	0.0020	0.12	0.0599	0.0024	648	18	661	11	591	79	**661**	**11.0**	2.01	244	100
1.0930	0.0410	0.1267	0.0026	0.19	0.0636	0.0025	753	19	768	15	701	84	**768**	**15.0**	1.99	117	87
1.4710	0.0900	0.1293	0.0047	0.91	0.0800	0.0024	907	38	782	27	1183	63	**782**	**27.0**	13.78	599	157
1.4860	0.0630	0.1546	0.0027	0.11	0.0693	0.0031	918	26	926	15	868	92	**868**	**92.0**	0.87	96	34
1.8900	0.1100	0.1856	0.0050	0.20	0.0754	0.0044	1072	38	1096	27	990	120	**990**	**120.0**	2.24	47	20
1.4880	0.0430	0.1470	0.0033	0.64	0.0727	0.0017	925	18	887	19	1000	47	**1000**	**47.0**	4.11	539	317
1.7130	0.0470	0.1682	0.0022	0.13	0.0732	0.0022	1011	17	1002	12	1010	59	**1010**	**59.0**	0.89	156	69
1.7440	0.0420	0.1708	0.0025	0.12	0.0739	0.0020	1023	15	1016	14	1029	55	**1029**	**55.0**	0.68	304	103
1.7200	0.0480	0.1667	0.0029	-0.04	0.0737	0.0024	1016	18	993	16	1040	67	**1040**	**67.0**	2.26	97	31
1.9400	0.1500	0.1838	0.0073	0.11	0.0785	0.0066	1111	52	1084	40	1060	160	**1060**	**160.0**	2.43	42	31
1.9570	0.0330	0.1871	0.0020	0.46	0.0748	0.0012	1100	11	1106	11	1072	30	**1072**	**30.0**	0.55	363	169
1.7250	0.0710	0.1687	0.0033	0.06	0.0755	0.0036	1020	29	1007	18	1085	96	**1085**	**96.0**	1.27	116	55

1.8030	0.0560	0.1731	0.0031	0.21	0.0760	0.0023	1050	19	1028	17	1092	66	**1092**	**66.0**	2.10	63	43
2.0260	0.0360	0.1927	0.0021	0.08	0.0762	0.0016	1124	12	1136	11	1094	41	**1094**	**41.0**	1.07	445	252
1.9450	0.0610	0.1849	0.0035	0.19	0.0769	0.0026	1093	22	1093	19	1107	70	**1107**	**70.0**	0.00	142	12
2.1360	0.0460	0.1930	0.0025	0.25	0.0798	0.0017	1160	15	1137	14	1188	41	**1188**	**41.0**	1.98	215	208
2.3030	0.0760	0.2097	0.0047	0.08	0.0795	0.0032	1209	23	1226	25	1190	79	**1190**	**79.0**	1.41	111	63
2.3500	0.0750	0.2111	0.0035	0.22	0.0808	0.0027	1230	24	1234	18	1208	67	**1208**	**67.0**	0.33	84	146
2.3260	0.0490	0.2095	0.0029	0.25	0.0824	0.0019	1219	15	1225	15	1241	45	**1241**	**45.0**	0.49	203	176
2.7650	0.0910	0.2352	0.0055	0.38	0.0850	0.0029	1349	25	1360	28	1288	68	**1288**	**68.0**	0.82	96	128
2.1910	0.0920	0.1784	0.0047	0.57	0.0892	0.0030	1175	29	1057	26	1403	64	**1403**	**64.0**	10.04	137	51
3.1030	0.0540	0.2435	0.0032	0.11	0.0908	0.0017	1434	13	1404	16	1459	38	**1459**	**38.0**	2.09	87	40
3.7900	0.1100	0.2623	0.0053	0.40	0.1039	0.0031	1590	23	1503	27	1704	53	**1704**	**53.0**	5.47	109	75
4.6000	0.0680	0.3075	0.0032	0.23	0.1076	0.0017	1747	12	1728	16	1759	29	**1759**	**29.0**	1.09	291	60
5.2800	0.0810	0.3436	0.0044	0.39	0.1105	0.0017	1863	13	1907	22	1807	28	**1807**	**28.0**	2.36	301	371
9.2800	0.1100	0.4259	0.0043	0.50	0.1571	0.0017	2368	11	2286	20	2424	18	**2424**	**18.0**	3.46	147	64
10.7300	0.1100	0.4190	0.0043	0.45	0.1859	0.0020	2502	10	2255	20	2705	18	**2705**	**18.0**	9.87	385	178

Rio Bermejo **Lat:** **-29.35045** **Lon:** **-68.24095**

Isotope Ratios							**Apparent Ages**								**Concentrations**		
$^{207}Pb/^{235}U$	± (2σ)	$^{206}Pb/^{238}U$	± (2σ)	Error Corr.	$^{207}Pb/^{206}Pb$	± (2σ)	$^{207}Pb/^{235}U$ (Ma)	± (Ma)	$^{206}Pb/^{238}U$ (Ma)	± (Ma)	$^{207}Pb/^{206}Pb$ (Ma)	± (Ma)	Plot Age (Ma)	± (Ma)	conc. (%)	U (ppm)	Th (ppm)
0.0010	0.0094	0.0004	0.0001	0.15	0.2100	0.2000	0.4	9.5	2.45	0.81	-	-	**1.9**	**0.8**	512.50	297	198
0.0086	0.0089	0.0010	0.0001	0.00	0.0700	0.1400	7.7	8.9	6.41	0.85	-	-	**6.2**	**0.9**	16.75	225	141
0.0145	0.0072	0.0014	0.0002	0.01	0.1110	0.0590	14.2	7.2	8.67	0.99	-	-	**8.0**	**1.0**	38.94	396	304
0.0070	0.0170	0.0014	0.0003	-0.09	0.0300	0.1700	4	17	9.4	1.6	-	-	**9.3**	**1.6**	135.00	114	97
0.0570	0.0110	0.0042	0.0003	0.05	0.1030	0.0270	55	11	27	1.8	-	-	**25.1**	**1.8**	50.91	282	88
0.0270	0.0130	0.0060	0.0003	0.04	0.0410	0.0210	25	13	38.6	2.2	-	-	**38.8**	**2.2**	54.40	153	54
0.1570	0.0330	0.0227	0.0010	-0.07	0.0470	0.0110	139	29	144.5	6.1	-	-	**144.5**	**6.1**	3.96	114	84
0.2350	0.0100	0.0299	0.0005	0.24	0.0524	0.0022	213.6	8.1	189.6	3.2	272	85	**189.6**	**3.2**	11.24	612	407
0.2590	0.0230	0.0344	0.0007	0.18	0.0553	0.0051	240	18	218	4.6	410	170	**218**	**4.6**	9.17	209	246
0.2580	0.0350	0.0371	0.0014	-0.03	0.0495	0.0065	228	28	234.5	8.6	160	240	**234.5**	**8.6**	2.85	147	114
0.2770	0.0130	0.0376	0.0008	0.11	0.0514	0.0025	247	11	237.9	4.6	250	100	**237.9**	**4.6**	3.68	366	177
0.2670	0.0500	0.0402	0.0022	0.04	0.0474	0.0091	233	41	254	14	20	330	**254**	**14.0**	9.01	131	92
0.2680	0.0630	0.0411	0.0017	-0.21	0.0460	0.0110	232	51	259	10	110	430	**259**	**10.0**	11.64	161	141
0.3170	0.0160	0.0423	0.0008	0.26	0.0527	0.0025	282	12	266.7	5.1	320	100	**266.7**	**5.1**	5.43	561	265
0.3070	0.0150	0.0423	0.0009	0.23	0.0529	0.0026	272	12	267	5.4	290	100	**267**	**5.4**	1.84	703	543
0.3070	0.0430	0.0425	0.0015	-0.07	0.0558	0.0082	277	32	268.4	9.4	350	240	**268.4**	**9.4**	3.10	118	38
0.3320	0.0250	0.0430	0.0009	-0.01	0.0544	0.0045	289	19	271.5	5.4	360	170	**271.5**	**5.4**	6.06	163	89
0.3580	0.0280	0.0446	0.0009	0.10	0.0573	0.0045	312	21	281.4	5.5	480	160	**281.4**	**5.5**	9.81	182	64
0.3400	0.0440	0.0451	0.0017	0.20	0.0547	0.0065	297	33	285	10	310	230	**285**	**10.0**	4.04	131	86
0.3290	0.0280	0.0457	0.0009	-0.02	0.0526	0.0045	288	22	287.8	5.7	270	170	**287.8**	**5.7**	0.07	197	123
0.3290	0.0320	0.0474	0.0015	-0.17	0.0509	0.0052	283	24	299.1	8.8	200	190	**299.1**	**8.8**	5.69	138	97
0.3750	0.0610	0.0484	0.0020	0.19	0.0579	0.0093	319	43	304	12	420	290	**304**	**12.0**	4.70	98	50
0.3870	0.0330	0.0489	0.0015	0.06	0.0555	0.0053	327	25	307.8	9.4	410	180	**307.8**	**9.4**	5.87	117	57
0.3790	0.0890	0.0495	0.0026	0.05	0.0610	0.0140	324	66	311	16	390	380	**311**	**16.0**	4.01	22	52
0.3650	0.0350	0.0521	0.0015	0.13	0.0488	0.0047	318	25	327.2	8.9	180	180	**327.2**	**8.9**	2.89	222	311
0.3940	0.0340	0.0525	0.0015	0.19	0.0532	0.0045	334	26	329.5	9.3	310	160	**329.5**	**9.3**	1.35	173	80
0.4130	0.0520	0.0531	0.0015	0.13	0.0548	0.0069	350	38	333.6	9.3	380	230	**333.6**	**9.3**	4.69	105	105
0.3820	0.0210	0.0535	0.0010	0.09	0.0516	0.0030	327	16	336.2	6.3	250	120	**336.2**	**6.3**	2.81	129	120
0.3920	0.0420	0.0544	0.0016	-0.26	0.0540	0.0063	333	33	341.4	9.6	250	210	**341.4**	**9.6**	2.52	84	60
0.4040	0.0170	0.0547	0.0008	0.16	0.0532	0.0023	344	12	343.4	5.1	307	89	**343.4**	**5.1**	0.17	461	164
0.4300	0.0360	0.0556	0.0019	0.39	0.0552	0.0043	368	24	349	11	390	170	**349**	**11.0**	5.16	747	52
0.4240	0.0190	0.0569	0.0010	0.06	0.0526	0.0027	363	14	356.4	6.1	350	100	**356.4**	**6.1**	1.82	417	198
0.4990	0.0220	0.0616	0.0009	0.13	0.0571	0.0025	410	15	385	5.7	450	93	**385**	**5.7**	6.10	334	147
0.5690	0.0210	0.0728	0.0012	0.18	0.0562	0.0020	455	13	452.8	7.4	450	80	**452.8**	**7.4**	0.48	551	52
0.6000	0.0360	0.0728	0.0016	0.07	0.0597	0.0037	478	24	452.8	9.7	610	130	**452.8**	**9.7**	5.27	45	22
0.6060	0.0390	0.0740	0.0014	0.23	0.0586	0.0038	476	24	459.8	8.2	560	130	**459.8**	**8.2**	3.40	255	116
0.5930	0.0710	0.0745	0.0029	0.05	0.0583	0.0075	489	45	463	17	440	250	**463**	**17.0**	5.32	55	32
0.6000	0.1000	0.0745	0.0032	0.21	0.0610	0.0110	468	63	463	19	560	340	**463**	**19.0**	1.07	88	64
0.5830	0.0590	0.0747	0.0029	0.23	0.0539	0.0054	457	37	464	18	330	180	**464**	**18.0**	1.53	101	23
0.5940	0.0250	0.0751	0.0011	0.14	0.0565	0.0023	472	16	466.9	6.8	446	88	**466.9**	**6.8**	1.08	496	330
0.5690	0.0300	0.0752	0.0014	0.26	0.0537	0.0028	459	19	467.2	8.4	370	100	**467.2**	**8.4**	1.79	74	62
0.6270	0.0960	0.0757	0.0036	0.28	0.0585	0.0081	479	59	470	21	420	270	**470**	**21.0**	1.88	80	46
0.6200	0.0280	0.0761	0.0014	0.26	0.0586	0.0026	490	18	472.9	8.6	524	93	**472.9**	**8.6**	3.49	346	88
0.5890	0.0360	0.0766	0.0020	0.23	0.0561	0.0036	465	24	476	12	420	130	**476**	**12.0**	2.37	217	208
0.5940	0.0680	0.0767	0.0020	0.13	0.0567	0.0066	467	45	478	12	340	220	**478**	**12.0**	2.36	86	8
0.6280	0.0460	0.0773	0.0024	0.49	0.0565	0.0035	491	29	480	14	460	140	**480**	**14.0**	2.24	441	210
0.6200	0.0410	0.0823	0.0018	0.24	0.0547	0.0037	487	26	510	11	360	130	**510**	**11.0**	4.72	101	104
0.6270	0.0390	0.0837	0.0018	0.15	0.0550	0.0034	495	24	518	11	360	120	**518**	**11.0**	4.65	81	41
0.7100	0.0600	0.0844	0.0023	0.28	0.0599	0.0048	534	35	524	14	500	170	**524**	**14.0**	1.87	179	70
0.7030	0.0410	0.0858	0.0019	0.14	0.0593	0.0035	544	23	531	12	540	130	**531**	**12.0**	2.39	127	68
0.7940	0.0760	0.0870	0.0031	0.04	0.0689	0.0071	585	44	537	18	750	210	**537**	**18.0**	8.21	77	26
0.7970	0.0220	0.0958	0.0012	0.29	0.0594	0.0016	594	12	589.7	7.3	562	60	**589.7**	**7.3**	0.72	415	71
0.9360	0.0790	0.0968	0.0042	0.21	0.0710	0.0062	675	43	595	25	840	190	**595**	**25.0**	11.85	43	5
0.8080	0.0650	0.0978	0.0027	-0.02	0.0605	0.0049	595	38	601	16	520	170	**601**	**16.0**	1.01	72	107

0.8340	0.0450	0.0978	0.0025	0.20	0.0622	0.0033	617	25	601	14	690	110	**601**	**14.0**	2.59	144	28
0.8480	0.0420	0.0987	0.0022	0.22	0.0624	0.0032	617	23	606	13	620	110	**606**	**13.0**	1.78	57	31
0.8920	0.0330	0.1057	0.0018	0.22	0.0613	0.0023	649	18	647	10	623	81	**647**	**10.0**	0.31	237	85
1.2350	0.0520	0.1196	0.0027	0.20	0.0734	0.0035	817	25	728	16	1028	96	**728**	**16.0**	10.89	215	50
1.1370	0.0360	0.1207	0.0019	0.35	0.0689	0.0021	773	17	734	11	885	65	**734**	**11.0**	5.05	222	51
1.3120	0.0570	0.1430	0.0028	0.06	0.0653	0.0031	852	24	861	16	770	93	**770**	**93.0**	1.06	73	65
1.4860	0.0960	0.1585	0.0044	0.30	0.0689	0.0045	929	36	948	24	850	130	**850**	**130.0**	2.05	106	78
1.9800	0.4000	0.1810	0.0120	0.19	0.0830	0.0170	1030	160	1071	66	870	470	**870**	**470.0**	3.98	11	7
1.6800	0.0750	0.1674	0.0035	0.27	0.0723	0.0031	994	28	1001	20	964	91	**964**	**91.0**	0.70	115	70
1.5800	0.1200	0.1513	0.0058	0.11	0.0737	0.0060	948	47	907	32	980	170	**980**	**170.0**	4.32	77	36
1.6510	0.0930	0.1652	0.0036	0.34	0.0726	0.0041	1000	37	985	20	990	120	**990**	**120.0**	1.50	44	18
1.7140	0.0900	0.1677	0.0040	0.03	0.0737	0.0042	1010	35	999	22	1000	130	**1000**	**130.0**	1.09	85	32
1.5030	0.0330	0.1429	0.0020	0.43	0.0736	0.0017	934	14	861	11	1026	45	**1026**	**45.0**	7.82	437	84
1.7680	0.0710	0.1747	0.0028	0.19	0.0743	0.0030	1036	26	1038	16	1030	82	**1030**	**82.0**	0.19	136	69
1.9280	0.0950	0.1831	0.0035	0.20	0.0745	0.0037	1083	34	1083	19	1040	110	**1040**	**110.0**	0.00	73	36
1.8880	0.0850	0.1796	0.0038	0.01	0.0745	0.0037	1076	30	1064	21	1041	96	**1041**	**96.0**	1.12	96	55
1.9580	0.0620	0.1882	0.0039	0.69	0.0745	0.0017	1097	21	1113	21	1050	46	**1050**	**46.0**	1.46	314	126
1.7620	0.0730	0.1720	0.0029	0.32	0.0739	0.0028	1031	26	1023	16	1052	72	**1052**	**72.0**	0.78	160	59
1.5430	0.0540	0.1495	0.0033	0.43	0.0747	0.0026	944	22	897	19	1054	70	**1054**	**70.0**	4.98	135	37
1.7480	0.0600	0.1677	0.0028	0.28	0.0746	0.0025	1026	23	999	16	1065	69	**1065**	**69.0**	2.63	192	94
1.8270	0.0610	0.1752	0.0038	0.34	0.0753	0.0027	1051	22	1040	21	1070	69	**1070**	**69.0**	1.05	244	94
2.0500	0.2200	0.1883	0.0073	0.24	0.0789	0.0086	1119	71	1110	40	1070	210	**1070**	**210.0**	0.80	29	31
1.7130	0.0830	0.1624	0.0035	0.31	0.0748	0.0034	1013	31	970	19	1080	90	**1080**	**90.0**	4.24	161	118
2.0900	0.1100	0.2028	0.0037	0.15	0.0761	0.0041	1143	37	1190	20	1080	110	**1080**	**110.0**	4.11	113	86
1.8300	0.1000	0.1735	0.0040	0.37	0.0764	0.0041	1042	36	1030	22	1080	110	**1080**	**110.0**	1.15	145	36
1.7060	0.0650	0.1595	0.0032	0.32	0.0767	0.0031	1012	25	955	18	1083	83	**1083**	**83.0**	5.63	131	54
1.4790	0.0600	0.1394	0.0033	0.12	0.0769	0.0030	921	24	841	18	1086	80	**1086**	**80.0**	8.69	258	49
2.0700	0.1100	0.1956	0.0050	0.15	0.0757	0.0043	1130	38	1154	28	1090	120	**1090**	**120.0**	2.12	90	31
1.8250	0.0720	0.1728	0.0036	0.30	0.0760	0.0028	1049	26	1027	20	1091	75	**1091**	**75.0**	2.10	301	272
1.5650	0.0730	0.1418	0.0025	-0.06	0.0823	0.0059	942	25	855	14	1113	94	**1113**	**94.0**	9.24	188	88
2.0280	0.0460	0.1892	0.0039	0.41	0.0772	0.0018	1125	15	1117	21	1125	49	**1125**	**49.0**	0.71	313	91
1.9320	0.0670	0.1782	0.0040	0.63	0.0775	0.0020	1092	23	1056	22	1130	51	**1130**	**51.0**	3.30	570	233
1.8940	0.0870	0.1765	0.0032	0.08	0.0771	0.0036	1080	31	1047	18	1138	99	**1138**	**99.0**	3.06	35	18
2.2740	0.0560	0.2085	0.0031	0.22	0.0781	0.0021	1203	17	1222	17	1141	55	**1141**	**55.0**	1.58	225	41
1.7250	0.0350	0.1557	0.0020	0.36	0.0777	0.0016	1020	14	933	11	1142	41	**1142**	**41.0**	8.53	181	42
1.9550	0.0940	0.1829	0.0036	0.12	0.0768	0.0037	1103	32	1082	20	1144	95	**1144**	**95.0**	1.90	94	41
1.9300	0.1100	0.1769	0.0036	0.12	0.0796	0.0046	1106	36	1049	20	1160	100	**1160**	**100.0**	5.15	116	59
2.0890	0.0700	0.1927	0.0041	0.30	0.0791	0.0026	1146	23	1135	22	1162	67	**1162**	**67.0**	0.96	268	71
2.2700	0.1100	0.2034	0.0045	0.21	0.0797	0.0041	1197	36	1193	24	1170	100	**1170**	**100.0**	0.33	108	18
1.8990	0.0950	0.1708	0.0046	0.55	0.0797	0.0031	1081	34	1016	25	1171	80	**1171**	**80.0**	6.01	180	59
1.9600	0.0860	0.1782	0.0043	0.32	0.0796	0.0035	1097	30	1056	24	1179	86	**1179**	**86.0**	3.74	99	49
2.1890	0.0750	0.1979	0.0035	0.29	0.0801	0.0026	1178	24	1166	19	1182	61	**1182**	**61.0**	1.02	58	15
2.6000	0.2000	0.2220	0.0083	0.11	0.0852	0.0071	1287	61	1295	43	1210	180	**1210**	**180.0**	0.62	36	30
2.1210	0.0810	0.1857	0.0033	0.13	0.0811	0.0034	1157	27	1097	18	1232	81	**1232**	**81.0**	5.19	43	31
1.6420	0.0660	0.1443	0.0029	0.30	0.0806	0.0031	993	25	870	16	1233	75	**1233**	**75.0**	12.39	140	122
2.0340	0.0690	0.1743	0.0034	0.32	0.0825	0.0027	1128	24	1035	19	1234	67	**1234**	**67.0**	8.24	130	68
2.3200	0.1400	0.2021	0.0050	0.18	0.0841	0.0053	1220	43	1185	27	1240	130	**1240**	**130.0**	2.87	50	17
2.3900	0.1100	0.2020	0.0045	0.22	0.0844	0.0040	1231	34	1188	23	1254	99	**1254**	**99.0**	3.49	88	37
2.4500	0.1400	0.2113	0.0060	0.42	0.0843	0.0042	1247	41	1235	32	1265	99	**1265**	**99.0**	0.96	172	69
2.0700	0.1500	0.1844	0.0064	0.46	0.0825	0.0059	1154	53	1090	35	1300	140	**1300**	**140.0**	5.55	101	33
2.7400	0.1100	0.2216	0.0044	0.43	0.0882	0.0032	1343	28	1292	23	1389	71	**1389**	**71.0**	3.80	115	41
3.0700	0.1100	0.2480	0.0048	0.26	0.0883	0.0030	1420	27	1427	25	1396	64	**1396**	**64.0**	0.49	115	38
3.2600	0.1200	0.2581	0.0053	0.15	0.0905	0.0037	1475	27	1479	27	1414	77	**1414**	**77.0**	0.27	108	102
2.4140	0.0730	0.1888	0.0042	0.22	0.0923	0.0030	1252	22	1114	23	1464	61	**1464**	**61.0**	11.02	142	38
4.0900	0.2700	0.2746	0.0085	0.30	0.1041	0.0059	1638	53	1563	43	1689	97	**1689**	**97.0**	4.58	85	46
4.7300	0.1100	0.3182	0.0038	0.34	0.1078	0.0024	1772	19	1780	19	1753	40	**1753**	**40.0**	0.45	211	559
5.5800	0.1400	0.3625	0.0060	0.48	0.1102	0.0025	1908	22	1992	28	1796	42	**1796**	**42.0**	4.40	43	31
5.2400	0.1600	0.3202	0.0068	0.58	0.1158	0.0032	1860	26	1789	33	1896	52	**1896**	**52.0**	3.82	120	95
12.9300	0.2500	0.4933	0.0086	0.60	0.1889	0.0033	2672	19	2582	37	2728	28	**2728**	**28.0**	3.37	128	54
~~1.9930~~	~~0.0790~~	~~0.1584~~	~~0.0034~~	~~0.21~~	~~0.0875~~	~~0.0034~~	~~1123~~	~~26~~	~~950~~	~~19~~	~~1373~~	~~76~~			~~15.41~~	~~133~~	~~200~~

Rio La Troya **Lat:** **29.52976** **Lon:** **68.50089**

Isotope Ratios							Apparent Ages								Concentrations		
$^{207}Pb/^{235}U$	± (2σ)	$^{206}Pb/^{238}U$	± (2σ)	Error Corr.	$^{207}Pb/^{206}Pb$	± (2σ)	$^{207}Pb/^{235}U$ (Ma)	± (Ma)	$^{206}Pb/^{238}U$ (Ma)	± (Ma)	$^{207}Pb/^{206}Pb$ (Ma)	± (Ma)	Plot Age (Ma)	± (Ma)	conc. (%)	U (ppm)	Th (ppm)
0.0270	0.0160	0.0029	0.0004	-0.07	0.0580	0.0630	24	15	18.4	2.6	-	-	**18.2**	**2.6**	23.33	145	165
0.0188	0.0031	0.0031	0.0001	-0.24	0.0440	0.0077	18.8	3.1	19.85	0.83	-	-	**19.9**	**0.8**	5.59	283	92
0.0202	0.0061	0.0034	0.0002	-0.02	0.0420	0.0140	19.8	6.1	21.9	1.2	-	-	**22.0**	**1.2**	10.61	542	495
0.0240	0.0039	0.0037	0.0002	-0.03	0.0534	0.0089	23.9	3.8	23.7	1.1	-	-	**23.3**	**1.1**	0.84	224	127
0.0206	0.0073	0.0038	0.0003	0.08	0.0400	0.0160	20.8	7.4	24.7	1.7	-	-	**24.9**	**1.7**	18.75	123	89
0.0245	0.0085	0.0047	0.0004	0.01	0.0570	0.0230	23.6	8.4	30	2.5	-	-	**29.6**	**2.5**	27.12	73	70
0.1120	0.0240	0.0047	0.0005	0.21	0.1980	0.0570	102	21	30.4	3.1	-	-	**24.7**	**3.1**	70.20	140	105
0.0530	0.0230	0.0050	0.0006	0.01	0.0580	0.0560	53	21	32.3	3.9	-	-	**31.9**	**3.9**	39.06	23	15

0.0317	0.0039	0.0051	0.0002	0.12	0.0448	0.0056	32.8	3.8	32.5	1.3	-	-	**32.5**	**1.3**	0.91	750	404
0.0460	0.0110	0.0051	0.0004	-0.08	0.0870	0.0270	46	11	32.9	2.5	-	-	**31.3**	**2.5**	28.48	192	147
0.0190	0.0130	0.0052	0.0004	0.06	0.0470	0.0240	23	13	33.5	2.8	-	-	**33.5**	**2.8**	45.65	188	110
0.0520	0.0250	0.0054	0.0005	-0.03	0.1140	0.0550	45	24	34.5	3.4	-	-	**31.6**	**3.4**	23.33	55	24
0.0358	0.0021	0.0054	0.0001	0.11	0.0473	0.0028	35.7	2.1	34.97	0.66	-	-	**34.9**	**0.7**	2.04	561	112
0.0494	0.0093	0.0057	0.0004	0.05	0.0650	0.0130	47.7	9	36.3	2.3	-	-	**35.5**	**2.3**	23.90	73	41
0.0826	0.0085	0.0120	0.0005	0.06	0.0520	0.0058	82	7.6	77	2.9	-	-	**77**	**2.9**	6.10	103	47
0.2530	0.0170	0.0337	0.0010	0.02	0.0539	0.0037	229	14	213.7	6.1	330	140	**213.7**	**6.1**	6.68	160	132
0.2580	0.0160	0.0343	0.0007	0.18	0.0543	0.0035	231	13	217.3	4.5	320	120	**217.3**	**4.5**	5.93	469	331
0.2920	0.0290	0.0369	0.0014	0.05	0.0573	0.0058	256	22	233.6	8.8	390	190	**233.6**	**8.8**	8.75	32	17
0.2730	0.0120	0.0377	0.0007	0.02	0.0525	0.0026	243.4	9.7	238.5	4.6	290	100	**238.5**	**4.6**	2.01	415	282
0.2970	0.0270	0.0390	0.0013	0.02	0.0559	0.0054	258	21	246.4	8.2	340	180	**246.4**	**8.2**	4.50	31	18
0.3100	0.0150	0.0391	0.0008	0.24	0.0574	0.0026	273	11	247.2	4.6	472	97	**247.2**	**4.6**	9.45	1026	1310
0.3350	0.0340	0.0400	0.0018	-0.21	0.0651	0.0078	294	28	253	11	570	230	**253**	**11.0**	13.95	25	15
0.3020	0.0110	0.0402	0.0008	0.37	0.0551	0.0020	267.4	8.3	254	4.9	391	78	**254**	**4.9**	5.01	207	224
0.3000	0.0100	0.0411	0.0008	0.38	0.0522	0.0016	267.4	7.6	259.7	5.1	283	67	**259.7**	**5.1**	2.88	1469	1254
0.2980	0.0110	0.0413	0.0008	0.09	0.0512	0.0021	263.4	8.9	261	4.8	245	83	**261**	**4.8**	0.91	147	121
0.3730	0.0380	0.0429	0.0016	0.10	0.0654	0.0069	315	28	270.6	9.9	620	200	**270.6**	**9.9**	14.10	30	19
0.3229	0.0097	0.0439	0.0009	0.08	0.0538	0.0020	283.4	7.5	276.7	5.7	328	75	**276.7**	**5.7**	2.36	167	90
0.3240	0.0250	0.0447	0.0013	-0.03	0.0515	0.0043	282	19	282	8.1	290	160	**282**	**8.1**	0.00	34	22
0.3270	0.0170	0.0448	0.0009	0.11	0.0528	0.0028	289	13	282.3	5.7	310	110	**282.3**	**5.7**	2.32	90	62
0.3170	0.0150	0.0451	0.0008	0.03	0.0525	0.0028	280	11	284.4	5.1	290	100	**284.4**	**5.1**	1.57	301	98
0.3194	0.0097	0.0455	0.0006	0.07	0.0503	0.0016	281.4	7.5	287.1	3.8	215	68	**287.1**	**3.8**	2.03	245	108
0.3520	0.0100	0.0457	0.0007	0.44	0.0558	0.0014	306	7.4	287.8	4.2	441	54	**287.8**	**4.2**	5.95	1172	655
0.3700	0.0240	0.0485	0.0016	0.04	0.0553	0.0040	321	17	305	10	370	140	**305**	**10.0**	4.98	84	38
0.3990	0.0200	0.0500	0.0016	0.45	0.0561	0.0027	339	14	314.5	9.8	437	98	**314.5**	**9.8**	7.23	328	48
0.3520	0.0200	0.0501	0.0012	0.06	0.0504	0.0030	306	15	315.2	7.4	220	120	**315.2**	**7.4**	3.01	79	51
0.3800	0.0140	0.0504	0.0011	0.20	0.0545	0.0022	325	11	316.8	6.8	376	85	**316.8**	**6.8**	2.52	118	82
0.3570	0.0170	0.0506	0.0011	0.01	0.0502	0.0027	307	13	317.9	6.5	220	110	**317.9**	**6.5**	3.55	305	211
0.3720	0.0130	0.0508	0.0008	0.05	0.0535	0.0021	320.5	9.3	319.4	4.6	343	83	**319.4**	**4.6**	0.34	147	32
0.3690	0.0100	0.0510	0.0007	-0.14	0.0527	0.0017	317.9	7.4	320.8	4.1	305	67	**320.8**	**4.1**	0.91	795	558
0.3720	0.0240	0.0516	0.0013	0.04	0.0530	0.0036	319	18	324.9	8.3	290	130	**324.9**	**8.3**	1.85	165	201
0.4440	0.0200	0.0550	0.0011	0.21	0.0573	0.0024	375	15	345.3	6.5	516	96	**345.3**	**6.5**	7.92	365	81
0.4890	0.0150	0.0631	0.0009	0.16	0.0558	0.0017	403	10	394.2	5.4	434	69	**394.2**	**5.4**	2.18	183	58
0.5800	0.0410	0.0713	0.0020	0.19	0.0593	0.0042	463	25	443	12	550	140	**443**	**12.0**	4.32	86	16
0.5600	0.0190	0.0728	0.0015	0.17	0.0557	0.0022	452	13	452.8	8.9	415	84	**452.8**	**8.9**	0.18	268	139
0.5720	0.0140	0.0739	0.0012	0.35	0.0562	0.0013	459.7	9	459.3	7.3	436	51	**459.3**	**7.3**	0.09	837	573
0.5880	0.0190	0.0741	0.0012	0.22	0.0572	0.0019	468	12	460.5	7.4	469	71	**460.5**	**7.4**	1.60	863	80
0.5680	0.0210	0.0741	0.0013	0.13	0.0555	0.0022	457	14	460.7	7.8	403	84	**460.7**	**7.8**	0.81	88	55
0.6330	0.0210	0.0798	0.0012	0.07	0.0582	0.0020	499	14	494.6	7.1	540	75	**494.6**	**7.1**	0.88	83	36
0.6570	0.0190	0.0816	0.0010	0.03	0.0581	0.0017	512	11	505.5	5.8	534	68	**505.5**	**5.8**	1.27	117	12
0.6780	0.0360	0.0851	0.0019	0.20	0.0576	0.0034	521	22	526	11	480	120	**526**	**11.0**	0.96	109	131
0.7110	0.0360	0.0859	0.0020	0.29	0.0589	0.0031	544	21	531	12	560	110	**531**	**12.0**	2.39	80	22
0.6740	0.0230	0.0865	0.0013	0.17	0.0557	0.0018	522	14	534.5	7.4	423	73	**534.5**	**7.4**	2.39	143	56
0.7550	0.0270	0.0877	0.0020	0.25	0.0618	0.0024	572	16	543	12	646	78	**543**	**12.0**	5.07	74	28
0.7250	0.0390	0.0879	0.0020	0.08	0.0595	0.0035	551	23	543	12	560	120	**543**	**12.0**	1.45	29	15
0.7750	0.0170	0.0938	0.0012	0.23	0.0593	0.0014	583.1	9.7	578.1	6.9	586	49	**578.1**	**6.9**	0.86	237	160
0.7820	0.0160	0.0964	0.0012	0.37	0.0588	0.0012	588.1	9.2	593.3	6.8	555	43	**593.3**	**6.8**	0.88	253	35
0.8900	0.1300	0.0983	0.0057	0.10	0.0650	0.0100	611	70	605	34	460	290	**605**	**34.0**	0.98	7	5
0.8970	0.0140	0.1053	0.0011	0.32	0.0614	0.0010	650.7	7.2	645.1	6.6	655	37	**645.1**	**6.6**	0.86	1328	979
0.9520	0.0160	0.1081	0.0012	0.46	0.0631	0.0011	678.1	8.4	661.6	7.2	704	37	**661.6**	**7.2**	2.43	1176	84
1.4120	0.0750	0.1494	0.0035	0.13	0.0682	0.0038	890	31	899	20	790	120	**790**	**120.0**	1.01	76	30
1.3870	0.0820	0.1505	0.0044	0.19	0.0680	0.0041	900	35	903	24	880	120	**880**	**120.0**	0.33	48	24
1.5790	0.0670	0.1600	0.0041	0.34	0.0708	0.0028	967	26	956	23	919	82	**919**	**82.0**	1.14	69	27
1.6630	0.0500	0.1681	0.0027	0.10	0.0720	0.0025	989	19	1001	15	974	70	**974**	**70.0**	1.21	44	23
1.6470	0.0520	0.1665	0.0030	0.09	0.0730	0.0027	988	20	992	16	987	78	**987**	**78.0**	0.40	158	61
1.7160	0.0220	0.1694	0.0016	0.17	0.0722	0.0011	1013.6	8.3	1008.8	8.8	1006	30	**1006**	**30.0**	0.47	404	98
1.8770	0.0290	0.1852	0.0023	0.51	0.0732	0.0010	1072	10	1095	12	1018	29	**1018**	**29.0**	2.15	1590	392
1.8350	0.0490	0.1799	0.0033	0.34	0.0735	0.0019	1057	18	1066	18	1021	54	**1021**	**54.0**	0.85	196	127
1.7740	0.0290	0.1724	0.0018	0.30	0.0735	0.0012	1036	10	1025.9	9.6	1029	32	**1029**	**32.0**	0.97	144	102
1.7350	0.0460	0.1666	0.0031	0.41	0.0734	0.0018	1023	18	993	17	1033	52	**1033**	**52.0**	2.93	142	49
1.7820	0.0300	0.1715	0.0018	0.25	0.0739	0.0013	1038	11	1020	10	1037	37	**1037**	**37.0**	1.73	181	80
1.5830	0.0640	0.1537	0.0037	0.14	0.0746	0.0029	963	24	921	21	1038	73	**1038**	**73.0**	4.36	49	23
1.6980	0.0500	0.1643	0.0027	0.25	0.0738	0.0023	1009	19	980	15	1043	62	**1043**	**62.0**	2.87	72	34
1.7440	0.0440	0.1682	0.0025	0.03	0.0753	0.0022	1021	16	1002	14	1062	59	**1062**	**59.0**	1.86	68	15
1.7590	0.0650	0.1661	0.0035	0.19	0.0750	0.0028	1025	23	990	19	1064	76	**1064**	**76.0**	3.41	95	35
1.8460	0.0370	0.1724	0.0021	0.16	0.0766	0.0017	1062	14	1026	12	1111	46	**1111**	**46.0**	3.39	333	78
1.8250	0.0360	0.1729	0.0021	0.24	0.0770	0.0016	1053	13	1028	12	1118	40	**1118**	**40.0**	2.37	377	80
1.8520	0.0740	0.1714	0.0042	0.29	0.0784	0.0031	1059	27	1021	23	1134	80	**1134**	**80.0**	3.59	23	7
1.7930	0.0610	0.1676	0.0035	-0.15	0.0776	0.0032	1040	22	998	20	1135	88	**1135**	**88.0**	4.04	78	22
1.9100	0.0730	0.1772	0.0036	0.12	0.0775	0.0031	1083	26	1053	20	1141	81	**1141**	**81.0**	2.77	53	21
2.0120	0.0350	0.1868	0.0020	0.11	0.0779	0.0014	1119	11	1104	11	1143	38	**1143**	**38.0**	1.34	571	310
2.1630	0.0350	0.2009	0.0022	0.15	0.0780	0.0014	1167	11	1180	12	1146	36	**1146**	**36.0**	1.11	426	389
1.8780	0.0800	0.1744	0.0037	0.23	0.0774	0.0033	1066	28	1037	20	1149	82	**1149**	**82.0**	2.72	88	46
1.8810	0.0930	0.1671	0.0053	-0.12	0.0806	0.0047	1073	32	996	29	1160	120	**1160**	**120.0**	7.18	41	8

1.9860	0.0350	0.1786	0.0024	0.42	0.0805	0.0013	1111	12	1060	13	1208	31	**1208**	**31.0**	4.59	118	55
1.9910	0.0420	0.1793	0.0028	0.30	0.0804	0.0019	1114	15	1063	15	1212	48	**1212**	**48.0**	4.58	129	64
2.1420	0.0460	0.1923	0.0025	0.30	0.0813	0.0018	1161	15	1133	14	1212	43	**1212**	**43.0**	2.41	71	37
1.9410	0.0910	0.1705	0.0041	0.14	0.0826	0.0040	1100	31	1014	22	1259	95	**1259**	**95.0**	7.82	25	15
2.3320	0.0490	0.2014	0.0025	0.21	0.0836	0.0019	1219	15	1183	13	1269	43	**1269**	**43.0**	2.95	241	70
1.9620	0.0590	0.1678	0.0029	0.43	0.0857	0.0026	1101	20	999	16	1321	58	**1321**	**58.0**	9.26	184	163
2.5870	0.0970	0.2197	0.0047	0.22	0.0862	0.0033	1296	28	1279	25	1324	70	**1324**	**70.0**	1.31	25	13
2.6230	0.0800	0.2195	0.0043	0.20	0.0864	0.0029	1310	23	1278	23	1330	66	**1330**	**66.0**	2.44	30	16
2.4000	0.0410	0.2051	0.0025	0.42	0.0859	0.0015	1243	12	1202	13	1335	34	**1335**	**34.0**	3.30	152	72
2.6640	0.0930	0.2254	0.0048	0.33	0.0866	0.0031	1320	26	1312	25	1343	69	**1343**	**69.0**	0.61	24	17
1.8620	0.0670	0.1575	0.0036	0.09	0.0861	0.0037	1073	24	942	20	1346	86	**1346**	**86.0**	12.21	55	27
2.4660	0.0860	0.2029	0.0040	0.19	0.0881	0.0033	1259	25	1192	22	1360	72	**1360**	**72.0**	5.32	96	43
2.8100	0.1100	0.2272	0.0056	-0.05	0.0910	0.0039	1354	28	1319	29	1420	84	**1420**	**84.0**	2.58	48	18
3.8660	0.0840	0.2403	0.0047	0.48	0.1172	0.0026	1604	18	1388	24	1912	38	**1912**	**38.0**	13.47	136	114
5.9800	0.0730	0.3327	0.0038	-0.12	0.1307	0.0022	1973	10	1850	19	2088	25	**2088**	**25.0**	6.23	221	140
10.3200	0.2400	0.4181	0.0085	0.85	0.1767	0.0019	2459	22	2252	38	2622	19	**2622**	**19.0**	8.42	115	85
137.7000	7.3000	1.2900	0.0740	0.87	0.7680	0.0190	4994	53	5320	200	4891	42	**4891**	**42.0**	6.53	23	5

Rio Jachal **Lat:** **30.30074** **Lon:** **68.69052**

Isotope Ratios							Apparent Ages								Concentrations		
$^{207}Pb/^{235}U$	± (2σ)	$^{206}Pb/^{238}U$	± (2σ)	Error Corr.	$^{207}Pb/^{206}Pb$	± (2σ)	$^{207}Pb/^{235}U$ (Ma)	± (Ma)	$^{206}Pb/^{238}U$ (Ma)	± (Ma)	$^{207}Pb/^{206}Pb$ (Ma)	± (Ma)	Plot Age (Ma)	± (Ma)	conc. (%)	U (ppm)	Th (ppm)
0.0081	0.0018	0.0009	0.0001	-0.12	0.0860	0.0220	8.2	1.8	6.11	0.46	-	-	**5.8**	**0.5**	25.49	1063	784
0.0110	0.0025	0.0010	0.0001	0.19	0.1000	0.0250	11	2.5	6.54	0.53	-	-	**6.1**	**0.5**	40.55	300	158
0.0182	0.0060	0.0017	0.0002	0.13	0.0840	0.0270	18.1	5.9	10.8	1.2	-	-	**10.3**	**1.2**	40.33	841	516
0.0100	0.0024	0.0017	0.0001	-0.11	0.0500	0.0130	10	2.5	10.84	0.65	-	-	**10.8**	**0.7**	8.40	230	126
0.0028	0.0056	0.0017	0.0002	0.17	0.0180	0.0500	3.2	5.8	11	1.3	-	-	**11.4**	**1.3**	243.75	64	59
0.0129	0.0014	0.0019	0.0001	-0.02	0.0518	0.0060	13	1.4	12.02	0.45	-	-	**11.9**	**0.5**	7.54	826	732
0.0117	0.0017	0.0019	0.0001	-0.14	0.0462	0.0088	11.8	1.7	12.39	0.61	-	-	**12.4**	**0.6**	5.00	1236	1200
0.0139	0.0018	0.0020	0.0001	-0.06	0.0510	0.0071	14	1.8	12.92	0.52	-	-	**12.8**	**0.5**	7.71	1196	2959
0.0119	0.0048	0.0020	0.0002	0.06	0.0490	0.0250	11.6	4.8	13.2	1.3	-	-	**13.1**	**1.3**	13.79	70	34
0.0114	0.0050	0.0021	0.0002	0.09	0.0950	0.0530	11.1	5	13.3	1.2	-	-	**12.5**	**1.2**	19.82	94	53
0.0143	0.0051	0.0021	0.0002	-0.05	0.0590	0.0260	14.6	5	13.4	1.1	-	-	**13.2**	**1.1**	8.22	88	69
0.0700	0.0160	0.0023	0.0003	0.00	0.2790	0.0940	65	15	14.6	2.2	-	-	**10.3**	**2.2**	77.54	26	8
0.0152	0.0023	0.0026	0.0001	0.16	0.0444	0.0067	15.3	2.3	16.75	0.78	-	-	**16.8**	**0.8**	9.48	831	671
0.0244	0.0067	0.0028	0.0002	0.05	0.0790	0.0240	24.5	6.7	17.7	1.4	-	-	**17.0**	**1.4**	27.76	56	32
0.0060	0.0130	0.0028	0.0004	0.02	0.0060	0.0550	3	13	18	2.8	-	-	**19.0**	**2.8**	500.00	110	95
0.0161	0.0053	0.0028	0.0002	-0.02	0.0480	0.0170	15.8	5.3	18.1	1.2	-	-	**18.1**	**1.2**	14.56	377	216
0.0181	0.0055	0.0029	0.0002	0.03	0.0380	0.0140	17.8	5.5	18.7	1.2	-	-	**18.9**	**1.2**	5.06	117	64
0.0210	0.0140	0.0029	0.0003	-0.15	0.1020	0.0550	20	14	18.9	2.2	-	-	**17.6**	**2.2**	5.50	126	93
0.0180	0.0018	0.0030	0.0001	-0.11	0.0422	0.0049	18	1.8	19.45	0.68	-	-	**19.5**	**0.7**	8.06	356	353
0.0205	0.0033	0.0031	0.0002	0.11	0.0491	0.0080	20.4	3.3	19.66	0.96	-	-	**19.6**	**1.0**	3.63	597	183
0.0241	0.0043	0.0031	0.0002	-0.02	0.0560	0.0110	23.9	4.3	20.1	1	-	-	**19.9**	**1.0**	15.90	120	75
0.0300	0.0120	0.0032	0.0004	-0.09	0.0820	0.0380	31	12	20.5	2.5	-	-	**19.6**	**2.5**	33.87	49	53
0.0198	0.0023	0.0032	0.0001	-0.04	0.0459	0.0057	19.8	2.3	20.51	0.73	-	-	**20.5**	**0.7**	3.59	325	171
0.0225	0.0054	0.0033	0.0002	0.25	0.0510	0.0120	22.2	5.4	20.9	1.3	-	-	**20.8**	**1.3**	5.86	75	33
0.0204	0.0045	0.0033	0.0002	-0.03	0.0480	0.0110	20.2	4.5	21.2	1.3	-	-	**21.2**	**1.3**	4.95	330	166
0.0294	0.0042	0.0034	0.0002	-0.18	0.0750	0.0140	29.2	4.1	21.6	1.1	-	-	**20.8**	**1.1**	26.03	167	306
0.0236	0.0017	0.0034	0.0001	-0.09	0.0505	0.0038	23.6	1.6	21.63	0.6	-	-	**21.5**	**0.6**	8.35	392	380
0.0100	0.0150	0.0034	0.0004	-0.02	0.0270	0.0480	10	15	21.8	2.8	-	-	**22.3**	**2.8**	118.00	34	38
0.0219	0.0024	0.0034	0.0001	-0.17	0.0476	0.0058	22.5	2.5	22.01	0.73	-	-	**22.0**	**0.7**	2.18	289	143
0.0243	0.0035	0.0037	0.0001	0.11	0.0492	0.0073	24.2	3.4	23.77	0.89	-	-	**23.7**	**0.9**	1.78	730	567
0.0646	0.0077	0.0038	0.0002	0.30	0.1250	0.0140	62.7	7.3	24.2	1.2	-	-	**21.9**	**1.2**	61.40	451	538
0.0226	0.0025	0.0038	0.0001	-0.03	0.0455	0.0055	22.6	2.5	24.36	0.88	-	-	**24.4**	**0.9**	7.79	812	690
0.0311	0.0057	0.0038	0.0002	0.20	0.0600	0.0110	30.8	5.6	24.4	1.5	-	-	**24.0**	**1.5**	20.78	98	47
0.0356	0.0074	0.0038	0.0003	0.00	0.0740	0.0160	35	7.2	24.4	2.1	-	-	**23.5**	**2.1**	30.29	111	81
0.0309	0.0021	0.0048	0.0001	0.03	0.0459	0.0032	30.9	2.1	31.07	0.69	-	-	**31.1**	**0.7**	0.55	425	385
0.0680	0.0120	0.0049	0.0004	0.11	0.1080	0.0220	65	11	31.6	2.7	-	-	**29.1**	**2.7**	51.38	180	92
0.0319	0.0070	0.0052	0.0003	-0.04	0.0510	0.0120	32	7	33.2	1.9	-	-	**33.1**	**1.9**	3.75	88	73
0.0365	0.0087	0.0052	0.0005	0.22	0.0690	0.0170	35.4	8.4	33.4	2.9	-	-	**32.4**	**2.9**	5.65	44	12
0.0355	0.0041	0.0053	0.0002	0.09	0.0471	0.0054	35.3	4	34.1	1.3	-	-	**34.1**	**1.3**	3.40	596	296
0.0432	0.0057	0.0059	0.0003	0.07	0.0517	0.0072	42.5	5.5	38	1.6	-	-	**37.8**	**1.6**	10.59	410	184
0.2168	0.0054	0.0301	0.0006	0.35	0.0514	0.0013	198.9	4.5	191.2	3.8	242	53	**191.2**	**3.8**	3.87	1540	767
0.2160	0.0057	0.0311	0.0006	0.46	0.0487	0.0012	198.7	4.8	197.5	3.8	157	54	**197.5**	**3.8**	0.60	1890	985
0.2480	0.0190	0.0321	0.0009	0.32	0.0531	0.0038	225	16	204.1	5.9	300	140	**204.1**	**5.9**	9.29	44	55
0.2530	0.0140	0.0344	0.0010	-0.03	0.0539	0.0038	229	11	218.3	5.9	350	150	**218.3**	**5.9**	4.67	835	283
0.2470	0.0110	0.0362	0.0009	0.11	0.0507	0.0026	223.7	9	229.2	5.4	218	98	**229.2**	**5.4**	2.46	93	50
0.2780	0.0140	0.0377	0.0009	0.02	0.0544	0.0031	248	11	238.2	5.7	390	110	**238.2**	**5.7**	3.95	326	561
0.2700	0.0130	0.0380	0.0008	0.08	0.0512	0.0024	243.1	9.8	240.7	4.9	245	95	**240.7**	**4.9**	0.99	342	221
0.2940	0.0190	0.0382	0.0010	0.16	0.0558	0.0038	260	15	241.6	6.5	380	130	**241.6**	**6.5**	7.08	47	62
0.2870	0.0130	0.0391	0.0008	0.16	0.0523	0.0025	256	11	246.9	4.8	330	100	**246.9**	**4.8**	3.55	87	86
0.3090	0.0110	0.0394	0.0007	0.22	0.0569	0.0021	272.8	8.6	248.8	4.3	483	81	**248.8**	**4.3**	8.80	271	217
0.2770	0.0270	0.0399	0.0013	-0.02	0.0538	0.0057	243	22	251.9	7.8	200	190	**251.9**	**7.8**	3.66	95	111
0.3190	0.0260	0.0403	0.0012	0.19	0.0579	0.0046	284	20	254.4	7.5	490	160	**254.4**	**7.5**	10.42	34	26

0.2810	0.0110	0.0403	0.0008	0.19	0.0505	0.0021	251.2	8.9	254.6	5.2	227	83	**254.6**	**5.2**	1.35	116	85
0.2830	0.0100	0.0407	0.0006	0.15	0.0502	0.0018	251.7	7.9	257.2	3.8	189	72	**257.2**	**3.8**	2.19	557	383
0.2890	0.0120	0.0409	0.0008	0.03	0.0513	0.0024	257.3	9.7	258.4	4.8	246	95	**258.4**	**4.8**	0.43	406	169
0.3000	0.0110	0.0409	0.0007	-0.02	0.0535	0.0023	266.3	8.8	258.5	4.2	319	87	**258.5**	**4.2**	2.93	501	373
0.3190	0.0220	0.0410	0.0011	0.22	0.0574	0.0040	285	16	258.7	6.8	480	130	**258.7**	**6.8**	9.23	64	48
0.2710	0.0220	0.0410	0.0013	0.20	0.0496	0.0042	245	18	259	7.8	150	150	**259**	**7.8**	5.71	124	96
0.3020	0.0160	0.0412	0.0009	0.16	0.0526	0.0028	267	12	259.9	5.7	290	100	**259.9**	**5.7**	2.66	367	252
0.3100	0.0200	0.0422	0.0010	-0.03	0.0536	0.0037	275	15	266.4	6.3	320	140	**266.4**	**6.3**	3.13	48	25
0.2990	0.0220	0.0429	0.0012	0.05	0.0515	0.0039	262	17	270.6	7.5	220	140	**270.6**	**7.5**	3.28	156	90
0.3390	0.0290	0.0473	0.0015	0.06	0.0539	0.0048	294	22	297.8	9.1	300	170	**297.8**	**9.1**	1.29	95	80
0.3620	0.0180	0.0495	0.0010	0.06	0.0542	0.0029	314	13	311.2	6.4	330	110	**311.2**	**6.4**	0.89	66	64
0.4880	0.0120	0.0598	0.0010	0.23	0.0568	0.0015	402.4	8.5	374	5.8	476	56	**374**	**5.8**	7.06	448	196
0.4750	0.0120	0.0615	0.0009	0.52	0.0555	0.0013	393.8	8.4	384.5	5.4	430	50	**384.5**	**5.4**	2.36	525	325
0.4960	0.0170	0.0621	0.0011	0.23	0.0570	0.0021	408	12	388.5	6.8	470	80	**388.5**	**6.8**	4.78	387	524
0.5060	0.0310	0.0656	0.0017	0.02	0.0575	0.0037	418	21	409	10	440	120	**409**	**10.0**	2.15	28	11
0.5320	0.0210	0.0695	0.0015	0.28	0.0557	0.0021	438	13	432.9	9.2	431	82	**432.9**	**9.2**	1.16	331	334
0.5900	0.0170	0.0716	0.0011	0.22	0.0592	0.0018	470	11	445.4	6.5	556	66	**445.4**	**6.5**	5.23	162	140
0.5670	0.0480	0.0721	0.0020	0.03	0.0574	0.0051	441	31	448	12	350	160	**448**	**12.0**	1.59	81	46
0.6120	0.0270	0.0738	0.0019	0.26	0.0586	0.0026	482	17	458	11	522	92	**458**	**11.0**	4.98	225	206
0.6220	0.0650	0.0814	0.0029	0.04	0.0572	0.0063	488	39	504	17	330	200	**504**	**17.0**	3.28	13	9
0.6900	0.0340	0.0832	0.0016	0.08	0.0589	0.0031	527	21	516	9.9	510	110	**516**	**9.9**	2.09	52	23
0.7060	0.0250	0.0844	0.0020	0.33	0.0603	0.0021	542	15	522	12	584	75	**522**	**12.0**	3.69	52	28
0.7050	0.0690	0.0879	0.0034	0.04	0.0588	0.0062	554	41	544	20	480	190	**544**	**20.0**	1.81	46	39
0.7380	0.0190	0.0885	0.0014	0.32	0.0606	0.0015	562	11	546.3	8	615	54	**546.3**	**8.0**	2.79	443	93
0.7090	0.0270	0.0886	0.0026	0.39	0.0573	0.0022	544	16	547	15	495	80	**547**	**15.0**	0.55	410	232
0.7840	0.0160	0.0930	0.0011	0.00	0.0606	0.0014	587.4	9.1	573.4	6.6	616	50	**573.4**	**6.6**	2.38	348	140
1.1290	0.0470	0.1243	0.0023	0.03	0.0653	0.0029	762	22	755	13	780	93	**755**	**13.0**	0.92	27	19
1.4220	0.0580	0.1495	0.0028	0.16	0.0686	0.0029	890	24	898	16	829	90	**829**	**90.0**	0.90	104	95
1.5680	0.0440	0.1620	0.0029	0.08	0.0694	0.0023	955	18	967	16	895	68	**895**	**68.0**	1.26	61	27
1.7100	0.1000	0.1684	0.0053	0.16	0.0720	0.0047	1012	37	1004	29	950	130	**950**	**130.0**	0.79	10	12
1.5620	0.0680	0.1591	0.0038	0.23	0.0716	0.0033	955	27	951	21	960	100	**960**	**100.0**	0.42	79	90
1.5180	0.0310	0.1500	0.0020	0.27	0.0722	0.0015	937	13	901	11	997	41	**997**	**41.0**	3.84	83	15
1.7170	0.0290	0.1688	0.0019	0.18	0.0740	0.0014	1014	11	1005	11	1035	39	**1035**	**39.0**	0.89	374	159
1.4630	0.0300	0.1442	0.0023	0.33	0.0742	0.0015	917	12	868	13	1044	42	**1044**	**42.0**	5.34	127	17
1.7140	0.0450	0.1638	0.0024	0.08	0.0749	0.0023	1011	17	977	13	1054	63	**1054**	**63.0**	3.36	37	10
1.6470	0.0390	0.1597	0.0026	0.09	0.0750	0.0020	990	15	955	14	1066	55	**1066**	**55.0**	3.54	51	21
1.9180	0.0370	0.1829	0.0022	0.28	0.0754	0.0016	1088	13	1083	12	1071	43	**1071**	**43.0**	0.46	282	205
1.6800	0.0420	0.1586	0.0024	0.27	0.0752	0.0018	999	16	949	14	1079	49	**1079**	**49.0**	5.01	237	81
1.7910	0.0720	0.1699	0.0033	0.05	0.0770	0.0032	1035	26	1011	18	1102	84	**1102**	**84.0**	2.32	26	8
1.9840	0.0400	0.1872	0.0024	0.25	0.0766	0.0016	1107	14	1106	13	1104	42	**1104**	**42.0**	0.09	318	104
1.9190	0.0300	0.1800	0.0021	0.32	0.0767	0.0014	1086	11	1067	11	1113	34	**1113**	**34.0**	1.75	119	42
2.0930	0.0510	0.1884	0.0035	0.29	0.0809	0.0022	1142	17	1114	19	1227	48	**1227**	**48.0**	2.45	46	47
2.0760	0.0730	0.1722	0.0026	0.39	0.0858	0.0027	1132	24	1024	14	1321	61	**1321**	**61.0**	9.54	414	291
2.8610	0.0650	0.2314	0.0034	0.17	0.0886	0.0023	1370	17	1341	18	1396	50	**1396**	**50.0**	2.12	64	22
3.2500	0.0860	0.2566	0.0044	0.30	0.0912	0.0024	1463	21	1471	22	1450	51	**1450**	**51.0**	0.55	87	63

HU-1.1 **Lat:** **-30.22601** **Lon:** **-68.39898**

Isotope Ratios							Apparent Ages								Concentrations		
$^{207}Pb/^{235}U$	± (2σ)	$^{206}Pb/^{238}U$	± (2σ)	Error Corr.	$^{207}Pb/^{206}Pb$	± (2σ)	$^{207}Pb/^{235}U$ (Ma)	± (Ma)	$^{206}Pb/^{238}U$ (Ma)	± (Ma)	$^{207}Pb/^{206}Pb$ (Ma)	± (Ma)	Plot Age (Ma)	± (Ma)	conc. (%)	U (ppm)	Th (ppm)
0.0440	0.0170	0.0025	0.0004	0.24	0.1310	0.0730	40	16	16.3	2.6	-	-	**14.6**	**2.6**	59.25	102	32
0.0380	0.0100	0.0029	0.0004	0.00	0.1280	0.0430	37	10	18.7	2.3	-	-	**16.7**	**2.3**	49.46	103	83
0.0276	0.0065	0.0031	0.0002	0.05	0.0610	0.0190	27.1	6.4	19.8	1.5	-	-	**19.4**	**1.5**	26.94	161	149
0.0229	0.0032	0.0032	0.0001	0.10	0.0548	0.0073	23.2	3.1	20.27	0.89	-	-	**20.1**	**0.9**	12.63	502	202
0.0236	0.0058	0.0034	0.0002	0.32	0.0530	0.0130	23.9	5.6	22	1.4	-	-	**21.8**	**1.4**	7.95	225	478
0.0208	0.0025	0.0035	0.0002	0.09	0.0425	0.0055	20.8	2.5	22.8	1	-	-	**22.9**	**1.0**	9.62	336	178
0.0339	0.0046	0.0036	0.0002	0.09	0.0730	0.0097	33.6	4.5	23.4	1.1	-	-	**22.6**	**1.1**	30.36	351	326
0.0250	0.0130	0.0037	0.0004	0.15	0.1040	0.0390	23	13	23.6	2.8	-	-	**21.9**	**2.8**	2.61	64	68
0.0550	0.0210	0.0038	0.0007	0.08	0.0290	0.0750	48	20	24.1	4.2	-	-	**24.7**	**4.2**	49.79	59	55
0.0424	0.0055	0.0038	0.0001	0.24	0.0803	0.0098	42.3	5.4	24.68	0.85	-	-	**23.7**	**0.9**	41.65	1004	652
0.0500	0.0170	0.0039	0.0004	0.18	0.0730	0.0420	46	16	25.1	2.5	-	-	**24.3**	**2.5**	45.43	177	88
0.0312	0.0074	0.0045	0.0003	-0.03	0.0530	0.0140	30.4	7.3	28.8	1.9	-	-	**28.6**	**1.9**	5.26	80	47
0.0450	0.0150	0.0048	0.0005	-0.05	0.1080	0.0390	42	15	31	3.1	-	-	**28.6**	**3.1**	26.19	50	21
0.0303	0.0094	0.0052	0.0004	-0.20	0.0710	0.0230	29.1	9.1	33.4	2.8	-	-	**32.4**	**2.8**	14.78	77	33
0.0427	0.0075	0.0053	0.0003	0.16	0.0620	0.0110	42.5	7.4	33.9	1.9	-	-	**33.2**	**1.9**	20.24	190	90
0.0518	0.0038	0.0068	0.0002	0.22	0.0554	0.0042	51.1	3.7	43.5	1.2	-	-	**43.0**	**1.2**	14.87	875	505
0.0532	0.0060	0.0075	0.0003	0.14	0.0515	0.0061	52.1	5.7	48.3	1.9	-	-	**48.0**	**1.9**	7.29	279	234
0.0585	0.0036	0.0089	0.0002	0.15	0.0462	0.0028	57.6	3.5	57.2	1.5	-	-	**57.2**	**1.5**	0.69	496	347
0.0740	0.0100	0.0103	0.0004	0.10	0.0551	0.0078	73	9.8	65.7	2.7	-	-	**65.7**	**2.7**	10.00	132	162
0.2000	0.0210	0.0259	0.0010	0.17	0.0556	0.0058	181	17	164.4	6.1	-	-	**164.4**	**6.1**	9.17	104	103
0.2660	0.0140	0.0336	0.0009	0.08	0.0562	0.0032	240	11	213	5.8	440	120	**213**	**5.8**	11.25	194	373
0.2454	0.0094	0.0352	0.0005	0.05	0.0503	0.0020	222.6	7.8	223	3.4	206	81	**223**	**3.4**	0.18	573	1072
0.2740	0.0150	0.0387	0.0010	0.05	0.0507	0.0031	247	11	244.6	5.9	230	110	**244.6**	**5.9**	0.97	207	120
0.3000	0.0140	0.0396	0.0008	0.03	0.0548	0.0027	266	11	250.7	5.1	360	100	**250.7**	**5.1**	5.75	240	217

0.3020	0.0120	0.0408	0.0008	0.37	0.0539	0.0020	268	9.1	257.9	5	354	79	**257.9**	**5.0**	3.77	779	448
0.3120	0.0130	0.0425	0.0008	0.24	0.0516	0.0021	274.8	9.9	268.5	4.7	263	86	**268.5**	**4.7**	2.29	400	319
0.2960	0.0120	0.0428	0.0009	0.05	0.0502	0.0023	261.9	9.7	269.8	5.3	184	91	**269.8**	**5.3**	3.02	117	52
0.2890	0.0190	0.0428	0.0011	-0.01	0.0486	0.0035	257	15	270.3	6.6	150	130	**270.3**	**6.6**	5.18	183	127
0.3050	0.0150	0.0449	0.0011	0.09	0.0487	0.0027	271	12	282.8	7.1	150	110	**282.8**	**7.1**	4.35	80	33
0.3250	0.0140	0.0451	0.0010	0.16	0.0520	0.0023	284	11	284	5.8	268	91	**284**	**5.8**	0.00	340	164
0.3600	0.0190	0.0452	0.0011	0.32	0.0568	0.0029	311	14	285	6.8	470	110	**285**	**6.8**	8.36	352	194
0.3420	0.0180	0.0459	0.0010	0.24	0.0539	0.0030	297	13	289.1	6.4	320	110	**289.1**	**6.4**	2.66	271	142
0.3350	0.0160	0.0466	0.0011	0.18	0.0521	0.0026	294	12	293.6	6.7	285	97	**293.6**	**6.7**	0.14	230	217
0.3600	0.0200	0.0472	0.0010	-0.07	0.0544	0.0032	312	15	297.3	5.9	380	120	**297.3**	**5.9**	4.71	226	151
0.3480	0.0110	0.0474	0.0008	0.21	0.0521	0.0018	302.8	8.5	299	5.2	275	72	**299**	**5.2**	1.25	225	77
0.3480	0.0120	0.0479	0.0008	0.23	0.0530	0.0018	302.8	8.9	301.5	5	297	69	**301.5**	**5.0**	0.43	588	279
0.3590	0.0200	0.0485	0.0010	0.13	0.0540	0.0032	312	15	305.7	5.9	360	110	**305.7**	**5.9**	2.02	252	134
0.3930	0.0170	0.0489	0.0009	0.33	0.0565	0.0022	334	12	308.2	5.9	449	80	**308.2**	**5.9**	7.72	276	214
0.3630	0.0160	0.0501	0.0012	0.29	0.0522	0.0023	314	12	315.3	7.1	292	86	**315.3**	**7.1**	0.41	105	40
0.4590	0.0160	0.0530	0.0010	0.16	0.0605	0.0022	383	11	332.7	6	618	76	**332.7**	**6.0**	13.13	910	1034
0.4310	0.0130	0.0577	0.0011	0.32	0.0539	0.0016	362.7	9.2	361.4	6.6	346	63	**361.4**	**6.6**	0.36	421	67
0.4140	0.0200	0.0578	0.0013	0.08	0.0523	0.0027	353	14	362	7.7	290	100	**362**	**7.7**	2.55	224	374
0.4770	0.0190	0.0612	0.0013	0.10	0.0576	0.0024	398	13	383	7.8	478	90	**383**	**7.8**	3.77	302	78
0.4770	0.0180	0.0622	0.0013	0.22	0.0549	0.0022	397	12	389	7.8	395	87	**389**	**7.8**	2.02	245	98
0.4770	0.0130	0.0624	0.0010	0.22	0.0558	0.0016	394.7	9.3	389.8	5.8	419	64	**389.8**	**5.8**	1.24	467	149
0.4860	0.0140	0.0634	0.0009	0.20	0.0557	0.0017	401.6	9.7	396.2	5.7	426	69	**396.2**	**5.7**	1.34	522	136
0.4820	0.0160	0.0637	0.0009	0.23	0.0549	0.0019	399	11	398	5.4	390	72	**398**	**5.4**	0.25	678	197
0.5120	0.0150	0.0638	0.0009	0.08	0.0573	0.0019	419	10	399.1	5.6	502	74	**399.1**	**5.6**	4.75	432	169
0.4730	0.0150	0.0640	0.0010	0.23	0.0541	0.0017	392	10	400	6.3	359	68	**400**	**6.3**	2.04	774	440
0.5370	0.0150	0.0698	0.0012	0.20	0.0553	0.0016	438.2	9.8	434.8	7.3	408	64	**434.8**	**7.3**	0.78	160	70
0.5460	0.0180	0.0698	0.0011	0.17	0.0558	0.0018	440	12	435	6.4	428	72	**435**	**6.4**	1.14	303	212
0.5910	0.0210	0.0718	0.0012	0.32	0.0596	0.0020	470	13	446.7	7.2	581	76	**446.7**	**7.2**	4.96	564	80
0.5910	0.0290	0.0720	0.0020	0.43	0.0608	0.0024	470	18	448	12	611	85	**448**	**12.0**	4.68	501	63
0.5780	0.0170	0.0728	0.0014	0.18	0.0568	0.0018	462	11	452.9	8.2	486	69	**452.9**	**8.2**	1.97	292	257
0.5670	0.0250	0.0733	0.0017	0.12	0.0561	0.0027	459	16	456	10	420	100	**456**	**10.0**	0.65	95	30
0.5890	0.0190	0.0735	0.0015	0.09	0.0590	0.0021	469	12	457.4	8.8	559	81	**457.4**	**8.8**	2.47	961	807
0.5660	0.0180	0.0745	0.0011	0.30	0.0550	0.0017	453	12	463.2	6.5	408	66	**463.2**	**6.5**	2.25	463	98
0.5850	0.0160	0.0751	0.0014	0.25	0.0559	0.0016	472	10	466.4	8.1	433	63	**466.4**	**8.1**	1.19	321	123
0.6090	0.0250	0.0754	0.0015	0.14	0.0591	0.0026	480	16	468.5	9	518	96	**468.5**	**9.0**	2.40	197	45
0.6120	0.0190	0.0759	0.0014	0.17	0.0577	0.0018	484	12	471.3	8.2	504	66	**471.3**	**8.2**	2.62	323	233
0.5920	0.0590	0.0760	0.0030	-0.12	0.0582	0.0069	461	38	472	18	460	220	**472**	**18.0**	2.39	57	46
0.6280	0.0450	0.0770	0.0021	0.01	0.0576	0.0044	499	27	479	12	480	150	**479**	**12.0**	4.01	60	33
0.6710	0.0380	0.0782	0.0020	0.14	0.0626	0.0040	522	23	485	12	630	130	**485**	**12.0**	7.09	99	66
0.6360	0.0390	0.0799	0.0023	0.12	0.0581	0.0039	491	24	495	14	440	130	**495**	**14.0**	0.81	37	19
0.6470	0.0710	0.0803	0.0032	-0.04	0.0590	0.0071	519	46	497	19	470	220	**497**	**19.0**	4.24	52	39
0.6150	0.0230	0.0803	0.0014	0.13	0.0553	0.0023	485	14	497.9	8.6	407	87	**497.9**	**8.6**	2.66	110	111
0.6570	0.0230	0.0808	0.0013	0.13	0.0590	0.0022	511	14	500.5	7.7	546	81	**500.5**	**7.7**	2.05	236	103
0.6250	0.0190	0.0808	0.0014	-0.03	0.0570	0.0021	494	12	500.6	8.2	477	78	**500.6**	**8.2**	1.34	290	104
0.6280	0.0350	0.0810	0.0021	0.12	0.0568	0.0035	492	22	502	12	420	120	**502**	**12.0**	2.03	150	100
0.6570	0.0300	0.0811	0.0016	0.04	0.0586	0.0029	508	19	502.7	9.6	490	100	**502.7**	**9.6**	1.04	186	96
0.6380	0.0190	0.0811	0.0011	0.19	0.0562	0.0016	499	12	502.8	6.5	443	64	**502.8**	**6.5**	0.76	496	278
0.6500	0.0210	0.0822	0.0014	0.23	0.0562	0.0018	510	12	508.9	8.1	455	68	**508.9**	**8.1**	0.22	162	76
0.6540	0.0240	0.0825	0.0015	0.22	0.0574	0.0022	508	15	510.9	9.1	474	81	**510.9**	**9.1**	0.57	197	98
0.7050	0.0200	0.0838	0.0016	0.00	0.0602	0.0019	542	12	518.8	9.8	612	69	**518.8**	**9.8**	4.28	130	37
0.6880	0.0160	0.0842	0.0013	0.33	0.0585	0.0014	532.1	9.7	521.1	7.8	571	52	**521.1**	**7.8**	2.07	450	59
0.6890	0.0190	0.0844	0.0013	0.10	0.0587	0.0017	530	12	522.2	7.7	542	64	**522.2**	**7.7**	1.47	367	209
0.6500	0.0180	0.0854	0.0013	0.16	0.0562	0.0017	511	11	528	8	442	65	**528**	**8.0**	3.33	396	153
0.6510	0.0200	0.0861	0.0014	0.20	0.0546	0.0017	509	13	532.2	8.4	377	69	**532.2**	**8.4**	4.56	238	222
0.6740	0.0480	0.0877	0.0029	-0.04	0.0563	0.0045	517	29	541	17	370	160	**541**	**17.0**	4.64	60	231
0.7130	0.0190	0.0881	0.0013	0.21	0.0593	0.0017	547	11	544.1	7.9	568	61	**544.1**	**7.9**	0.53	445	148
0.7720	0.0270	0.0909	0.0015	0.14	0.0612	0.0023	579	15	560.7	8.6	620	77	**560.7**	**8.6**	3.16	327	183
0.7520	0.0190	0.0910	0.0015	0.04	0.0595	0.0016	569	11	561	9.1	578	61	**561**	**9.1**	1.41	501	168
0.7810	0.0250	0.0916	0.0016	0.37	0.0607	0.0018	586	14	564.6	9.3	610	65	**564.6**	**9.3**	3.65	215	334
0.7840	0.0750	0.0917	0.0032	0.06	0.0603	0.0060	584	42	567	19	580	190	**567**	**19.0**	2.91	20	8
0.7720	0.0190	0.0923	0.0012	0.20	0.0598	0.0016	582	11	568.8	7.1	589	56	**568.8**	**7.1**	2.27	466	361
0.7700	0.1500	0.0932	0.0063	0.05	0.0600	0.0140	591	83	573	37	530	370	**573**	**37.0**	3.05	30	15
0.8040	0.0330	0.0930	0.0019	0.05	0.0614	0.0027	594	18	573	11	613	93	**573**	**11.0**	3.54	142	59
0.8100	0.0270	0.0957	0.0018	0.21	0.0608	0.0021	605	15	590	10	613	75	**590**	**10.0**	2.48	90	36
0.8150	0.0190	0.0963	0.0016	0.40	0.0612	0.0015	605	11	592.3	9.3	644	51	**592.3**	**9.3**	2.10	588	254
0.8600	0.0310	0.0992	0.0023	0.42	0.0629	0.0019	630	17	612	14	671	67	**612**	**14.0**	2.86	425	107
0.8770	0.0250	0.0999	0.0013	0.32	0.0633	0.0017	639	13	613.8	7.6	706	59	**613.8**	**7.6**	3.94	506	106
0.8320	0.0210	0.1002	0.0015	0.29	0.0602	0.0016	616	12	615.3	8.5	605	58	**615.3**	**8.5**	0.11	134	96
0.8810	0.0240	0.1017	0.0019	0.16	0.0637	0.0020	642	13	624	11	740	62	**624**	**11.0**	2.80	278	88
0.9420	0.0480	0.1030	0.0025	0.09	0.0664	0.0039	665	25	631	15	730	120	**631**	**15.0**	5.11	42	14
0.8380	0.0470	0.1040	0.0027	0.23	0.0594	0.0034	616	26	637	16	560	120	**637**	**16.0**	3.41	64	30
0.9180	0.0290	0.1099	0.0017	0.23	0.0619	0.0020	660	16	672.1	9.7	658	70	**672.1**	**9.7**	1.83	251	102
1.0440	0.0380	0.1119	0.0025	0.35	0.0675	0.0022	723	19	684	14	839	72	**684**	**14.0**	5.39	127	78
1.0840	0.0540	0.1129	0.0030	0.33	0.0691	0.0033	743	27	692	17	868	95	**692**	**17.0**	6.86	172	80
1.1060	0.0490	0.1165	0.0029	0.35	0.0683	0.0030	748	24	710	17	854	91	**710**	**17.0**	5.08	81	87

1.0780	0.0480	0.1195	0.0026	0.08	0.0654	0.0032	739	24	729	15	750	100	**729**	**15.0**	1.35	109	200
1.1760	0.0410	0.1272	0.0025	0.26	0.0665	0.0022	792	19	771	15	819	66	**771**	**15.0**	2.65	95	27
1.2080	0.0280	0.1279	0.0021	0.25	0.0688	0.0018	803	13	777	12	885	53	**777**	**12.0**	3.24	191	95
1.2560	0.0280	0.1335	0.0019	0.29	0.0675	0.0016	824	13	809	11	855	49	**855**	**49.0**	1.82	296	168
1.5740	0.0410	0.1595	0.0027	0.20	0.0716	0.0020	961	16	957	15	961	58	**961**	**58.0**	0.42	159	115
1.4940	0.0330	0.1493	0.0024	0.35	0.0711	0.0016	927	14	898	13	969	48	**969**	**48.0**	3.13	195	84
1.3940	0.0450	0.1420	0.0030	0.34	0.0715	0.0023	890	20	855	17	976	66	**976**	**66.0**	3.93	164	85
1.9500	0.1400	0.1913	0.0065	-0.02	0.0735	0.0054	1098	43	1126	35	1010	150	**1010**	**150.0**	2.55	41	81
1.5530	0.0320	0.1538	0.0022	0.44	0.0726	0.0014	952	13	922	13	1011	40	**1011**	**40.0**	3.15	237	56
1.7010	0.0500	0.1679	0.0028	0.18	0.0729	0.0022	1011	18	1000	16	1012	63	**1012**	**63.0**	1.09	140	85
1.7240	0.0500	0.1695	0.0037	-0.01	0.0730	0.0026	1017	19	1008	20	1014	73	**1014**	**73.0**	0.88	52	24
1.5000	0.0270	0.1469	0.0020	0.46	0.0741	0.0013	928	11	884	12	1036	37	**1036**	**37.0**	4.74	572	44
1.7030	0.0380	0.1676	0.0019	0.36	0.0741	0.0015	1010	14	998	11	1047	40	**1047**	**40.0**	1.19	501	329
1.7830	0.0680	0.1744	0.0035	0.20	0.0740	0.0027	1040	26	1036	19	1047	69	**1047**	**69.0**	0.38	247	90
1.7810	0.0620	0.1727	0.0035	0.02	0.0741	0.0029	1034	22	1030	20	1047	77	**1047**	**77.0**	0.39	41	14
1.7530	0.0490	0.1677	0.0030	0.18	0.0754	0.0022	1025	18	999	17	1058	59	**1058**	**59.0**	2.54	191	129
1.6790	0.0360	0.1620	0.0021	0.18	0.0749	0.0018	1004	13	968	12	1058	48	**1058**	**48.0**	3.59	280	111
1.6850	0.0520	0.1620	0.0031	0.24	0.0757	0.0026	1004	20	967	17	1063	69	**1063**	**69.0**	3.69	128	52
1.7880	0.0340	0.1700	0.0020	0.32	0.0750	0.0014	1039	12	1012	11	1070	37	**1070**	**37.0**	2.60	163	97
1.7990	0.0380	0.1735	0.0025	0.34	0.0754	0.0016	1046	13	1031	13	1073	43	**1073**	**43.0**	1.43	295	122
1.7140	0.0440	0.1641	0.0033	0.36	0.0750	0.0020	1013	16	979	18	1076	56	**1076**	**56.0**	3.36	177	101
1.9120	0.0430	0.1789	0.0024	0.23	0.0763	0.0018	1084	15	1061	13	1090	48	**1090**	**48.0**	2.12	223	123
1.8280	0.0320	0.1734	0.0021	0.21	0.0759	0.0015	1057	11	1030	11	1091	38	**1091**	**38.0**	2.55	394	154
2.0430	0.0740	0.1920	0.0038	-0.02	0.0772	0.0031	1125	25	1132	21	1112	80	**1112**	**80.0**	0.62	119	39
1.7320	0.0610	0.1647	0.0033	0.28	0.0774	0.0028	1013	23	982	18	1128	72	**1128**	**72.0**	3.06	110	79
1.9490	0.0540	0.1788	0.0026	0.25	0.0776	0.0021	1099	18	1060	14	1140	52	**1140**	**52.0**	3.55	200	101
1.8660	0.0520	0.1680	0.0026	0.24	0.0791	0.0021	1069	18	1001	15	1164	55	**1164**	**55.0**	6.36	240	95
1.9880	0.0460	0.1837	0.0026	0.23	0.0783	0.0019	1111	16	1089	14	1168	48	**1168**	**48.0**	1.98	259	100
1.9730	0.0480	0.1799	0.0030	0.30	0.0791	0.0017	1103	16	1066	17	1175	43	**1175**	**43.0**	3.35	261	178
1.9410	0.0360	0.1723	0.0020	0.47	0.0814	0.0015	1093	12	1024	11	1223	35	**1223**	**35.0**	6.31	951	519
2.3810	0.0440	0.2136	0.0033	0.18	0.0811	0.0018	1236	13	1247	17	1224	45	**1224**	**45.0**	0.89	196	180
2.3620	0.0510	0.2066	0.0034	0.20	0.0825	0.0019	1233	15	1210	18	1251	47	**1251**	**47.0**	1.87	172	147
2.5570	0.0650	0.2149	0.0043	0.34	0.0855	0.0024	1287	19	1253	23	1321	53	**1321**	**53.0**	2.64	133	122
2.3440	0.0560	0.1997	0.0031	0.18	0.0857	0.0020	1223	17	1173	17	1326	46	**1326**	**46.0**	4.09	206	92
3.6470	0.0590	0.2700	0.0036	0.40	0.0974	0.0017	1562	13	1542	18	1575	32	**1575**	**32.0**	1.28	251	199
4.4700	0.1600	0.2889	0.0072	0.31	0.1124	0.0040	1719	29	1633	36	1824	66	**1824**	**66.0**	5.00	43	32
4.7500	0.1100	0.3068	0.0052	0.37	0.1119	0.0026	1775	20	1723	26	1826	45	**1826**	**45.0**	2.93	43	98
5.9490	0.0710	0.3508	0.0041	0.39	0.1228	0.0015	1967	10	1937	19	1994	23	**1994**	**23.0**	1.53	250	301
6.3400	0.1000	0.3454	0.0054	0.42	0.1316	0.0021	2025	14	1914	25	2118	28	**2118**	**28.0**	5.48	96	78
12.1400	0.1700	0.4764	0.0066	0.51	0.1836	0.0026	2615	13	2515	29	2682	24	**2682**	**24.0**	3.82	201	203
12.4000	0.1300	0.4853	0.0059	0.50	0.1853	0.0020	2635	10	2551	25	2701	17	**2701**	**17.0**	3.19	151	112
11.6800	0.1600	0.4199	0.0057	0.61	0.1988	0.0024	2579	13	2261	26	2817	20	**2817**	**20.0**	12.33	289	161

HU2.1 **Lat:** **-30.22192** **Lon:** **-68.39499**

Isotope Ratios							**Apparent Ages**								**Concentrations**	
$^{207}Pb/^{235}U$	± (2σ)	$^{206}Pb/^{238}U$	± (2σ)	Error Corr.	$^{207}Pb/^{206}Pb$	± (2σ)	$^{207}Pb/^{235}U$ (Ma)	± (Ma)	$^{206}Pb/^{238}U$ (Ma)	± (Ma)	$^{207}Pb/^{206}Pb$ (Ma)	± (Ma)	Plot Age (Ma)	± (Ma)	conc. (%)	U (ppm)	Th (ppm)
0.0260	0.0066	0.0029	0.0002	0.12	0.0650	0.0180	25.4	6.5	18.8	1	-	-	**18.4**	**1.0**	25.98	48	42
0.0180	0.0075	0.0031	0.0002	-0.02	0.0550	0.0220	17.3	7.5	20.2	1.3	-	-	**19.9**	**1.3**	16.76	33	26
0.0242	0.0030	0.0035	0.0001	-0.02	0.0489	0.0065	24.1	3	22.68	0.7	-	-	**22.6**	**0.7**	5.89	606	267
0.0236	0.0099	0.0038	0.0003	0.16	0.0370	0.0210	22.2	9.8	24.1	1.7	-	-	**24.4**	**1.7**	8.56	25	19
0.0300	0.0054	0.0039	0.0002	0.08	0.0590	0.0110	30.7	5.3	24.8	1	-	-	**24.4**	**1.0**	19.22	152	56
0.0220	0.0130	0.0039	0.0003	-0.06	0.0690	0.0340	21	13	25.3	1.7	-	-	**24.6**	**1.7**	20.48	62	31
0.0343	0.0029	0.0051	0.0001	0.05	0.0482	0.0041	34.1	2.8	32.99	0.74	-	-	**32.9**	**0.7**	3.26	797	658
0.0890	0.0310	0.0053	0.0006	-0.04	0.2240	0.0970	77	28	34.3	3.6	-	-	**26.7**	**3.6**	55.45	48	99
0.0623	0.0031	0.0092	0.0002	0.07	0.0497	0.0027	61.2	2.9	59	1.3	-	-	**59**	**1.3**	3.59	508	690
0.2400	0.0140	0.0333	0.0007	0.13	0.0517	0.0031	216	11	211.1	4.6	250	120	**211.1**	**4.6**	2.27	207	365
0.3280	0.0130	0.0430	0.0007	0.08	0.0550	0.0022	288	10	271.5	4.3	402	87	**271.5**	**4.3**	5.73	381	452
0.3200	0.0110	0.0432	0.0006	0.22	0.0539	0.0018	280.8	8.2	272.5	3.8	343	69	**272.5**	**3.8**	2.96	89	48
0.3220	0.0130	0.0440	0.0006	0.18	0.0533	0.0021	284.6	9.9	277.4	4	313	83	**277.4**	**4.0**	2.53	-1036	-365
0.3770	0.0260	0.0445	0.0010	0.04	0.0612	0.0045	321	20	280.5	6.2	560	150	**280.5**	**6.2**	12.62	17	6
0.3180	0.0120	0.0449	0.0008	0.12	0.0510	0.0019	280.3	9.1	283.1	4.8	236	79	**283.1**	**4.8**	1.00	236	135
0.3060	0.0160	0.0452	0.0008	0.22	0.0498	0.0026	275	13	284.8	4.9	170	100	**284.8**	**4.9**	3.56	31	13
0.3960	0.0200	0.0480	0.0010	0.39	0.0600	0.0030	336	14	303.1	6.5	580	110	**303.1**	**6.5**	9.79	54	23
0.3910	0.0270	0.0517	0.0015	0.12	0.0562	0.0041	337	21	324.8	9	370	150	**324.8**	**9.0**	3.62	109	88
0.4260	0.0230	0.0547	0.0012	0.20	0.0559	0.0030	359	16	343	7.4	420	110	**343**	**7.4**	4.46	26	7
0.4230	0.0110	0.0551	0.0010	0.22	0.0561	0.0016	358.7	7.8	345.6	5.9	438	60	**345.6**	**5.9**	3.65	524	351
0.4330	0.0130	0.0571	0.0009	0.16	0.0541	0.0017	364.5	9.4	357.9	5.5	371	69	**357.9**	**5.5**	1.81	342	228
0.4990	0.0270	0.0579	0.0010	0.43	0.0621	0.0031	407	18	362.7	6	610	100	**362.7**	**6.0**	10.88	54	29
0.4430	0.0300	0.0616	0.0013	0.14	0.0512	0.0036	369	22	385	8.1	260	140	**385**	**8.1**	4.34	14	7
0.5010	0.0270	0.0616	0.0012	0.17	0.0587	0.0032	409	18	385.3	7.2	510	120	**385.3**	**7.2**	5.79	22	9
0.5270	0.0130	0.0644	0.0009	0.34	0.0588	0.0015	429	8.7	402	5.6	550	54	**402**	**5.6**	6.29	240	127
0.5750	0.0170	0.0652	0.0011	0.43	0.0638	0.0017	459	11	407.3	6.6	716	59	**407.3**	**6.6**	11.26	345	160
0.5160	0.0250	0.0677	0.0011	0.20	0.0554	0.0027	422	17	422.3	6.5	390	100	**422.3**	**6.5**	0.07	168	26

0.6100	0.0210	0.0692	0.0012	0.11	0.0648	0.0025	489	14	431.2	7.4	743	82	**431.2**	**7.4**	11.82	229	167
0.5490	0.0150	0.0695	0.0010	0.38	0.0576	0.0016	443	10	432.9	6	514	61	**432.9**	**6.0**	2.28	442	714
0.5260	0.0130	0.0695	0.0010	0.41	0.0543	0.0013	427.8	9	433.2	5.8	372	53	**433.2**	**5.8**	1.26	593	378
0.5320	0.0210	0.0700	0.0012	0.13	0.0553	0.0022	432	14	435.8	7.1	386	85	**435.8**	**7.1**	0.88	60	35
0.6040	0.0260	0.0703	0.0013	0.24	0.0616	0.0025	482	16	438.5	7.8	643	84	**438.5**	**7.8**	9.02	24	10
0.5570	0.0190	0.0711	0.0010	0.15	0.0568	0.0020	451	12	442.5	6.3	471	73	**442.5**	**6.3**	1.88	217	207
0.5690	0.0210	0.0726	0.0011	-0.05	0.0580	0.0022	456	13	451.9	6.5	506	81	**451.9**	**6.5**	0.90	317	118
0.5770	0.0200	0.0729	0.0010	0.10	0.0569	0.0020	462	13	453.4	6	484	77	**453.4**	**6.0**	1.86	107	66
0.6860	0.0390	0.0733	0.0016	0.69	0.0662	0.0029	529	24	455.5	9.4	792	93	**455.5**	**9.4**	13.89	45	37
0.5950	0.0370	0.0748	0.0017	0.09	0.0585	0.0038	466	24	464	10	460	130	**464**	**10.0**	0.43	10	7
0.6090	0.0300	0.0759	0.0014	0.18	0.0584	0.0029	479	19	471.3	8.6	510	100	**471.3**	**8.6**	1.61	68	37
0.6160	0.0210	0.0764	0.0013	0.12	0.0584	0.0022	485	13	474.3	7.6	514	83	**474.3**	**7.6**	2.21	42	20
0.6020	0.0950	0.0783	0.0030	0.07	0.0570	0.0096	488	62	485	18	440	270	**485**	**18.0**	0.61	41	73
0.6270	0.0280	0.0785	0.0016	0.26	0.0579	0.0024	495	17	487.9	9.6	488	88	**487.9**	**9.6**	1.43	52	33
0.6780	0.0350	0.0788	0.0015	0.30	0.0620	0.0032	528	20	489	9.1	640	110	**489**	**9.1**	7.39	20	12
0.6560	0.0300	0.0790	0.0016	0.08	0.0598	0.0029	507	18	489.9	9.4	550	100	**489.9**	**9.4**	3.37	51	20
0.6030	0.0290	0.0795	0.0014	0.14	0.0555	0.0027	478	19	493.1	8.3	400	100	**493.1**	**8.3**	3.16	18	10
0.6980	0.0400	0.0797	0.0014	-0.03	0.0639	0.0039	529	24	494.4	8.5	640	120	**494.4**	**8.5**	6.54	17	6
0.6430	0.0270	0.0804	0.0013	0.12	0.0571	0.0025	505	16	498.2	7.9	482	92	**498.2**	**7.9**	1.35	138	60
0.6410	0.0170	0.0804	0.0011	0.32	0.0567	0.0015	501	11	499.1	6.9	481	59	**499.1**	**6.9**	0.38	297	117
0.7400	0.0290	0.0809	0.0015	0.38	0.0653	0.0023	558	17	501.3	9.1	773	73	**501.3**	**9.1**	10.16	39	14
0.7180	0.0600	0.0815	0.0022	0.18	0.0637	0.0056	534	37	505	13	630	170	**505**	**13.0**	5.43	21	19
0.6650	0.0260	0.0817	0.0014	0.19	0.0581	0.0023	514	16	506.4	8.3	506	88	**506.4**	**8.3**	1.48	34	11
0.6560	0.0200	0.0819	0.0011	0.18	0.0578	0.0019	516	12	507	6.7	514	70	**507**	**6.7**	1.74	191	290
0.7260	0.0410	0.0828	0.0020	0.14	0.0630	0.0038	547	24	512	12	640	130	**512**	**12.0**	6.40	50	21
0.6550	0.0300	0.0831	0.0017	0.28	0.0575	0.0025	513	18	514.6	9.8	511	91	**514.6**	**9.8**	0.31	62	11
0.7120	0.0240	0.0832	0.0012	0.16	0.0614	0.0022	546	14	514.9	7.4	675	76	**514.9**	**7.4**	5.70	469	464
0.7210	0.0270	0.0841	0.0017	0.19	0.0638	0.0025	549	16	520.5	9.9	706	84	**520.5**	**9.9**	5.19	34	38
0.6940	0.0290	0.0842	0.0013	0.17	0.0592	0.0026	536	17	521	7.9	560	92	**521**	**7.9**	2.80	124	59
0.7150	0.0460	0.0842	0.0020	0.20	0.0620	0.0040	540	27	521	12	600	130	**521**	**12.0**	3.52	19	12
0.7170	0.0470	0.0841	0.0021	0.21	0.0605	0.0040	543	27	524	12	550	130	**524**	**12.0**	3.50	37	7
0.7100	0.0250	0.0853	0.0010	0.27	0.0601	0.0019	544	15	527.7	6.1	610	72	**527.7**	**6.1**	3.00	396	408
0.7170	0.0180	0.0860	0.0011	0.28	0.0599	0.0015	548	11	531.9	6.6	603	55	**531.9**	**6.6**	2.94	433	388
0.7000	0.0310	0.0861	0.0017	0.05	0.0596	0.0029	538	18	532	10	560	100	**532**	**10.0**	1.12	157	116
0.6930	0.0580	0.0858	0.0025	0.17	0.0578	0.0047	538	35	532	15	510	160	**532**	**15.0**	1.12	35	75
0.7510	0.0200	0.0866	0.0012	0.20	0.0619	0.0017	569	12	535.5	7.2	670	56	**535.5**	**7.2**	5.89	497	193
0.7350	0.0260	0.0865	0.0013	0.27	0.0612	0.0020	556	15	535.7	8.1	621	68	**535.7**	**8.1**	3.65	42	16
0.7430	0.0250	0.0874	0.0017	0.32	0.0611	0.0021	563	15	541	9.8	643	75	**541**	**9.8**	3.91	-641	-275
0.7530	0.0310	0.0882	0.0015	0.12	0.0614	0.0027	567	18	544.5	9.2	626	97	**544.5**	**9.2**	3.97	98	39
0.7250	0.0260	0.0886	0.0015	0.37	0.0595	0.0021	555	15	547.1	9.1	576	75	**547.1**	**9.1**	1.42	25	34
0.7380	0.0160	0.0893	0.0012	0.43	0.0595	0.0012	559.7	9.4	551	7	579	45	**551**	**7.0**	1.55	214	127
0.7120	0.0300	0.0894	0.0015	0.25	0.0573	0.0024	548	17	551.5	9	477	89	**551.5**	**9.0**	0.64	23	9
0.7170	0.0200	0.0897	0.0015	0.14	0.0579	0.0018	548	12	553.5	9.1	518	70	**553.5**	**9.1**	1.00	36	18
0.8670	0.0280	0.0907	0.0015	0.56	0.0698	0.0019	633	15	559.3	9	909	54	**559.3**	**9.0**	11.64	57	16
0.7650	0.0450	0.0922	0.0023	0.17	0.0601	0.0037	575	26	568	14	600	120	**568**	**14.0**	1.22	9	2
0.7870	0.0280	0.0938	0.0015	0.22	0.0605	0.0023	591	17	577.8	8.9	598	83	**577.8**	**8.9**	2.23	23	18
0.8710	0.0290	0.0944	0.0014	0.50	0.0670	0.0020	634	15	581.3	8.3	830	62	**581.3**	**8.3**	8.31	56	22
0.8320	0.0280	0.0951	0.0019	0.22	0.0630	0.0022	616	16	585	11	707	73	**585**	**11.0**	5.03	25	8
0.9350	0.0210	0.0959	0.0016	0.26	0.0711	0.0017	668	11	590.9	9.3	942	48	**590.9**	**9.3**	11.54	297	117
0.7700	0.0320	0.0962	0.0017	0.09	0.0588	0.0027	579	18	592	9.8	505	96	**592**	**9.8**	2.25	122	219
0.8350	0.0240	0.0987	0.0015	0.25	0.0617	0.0018	617	13	606.3	9.1	655	60	**606.3**	**9.1**	1.73	43	16
0.8430	0.0340	0.0995	0.0017	0.22	0.0614	0.0026	622	18	611.1	9.8	658	87	**611.1**	**9.8**	1.75	24	6
0.8490	0.0530	0.1004	0.0024	0.01	0.0617	0.0042	621	30	616	14	640	140	**616**	**14.0**	0.81	11	21
0.9600	0.0540	0.1074	0.0023	0.05	0.0643	0.0038	682	29	658	13	730	120	**658**	**13.0**	3.52	11	2
1.0120	0.0510	0.1163	0.0028	0.08	0.0642	0.0034	713	26	709	16	700	110	**709**	**16.0**	0.56	11	2
1.1450	0.0510	0.1190	0.0029	0.43	0.0702	0.0026	769	24	724	17	914	77	**724**	**17.0**	5.85	18	8
1.1670	0.0240	0.1211	0.0019	0.45	0.0699	0.0013	786	11	737	11	923	39	**737**	**11.0**	6.23	275	128
1.1260	0.0250	0.1216	0.0017	0.43	0.0670	0.0014	765	12	739.4	9.7	850	43	**739.4**	**9.7**	3.35	379	394
1.3290	0.0340	0.1446	0.0021	0.24	0.0674	0.0018	857	15	870	12	826	58	**826**	**58.0**	1.52	40	21
1.4320	0.0800	0.1500	0.0038	0.15	0.0696	0.0040	892	35	900	21	840	120	**840**	**120.0**	0.90	19	15
1.4070	0.0400	0.1473	0.0023	0.36	0.0676	0.0018	893	18	886	13	870	56	**870**	**56.0**	0.78	31	17
1.3400	0.0270	0.1416	0.0015	0.50	0.0685	0.0013	861	12	853.5	8.3	877	41	**877**	**41.0**	0.87	585	277
1.7600	0.1200	0.1681	0.0050	0.03	0.0744	0.0055	1023	44	1000	28	980	150	**980**	**150.0**	2.25	4	3
1.7700	0.1100	0.1756	0.0041	0.07	0.0734	0.0047	1048	40	1044	23	990	130	**990**	**130.0**	0.38	11	3
1.7370	0.0920	0.1710	0.0039	0.25	0.0745	0.0039	1018	34	1019	22	1000	110	**1000**	**110.0**	0.10	6	3
1.5750	0.0350	0.1570	0.0022	0.26	0.0730	0.0016	961	13	940	12	1007	44	**1007**	**44.0**	2.19	271	109
1.6330	0.0350	0.1610	0.0021	0.35	0.0730	0.0015	983	13	962	12	1021	43	**1021**	**43.0**	2.14	188	57
1.7510	0.0420	0.1700	0.0026	0.30	0.0740	0.0018	1026	15	1013	14	1034	48	**1034**	**48.0**	1.27	356	-254
1.6780	0.0390	0.1633	0.0024	0.32	0.0740	0.0017	1000	14	975	13	1044	45	**1044**	**45.0**	2.50	35	6
1.7560	0.0560	0.1692	0.0027	0.22	0.0756	0.0024	1027	21	1007	15	1048	67	**1048**	**67.0**	1.95	51	29
1.7710	0.0580	0.1710	0.0022	0.35	0.0739	0.0024	1038	21	1018	12	1054	63	**1054**	**63.0**	1.93	-1400	-284
1.6080	0.0480	0.1553	0.0027	0.41	0.0739	0.0018	971	18	930	15	1056	53	**1056**	**53.0**	4.22	44	13
1.6500	0.0680	0.1596	0.0031	0.28	0.0746	0.0030	993	26	954	17	1062	81	**1062**	**81.0**	3.93	67	36
1.8120	0.0410	0.1698	0.0029	0.33	0.0760	0.0019	1048	15	1010	16	1095	50	**1095**	**50.0**	3.63	30	9
1.8300	0.0340	0.1754	0.0023	0.45	0.0759	0.0013	1054	12	1041	13	1095	34	**1095**	**34.0**	1.23	68	12

1.7780	0.0480	0.1682	0.0025	0.15	0.0764	0.0024	1038	18	1002	14	1097	64	**1097**	**64.0**	3.47	26	9
1.8800	0.1200	0.1750	0.0048	0.10	0.0762	0.0051	1066	41	1041	26	1110	130	**1110**	**130.0**	2.35	32	18
1.8430	0.0660	0.1704	0.0037	0.28	0.0776	0.0027	1063	24	1015	20	1156	69	**1156**	**69.0**	4.52	14	9
1.9550	0.0660	0.1754	0.0031	0.08	0.0802	0.0030	1098	23	1041	17	1176	78	**1176**	**78.0**	5.19	70	28
2.1350	0.0590	0.1917	0.0032	0.20	0.0811	0.0024	1157	19	1130	17	1206	61	**1206**	**61.0**	2.33	23	16
2.2610	0.0570	0.2009	0.0031	0.42	0.0811	0.0019	1200	18	1179	17	1207	48	**1207**	**48.0**	1.75	79	51
1.9290	0.0830	0.1694	0.0034	0.32	0.0820	0.0035	1087	29	1008	19	1212	88	**1212**	**88.0**	7.27	10	5
1.8250	0.0560	0.1608	0.0026	0.39	0.0823	0.0023	1048	20	961	14	1234	52	**1234**	**52.0**	8.30	64	25
2.3120	0.0440	0.2025	0.0026	0.43	0.0823	0.0014	1213	14	1188	14	1249	33	**1249**	**33.0**	2.06	900	-105
2.2200	0.0860	0.1971	0.0037	0.14	0.0827	0.0032	1189	28	1159	20	1250	77	**1250**	**77.0**	2.52	33	17
1.7820	0.0820	0.1526	0.0023	0.50	0.0838	0.0034	1029	29	915	13	1267	77	**1267**	**77.0**	11.08	73	50
2.0670	0.0490	0.1679	0.0030	0.65	0.0897	0.0017	1139	16	1000	17	1422	35	**1422**	**35.0**	12.20	155	63
3.2660	0.0420	0.2504	0.0026	0.55	0.0953	0.0011	1472	10	1440	13	1533	22	**1533**	**22.0**	2.17	503	333
2.7500	0.1200	0.1942	0.0032	0.22	0.1009	0.0041	1332	31	1143	17	1629	78	**1629**	**78.0**	14.19	13	5
4.3030	0.0900	0.2762	0.0039	0.56	0.1127	0.0020	1691	17	1571	20	1852	34	**1852**	**34.0**	7.10	29	28
5.6190	0.0650	0.3313	0.0033	0.60	0.1232	0.0012	1918	10	1848	16	2000	18	**2000**	**18.0**	3.65	665	297
5.2700	0.1800	0.2833	0.0042	0.43	0.1318	0.0041	1857	29	1609	21	2106	54	**2106**	**54.0**	13.35	-322	-189
5.0500	0.1200	0.2791	0.0047	0.45	0.1325	0.0028	1829	20	1588	23	2131	36	**2131**	**36.0**	13.18	22	35
9.4500	0.1400	0.4291	0.0056	0.70	0.1592	0.0019	2383	14	2303	25	2442	21	**2442**	**21.0**	3.36	32	52
10.3700	0.1400	0.4130	0.0052	0.60	0.1834	0.0024	2466	13	2227	24	2683	22	**2683**	**22.0**	9.69	45	19
12.2900	0.2200	0.4390	0.0075	0.67	0.2026	0.0032	2625	17	2347	34	2845	25	**2845**	**25.0**	10.59	28	25
18.6100	0.2200	0.5466	0.0066	0.72	0.2456	0.0023	3022	11	2809	28	3154	15	**3154**	**15.0**	7.05	355	47
21.6200	0.3900	0.5912	0.0092	0.55	0.2643	0.0040	3169	18	2995	37	3271	24	**3271**	**24.0**	5.49	47	38

HU-5 **Lat:** **-30.21285** **Lon:** **-68.38584**

Isotope Ratios							**Apparent Ages**								**Concentrations**		
$^{207}Pb/^{235}U$	**± (2σ)**	**$^{206}Pb/^{238}U$**	**± (2σ)**	**Error Corr.**	**$^{207}Pb/^{206}Pb$**	**± (2σ)**	**$^{207}Pb/^{235}U$ (Ma)**	**± (Ma)**	**$^{206}Pb/^{238}U$ (Ma)**	**± (Ma)**	**$^{207}Pb/^{206}Pb$ (Ma)**	**± (Ma)**	**Plot Age (Ma)**	**± (Ma)**	**conc. (%)**	**U (ppm)**	**Th (ppm)**
0.0091	0.0082	0.0024	0.0002	0.05	0.0310	0.0310	8.3	8.2	15.7	1.1	-	-	**16.0**	**1.1**	89.16	321	166
0.0110	0.0230	0.0033	0.0004	-0.09	0.2000	1.2000	9	24	21	2.3	-	-	**17.0**	**2.3**	133.33	55	39
0.0150	0.0170	0.0039	0.0004	-0.16	0.0300	0.0840	11	18	24.8	2.4	-	-	**25.4**	**2.4**	125.45	121	116
0.0280	0.0120	0.0040	0.0003	-0.15	0.0000	0.1500	26	12	25.6	1.8	-	-	**27.0**	**1.8**	1.54	160	125
0.0120	0.0240	0.0040	0.0005	-0.05	0.0870	0.0860	11	25	26	2.9	-	-	**24.7**	**2.9**	136.36	13	10
0.0306	0.0076	0.0045	0.0002	-0.09	0.0530	0.0130	30.6	7.4	29.1	1.1	-	-	**28.9**	**1.1**	4.90	284	434
0.0340	0.0400	0.0047	0.0006	-0.01	-0.1700	0.3200	22	38	30	3.8	-	-	**38.1**	**3.8**	36.36	45	41
0.0330	0.0360	0.0050	0.0007	-0.02	0.1220	0.0870	26	36	31.9	4.6	-	-	**28.9**	**4.6**	22.69	19	5
0.0921	0.0049	0.0138	0.0003	-0.03	0.0488	0.0026	89.7	4.4	88.1	1.6	-	-	**88.1**	**1.6**	1.78	163	114
0.3560	0.0200	0.0422	0.0013	-0.31	0.0604	0.0042	312	15	266.4	8	600	150	**266.4**	**8.0**	14.62	178	130
0.3640	0.0270	0.0437	0.0011	0.07	0.0598	0.0046	314	21	276.3	6.7	540	160	**276.3**	**6.7**	12.01	24	28
0.3090	0.0210	0.0456	0.0009	0.27	0.0502	0.0033	272	16	287.2	5.3	210	130	**287.2**	**5.3**	5.59	264	172
0.3370	0.0190	0.0457	0.0008	0.08	0.0524	0.0029	293	14	288.2	4.8	320	120	**288.2**	**4.8**	1.64	535	512
0.3210	0.0350	0.0464	0.0014	0.02	0.0500	0.0059	276	28	292.3	8.4	180	200	**292.3**	**8.4**	5.91	87	60
0.3640	0.0150	0.0483	0.0009	0.30	0.0537	0.0021	315	11	304	5.4	332	87	**304**	**5.4**	3.49	1126	1560
0.4220	0.0310	0.0546	0.0012	0.11	0.0571	0.0043	356	22	343.6	7.3	420	140	**343.6**	**7.3**	3.48	134	106
0.4110	0.0250	0.0553	0.0011	0.18	0.0538	0.0032	353	18	346.9	6.5	360	120	**346.9**	**6.5**	1.73	229	186
0.4660	0.0400	0.0589	0.0024	0.11	0.0595	0.0054	383	27	369	14	480	180	**369**	**14.0**	3.66	196	99
0.4860	0.0210	0.0615	0.0013	0.39	0.0577	0.0023	401	14	384.6	7.8	497	86	**384.6**	**7.8**	4.09	303	246
0.4580	0.0150	0.0629	0.0009	0.07	0.0520	0.0018	382	11	393.3	5.2	289	74	**393.3**	**5.2**	2.96	898	834
0.5000	0.0200	0.0663	0.0010	0.34	0.0548	0.0020	410	13	413.7	6.3	387	77	**413.7**	**6.3**	0.90	415	199
0.5170	0.0670	0.0687	0.0022	0.04	0.0549	0.0074	405	46	428	13	270	240	**428**	**13.0**	5.68	48	25
0.5190	0.0270	0.0699	0.0012	0.18	0.0539	0.0028	420	18	435.2	7.4	320	100	**435.2**	**7.4**	3.62	18	4
0.5660	0.0240	0.0701	0.0017	0.50	0.0598	0.0023	455	15	437	10	587	83	**437**	**10.0**	3.96	111	41
0.5620	0.0190	0.0703	0.0011	0.14	0.0573	0.0021	451	12	437.9	6.4	491	76	**437.9**	**6.4**	2.90	122	192
0.5690	0.0230	0.0712	0.0010	0.22	0.0568	0.0023	459	15	443	5.9	456	85	**443**	**5.9**	3.49	172	47
0.5620	0.0260	0.0711	0.0012	-0.07	0.0568	0.0029	451	17	443.5	7.3	450	110	**443.5**	**7.3**	1.66	427	455
0.5640	0.0300	0.0715	0.0015	0.01	0.0561	0.0031	454	20	445.1	8.8	470	110	**445.1**	**8.8**	1.96	118	107
0.5480	0.0160	0.0720	0.0010	0.24	0.0557	0.0018	443	11	448	6.2	425	70	**448**	**6.2**	1.13	713	455
0.5820	0.0280	0.0739	0.0012	0.07	0.0582	0.0028	465	18	459.2	7.2	520	100	**459.2**	**7.2**	1.25	241	109
0.5910	0.0310	0.0743	0.0015	0.18	0.0571	0.0030	470	20	462.7	9.2	480	110	**462.7**	**9.2**	1.55	266	139
0.5630	0.0240	0.0750	0.0011	0.26	0.0542	0.0022	455	16	466.2	6.9	376	91	**466.2**	**6.9**	2.46	45	6
0.5730	0.0280	0.0761	0.0014	0.06	0.0548	0.0029	455	18	472.8	8.3	400	110	**472.8**	**8.3**	3.91	37	21
0.5980	0.0170	0.0776	0.0010	0.25	0.0567	0.0016	477	11	481.8	6	475	64	**481.8**	**6.0**	1.01	418	400
0.6910	0.0480	0.0777	0.0023	0.52	0.0663	0.0040	531	28	482	14	770	120	**482**	**14.0**	9.23	286	246
0.6320	0.0240	0.0789	0.0011	0.18	0.0565	0.0021	495	15	489.6	6.8	453	80	**489.6**	**6.8**	1.09	100	21
0.6100	0.0250	0.0796	0.0011	-0.01	0.0561	0.0023	485	16	493.5	6.5	445	91	**493.5**	**6.5**	1.75	52	36
0.7780	0.0720	0.0803	0.0017	0.25	0.0709	0.0062	571	41	497	10	850	180	**497**	**10.0**	12.96	140	142
0.6220	0.0250	0.0810	0.0011	0.17	0.0557	0.0022	488	16	502.8	6.5	403	84	**502.8**	**6.5**	3.03	267	69
0.7000	0.0400	0.0812	0.0018	0.20	0.0621	0.0036	541	24	503	11	640	120	**503**	**11.0**	7.02	234	313
0.6870	0.0370	0.0813	0.0021	0.38	0.0608	0.0030	528	22	504	12	590	110	**504**	**12.0**	4.55	61	39
0.5940	0.0370	0.0814	0.0015	0.27	0.0551	0.0034	474	23	504.1	8.9	370	120	**504.1**	**8.9**	6.35	234	222
0.6270	0.0170	0.0818	0.0012	0.10	0.0560	0.0017	494	11	506.9	7	448	66	**506.9**	**7.0**	2.61	738	825
0.6540	0.0430	0.0830	0.0016	-0.02	0.0580	0.0040	518	25	513.6	9.4	450	130	**513.6**	**9.4**	0.85	131	63
0.6430	0.0570	0.0832	0.0022	-0.02	0.0576	0.0056	498	37	515	13	430	180	**515**	**13.0**	3.41	97	52
0.6610	0.0240	0.0835	0.0011	0.12	0.0582	0.0021	516	15	517	6.5	518	80	**517**	**6.5**	0.19	388	319

0.6690	0.0690	0.0840	0.0021	0.15	0.0577	0.0058	508	43	520	13	390	190	**520**	**13.0**	2.36	44	24
0.6740	0.0200	0.0842	0.0013	0.29	0.0576	0.0017	522	12	520.8	7.5	513	64	**520.8**	**7.5**	0.23	1136	212
0.6840	0.0230	0.0845	0.0012	0.19	0.0588	0.0021	529	14	522.6	7.2	546	79	**522.6**	**7.2**	1.21	753	241
0.7130	0.0290	0.0853	0.0012	0.32	0.0607	0.0023	546	18	527.4	7.3	609	86	**527.4**	**7.3**	3.41	410	243
0.6870	0.0380	0.0854	0.0016	0.30	0.0578	0.0030	531	23	529.2	9.1	540	110	**529.2**	**9.1**	0.34	157	293
0.7050	0.0190	0.0865	0.0011	0.46	0.0594	0.0014	540	11	534.6	6.4	568	54	**534.6**	**6.4**	1.00	476	56
0.6970	0.0330	0.0867	0.0015	-0.04	0.0579	0.0029	534	20	536	8.7	480	110	**536**	**8.7**	0.37	367	242
0.6810	0.0170	0.0870	0.0010	0.23	0.0578	0.0014	527.2	9.9	537.5	6.1	514	55	**537.5**	**6.1**	1.95	1593	450
0.7240	0.0190	0.0875	0.0013	0.39	0.0604	0.0015	553	12	540.6	7.7	598	55	**540.6**	**7.7**	2.24	808	151
0.7300	0.0320	0.0876	0.0015	0.22	0.0606	0.0027	554	19	541.1	9.2	569	95	**541.1**	**9.2**	2.33	235	125
0.7240	0.0430	0.0877	0.0014	0.02	0.0596	0.0036	549	25	541.9	8.5	540	130	**541.9**	**8.5**	1.29	115	44
0.7700	0.0360	0.0893	0.0017	0.03	0.0618	0.0030	577	21	551	10	650	100	**551**	**10.0**	4.51	131	70
0.7170	0.0230	0.0904	0.0015	0.24	0.0579	0.0018	550	13	557.7	9	521	70	**557.7**	**9.0**	1.40	123	8
0.7670	0.0310	0.0904	0.0017	0.30	0.0609	0.0025	580	17	558	10	603	87	**558**	**10.0**	3.79	109	36
0.7420	0.0460	0.0909	0.0018	0.24	0.0577	0.0034	560	26	562	11	460	120	**562**	**11.0**	0.36	39	48
0.8000	0.0390	0.0928	0.0019	-0.02	0.0616	0.0033	591	22	572	11	620	110	**572**	**11.0**	3.21	24	25
0.7740	0.0190	0.0929	0.0013	0.39	0.0611	0.0014	582	11	572.5	7.6	625	50	**572.5**	**7.6**	1.63	146	82
0.7620	0.0190	0.0934	0.0010	0.42	0.0594	0.0014	573	11	575.3	5.9	574	53	**575.3**	**5.9**	0.40	502	257
0.8010	0.0440	0.0948	0.0018	0.23	0.0617	0.0034	601	26	583	11	620	120	**583**	**11.0**	3.00	25	23
0.8420	0.0430	0.0948	0.0014	0.30	0.0649	0.0030	620	23	583.4	8.4	745	99	**583.4**	**8.4**	5.90	225	291
0.7780	0.0800	0.0953	0.0025	-0.08	0.0627	0.0069	591	46	587	15	550	200	**587**	**15.0**	0.68	11	4
0.8170	0.0420	0.0971	0.0017	0.11	0.0595	0.0031	609	23	597	10	590	110	**597**	**10.0**	1.97	153	47
0.8790	0.0330	0.1010	0.0032	0.50	0.0642	0.0024	638	18	620	18	781	74	**620**	**18.0**	2.82	855	925
0.8500	0.0340	0.1013	0.0020	0.35	0.0601	0.0022	626	19	622	12	622	83	**622**	**12.0**	0.64	193	185
0.9530	0.0270	0.1073	0.0014	0.18	0.0660	0.0019	677	14	657	8.3	785	64	**657**	**8.3**	2.95	287	181
1.0510	0.0200	0.1187	0.0013	0.23	0.0636	0.0011	730.2	9.9	722.8	7.4	719	39	**722.8**	**7.4**	1.01	674	496
1.2550	0.0500	0.1218	0.0022	0.40	0.0743	0.0028	826	22	741	13	1052	75	**741**	**13.0**	10.29	234	63
1.7900	0.0450	0.1764	0.0022	0.26	0.0729	0.0018	1040	17	1047	12	989	50	**989**	**50.0**	0.67	560	344
1.7000	0.1100	0.1659	0.0038	0.09	0.0749	0.0051	991	44	989	21	990	140	**990**	**140.0**	0.20	71	43
1.6120	0.0310	0.1598	0.0016	0.23	0.0733	0.0014	976	12	955.3	8.8	1018	39	**1018**	**39.0**	2.12	100	39
1.6780	0.0530	0.1657	0.0028	0.27	0.0736	0.0021	997	20	988	15	1037	61	**1037**	**61.0**	0.90	28	23
1.6320	0.0770	0.1624	0.0032	0.29	0.0743	0.0035	980	30	970	18	1037	96	**1037**	**96.0**	1.02	130	94
1.4490	0.0860	0.1400	0.0033	0.17	0.0755	0.0043	912	36	846	19	1050	120	**1050**	**120.0**	7.24	106	37
1.7550	0.0610	0.1685	0.0026	0.18	0.0754	0.0026	1027	23	1003	14	1076	72	**1076**	**72.0**	2.34	224	113
1.7600	0.0370	0.1687	0.0023	0.26	0.0754	0.0017	1028	14	1005	13	1079	43	**1079**	**43.0**	2.24	284	316
1.7800	0.1100	0.1687	0.0037	0.09	0.0760	0.0046	1032	39	1006	20	1080	130	**1080**	**130.0**	2.52	69	23
1.6950	0.0400	0.1663	0.0022	0.40	0.0759	0.0017	1007	15	992	12	1087	45	**1087**	**45.0**	1.49	574	351
1.8330	0.0470	0.1765	0.0022	0.34	0.0763	0.0017	1059	16	1047	12	1101	46	**1101**	**46.0**	1.13	531	754
1.7580	0.0590	0.1655	0.0025	0.11	0.0766	0.0026	1031	22	987	14	1115	67	**1115**	**67.0**	4.27	103	37
1.9100	0.1000	0.1799	0.0039	0.27	0.0791	0.0043	1084	36	1066	22	1120	110	**1120**	**110.0**	1.66	94	48
1.9040	0.0580	0.1779	0.0026	0.11	0.0779	0.0023	1082	19	1056	14	1140	62	**1140**	**62.0**	2.40	148	127
1.9800	0.0480	0.1836	0.0023	0.38	0.0786	0.0017	1108	16	1086	12	1151	44	**1151**	**44.0**	1.99	415	223
1.8430	0.0580	0.1671	0.0029	0.31	0.0780	0.0025	1064	21	995	16	1176	64	**1176**	**64.0**	6.48	76	21
1.7380	0.0600	0.1590	0.0025	0.30	0.0806	0.0029	1016	22	951	14	1179	67	**1179**	**67.0**	6.40	695	790
1.9490	0.0380	0.1793	0.0023	0.37	0.0804	0.0016	1100	12	1063	13	1201	40	**1201**	**40.0**	3.36	228	112
2.0690	0.0780	0.1847	0.0032	0.28	0.0804	0.0028	1138	26	1092	17	1213	74	**1213**	**74.0**	4.04	21	7
2.0670	0.0810	0.1825	0.0035	0.35	0.0819	0.0031	1131	27	1080	19	1237	75	**1237**	**75.0**	4.51	195	88
1.8000	0.1100	0.1557	0.0025	0.49	0.0835	0.0043	1027	38	932	14	1260	100	**1260**	**100.0**	9.25	119	63
2.5540	0.0560	0.2201	0.0029	0.58	0.0834	0.0014	1289	16	1282	15	1283	33	**1283**	**33.0**	0.54	356	224
2.5950	0.0730	0.2149	0.0031	0.34	0.0855	0.0024	1293	21	1254	16	1340	53	**1340**	**53.0**	3.02	22	7
4.1400	0.0690	0.2870	0.0038	0.41	0.1047	0.0016	1662	14	1626	19	1700	29	**1700**	**29.0**	2.17	377	317
5.6900	0.1000	0.3401	0.0038	0.35	0.1203	0.0019	1927	16	1890	18	1963	29	**1963**	**29.0**	1.92	66	61
8.6100	0.1300	0.4012	0.0048	0.57	0.1575	0.0020	2299	14	2174	22	2427	21	**2427**	**21.0**	5.44	245	120
11.7800	0.1600	0.4596	0.0049	0.56	0.1875	0.0022	2586	13	2437	22	2721	20	**2721**	**20.0**	5.76	229	156

HU-6.1 **Lat:** **-30.20465** **Lon:** **-68.37821**

Isotope Ratios							**Apparent Ages**								**Concentrations**		
$^{207}Pb/^{235}U$	**±**	**$^{206}Pb/^{238}U$**	**±**	**Error**	**$^{207}Pb/^{206}Pb$**	**±**	**$^{207}Pb/^{235}U$**	**±**	**$^{206}Pb/^{238}U$**	**±**	**$^{207}Pb/^{206}Pb$**	**±**	**Plot Age**	**±**	**conc.**	**U**	**Th**
	(2σ)		**(2σ)**	**Corr.**		**(2σ)**	**(Ma)**	**(Ma)**	**(Ma)**	**(Ma)**	**(Ma)**	**(Ma)**	**(Ma)**	**(Ma)**	**(%)**	**(ppm)**	**(ppm)**
0.0052	0.0005	0.0009	0.0000	0.18	0.0441	0.0052	5.25	0.52	5.48	0.16	-	-	**5.5**	**0.2**	1.05	460	283
0.0148	0.0028	0.0021	0.0000	-0.04	0.0506	0.0095	14.9	2.8	13.56	0.27	-	-	**13.5**	**0.4**	0.91	185	89
0.0156	0.0048	0.0025	0.0001	0.06	0.0440	0.0130	15.7	4.8	16.16	0.39	-	-	**16.2**	**0.6**	1.03	63	70
0.0212	0.0022	0.0034	0.0001	0.03	0.0450	0.0045	21.3	2.2	21.66	0.52	-	-	**21.7**	**0.7**	1.02	151	181
0.0261	0.0034	0.0037	0.0001	-0.01	0.0494	0.0072	26.2	3.3	24.01	0.53	-	-	**23.9**	**0.7**	0.92	88	57
0.0238	0.0014	0.0040	0.0001	0.47	0.0431	0.0021	23.9	1.4	25.5	0.52	-	-	**25.6**	**0.7**	1.07	380	123
0.0283	0.0034	0.0043	0.0001	0.44	0.0484	0.0057	28.3	3.3	27.34	0.61	-	-	**27.3**	**0.8**	0.97	80	79
0.0315	0.0020	0.0052	0.0001	0.13	0.0436	0.0026	31.5	2	33.26	0.54	-	-	**33.4**	**0.9**	1.06	216	150
0.0352	0.0026	0.0055	0.0001	0.15	0.0463	0.0035	35.1	2.5	35.51	0.65	-	-	**35.5**	**1.0**	1.01	67	25
0.0365	0.0090	0.0057	0.0002	0.06	0.0500	0.0130	38.4	8.1	36.5	1.1	-	-	**36.3**	**1.4**	1.00	41	41
0.2602	0.0044	0.0368	0.0004	0.53	0.0508	0.0007	234.8	3.5	233	2.4	231	32	**233.0**	**5.2**	0.99	454	331
0.2707	0.0075	0.0383	0.0003	0.62	0.0510	0.0010	243.1	6	242.3	2	249	51	**242.4**	**5.2**	1.00	112	91
0.2933	0.0063	0.0411	0.0008	0.94	0.0519	0.0004	261.1	4.9	259.9	4.8	281	17	**259.7**	**7.0**	1.00	373	486
0.3101	0.0053	0.0428	0.0004	0.55	0.0522	0.0008	274.2	4.1	270.3	2.1	293	34	**270.2**	**5.8**	0.99	394	189

0.3262	0.0073	0.0457	0.0007	0.31	0.0520	0.0012	286.6	5.6	287.7	4.4	296	46	**287.8**	**7.2**	1.00	105	76
0.3354	0.0069	0.0460	0.0007	0.80	0.0530	0.0007	293.6	5.3	289.6	4.3	325	29	**289.3**	**7.2**	0.99	348	183
0.3378	0.0096	0.0462	0.0007	0.31	0.0532	0.0015	295.4	7.3	291.4	4.2	333	64	**291.0**	**7.1**	0.99	112	65
0.3440	0.0042	0.0471	0.0004	0.45	0.0532	0.0005	300.2	3.1	296.7	2.3	335	22	**296.4**	**6.3**	0.99	272	126
0.3410	0.0110	0.0479	0.0008	0.76	0.0513	0.0010	297.8	8	301.6	4.6	262	49	**302.1**	**7.6**	1.01	161	62
0.3690	0.0150	0.0483	0.0006	0.74	0.0551	0.0019	318	11	303.9	3.9	406	73	**302.8**	**7.2**	0.95	237	127
0.3658	0.0070	0.0492	0.0006	0.57	0.0532	0.0009	316.5	5.2	309.3	3.7	342	39	**309.1**	**7.1**	0.98	245	91
0.3503	0.0085	0.0493	0.0005	0.74	0.0516	0.0009	304.8	6.4	310.3	3.3	266	39	**310.7**	**7.0**	1.02	106	45
0.3614	0.0090	0.0497	0.0011	0.82	0.0526	0.0007	313.2	6.7	312.4	6.5	317	33	**312.7**	**9.2**	1.00	208	108
0.3596	0.0074	0.0498	0.0006	0.63	0.0521	0.0009	311.8	5.5	313.1	3.7	289	39	**313.3**	**7.2**	1.00	211	121
0.3710	0.0130	0.0511	0.0009	0.64	0.0531	0.0017	322	10	321.1	5.7	324	72	**321.0**	**8.5**	1.00	80	35
0.3830	0.0140	0.0517	0.0009	0.65	0.0541	0.0015	329	11	324.9	5.3	369	62	**324.5**	**8.4**	0.99	87	61
0.3810	0.0079	0.0519	0.0006	0.43	0.0528	0.0009	327.7	5.8	326.1	3.5	317	41	**326.1**	**7.3**	0.99	132	72
0.4228	0.0090	0.0566	0.0008	0.63	0.0538	0.0009	357.9	6.4	354.8	4.9	361	37	**354.7**	**8.6**	0.99	217	43
0.4656	0.0099	0.0606	0.0009	0.83	0.0552	0.0007	388.1	6.9	379.2	5.3	419	26	**378.7**	**9.2**	0.98	253	265
0.4820	0.0100	0.0635	0.0011	0.30	0.0547	0.0011	399.1	6.8	396.9	6.5	399	45	**396.8**	**10.3**	0.99	83	42
0.4850	0.0085	0.0638	0.0009	0.76	0.0546	0.0006	401.4	5.8	398.8	5.6	395	25	**398.9**	**9.7**	0.99	250	251
0.4918	0.0070	0.0646	0.0007	0.78	0.0550	0.0006	406.1	4.8	403.2	4.5	412	22	**403.1**	**9.1**	0.99	597	217
0.4970	0.0077	0.0647	0.0006	0.55	0.0557	0.0007	409.6	5.2	404	3.4	438	27	**403.6**	**8.7**	0.99	195	174
0.4854	0.0092	0.0649	0.0008	0.86	0.0541	0.0006	401.7	6.3	405.6	5.1	374	26	**406.0**	**9.5**	1.01	353	372
0.4968	0.0088	0.0652	0.0005	0.70	0.0551	0.0008	409.5	6	407.2	3.2	420	30	**407.1**	**8.6**	0.99	242	72
0.4951	0.0095	0.0654	0.0010	0.91	0.0547	0.0004	408.3	6.4	408.6	6.1	399	17	**408.5**	**10.1**	1.00	521	336
0.5001	0.0084	0.0655	0.0008	0.68	0.0550	0.0006	411.7	5.7	408.9	4.7	413	25	**408.9**	**9.3**	0.99	249	112
0.5073	0.0071	0.0659	0.0004	0.43	0.0559	0.0007	416.6	4.8	411.2	2.4	449	29	**410.7**	**8.5**	0.99	206	152
0.5020	0.0140	0.0680	0.0012	0.69	0.0536	0.0011	413	9.3	424.3	7.2	352	47	**425.0**	**11.1**	1.03	114	74
0.5683	0.0081	0.0726	0.0008	0.36	0.0559	0.0008	456.8	5.2	451.7	4.9	445	31	**451.8**	**10.2**	0.99	256	332
0.7040	0.0520	0.0740	0.0010	0.61	0.0674	0.0041	539	31	460.3	5.9	850	130	**453.7**	**11.0**	0.85	351	54
0.5630	0.0130	0.0733	0.0007	0.32	0.0553	0.0014	453.4	8.7	456.1	4	419	56	**456.5**	**9.9**	1.01	73	2
0.5745	0.0092	0.0748	0.0008	0.68	0.0555	0.0006	460.8	5.9	464.8	5	433	25	**465.3**	**10.4**	1.01	180	210
0.5830	0.0100	0.0755	0.0008	0.42	0.0560	0.0008	466.4	6.4	468.9	4.8	452	33	**469.1**	**10.5**	1.01	211	162
0.5870	0.0120	0.0755	0.0011	0.90	0.0561	0.0005	470.4	8.3	469.2	6.4	457	19	**469.4**	**11.4**	1.00	277	228
0.5810	0.0140	0.0756	0.0011	0.43	0.0553	0.0011	464.8	9.2	470	6.8	423	43	**470.5**	**11.4**	1.01	88	50
0.5922	0.0082	0.0757	0.0007	0.51	0.0559	0.0006	472.2	5.2	470.5	3.9	448	23	**470.9**	**10.1**	1.00	288	124
0.5900	0.0130	0.0761	0.0014	0.65	0.0566	0.0011	470.4	8.1	472.9	8.2	473	43	**472.8**	**12.6**	1.00	113	96
0.5961	0.0076	0.0768	0.0007	0.67	0.0558	0.0006	474.7	4.8	476.7	4.1	444	25	**477.2**	**10.3**	1.00	337	221
0.5970	0.0150	0.0774	0.0008	0.57	0.0559	0.0009	477.8	8.9	480.8	4.7	458	41	**481.3**	**10.6**	1.01	65	41
0.6140	0.0210	0.0778	0.0010	0.72	0.0563	0.0014	485	13	483.2	6	460	53	**483.3**	**11.3**	0.99	68	42
0.6050	0.0130	0.0788	0.0014	0.51	0.0558	0.0009	482.1	7.6	489.1	8.4	442	35	**489.7**	**12.8**	1.02	147	86
0.6110	0.0180	0.0789	0.0016	0.81	0.0564	0.0009	484	12	489.5	9.4	464	36	**489.9**	**13.7**	1.01	158	101
0.6370	0.0140	0.0796	0.0012	0.57	0.0575	0.0011	500.1	8.6	493.7	7.1	507	42	**493.5**	**12.1**	0.99	123	83
0.6290	0.0150	0.0800	0.0008	0.62	0.0567	0.0010	495.3	9	496.2	4.5	488	41	**496.5**	**10.8**	1.00	131	16
0.6308	0.0094	0.0807	0.0006	0.59	0.0568	0.0006	496.5	5.9	500	3.5	481	23	**500.3**	**10.4**	1.01	212	157
0.6390	0.0140	0.0813	0.0009	0.59	0.0570	0.0010	501.3	8.7	503.5	5.4	489	41	**503.8**	**11.3**	1.00	121	112
0.6560	0.0110	0.0819	0.0009	0.82	0.0575	0.0006	512.4	6.5	507.6	5.3	511	23	**507.6**	**11.3**	0.99	174	86
0.6450	0.0230	0.0819	0.0009	0.40	0.0573	0.0016	505	14	507.6	5.4	496	64	**507.7**	**11.4**	1.00	38	21
0.6630	0.0160	0.0830	0.0010	0.44	0.0579	0.0014	516.3	9.8	513.9	5.8	533	48	**513.7**	**11.7**	1.00	81	45
0.6747	0.0089	0.0834	0.0008	0.85	0.0582	0.0004	523.5	5.4	516.5	5	536	15	**516.2**	**11.3**	0.99	419	200
0.6820	0.0180	0.0843	0.0009	0.46	0.0584	0.0013	527	11	521.8	5.6	553	47	**521.3**	**11.7**	0.99	41	2
0.6880	0.0120	0.0847	0.0008	0.81	0.0580	0.0007	531.5	7.1	523.9	4.6	536	25	**523.8**	**11.3**	0.99	172	200
0.6740	0.0120	0.0849	0.0012	0.45	0.0570	0.0010	523	7.4	525.5	7.3	488	40	**525.9**	**12.6**	1.00	105	68
0.6990	0.0160	0.0853	0.0009	0.39	0.0591	0.0011	538	9.3	527.9	5.5	579	35	**527.2**	**11.8**	0.98	90	17
0.6920	0.0140	0.0859	0.0011	0.68	0.0582	0.0009	533.9	8.7	531.1	6.4	537	34	**531.1**	**12.4**	0.99	154	37
0.6870	0.0130	0.0866	0.0019	0.86	0.0570	0.0007	530.9	8	536	11	489	25	**536.2**	**15.6**	1.01	316	175
0.6990	0.0170	0.0868	0.0012	0.27	0.0579	0.0014	538	10	536.3	7.2	523	54	**536.8**	**12.8**	1.00	51	41
0.7047	0.0086	0.0869	0.0009	0.77	0.0584	0.0004	541.6	5.1	537.4	5	544	13	**537.3**	**11.7**	0.99	337	98
0.7083	0.0099	0.0873	0.0010	0.69	0.0588	0.0007	543.7	5.9	539.3	5.8	560	25	**538.9**	**12.1**	0.99	149	38
0.6984	0.0090	0.0873	0.0010	0.64	0.0574	0.0006	537.8	5.4	539.6	5.9	512	23	**540.1**	**12.2**	1.00	169	49
0.7240	0.0210	0.0882	0.0011	0.47	0.0588	0.0015	552	13	545	6.4	555	58	**544.6**	**12.6**	0.99	46	57
0.7250	0.0110	0.0887	0.0009	0.54	0.0587	0.0009	553.5	6.7	548.1	5.2	553	32	**548.0**	**12.0**	0.99	138	91
0.7190	0.0100	0.0893	0.0009	0.57	0.0580	0.0006	551.5	5.3	551.6	5.3	531	22	**552.0**	**12.1**	1.00	179	10
0.7400	0.0190	0.0898	0.0018	0.95	0.0599	0.0006	562	11	555	11	600	20	**553.5**	**15.3**	0.99	443	99
0.7270	0.0270	0.0899	0.0016	0.83	0.0586	0.0016	554	16	554.8	9.6	548	60	**555.0**	**14.6**	1.00	25	18
0.7290	0.0160	0.0900	0.0017	0.76	0.0588	0.0009	555.6	9.7	555	10	558	34	**555.5**	**15.0**	1.00	102	180
0.7388	0.0081	0.0901	0.0008	0.86	0.0588	0.0003	561.7	4.7	556.2	4.7	561	11	**556.2**	**11.9**	0.99	409	99
0.7340	0.0130	0.0902	0.0007	0.64	0.0584	0.0008	558.6	7.7	556.9	4.3	559	29	**557.1**	**11.8**	1.00	122	117
0.7300	0.0190	0.0904	0.0009	0.56	0.0584	0.0012	556	11	557.6	5.3	543	45	**557.9**	**12.2**	1.00	50	61
0.7470	0.0140	0.0906	0.0015	0.86	0.0594	0.0007	566.2	8	559	8.6	581	24	**558.6**	**14.2**	0.99	159	27
0.7460	0.0120	0.0907	0.0009	0.57	0.0593	0.0009	565.6	7.3	559.8	5.3	575	34	**559.5**	**12.2**	0.99	125	96
0.7470	0.0110	0.0909	0.0007	0.35	0.0595	0.0009	568	6.1	560.9	4.3	582	31	**560.5**	**11.8**	0.99	192	106
0.7490	0.0350	0.0914	0.0012	0.01	0.0592	0.0028	567	20	563.8	7.1	590	110	**563.6**	**13.3**	0.99	16	29
0.7370	0.0150	0.0914	0.0015	0.91	0.0586	0.0005	560.6	8.6	563.6	8.6	553	18	**564.0**	**14.3**	1.01	399	449
0.7480	0.0110	0.0918	0.0016	0.83	0.0591	0.0006	566.8	6.5	566	9.2	568	21	**566.1**	**14.7**	1.00	211	186
0.7380	0.0100	0.0918	0.0008	0.54	0.0588	0.0007	561.3	5.8	566.4	4.4	571	26	**566.5**	**12.0**	1.01	143	57
0.7530	0.0120	0.0919	0.0008	-0.02	0.0586	0.0012	569.7	7.2	566.7	4.9	562	42	**566.9**	**12.2**	0.99	82	77
0.7630	0.0200	0.0921	0.0011	0.34	0.0594	0.0013	575	11	567.7	6.5	593	54	**567.7**	**13.0**	0.99	30	104

0.7530	0.0130	0.0925	0.0010	0.73	0.0591	0.0006	573.5	6	570.3	6	575	25	**570.3**	**12.7**	1.00	222	146
0.7660	0.0190	0.0938	0.0009	0.46	0.0590	0.0014	580	12	577.7	5.5	565	52	**577.9**	**12.7**	1.00	66	131
0.7780	0.0200	0.0942	0.0011	0.31	0.0595	0.0014	584	11	580.2	6.7	595	48	**580.2**	**13.2**	0.99	84	19
0.7720	0.0100	0.0943	0.0008	0.51	0.0592	0.0006	581	6	580.6	4.9	573	22	**580.8**	**12.4**	1.00	292	135
0.7760	0.0210	0.0949	0.0013	0.58	0.0588	0.0013	583	12	584.6	7.4	556	47	**584.9**	**13.9**	1.00	28	5
0.7740	0.0170	0.0954	0.0020	0.86	0.0590	0.0007	582	9.9	587	12	565	27	**587.8**	**16.6**	1.01	255	185
0.7780	0.0250	0.0958	0.0013	0.42	0.0587	0.0017	584	14	589.9	7.4	551	64	**590.4**	**14.0**	1.01	31	17
0.7980	0.0210	0.0968	0.0013	0.51	0.0592	0.0012	595	12	595.9	7.8	572	45	**596.1**	**14.1**	1.00	55	124
0.8180	0.0210	0.0975	0.0015	0.91	0.0609	0.0008	607	12	599.6	9	634	27	**599.0**	**14.8**	0.99	124	5
0.8690	0.0220	0.1018	0.0018	0.88	0.0619	0.0009	638	11	625	10	668	30	**624.0**	**16.3**	0.98	124	81
0.8630	0.0140	0.1018	0.0007	0.39	0.0615	0.0009	631.7	7.6	624.9	4.2	656	32	**624.2**	**13.0**	0.99	174	48
0.8730	0.0430	0.1020	0.0025	0.78	0.0605	0.0020	636	23	626	15	651	71	**626.2**	**19.4**	0.98	27	0
0.8810	0.0120	0.1029	0.0012	0.89	0.0617	0.0005	641.2	6.7	631.4	6.8	665	16	**630.6**	**14.3**	0.98	178	50
0.8780	0.0180	0.1029	0.0006	0.45	0.0619	0.0010	639.8	9.5	631.6	3.4	670	36	**630.7**	**12.8**	0.99	53	96
0.9040	0.0180	0.1034	0.0014	0.71	0.0634	0.0008	653.8	9.4	634.2	7.9	719	26	**632.3**	**14.9**	0.97	220	9
0.8810	0.0110	0.1035	0.0010	0.80	0.0622	0.0004	641.4	5.9	635.1	6	680	15	**633.8**	**13.8**	0.99	416	198
0.9490	0.0190	0.1053	0.0023	0.99	0.0655	0.0003	677.6	9.7	645	13	789	8.5	**641.9**	**18.5**	0.95	767	275
1.0100	0.0380	0.1087	0.0032	0.90	0.0671	0.0011	708	19	665	19	847	35	**660.7**	**22.9**	0.94	89	13
0.9180	0.0170	0.1086	0.0013	0.91	0.0603	0.0004	660.9	8.9	664.5	7.3	615	14	**665.8**	**15.1**	1.01	269	82
1.0250	0.0210	0.1108	0.0009	0.34	0.0673	0.0014	716	10	677.2	5.3	844	42	**672.7**	**14.3**	0.95	43	28
0.9840	0.0160	0.1131	0.0016	0.87	0.0632	0.0005	695.4	8	690.7	9.5	713	16	**690.1**	**16.5**	0.99	416	91
0.9770	0.0170	0.1140	0.0016	0.78	0.0621	0.0007	692.1	8.5	695.7	9.2	676	25	**696.4**	**16.6**	1.01	122	26
1.0890	0.0350	0.1200	0.0024	0.96	0.0645	0.0005	747	17	730	14	762	20	**729.8**	**20.1**	0.98	286	53
1.0940	0.0190	0.1212	0.0009	0.43	0.0652	0.0011	750.1	9.3	737.4	5.3	778	35	**736.2**	**15.4**	0.98	73	22
1.2080	0.0200	0.1298	0.0017	0.56	0.0671	0.0011	803.9	9.4	786.6	9.5	839	36	**785.0**	**18.3**	0.98	60	53
1.1700	0.0330	0.1305	0.0021	0.66	0.0645	0.0014	786	16	793	11	755	45	**791.7**	**19.8**	1.01	62	37
1.2390	0.0230	0.1366	0.0013	0.23	0.0660	0.0013	818	10	825.4	7.6	803	41	**826.0**	**17.9**	1.01	65	25
1.3030	0.0210	0.1388	0.0013	0.39	0.0672	0.0009	846.7	9.2	837.9	7.4	841	28	**837.7**	**18.0**	0.99	66	43
1.2990	0.0190	0.1403	0.0011	0.75	0.0668	0.0006	845	8.4	846.3	6.2	831	19	**846.9**	**17.7**	1.00	180	122
1.4000	0.0300	0.1470	0.0029	0.82	0.0699	0.0011	888	13	884	16	924	31	**882.6**	**24.1**	0.99	32	19
1.5010	0.0220	0.1524	0.0017	0.41	0.0711	0.0010	930.5	8.7	914.5	9.7	960	29	**912.6**	**20.4**	0.98	47	39
1.7080	0.0340	0.1646	0.0023	0.80	0.0754	0.0010	1011	13	982	13	1079	27	**977.9**	**23.2**	0.97	67	19
1.6440	0.0180	0.1649	0.0016	0.43	0.0724	0.0007	987.2	6.9	983.7	8.7	997	20	**983.3**	**21.3**	1.00	102	58
1.6550	0.0180	0.1655	0.0016	0.83	0.0722	0.0005	991.2	6.9	987.4	8.9	990	14	**987.1**	**21.3**	1.00	175	3
1.7540	0.0220	0.1687	0.0020	0.88	0.0744	0.0005	1028.4	8	1005	11	1053	13	**1003**	**22.7**	0.98	427	155
1.7310	0.0240	0.1689	0.0028	0.55	0.0747	0.0012	1019.8	8.8	1006	15	1059	31	**1003**	**25.4**	0.99	48	34
1.7550	0.0200	0.1704	0.0019	0.76	0.0742	0.0005	1028.9	7.4	1014	10	1047	13	**1013**	**22.6**	0.99	183	105
1.7110	0.0320	0.1710	0.0034	0.94	0.0723	0.0005	1012	12	1018	19	994	13	**1019**	**27.9**	1.00	231	54
1.7690	0.0210	0.1723	0.0016	0.69	0.0737	0.0007	1034	7.7	1024.8	8.9	1032	18	**1024**	**22.0**	0.99	99	32
1.7730	0.0270	0.1732	0.0020	0.83	0.0745	0.0006	1035.3	9.9	1030	11	1058	16	**1029**	**23.1**	0.99	169	66
1.7720	0.0240	0.1747	0.0019	0.83	0.0734	0.0007	1035	8.8	1038	10	1024	19	**1039**	**23.0**	1.00	118	64
1.7820	0.0230	0.1751	0.0022	0.82	0.0734	0.0006	1038.9	8.5	1040	12	1023	17	**1041**	**23.9**	1.00	154	61
1.7910	0.0170	0.1763	0.0013	0.87	0.0734	0.0004	1042.3	6.2	1046.9	7	1024	10	**1048**	**21.7**	1.00	273	130
1.8220	0.0260	0.1770	0.0027	0.86	0.0745	0.0007	1053.3	9.5	1050	15	1054	18	**1050**	**25.7**	1.00	162	68
1.8450	0.0400	0.1777	0.0033	0.69	0.0749	0.0010	1061	14	1054	18	1064	27	**1054**	**28.0**	0.99	25	6
1.8440	0.0290	0.1785	0.0013	0.47	0.0746	0.0011	1061	10	1058.8	7.1	1056	30	**1059**	**22.0**	1.00	69	31
1.8630	0.0350	0.1790	0.0027	0.81	0.0761	0.0009	1068	12	1061	15	1098	24	**1060**	**25.8**	0.99	91	28
1.8700	0.0320	0.1791	0.0025	0.79	0.0757	0.0008	1073	11	1062	13	1086	20	**1061**	**25.2**	0.99	142	65
1.8580	0.0130	0.1791	0.0009	0.39	0.0754	0.0006	1066.3	4.8	1062.2	5.1	1079	16	**1061**	**21.4**	1.00	263	62
1.8350	0.0380	0.1787	0.0030	0.90	0.0736	0.0008	1058	13	1060	17	1031	21	**1061**	**26.9**	1.00	94	35
1.8430	0.0150	0.1790	0.0015	0.76	0.0742	0.0004	1061	5.5	1061.4	8.4	1050	11	**1062**	**22.4**	1.00	267	80
1.8790	0.0210	0.1797	0.0018	0.61	0.0756	0.0008	1073.4	7.4	1065	10	1083	22	**1064**	**23.2**	0.99	121	36
1.8660	0.0400	0.1800	0.0031	0.78	0.0757	0.0008	1068	14	1067	17	1092	25	**1066**	**27.4**	1.00	35	20
1.8800	0.0270	0.1825	0.0032	0.90	0.0755	0.0007	1074	9.4	1080	17	1082	18	**1081**	**27.9**	1.01	120	85
1.9200	0.0320	0.1831	0.0028	0.90	0.0758	0.0005	1088	11	1084	15	1094	14	**1084**	**26.5**	1.00	187	78
1.9240	0.0310	0.1829	0.0020	0.48	0.0750	0.0012	1089	11	1083	11	1081	31	**1084**	**24.1**	0.99	64	25
1.9420	0.0320	0.1834	0.0031	0.91	0.0765	0.0007	1095	11	1085	17	1108	19	**1084**	**27.6**	0.99	128	58
1.9640	0.0300	0.1838	0.0023	0.77	0.0766	0.0008	1103	10	1088	12	1110	20	**1087**	**24.9**	0.99	87	43
1.9510	0.0510	0.1849	0.0021	0.72	0.0757	0.0013	1098	17	1094	11	1087	33	**1094**	**24.5**	1.00	39	20
1.9550	0.0310	0.1859	0.0028	0.96	0.0761	0.0004	1100	11	1099	15	1098	10	**1099**	**26.8**	1.00	367	137
2.0790	0.0500	0.1901	0.0039	0.90	0.0785	0.0009	1141	16	1128	23	1159	23	**1120**	**31.2**	0.98	73	44
1.9860	0.0380	0.1896	0.0022	0.77	0.0753	0.0008	1110	13	1119	12	1076	22	**1122**	**25.2**	1.01	110	136
2.0860	0.0170	0.1925	0.0015	0.81	0.0786	0.0003	1144.2	5.5	1134.7	8.2	1165	7.7	**1133**	**23.7**	0.99	325	64
2.1490	0.0310	0.1950	0.0016	0.46	0.0791	0.0011	1164	10	1148.4	8.7	1172	28	**1147**	**24.2**	0.99	67	35
2.1010	0.0320	0.1975	0.0022	0.70	0.0769	0.0008	1149	11	1162	12	1118	21	**1164**	**26.0**	1.01	108	30
2.2120	0.0370	0.1995	0.0030	0.91	0.0804	0.0004	1185	12	1173	16	1206	10	**1171**	**28.5**	0.99	235	119
2.2410	0.0210	0.2032	0.0019	0.92	0.0796	0.0003	1193.8	6.6	1192	10	1186	8.1	**1193**	**25.7**	1.00	411	137
2.4440	0.0420	0.2121	0.0020	0.71	0.0822	0.0008	1255	12	1240	11	1250	20	**1239**	**26.8**	0.99	61	35
2.6100	0.0370	0.2221	0.0024	0.85	0.0846	0.0007	1303	10	1293	13	1310	15	**1292**	**28.7**	0.99	87	39
2.6210	0.0380	0.2241	0.0024	0.65	0.0847	0.0010	1306	11	1304	13	1308	23	**1303**	**28.9**	1.00	44	38
2.8370	0.0360	0.2332	0.0028	0.85	0.0874	0.0006	1365.1	9.6	1355	16	1369	13	**1350**	**30.8**	0.99	82	61
3.0360	0.0640	0.2354	0.0044	0.88	0.0935	0.0010	1420	15	1363	23	1497	19	**1351**	**36.2**	0.96	76	22
2.8110	0.0460	0.2346	0.0039	0.94	0.0866	0.0006	1358	12	1359	20	1354	12	**1359**	**34.6**	1.00	220	73
2.8890	0.0420	0.2373	0.0028	0.78	0.0876	0.0009	1379	11	1372	15	1374	20	**1373**	**31.3**	1.00	53	110
3.0620	0.0360	0.2476	0.0025	0.79	0.0888	0.0006	1423	8.9	1426	13	1400	13	**1400**	**40.4**	1.00	103	98

3.0660	0.0320	0.2478	0.0018	0.68	0.0893	0.0006	1424.2	7.9	1426.8	9.3	1410	14	**1410**	**40.7**	1.00	114	65
3.0670	0.0450	0.2472	0.0025	0.83	0.0896	0.0008	1424	11	1424	13	1416	17	**1416**	**41.9**	1.00	115	51
3.0430	0.0750	0.2471	0.0044	0.88	0.0901	0.0009	1418	19	1423	23	1427	19	**1427**	**42.5**	1.00	60	40
3.6440	0.0330	0.2716	0.0021	0.87	0.0969	0.0005	1559	7.2	1549	10	1566	9.5	**1566**	**38.7**	0.99	197	77
4.0770	0.0430	0.2916	0.0036	0.71	0.1008	0.0008	1649.6	8.5	1650	18	1638	15	**1638**	**40.2**	1.00	73	57
4.1260	0.0690	0.2909	0.0044	0.96	0.1025	0.0005	1663	12	1646	22	1670	9.6	**1671**	**38.2**	0.99	188	48
4.9950	0.0620	0.3239	0.0036	0.75	0.1118	0.0009	1818	11	1809	18	1829	14	**1829**	**39.0**	0.99	67	103
5.1710	0.0610	0.3244	0.0030	0.85	0.1146	0.0006	1848	10	1815	12	1876	11	**1874**	**37.4**	0.98	91	93
7.6800	0.1100	0.4084	0.0053	0.92	0.1362	0.0007	2194	12	2207	24	2179	8.8	**2179**	**35.9**	1.01	76	72
12.5500	0.1700	0.5044	0.0078	0.97	0.1797	0.0007	2646	13	2632	33	2650	6.8	**2650**	**33.9**	0.99	112	117
11.6900	0.2200	0.4632	0.0076	0.96	0.1823	0.0010	2579	17	2453	33	2674	9.1	**2674**	**34.3**	0.95	77	91
12.2300	0.1900	0.4765	0.0079	0.98	0.1858	0.0006	2622	15	2511	34	2705	5.5	**2705**	**33.5**	0.96	138	89
12.2900	0.2700	0.4713	0.0095	0.99	0.1885	0.0007	2626	21	2489	42	2729	6.3	**2729**	**33.5**	0.95	244	111
13.5000	0.1200	0.5121	0.0046	0.96	0.1903	0.0005	2717.4	9.1	2665	20	2745	4.3	**2745**	**33.2**	0.98	232	212
13.8300	0.1300	0.5219	0.0048	0.95	0.1906	0.0006	2737.8	8.8	2707	20	2748	4.8	**2748**	**33.2**	0.99	170	149
13.8500	0.1800	0.5230	0.0068	0.99	0.1916	0.0004	2739	12	2712	29	2756	3.6	**2756**	**33.0**	0.99	388	233
11.9600	0.1700	0.4320	0.0050	0.87	0.1999	0.0013	2601	13	2315	22	2826	11	**2825**	**34.3**	0.89	111	12
14.9900	0.4100	0.4750	0.0110	0.97	0.2273	0.0019	2813	26	2506	46	3033	13	**3033**	**34.7**	0.89	170	75

HU-8 **Lat:** **-30.20393** **Lon:** **-68.37521**

Isotope Ratios							**Apparent Ages**								**Concentrations**		
$^{207}Pb/^{235}U$	**± (2σ)**	**$^{206}Pb/^{238}U$**	**± (2σ)**	**Error Corr.**	**$^{207}Pb/^{206}Pb$**	**± (2σ)**	**$^{207}Pb/^{235}U$ (Ma)**	**± (Ma)**	**$^{206}Pb/^{238}U$ (Ma)**	**± (Ma)**	**$^{207}Pb/^{206}Pb$ (Ma)**	**± (Ma)**	**Plot Age (Ma)**	**± (Ma)**	**conc. (%)**	**U (ppm)**	**Th (ppm)**
-0.0180	0.0220	0.0014	0.0005	-0.02	-0.1600	0.4000	-26	22	8.8	3.5	-	-	**11.0**	**3.5**	133.85	56	42
0.0141	0.0055	0.0028	0.0003	0.14	0.0380	0.0190	13.8	5.5	17.8	1.6	-	-	**18.0**	**1.6**	28.99	140	41
0.0272	0.0033	0.0034	0.0002	0.11	0.0571	0.0076	27.7	3.2	21.9	0.95	-	-	**21.6**	**1.0**	20.94	1032	825
0.0175	0.0054	0.0035	0.0003	0.02	0.0400	0.0150	17.3	5.4	22.7	1.6	-	-	**23.0**	**1.6**	31.21	263	89
0.0470	0.0110	0.0037	0.0004	0.05	0.1020	0.0330	45	11	23.9	2.5	-	-	**22.3**	**2.5**	46.89	159	106
0.0271	0.0049	0.0037	0.0002	0.04	0.0550	0.0110	26.8	4.9	24.1	1.5	-	-	**23.8**	**1.5**	10.07	584	381
0.0290	0.0180	0.0047	0.0006	-0.03	0.0180	0.0550	24	18	29.9	4.1	-	-	**31.0**	**4.1**	24.58	75	50
0.0334	0.0072	0.0052	0.0003	0.09	0.0448	0.0098	32.6	7	33.3	1.7	-	-	**33.4**	**1.7**	2.15	470	128
0.0220	0.0230	0.0052	0.0007	0.23	0.1150	0.0560	16	23	33.6	4.4	-	-	**30.7**	**4.4**	110.00	77	29
0.0402	0.0085	0.0056	0.0003	-0.06	0.0540	0.0120	40.7	8.4	36	2.1	-	-	**35.5**	**2.1**	11.55	488	203
0.0190	0.0100	0.0064	0.0004	0.09	0.0240	0.0120	18	10	41	2.6	-	-	**42.2**	**2.6**	127.78	248	150
0.2500	0.0200	0.0347	0.0012	0.27	0.0523	0.0042	225	16	219.9	7.8	270	150	**219.9**	**7.8**	2.27	338	307
0.3130	0.0120	0.0450	0.0010	0.21	0.0509	0.0022	276.2	9.7	283.5	6.1	221	86	**283.5**	**6.1**	2.64	392	251
0.3760	0.0210	0.0475	0.0011	-0.05	0.0566	0.0035	321	15	298.9	6.7	430	120	**298.9**	**6.7**	6.88	264	195
0.3700	0.0490	0.0506	0.0020	-0.03	0.0536	0.0077	313	37	318	12	240	250	**318**	**12.0**	1.60	64	74
0.4190	0.0170	0.0551	0.0012	0.04	0.0549	0.0023	354	12	345.7	7.6	382	87	**345.7**	**7.6**	2.34	438	211
0.4410	0.0200	0.0555	0.0014	-0.05	0.0568	0.0031	371	15	348.1	8.4	440	120	**348.1**	**8.4**	6.17	194	211
0.4300	0.0190	0.0594	0.0013	0.21	0.0516	0.0024	361	14	371.6	8	261	94	**371.6**	**8.0**	2.94	91	62
0.4660	0.0200	0.0625	0.0013	0.05	0.0533	0.0023	389	14	390.4	7.9	331	90	**390.4**	**7.9**	0.36	406	439
0.5130	0.0260	0.0638	0.0016	0.26	0.0563	0.0031	420	17	398.4	9.6	470	110	**398.4**	**9.6**	5.14	211	253
0.4860	0.0190	0.0639	0.0013	0.18	0.0553	0.0021	400	13	399.2	8.1	389	81	**399.2**	**8.1**	0.20	296	283
0.5110	0.0220	0.0655	0.0015	0.38	0.0568	0.0023	420	15	409.1	9.1	487	87	**409.1**	**9.1**	2.60	458	182
0.5170	0.0140	0.0685	0.0009	0.29	0.0542	0.0015	422.1	9.6	426.9	5.7	365	62	**426.9**	**5.7**	1.14	1164	56
0.5610	0.0170	0.0698	0.0012	0.37	0.0570	0.0018	451	11	434.8	7.2	471	68	**434.8**	**7.2**	3.59	553	407
0.5390	0.0150	0.0699	0.0011	0.00	0.0556	0.0018	439	10	435.4	6.8	424	70	**435.4**	**6.8**	0.82	693	144
0.5660	0.0250	0.0702	0.0017	0.30	0.0585	0.0026	454	16	437	10	522	96	**437**	**10.0**	3.74	132	68
0.5550	0.0270	0.0703	0.0015	0.26	0.0577	0.0030	447	18	438.1	9.2	480	110	**438.1**	**9.2**	1.99	275	213
0.5330	0.0160	0.0704	0.0012	0.14	0.0555	0.0018	433	11	438.6	7.1	436	71	**438.6**	**7.1**	1.29	716	215
0.5650	0.0430	0.0705	0.0022	0.18	0.0582	0.0046	453	27	439	13	470	150	**439**	**13.0**	3.09	84	31
0.5620	0.0230	0.0706	0.0014	0.06	0.0579	0.0026	454	15	440.8	8.1	473	95	**440.8**	**8.1**	2.91	437	233
0.5800	0.0400	0.0707	0.0023	0.02	0.0585	0.0045	463	26	441	14	520	150	**441**	**14.0**	4.75	97	45
0.5360	0.0190	0.0709	0.0015	0.12	0.0558	0.0022	434	12	441.6	8.8	422	84	**441.6**	**8.8**	1.75	478	284
0.5900	0.0280	0.0720	0.0015	0.17	0.0592	0.0029	467	18	449.2	8.9	530	110	**449.2**	**8.9**	3.81	324	119
0.5860	0.0300	0.0728	0.0021	0.12	0.0587	0.0033	465	19	452	12	500	110	**452**	**12.0**	2.80	133	77
0.6080	0.0150	0.0751	0.0011	0.15	0.0573	0.0016	482.8	9.2	466.8	6.6	497	64	**466.8**	**6.6**	3.31	747	221
0.6360	0.0350	0.0760	0.0021	0.12	0.0613	0.0036	499	22	472	13	590	120	**472**	**13.0**	5.41	228	161
0.6680	0.0290	0.0767	0.0017	0.32	0.0625	0.0026	518	17	476	10	662	90	**476**	**10.0**	8.11	373	226
0.6250	0.0290	0.0780	0.0020	0.42	0.0595	0.0027	492	18	484	12	562	96	**484**	**12.0**	1.63	495	612
0.6400	0.0270	0.0788	0.0018	0.06	0.0595	0.0026	502	17	489	11	562	97	**489**	**11.0**	2.59	276	101
0.6470	0.0220	0.0811	0.0015	0.16	0.0580	0.0023	507	14	502.6	9.2	507	84	**502.6**	**9.2**	0.87	307	163
0.7060	0.0240	0.0818	0.0012	0.13	0.0626	0.0022	539	14	506.8	7.3	654	76	**506.8**	**7.3**	5.97	492	219
0.7330	0.0250	0.0829	0.0016	0.18	0.0631	0.0023	555	14	513	9.4	694	74	**513**	**9.4**	7.57	294	211
0.6720	0.0650	0.0831	0.0034	-0.05	0.0590	0.0061	512	39	514	20	450	190	**514**	**20.0**	0.39	53	22
0.6890	0.0230	0.0829	0.0016	0.24	0.0594	0.0020	529	14	514	9.4	584	76	**514**	**9.4**	2.84	282	114
0.6770	0.0190	0.0836	0.0013	0.20	0.0583	0.0017	524	12	517.6	7.6	516	64	**517.6**	**7.6**	1.22	462	231
0.6760	0.0190	0.0835	0.0014	0.14	0.0577	0.0018	522	11	517.8	8	495	67	**517.8**	**8.0**	0.80	470	121
0.6660	0.0270	0.0835	0.0018	0.16	0.0583	0.0024	516	16	518	11	526	86	**518**	**11.0**	0.39	313	183
0.7080	0.0170	0.0845	0.0014	0.26	0.0597	0.0016	544	10	522.8	8.2	580	57	**522.8**	**8.2**	3.90	546	225
0.6860	0.0130	0.0845	0.0014	0.31	0.0579	0.0013	530.3	7.8	523.6	7.9	548	47	**523.6**	**7.9**	1.26	694	56
0.6370	0.0360	0.0850	0.0019	0.18	0.0548	0.0033	497	22	526	11	350	120	**526**	**11.0**	5.84	227	139

0.7060	0.0210	0.0858	0.0015	0.28	0.0586	0.0017	541	13	531.2	9	560	63	**531.2**	**9.0**	1.81	426	280
0.6990	0.0150	0.0861	0.0013	0.36	0.0581	0.0013	536.8	9.1	532.4	7.9	529	47	**532.4**	**7.9**	0.82	369	234
0.7170	0.0330	0.0862	0.0018	0.21	0.0603	0.0027	551	19	533	11	556	98	**533**	**11.0**	3.27	307	378
0.6980	0.0160	0.0864	0.0013	0.17	0.0585	0.0014	537.3	9.3	534.1	7.6	543	50	**534.1**	**7.6**	0.60	846	389
0.7500	0.0290	0.0872	0.0017	0.26	0.0610	0.0024	566	17	539	10	647	87	**539**	**10.0**	4.77	284	112
0.7060	0.0210	0.0874	0.0016	0.30	0.0590	0.0018	546	13	539.9	9.3	565	66	**539.9**	**9.3**	1.12	559	1057
0.6850	0.0320	0.0879	0.0020	0.12	0.0577	0.0029	527	19	542	12	450	100	**542**	**12.0**	2.85	128	83
0.7600	0.0220	0.0878	0.0016	0.27	0.0613	0.0019	573	13	542.4	9.6	638	68	**542.4**	**9.6**	5.34	294	179
0.7390	0.0230	0.0876	0.0014	0.45	0.0604	0.0017	563	13	543.1	8.6	600	63	**543.1**	**8.6**	3.53	770	706
0.7390	0.0190	0.0884	0.0016	0.25	0.0607	0.0016	561	11	546	9.7	619	56	**546**	**9.7**	2.67	926	134
0.7160	0.0330	0.0890	0.0017	-0.04	0.0577	0.0028	548	19	550	10	480	100	**550**	**10.0**	0.36	144	79
0.7080	0.0230	0.0891	0.0014	0.07	0.0585	0.0021	547	14	550	8	510	78	**550**	**8.0**	0.55	498	219
0.7950	0.0510	0.0893	0.0031	-0.19	0.0645	0.0051	596	29	553	18	710	160	**553**	**18.0**	7.21	106	131
0.7820	0.0400	0.0899	0.0024	0.17	0.0633	0.0034	586	23	555	14	630	120	**555**	**14.0**	5.29	108	68
0.7990	0.0240	0.0903	0.0017	0.16	0.0646	0.0023	597	13	557	10	740	72	**557**	**10.0**	6.70	562	342
0.7430	0.0300	0.0907	0.0022	0.14	0.0590	0.0027	563	17	559	13	546	97	**559**	**13.0**	0.71	310	94
0.7390	0.0390	0.0908	0.0025	-0.05	0.0593	0.0037	559	23	560	15	530	120	**560**	**15.0**	0.18	153	17
0.7470	0.0250	0.0918	0.0018	0.28	0.0586	0.0019	564	15	567	11	518	72	**567**	**11.0**	0.53	385	164
0.7390	0.0330	0.0922	0.0021	-0.01	0.0584	0.0028	561	19	568	12	490	100	**568**	**12.0**	1.25	269	130
0.7900	0.0170	0.0945	0.0011	0.15	0.0613	0.0015	594.1	9.4	581.8	6.7	645	53	**581.8**	**6.7**	2.07	771	316
0.7830	0.0240	0.0945	0.0015	0.27	0.0598	0.0020	584	14	582.1	8.8	567	70	**582.1**	**8.8**	0.33	343	85
0.8220	0.0330	0.0945	0.0021	0.14	0.0621	0.0027	606	18	583	12	625	94	**583**	**12.0**	3.80	185	299
0.8050	0.0230	0.0950	0.0018	0.27	0.0605	0.0018	599	13	585	11	612	62	**585**	**11.0**	2.34	462	191
0.8030	0.0270	0.0972	0.0017	0.17	0.0609	0.0022	600	15	597.8	9.7	622	77	**597.8**	**9.7**	0.37	438	303
0.8180	0.0210	0.1018	0.0016	0.31	0.0591	0.0016	605	11	624.6	9.3	548	58	**624.6**	**9.3**	3.24	585	265
0.8800	0.0270	0.1028	0.0020	0.29	0.0628	0.0021	640	14	631	12	667	73	**631**	**12.0**	1.41	457	348
1.4200	0.1500	0.1432	0.0066	0.07	0.0734	0.0087	900	64	867	36	810	230	**810**	**230.0**	3.67	16	3
1.3090	0.0400	0.1375	0.0032	0.31	0.0689	0.0022	847	17	832	18	903	62	**903**	**62.0**	1.77	139	32
1.4630	0.0440	0.1508	0.0027	0.52	0.0701	0.0018	918	18	907	15	927	54	**927**	**54.0**	1.20	447	229
1.6400	0.0460	0.1656	0.0037	0.37	0.0712	0.0020	985	18	987	20	953	56	**953**	**56.0**	0.20	259	50
1.3510	0.0530	0.1390	0.0034	0.55	0.0709	0.0022	870	22	838	19	954	64	**954**	**64.0**	3.68	131	36
1.4730	0.0480	0.1458	0.0026	0.32	0.0717	0.0023	919	20	877	15	981	68	**981**	**68.0**	4.57	295	19
1.4950	0.0420	0.1517	0.0030	0.15	0.0725	0.0023	929	17	910	17	992	65	**992**	**65.0**	2.05	279	86
1.8240	0.0770	0.1821	0.0060	0.19	0.0727	0.0034	1051	28	1078	32	1013	95	**1013**	**95.0**	2.57	315	226
1.7080	0.0560	0.1676	0.0035	0.37	0.0732	0.0024	1010	21	1000	19	1020	67	**1020**	**67.0**	0.99	251	203
1.5800	0.0340	0.1549	0.0022	0.31	0.0737	0.0017	960	13	928	12	1030	46	**1030**	**46.0**	3.33	712	281
1.6010	0.0380	0.1541	0.0028	0.33	0.0735	0.0017	969	15	923	15	1032	46	**1032**	**46.0**	4.75	493	289
1.8430	0.0770	0.1753	0.0046	0.13	0.0744	0.0033	1061	26	1040	25	1048	95	**1048**	**95.0**	1.98	171	109
1.7610	0.0480	0.1685	0.0030	0.24	0.0748	0.0022	1029	18	1005	17	1048	60	**1048**	**60.0**	2.33	216	88
1.7590	0.0540	0.1694	0.0034	0.21	0.0754	0.0025	1033	20	1008	19	1050	67	**1050**	**67.0**	2.42	132	59
1.6950	0.0700	0.1668	0.0061	0.63	0.0746	0.0024	1005	26	994	34	1051	65	**1051**	**65.0**	1.09	438	28
1.6990	0.0620	0.1637	0.0034	0.08	0.0738	0.0030	1006	23	977	19	1056	81	**1056**	**81.0**	2.88	121	32
1.7200	0.1200	0.1653	0.0058	0.33	0.0789	0.0055	1013	47	984	32	1060	150	**1060**	**150.0**	2.86	31	23
1.7410	0.0410	0.1674	0.0025	0.24	0.0751	0.0020	1024	15	997	14	1073	53	**1073**	**53.0**	2.64	105	49
1.7320	0.0420	0.1659	0.0035	0.16	0.0757	0.0020	1019	15	989	19	1073	55	**1073**	**55.0**	2.94	178	47
1.7040	0.0510	0.1674	0.0031	0.17	0.0754	0.0025	1014	19	999	17	1080	67	**1080**	**67.0**	1.48	318	349
1.8820	0.0490	0.1800	0.0027	0.29	0.0755	0.0020	1080	16	1070	15	1083	51	**1083**	**51.0**	0.93	409	230
1.7730	0.0470	0.1698	0.0029	0.19	0.0765	0.0022	1031	18	1010	16	1087	59	**1087**	**59.0**	2.04	342	148
1.8600	0.1700	0.1822	0.0076	0.10	0.0766	0.0075	1093	55	1079	41	1090	170	**1090**	**170.0**	1.28	16	8
1.7000	0.0390	0.1605	0.0026	0.28	0.0761	0.0019	1013	15	959	15	1094	49	**1094**	**49.0**	5.33	397	162
1.8490	0.0590	0.1710	0.0039	0.25	0.0768	0.0026	1062	21	1017	21	1099	69	**1099**	**69.0**	4.24	152	34
1.7410	0.0470	0.1643	0.0026	0.24	0.0768	0.0022	1021	18	980	14	1110	56	**1110**	**56.0**	4.02	318	162
2.0570	0.0510	0.1927	0.0034	0.05	0.0771	0.0022	1137	17	1135	18	1123	57	**1123**	**57.0**	0.18	178	128
1.8830	0.0590	0.1753	0.0037	0.16	0.0779	0.0029	1075	21	1040	20	1139	72	**1139**	**72.0**	3.26	105	28
1.9100	0.0540	0.1774	0.0037	0.17	0.0789	0.0025	1086	19	1052	20	1151	66	**1151**	**66.0**	3.13	135	80
1.9810	0.0620	0.1781	0.0036	0.12	0.0801	0.0027	1111	20	1056	19	1168	67	**1168**	**67.0**	4.95	147	35
2.1030	0.0410	0.1930	0.0031	0.13	0.0792	0.0020	1149	14	1137	17	1169	49	**1169**	**49.0**	1.04	436	135
2.2060	0.0570	0.1961	0.0038	0.19	0.0812	0.0022	1180	18	1157	20	1217	55	**1217**	**55.0**	1.95	160	172
1.6770	0.0550	0.1498	0.0028	0.17	0.0809	0.0029	998	21	901	15	1219	74	**1219**	**74.0**	9.72	124	50
2.0020	0.0680	0.1765	0.0036	0.12	0.0829	0.0030	1118	22	1049	19	1254	69	**1254**	**69.0**	6.17	139	50
1.7740	0.0780	0.1496	0.0035	0.05	0.0823	0.0040	1034	28	898	20	1268	98	**1268**	**98.0**	13.15	75	20
1.7730	0.0470	0.1541	0.0030	0.34	0.0839	0.0021	1039	16	923	17	1282	49	**1282**	**49.0**	11.16	383	547
2.6070	0.0670	0.2229	0.0035	0.23	0.0844	0.0024	1297	19	1297	18	1317	55	**1317**	**55.0**	0.00	273	212
2.6040	0.0650	0.2197	0.0033	0.25	0.0866	0.0023	1306	18	1280	17	1346	52	**1346**	**52.0**	1.99	281	89
2.9060	0.0650	0.2371	0.0041	0.32	0.0879	0.0020	1383	17	1371	21	1371	44	**1371**	**44.0**	0.87	168	38
1.9500	0.1000	0.1590	0.0042	0.28	0.0852	0.0044	1096	35	950	23	1380	100	**1380**	**100.0**	13.32	57	13
3.5330	0.0690	0.2642	0.0045	0.33	0.0967	0.0020	1535	16	1510	23	1551	38	**1551**	**38.0**	1.63	316	318
3.7500	0.0520	0.2639	0.0035	0.46	0.1034	0.0013	1583	11	1509	18	1682	25	**1682**	**25.0**	4.67	598	131
4.0920	0.0740	0.2826	0.0034	0.32	0.1054	0.0020	1653	15	1604	17	1723	34	**1723**	**34.0**	2.96	399	219
4.7540	0.0920	0.3121	0.0049	0.40	0.1092	0.0020	1773	16	1750	24	1786	33	**1786**	**33.0**	1.30	269	126
11.9100	0.1900	0.4599	0.0083	0.57	0.1863	0.0027	2594	15	2440	37	2709	24	**2709**	**24.0**	5.94	206	71
12.0300	0.1300	0.4559	0.0059	0.43	0.1900	0.0024	2606	10	2420	26	2743	21	**2743**	**21.0**	7.14	382	227
17.3900	0.2000	0.5153	0.0065	0.65	0.2433	0.0023	2956	11	2681	27	3145	16	**3145**	**16.0**	9.30	722	208
20.8600	0.6000	0.5970	0.0160	0.24	0.2507	0.0084	3129	28	3015	64	3177	54	**3177**	**54.0**	3.64	18	4

HU-8.1 Lat: -30.20376 Lon: -68.37443

Isotope Ratios							Apparent Ages								Concentrations		
$^{207}Pb/^{235}U$	± (2σ)	$^{206}Pb/^{238}U$	± (2σ)	Error Corr.	$^{207}Pb/^{206}Pb$	± (2σ)	$^{207}Pb/^{235}U$ (Ma)	± (Ma)	$^{206}Pb/^{238}U$ (Ma)	± (Ma)	$^{207}Pb/^{206}Pb$ (Ma)	± (Ma)	Plot Age (Ma)	± (Ma)	conc. (%)	U (ppm)	Th (ppm)
0.0258	0.0024	0.0031	0.0001	0.04	0.0598	0.0057	25.8	2.4	20.21	0.46	-	-	**19.9**	**0.6**	0.78	104	32
0.0253	0.0038	0.0034	0.0001	-0.01	0.0545	0.0081	25.4	3.7	21.65	0.9	-	-	**21.4**	**1.0**	0.85	81	59
0.0218	0.0021	0.0035	0.0001	0.14	0.0470	0.0046	21.9	2	22.33	0.69	-	-	**22.3**	**0.8**	1.02	183	212
0.0229	0.0032	0.0037	0.0001	0.43	0.0472	0.0067	23	3.2	23.52	0.62	-	-	**23.5**	**0.8**	1.02	77	83
0.0382	0.0030	0.0056	0.0002	-0.24	0.0483	0.0045	38.1	2.9	36.1	1.1	-	-	**36.1**	**1.3**	0.95	78	53
0.0429	0.0053	0.0057	0.0002	0.98	0.0549	0.0065	42.6	5.1	36.5	1.1	-	-	**36.1**	**1.3**	0.86	283	274
0.0367	0.0054	0.0057	0.0002	0.29	0.0462	0.0066	36.6	5.3	36.8	1.1	-	-	**36.9**	**1.4**	1.01	71	89
0.2950	0.0062	0.0418	0.0006	0.78	0.0513	0.0007	262.4	4.9	264.2	3.9	251	31	**264.3**	**6.5**	1.01	213	167
0.3082	0.0063	0.0428	0.0006	0.92	0.0520	0.0004	272.7	4.9	270	3.9	283	19	**269.9**	**6.6**	0.99	727	223
0.3230	0.0120	0.0448	0.0012	0.83	0.0531	0.0011	284.4	8.9	282.4	7.2	328	49	**282.1**	**9.3**	0.99	81	42
0.3329	0.0099	0.0465	0.0009	0.42	0.0517	0.0014	291.6	7.5	293.1	5.6	268	63	**293.3**	**8.1**	1.00	78	59
0.3489	0.0062	0.0468	0.0009	0.87	0.0530	0.0006	303.9	4.7	295	5.6	327	25	**294.7**	**8.2**	0.97	183	241
0.3350	0.0110	0.0469	0.0011	0.77	0.0516	0.0010	294.8	7.7	295.5	7	263	46	**295.7**	**9.0**	1.01	146	97
0.3340	0.0067	0.0469	0.0012	0.81	0.0515	0.0007	292.6	5.1	295.3	7.2	261	29	**295.7**	**9.5**	1.01	295	162
0.3388	0.0088	0.0470	0.0011	0.96	0.0524	0.0004	296.1	6.7	296	7.1	303	16	**296.0**	**9.0**	1.00	636	311
0.3390	0.0070	0.0472	0.0010	0.80	0.0524	0.0008	296.4	5.3	297.5	6.1	301	33	**297.4**	**8.5**	1.00	219	92
0.3442	0.0094	0.0473	0.0010	0.87	0.0529	0.0007	300.2	7.1	297.8	6.3	322	31	**297.7**	**8.6**	0.99	353	356
0.3460	0.0110	0.0473	0.0011	0.86	0.0528	0.0009	301.5	8.3	298.2	6.7	318	38	**297.7**	**9.0**	0.99	227	114
0.3428	0.0088	0.0479	0.0012	0.77	0.0537	0.0010	301.2	5.6	301.4	7.5	355	41	**301.1**	**9.5**	1.01	175	53
0.3448	0.0098	0.0480	0.0008	0.73	0.0529	0.0009	300.7	7.4	302	4.8	323	40	**301.8**	**7.7**	1.00	150	71
0.3420	0.0100	0.0480	0.0011	0.84	0.0521	0.0008	298.5	7.6	302	6.6	287	36	**302.3**	**9.1**	1.01	206	108
0.3540	0.0120	0.0487	0.0011	0.89	0.0525	0.0008	307.3	9.1	306.7	6.9	315	37	**306.5**	**9.1**	1.00	184	87
0.3520	0.0110	0.0490	0.0011	0.88	0.0519	0.0008	306.3	8.2	308.6	6.9	288	34	**308.6**	**9.2**	1.01	310	116
0.3731	0.0059	0.0510	0.0011	0.80	0.0532	0.0008	321.9	4.4	320.4	6.8	337	35	**320.5**	**9.3**	1.00	197	127
0.3870	0.0230	0.0518	0.0014	0.33	0.0534	0.0029	336	18	325.7	8.5	320	120	**325.4**	**10.8**	0.98	18	8
0.3820	0.0160	0.0521	0.0014	0.79	0.0535	0.0010	328	12	327.4	8.7	349	42	**327.2**	**10.8**	1.00	63	37
0.4330	0.0120	0.0582	0.0014	0.88	0.0547	0.0008	365.3	8.4	364.9	8.8	399	33	**364.3**	**11.2**	1.00	142	151
0.4460	0.0110	0.0601	0.0013	0.93	0.0541	0.0004	376	8.2	376.1	7.9	376	18	**376.2**	**10.9**	1.00	391	143
0.4460	0.0130	0.0601	0.0012	0.54	0.0532	0.0017	374.5	8.8	376	7.5	331	72	**376.7**	**10.5**	1.00	59	35
0.4913	0.0097	0.0651	0.0013	0.96	0.0547	0.0004	405.7	6.6	406.4	8.1	401	17	**406.6**	**11.3**	1.00	423	147
0.4930	0.0170	0.0652	0.0016	0.90	0.0551	0.0009	407	11	407.4	9.9	424	39	**407.1**	**12.7**	1.00	153	102
0.5840	0.0170	0.0744	0.0015	0.67	0.0568	0.0014	467	11	462.4	8.8	478	55	**462.3**	**12.9**	0.99	88	81
0.5780	0.0120	0.0753	0.0014	0.88	0.0562	0.0006	462.8	7.9	467.9	8.6	460	25	**468.1**	**12.5**	1.01	234	92
0.6060	0.0210	0.0756	0.0017	0.87	0.0569	0.0010	481	13	470	9.9	496	41	**469.5**	**13.9**	0.98	73	55
0.5830	0.0130	0.0756	0.0015	0.77	0.0564	0.0008	466.4	8.5	470	9	466	33	**469.8**	**13.0**	1.01	173	160
0.5900	0.0130	0.0760	0.0012	0.66	0.0565	0.0010	470.6	8.2	472	7.3	469	40	**472.2**	**11.8**	1.00	111	34
0.5950	0.0160	0.0762	0.0022	0.83	0.0567	0.0010	474	10	474	13	478	39	**473.3**	**16.3**	1.00	151	94
0.5899	0.0099	0.0767	0.0010	0.87	0.0559	0.0005	470.7	6.4	476.4	6	448	21	**476.8**	**11.2**	1.01	381	119
0.6080	0.0150	0.0769	0.0016	0.84	0.0578	0.0008	484.6	9.1	477.6	9.4	521	29	**476.9**	**13.5**	0.99	154	73
0.6040	0.0140	0.0773	0.0017	0.71	0.0572	0.0012	479.6	8.7	480	10	495	46	**479.7**	**14.0**	1.00	114	89
0.6060	0.0160	0.0774	0.0021	0.72	0.0573	0.0013	481	10	480	12	499	51	**480.2**	**15.9**	1.00	45	32
0.5990	0.0180	0.0779	0.0025	0.95	0.0565	0.0007	480	13	483	15	473	27	**483.7**	**17.9**	1.01	286	166
0.6540	0.0240	0.0786	0.0012	0.72	0.0608	0.0018	510	15	488	7.1	626	63	**485.4**	**12.0**	0.95	378	12
0.6170	0.0190	0.0789	0.0017	0.81	0.0574	0.0009	488	12	489.8	9.9	503	35	**489.3**	**14.1**	1.00	127	64
0.6120	0.0120	0.0797	0.0010	0.83	0.0563	0.0005	484.9	7.4	494.5	6.2	465	21	**494.8**	**11.5**	1.02	124	21
0.6630	0.0150	0.0817	0.0010	0.18	0.0588	0.0012	516.5	9	506.4	6.1	556	43	**505.4**	**11.6**	0.98	60	48
0.6440	0.0140	0.0816	0.0018	0.85	0.0578	0.0007	504.5	8.4	506	11	522	26	**505.4**	**14.7**	1.00	149	8
0.6970	0.0130	0.0823	0.0013	0.88	0.0615	0.0006	536.9	7.9	509.7	8	656	23	**507.2**	**12.7**	0.95	116	54
0.6640	0.0150	0.0820	0.0018	0.96	0.0583	0.0003	516.7	9.3	508	10	541	12	**507.5**	**14.8**	0.98	292	206
0.6620	0.0130	0.0821	0.0009	0.81	0.0579	0.0008	515.9	8.1	508.4	5.5	526	30	**508.1**	**11.5**	0.99	208	143
0.6580	0.0250	0.0837	0.0034	0.93	0.0581	0.0008	513	15	518	20	533	31	**517.9**	**22.9**	1.01	184	83
0.6920	0.0140	0.0842	0.0016	0.89	0.0597	0.0006	533.8	8.4	521	9.6	593	21	**519.9**	**14.0**	0.98	189	48
0.6650	0.0170	0.0841	0.0015	0.96	0.0573	0.0005	517	10	520.4	8.8	502	20	**520.9**	**13.7**	1.01	197	3
0.6840	0.0150	0.0844	0.0015	0.94	0.0589	0.0004	528.7	9.3	522.3	9	564	15	**521.6**	**13.7**	0.99	508	88
0.6830	0.0170	0.0847	0.0018	0.82	0.0585	0.0010	528	10	524	10	555	33	**523.7**	**15.0**	0.99	120	60
0.6770	0.0190	0.0847	0.0020	0.92	0.0583	0.0006	525	12	524	12	539	24	**523.8**	**15.9**	1.00	98	72
0.6830	0.0160	0.0848	0.0016	0.80	0.0590	0.0010	528.5	9.7	524.9	9.7	576	39	**523.9**	**14.1**	0.99	112	75
0.6660	0.0180	0.0849	0.0016	0.77	0.0573	0.0010	518	11	525	9.7	509	37	**525.7**	**14.2**	1.01	73	46
0.6840	0.0140	0.0851	0.0012	0.80	0.0583	0.0007	528.8	8.4	526.4	7.1	541	27	**526.2**	**12.6**	0.99	142	26
0.6870	0.0140	0.0855	0.0018	0.92	0.0584	0.0006	530.5	8.4	529	10	545	21	**528.6**	**15.0**	1.00	278	346
0.6860	0.0220	0.0862	0.0023	0.91	0.0580	0.0010	530	13	533	14	527	36	**533.1**	**17.4**	1.00	78	40
0.6901	0.0086	0.0863	0.0013	0.88	0.0584	0.0006	532.8	5.2	533.8	7.8	542	21	**533.5**	**13.1**	1.00	350	42
0.6890	0.0180	0.0865	0.0018	0.93	0.0581	0.0005	532	11	534	11	531	17	**534.9**	**15.1**	1.00	234	197
0.6690	0.0140	0.0867	0.0021	0.78	0.0566	0.0008	519.9	8.5	536	12	484	33	**537.1**	**16.5**	1.03	133	61
0.6930	0.0120	0.0869	0.0014	0.85	0.0577	0.0005	534.4	7.2	537.3	8.5	517	18	**537.5**	**13.5**	1.00	182	115
0.6970	0.0160	0.0870	0.0017	0.87	0.0584	0.0006	537	9.5	538	10	545	24	**537.6**	**14.7**	1.00	205	94
0.7070	0.0170	0.0872	0.0018	0.81	0.0595	0.0012	543	10	539	11	582	43	**538.1**	**15.2**	0.99	89	57
0.7020	0.0120	0.0878	0.0014	0.90	0.0589	0.0004	539.7	7	542.2	8.4	563	16	**542.1**	**13.6**	1.00	351	196
0.7050	0.0180	0.0877	0.0018	0.91	0.0578	0.0005	541	11	542	11	527	18	**542.3**	**15.2**	1.00	225	149

0.7130	0.0220	0.0879	0.0025	0.89	0.0589	0.0008	546	13	543	15	561	28	**542.8**	**18.5**	0.99	202	439
0.7180	0.0210	0.0881	0.0014	0.81	0.0592	0.0009	549	13	544.2	8	573	32	**543.7**	**13.6**	0.99	101	75
0.6930	0.0160	0.0880	0.0014	0.81	0.0573	0.0007	534.7	9.6	543.7	8	503	28	**544.4**	**13.6**	1.02	93	60
0.7080	0.0140	0.0885	0.0018	0.84	0.0580	0.0007	543.4	8.4	546	11	528	26	**547.0**	**15.3**	1.01	178	157
0.7130	0.0200	0.0889	0.0025	0.95	0.0583	0.0006	546	12	549	15	541	21	**549.2**	**18.5**	1.00	272	164
0.7050	0.0210	0.0894	0.0018	0.82	0.0581	0.0009	542	12	552	11	538	37	**552.4**	**15.3**	1.02	120	403
0.7350	0.0200	0.0898	0.0019	0.66	0.0591	0.0014	559	12	554	11	568	52	**554.0**	**15.8**	0.99	72	36
0.7210	0.0230	0.0900	0.0022	0.86	0.0591	0.0010	551	13	555	13	569	36	**555.2**	**17.2**	1.01	53	20
0.7590	0.0350	0.0902	0.0021	0.67	0.0602	0.0018	572	20	556	13	604	65	**555.7**	**16.7**	0.97	18	10
0.7290	0.0160	0.0901	0.0019	0.93	0.0590	0.0005	555.7	9.2	556	11	567	18	**555.9**	**15.8**	1.00	369	31
0.7200	0.0160	0.0900	0.0020	0.92	0.0581	0.0005	550.2	9.6	555	12	531	17	**556.0**	**16.3**	1.01	471	31
0.7420	0.0170	0.0907	0.0015	0.73	0.0592	0.0010	563.2	9.8	559.5	9	573	35	**559.4**	**14.2**	0.99	94	94
0.7490	0.0230	0.0910	0.0021	0.79	0.0592	0.0011	567	13	561	12	582	37	**561.2**	**16.8**	0.99	60	24
0.7690	0.0320	0.0912	0.0022	0.65	0.0602	0.0021	579	18	563	13	601	76	**561.7**	**17.3**	0.97	36	6
0.7620	0.0240	0.0918	0.0017	0.60	0.0599	0.0016	575	14	566.2	9.9	595	58	**565.5**	**15.1**	0.98	34	33
0.7360	0.0170	0.0920	0.0019	0.87	0.0588	0.0007	559.6	9.7	567	11	566	24	**567.5**	**16.0**	1.01	136	31
0.7360	0.0170	0.0922	0.0020	0.77	0.0587	0.0010	559.7	9.7	569	12	553	37	**568.8**	**16.4**	1.02	96	53
0.7590	0.0210	0.0923	0.0014	0.74	0.0590	0.0010	573	12	569.1	8.2	566	37	**569.2**	**14.0**	0.99	120	279
0.7520	0.0180	0.0926	0.0024	0.87	0.0591	0.0010	569	10	571	14	568	35	**570.9**	**18.3**	1.00	163	49
0.7400	0.0130	0.0927	0.0020	0.72	0.0583	0.0010	562.4	7.3	571	12	538	38	**572.1**	**16.5**	1.02	70	8
0.7540	0.0180	0.0928	0.0019	0.85	0.0589	0.0007	570	11	572	11	562	24	**572.3**	**16.0**	1.00	94	88
0.7510	0.0210	0.0928	0.0019	0.60	0.0581	0.0011	568	12	572	11	541	38	**572.8**	**16.0**	1.01	85	49
0.7720	0.0260	0.0934	0.0032	0.95	0.0598	0.0006	580	15	576	19	594	22	**575.2**	**22.3**	0.99	194	345
0.7760	0.0180	0.0943	0.0021	0.97	0.0598	0.0004	583	10	581	12	597	16	**580.6**	**17.0**	1.00	506	528
0.7500	0.0250	0.0949	0.0017	0.60	0.0584	0.0019	568	14	584	10	553	65	**585.2**	**15.4**	1.03	26	42
0.7840	0.0240	0.0959	0.0025	0.82	0.0588	0.0009	588	13	590	15	557	34	**591.0**	**19.0**	1.00	90	38
0.7960	0.0180	0.0967	0.0020	0.96	0.0599	0.0004	594	10	595	12	601	14	**594.9**	**16.7**	1.00	503	347
0.7990	0.0240	0.0971	0.0026	0.95	0.0597	0.0007	596	13	597	15	599	24	**597.5**	**19.5**	1.00	197	65
0.8230	0.0310	0.0980	0.0022	0.73	0.0610	0.0016	609	17	603	13	634	57	**601.9**	**17.7**	0.99	34	53
0.8180	0.0280	0.0998	0.0027	0.95	0.0599	0.0007	606	16	613	16	599	23	**613.5**	**20.1**	1.01	161	60
0.8500	0.0250	0.1008	0.0025	0.95	0.0616	0.0005	624	13	619	15	659	19	**618.2**	**19.2**	0.99	124	87
0.8350	0.0240	0.1008	0.0023	0.89	0.0609	0.0009	616	13	619	14	634	32	**618.7**	**18.3**	1.00	144	59
0.8630	0.0250	0.1015	0.0026	0.95	0.0615	0.0005	631	13	623	15	656	18	**622.5**	**19.7**	0.99	228	87
0.8600	0.0220	0.1021	0.0020	0.92	0.0610	0.0006	632	11	627	12	640	22	**626.4**	**17.1**	0.99	296	84
0.8770	0.0200	0.1025	0.0019	0.67	0.0623	0.0011	639	11	629	11	682	37	**627.8**	**16.7**	0.98	68	92
0.8620	0.0190	0.1030	0.0022	0.97	0.0611	0.0003	631	11	632	13	643	12	**631.7**	**18.0**	1.00	565	16
0.8730	0.0180	0.1038	0.0020	0.91	0.0613	0.0007	636.7	9.8	637	11	648	24	**636.4**	**17.2**	1.00	170	98
0.8730	0.0250	0.1039	0.0024	0.97	0.0611	0.0005	637	13	637	14	646	18	**637.1**	**19.0**	1.00	573	20
0.8840	0.0220	0.1045	0.0021	0.93	0.0614	0.0006	643	12	641	12	652	21	**640.5**	**17.7**	1.00	215	65
0.9020	0.0190	0.1075	0.0024	0.84	0.0615	0.0008	653	10	658	14	656	27	**658.2**	**19.2**	1.01	145	82
0.9660	0.0230	0.1131	0.0023	0.98	0.0624	0.0003	686	12	691	13	692	11	**690.8**	**19.2**	1.01	271	225
1.1750	0.0260	0.1287	0.0025	0.93	0.0663	0.0006	789	12	780	14	816	18	**779.3**	**21.1**	0.99	216	90
1.2080	0.0640	0.1290	0.0060	0.98	0.0679	0.0007	802	29	782	34	863	20	**779.6**	**38.3**	0.97	175	143
1.2230	0.0300	0.1294	0.0030	0.97	0.0689	0.0005	811	14	785	17	896	15	**780.9**	**23.3**	0.97	207	62
1.1450	0.0180	0.1293	0.0017	0.81	0.0650	0.0008	774.6	8.4	783.9	9.5	772	26	**784.2**	**18.3**	1.01	146	30
1.1430	0.0350	0.1300	0.0030	0.93	0.0646	0.0007	773	16	788	17	760	22	**788.7**	**23.4**	1.02	316	123
1.2880	0.0230	0.1377	0.0023	0.69	0.0679	0.0010	840	10	835	12	863	31	**830.5**	**21.1**	0.99	61	22
1.3290	0.0280	0.1390	0.0022	0.91	0.0704	0.0006	861	13	839	12	938	17	**835.5**	**20.7**	0.98	142	96
1.3130	0.0330	0.1411	0.0026	0.90	0.0689	0.0009	851	15	851	15	895	27	**849.3**	**22.5**	1.00	50	38
1.3710	0.0390	0.1429	0.0034	0.87	0.0704	0.0012	876	17	861	19	937	35	**858.1**	**25.9**	0.98	40	26
1.5230	0.0490	0.1534	0.0041	0.93	0.0722	0.0009	939	20	920	23	991	24	**917.1**	**29.7**	0.98	144	42
1.5280	0.0350	0.1544	0.0038	0.94	0.0724	0.0005	941	14	926	21	997	14	**922.6**	**28.4**	0.98	154	170
1.4840	0.0350	0.1543	0.0031	0.93	0.0705	0.0006	923	15	925	18	941	18	**924.3**	**25.5**	1.00	131	67
1.4620	0.0310	0.1541	0.0032	0.96	0.0694	0.0004	914	13	924	18	909	11	**924.5**	**25.9**	1.01	276	166
1.4910	0.0410	0.1545	0.0036	0.98	0.0702	0.0005	930	16	926	20	933	14	**925.8**	**27.6**	1.00	375	152
1.6160	0.0360	0.1581	0.0032	0.97	0.0736	0.0004	976	14	946	18	1029	12	**942.7**	**26.1**	0.97	292	100
1.9580	0.0330	0.1651	0.0024	0.98	0.0864	0.0004	1101	11	985	13	1348	9.7	**966.9**	**23.3**	0.89	454	117
1.5950	0.0610	0.1622	0.0055	0.99	0.0720	0.0006	972	26	969	31	986	18	**968.2**	**37.0**	1.00	356	152
1.5830	0.0250	0.1629	0.0029	0.82	0.0713	0.0008	963.3	9.9	973	16	965	23	**973.2**	**25.3**	1.01	122	44
1.6450	0.0500	0.1635	0.0044	0.98	0.0729	0.0005	987	19	976	24	1009	13	**974.7**	**31.7**	0.99	117	116
1.6940	0.0530	0.1659	0.0039	0.92	0.0741	0.0008	1005	20	989	22	1043	22	**987.0**	**29.6**	0.98	60	30
1.6220	0.0300	0.1662	0.0027	0.72	0.0709	0.0011	979	12	991	15	954	31	**992.8**	**24.9**	1.01	93	23
1.6860	0.0870	0.1673	0.0076	0.98	0.0736	0.0009	1001	32	997	42	1029	26	**995.7**	**47.9**	0.99	166	31
1.6540	0.0460	0.1676	0.0044	0.96	0.0715	0.0005	990	18	999	24	972	13	**1000.1**	**32.0**	1.01	184	49
1.7370	0.0480	0.1700	0.0038	0.93	0.0746	0.0008	1022	18	1012	21	1058	21	**1009.9**	**29.4**	0.99	128	51
1.7470	0.0400	0.1727	0.0039	0.93	0.0736	0.0005	1025	15	1027	21	1031	12	**1026.8**	**30.1**	1.00	179	57
1.7530	0.0320	0.1733	0.0033	0.93	0.0735	0.0004	1028	12	1030	18	1027	9.9	**1030.4**	**27.6**	1.00	160	47
1.7760	0.0440	0.1763	0.0042	0.86	0.0735	0.0011	1036	16	1047	23	1025	29	**1047.7**	**31.7**	1.01	80	53
1.8320	0.0260	0.1778	0.0025	0.87	0.0750	0.0006	1056.9	9.2	1055	14	1070	14	**1054.3**	**25.1**	1.00	210	117
1.8500	0.0540	0.1790	0.0043	0.97	0.0747	0.0006	1063	19	1061	24	1060	15	**1061.6**	**32.2**	1.00	156	125
1.8460	0.0340	0.1799	0.0031	0.88	0.0754	0.0008	1065	14	1066	17	1077	22	**1065.8**	**27.4**	1.00	78	24
1.9040	0.0360	0.1813	0.0028	0.63	0.0756	0.0013	1082	13	1074	15	1084	34	**1073.5**	**26.4**	0.99	33	16
1.9220	0.0430	0.1817	0.0033	0.95	0.0766	0.0005	1088	15	1076	18	1114	15	**1074.4**	**28.2**	0.99	112	45
1.8810	0.0430	0.1825	0.0051	0.97	0.0753	0.0004	1074	15	1087	26	1075	11	**1080.9**	**36.0**	1.01	348	98
1.8830	0.0630	0.1827	0.0058	0.99	0.0751	0.0004	1074	22	1081	31	1071	10	**1082.2**	**39.4**	1.01	166	49

1.9070	0.0550	0.1833	0.0032	0.84	0.0750	0.0010	1083	19	1085	18	1066	28	**1085.8**	**28.0**	1.00	52	22
1.9370	0.0500	0.1839	0.0029	0.87	0.0773	0.0011	1093	18	1088	16	1127	29	**1086.1**	**26.9**	0.99	69	42
1.8990	0.0300	0.1841	0.0026	0.78	0.0755	0.0007	1080	11	1089	14	1082	19	**1089.7**	**26.0**	1.01	75	21
1.9160	0.0360	0.1844	0.0028	0.85	0.0763	0.0008	1089	13	1091	15	1103	21	**1090.3**	**26.6**	1.00	103	22
1.9610	0.0450	0.1861	0.0044	0.94	0.0762	0.0005	1106	17	1100	24	1099	14	**1100.2**	**33.1**	1.00	125	45
1.9460	0.0530	0.1863	0.0054	0.96	0.0758	0.0006	1096	18	1101	29	1089	16	**1101.9**	**37.7**	1.00	85	33
1.9550	0.0430	0.1866	0.0030	0.92	0.0764	0.0007	1100	15	1103	16	1105	18	**1102.8**	**27.5**	1.00	84	43
1.9700	0.0560	0.1876	0.0045	0.82	0.0779	0.0014	1104	19	1108	24	1151	32	**1106.4**	**33.6**	1.00	35	9
1.9590	0.0480	0.1885	0.0046	0.92	0.0756	0.0007	1101	16	1113	25	1085	19	**1114.8**	**34.2**	1.01	104	46
2.0130	0.0440	0.1890	0.0041	0.87	0.0769	0.0007	1123	14	1116	23	1117	19	**1115.9**	**32.0**	1.00	84	41
2.0390	0.0480	0.1915	0.0042	0.94	0.0777	0.0007	1128	16	1129	23	1139	18	**1128.9**	**32.6**	1.00	183	94
2.0280	0.0340	0.1923	0.0032	0.97	0.0768	0.0004	1125	11	1134	17	1115	9	**1134.9**	**28.7**	1.01	247	327
2.1230	0.0530	0.1935	0.0040	0.80	0.0800	0.0010	1156	17	1140	22	1202	21	**1137.0**	**31.8**	0.99	65	46
3.9200	0.1500	0.2104	0.0057	0.98	0.1367	0.0017	1616	31	1231	30	2186	22	**1142.6**	**38.2**	0.76	272	229
2.1070	0.0510	0.1950	0.0055	0.91	0.0784	0.0008	1150	17	1148	30	1161	22	**1147.9**	**38.6**	1.00	97	35
2.0890	0.0560	0.1960	0.0042	0.92	0.0780	0.0009	1144	18	1154	22	1152	25	**1154.2**	**32.9**	1.01	79	82
2.1340	0.0470	0.1984	0.0054	0.96	0.0784	0.0006	1160	15	1166	29	1156	15	**1167.3**	**38.3**	1.01	157	54
2.1220	0.0790	0.1989	0.0073	0.98	0.0781	0.0005	1154	26	1169	39	1149	14	**1170.6**	**47.5**	1.01	105	33
2.2630	0.0630	0.2029	0.0051	0.92	0.0800	0.0009	1200	20	1191	27	1195	21	**1190.6**	**37.2**	0.99	55	33
2.4020	0.0650	0.2142	0.0061	0.98	0.0822	0.0005	1242	19	1258	34	1250	12	**1251.2**	**42.4**	1.01	405	84
2.6200	0.0510	0.2190	0.0034	0.92	0.0864	0.0009	1306	14	1276	18	1346	19	**1271.5**	**31.4**	0.98	108	49
2.7780	0.0550	0.2369	0.0049	0.91	0.0855	0.0008	1349	15	1370	26	1327	18	**1374.1**	**38.7**	1.02	76	97
2.9820	0.0550	0.2461	0.0054	0.98	0.0889	0.0005	1403	14	1418	28	1402	10	**1402.0**	**39.6**	1.01	221	159
4.2070	0.0670	0.2974	0.0044	0.97	0.1029	0.0004	1675	13	1678	22	1676	7.7	**1676.4**	**37.8**	1.00	284	114
4.0590	0.0930	0.2833	0.0062	0.98	0.1039	0.0006	1645	19	1608	31	1695	11	**1694.7**	**38.4**	0.98	262	128
3.7990	0.0640	0.2653	0.0045	0.98	0.1044	0.0004	1592	14	1517	23	1704	6.8	**1703.8**	**37.4**	0.95	527	211
4.7350	0.0870	0.3164	0.0046	0.92	0.1087	0.0006	1773	16	1772	22	1777	11	**1777.4**	**38.0**	1.00	37	69
4.7300	0.1200	0.3156	0.0073	0.98	0.1095	0.0005	1771	22	1768	36	1791	8.7	**1791.1**	**37.4**	1.00	329	110
5.2500	0.1300	0.3360	0.0075	0.99	0.1146	0.0004	1865	20	1867	36	1873	6.5	**1872.8**	**36.6**	1.00	336	50
5.5200	0.1100	0.3475	0.0075	0.94	0.1160	0.0008	1903	17	1922	36	1901	12	**1895.2**	**37.9**	1.01	87	86
6.3600	0.1600	0.3668	0.0093	0.99	0.1260	0.0004	2025	22	2014	44	2044	6	**2042.2**	**35.8**	0.99	270	99
12.2400	0.2800	0.4890	0.0100	0.99	0.1819	0.0007	2622	22	2566	43	2670	6.4	**2670.3**	**33.7**	0.98	212	113
11.6800	0.8100	0.4530	0.0210	0.91	0.1874	0.0065	2569	65	2406	93	2714	59	**2719.5**	**66.0**	0.93	154	104
11.9000	0.2200	0.4448	0.0084	0.98	0.1938	0.0006	2596	17	2381	40	2775	5.4	**2774.9**	**33.2**	0.91	185	69
16.7700	0.3300	0.5450	0.0100	0.99	0.2233	0.0007	2921	19	2803	43	3005	4.7	**3004.9**	**32.5**	0.96	145	62
27.6600	0.4600	0.6940	0.0110	0.99	0.2900	0.0009	3411	18	3397	42	3418	4.6	**3417.6**	**31.4**	1.00	219	153
28.7000	1.1000	0.6040	0.0230	0.98	0.3455	0.0026	3440	36	3042	91	3687	12	**3687.3**	**32.6**	0.88	244	52

HU-9 **Lat:** **-30.20586** **Lon:** **-68.36574**

Isotope Ratios							Apparent Ages									Concentrations	
$^{207}Pb/^{235}U$	± (2σ)	$^{206}Pb/^{238}U$	± (2σ)	Error Corr.	$^{207}Pb/^{206}Pb$	± (2σ)	$^{207}Pb/^{235}U$ (Ma)	± (Ma)	$^{206}Pb/^{238}U$ (Ma)	± (Ma)	$^{207}Pb/^{206}Pb$ (Ma)	± (Ma)	Plot Age (Ma)	± (Ma)	conc. (%)	U (ppm)	Th (ppm)
0.0142	0.0045	0.0020	0.0001	0.35	0.0570	0.0170	14.1	4.5	12.63	0.73	-	-	**12.5**	**0.7**	10.43	76	40
0.0220	0.0110	0.0029	0.0002	0.03	0.0640	0.0270	24	11	18.7	1.3	-	-	**18.3**	**1.3**	22.08	26	18
0.0203	0.0039	0.0030	0.0001	0.27	0.0483	0.0093	20.7	3.8	19.39	0.69	-	-	**19.3**	**0.7**	6.33	499	354
0.0350	0.0190	0.0033	0.0003	-0.04	0.0720	0.0480	32	18	21.3	2.1	-	-	**20.6**	**2.1**	33.44	18	27
0.0228	0.0009	0.0033	0.0001	0.53	0.0497	0.0015	22.9	0.9	21.4	0.5	-	-	**21.4**	**0.5**	6.46	439	117
0.0247	0.0030	0.0034	0.0001	-0.08	0.0526	0.0068	24.7	2.9	21.69	0.58	-	-	**21.5**	**0.6**	12.19	210	134
0.0600	0.0130	0.0035	0.0002	0.37	0.1110	0.0260	58	13	22.4	1.4	-	-	**20.6**	**1.4**	61.38	62	124
0.0330	0.0150	0.0037	0.0003	-0.11	0.0620	0.0380	32	14	23.5	1.7	-	-	**23.0**	**1.7**	26.56	11	3
0.0270	0.0140	0.0038	0.0003	-0.04	0.0630	0.0380	29	13	24.7	1.7	-	-	**24.2**	**1.7**	14.83	94	49
0.0350	0.0260	0.0044	0.0004	0.04	0.0700	0.0640	27	25	28.5	2.8	-	-	**27.7**	**2.8**	5.56	58	18
0.0420	0.0220	0.0045	0.0004	0.14	0.1040	0.0640	36	21	28.7	2.7	-	-	**26.7**	**2.7**	20.28	68	47
0.0480	0.0190	0.0048	0.0004	-0.14	0.0640	0.0270	47	18	30.9	2.3	-	-	**30.3**	**2.3**	34.26	90	65
0.0370	0.0380	0.0049	0.0006	0.20	0.0900	0.1200	19	37	31.7	3.5	-	-	**30.0**	**3.5**	66.84	11	8
0.0315	0.0063	0.0050	0.0002	0.10	0.0465	0.0091	31	6.1	32.1	1.1	-	-	**32.2**	**1.1**	3.55	32	9
0.0351	0.0018	0.0052	0.0001	0.34	0.0487	0.0020	35.0	1.8	33.6	0.9	-	-	**33.5**	**0.9**	4.05	156	67
0.0500	0.0110	0.0053	0.0003	0.16	0.0740	0.0150	48	10	34	1.7	-	-	**32.8**	**1.7**	29.17	61	50
0.0370	0.0016	0.0054	0.0001	0.30	0.0494	0.0016	36.9	1.5	34.9	0.9	-	-	**34.8**	**0.9**	5.33	214	134
0.0389	0.0018	0.0059	0.0002	0.21	0.0478	0.0019	38.8	1.8	38.0	1.0	-	-	**37.9**	**1.0**	2.10	154	56
0.0912	0.0031	0.0133	0.0003	0.82	0.0496	0.0011	88.6	3.0	85.3	2.2	-	-	**85.3**	**2.2**	3.71	484	214
0.1008	0.0052	0.0142	0.0004	0.38	0.0515	0.0023	97.5	5.1	90.9	2.3	-	-	**90.9**	**2.3**	6.82	46	10
0.2011	0.0074	0.0292	0.0005	0.13	0.0500	0.0020	185.8	6.3	185.4	3	-	-	**185.4**	**3.0**	0.22	1277	425
0.2250	0.0180	0.0316	0.0007	0.00	0.0488	0.0040	204	15	200.3	4.4	140	160	**200.3**	**4.4**	1.81	285	123
0.2470	0.0260	0.0328	0.0008	0.10	0.0530	0.0056	221	21	208.3	5.2	270	200	**208.3**	**5.2**	5.75	138	180
0.2590	0.0200	0.0333	0.0007	0.05	0.0537	0.0044	230	16	211.4	4.5	360	150	**211.4**	**4.5**	8.09	34	42
0.2631	0.0091	0.0374	0.0010	0.88	0.0510	0.0011	237.2	8.2	236.9	6.2	239.9	51.8	**236.9**	**6.2**	0.12	281	178
0.2780	0.0170	0.0381	0.0007	-0.05	0.0535	0.0037	247	14	241.1	4	310	130	**241.1**	**4.0**	2.39	264	71
0.2725	0.0097	0.0385	0.0010	0.87	0.0514	0.0012	244.7	8.7	243.3	6.6	257.9	53.4	**243.3**	**6.6**	0.56	146	80
0.2759	0.0099	0.0389	0.0011	0.80	0.0514	0.0012	247.4	8.9	246.1	6.7	260.1	54.1	**246.1**	**6.7**	0.54	140	88
0.2880	0.0150	0.0396	0.0007	0.06	0.0510	0.0029	256	12	250.1	4.4	250	110	**250.1**	**4.4**	2.30	223	71
0.2990	0.0190	0.0400	0.0007	0.22	0.0531	0.0032	264	14	252.5	4.3	300	120	**252.5**	**4.3**	4.36	88	84
0.3010	0.0130	0.0403	0.0006	0.21	0.0521	0.0025	266	10	254.6	3.9	283	96	**254.6**	**3.9**	4.29	410	187

0.2770	0.0180	0.0403	0.0009	0.04	0.0490	0.0033	247	14	254.7	5.3	130	130	**254.7**	**5.3**	3.12	293	103
0.2750	0.0150	0.0407	0.0008	0.31	0.0481	0.0024	246	12	257.4	5.2	110	100	**257.4**	**5.2**	4.63	692	138
0.3010	0.0130	0.0411	0.0007	0.17	0.0520	0.0022	266	10	259.5	4	286	92	**259.5**	**4.0**	2.44	416	137
0.2930	0.0130	0.0417	0.0007	0.24	0.0512	0.0023	260	10	263.1	4.2	229	92	**263.1**	**4.2**	1.19	50	18
0.2920	0.0120	0.0419	0.0007	0.30	0.0496	0.0020	259.5	9.5	264.5	4.6	170	83	**264.5**	**4.6**	1.93	641	110
0.3000	0.0180	0.0419	0.0008	0.08	0.0520	0.0032	267	14	264.5	4.7	270	120	**264.5**	**4.7**	0.94	392	97
0.3064	0.0099	0.0420	0.0010	0.92	0.0529	0.0011	271.4	8.8	265.1	6.5	325.8	47.7	**265.1**	**6.5**	2.31	539	240
0.2980	0.0140	0.0426	0.0008	0.49	0.0507	0.0019	264	11	268.8	5.2	218	81	**268.8**	**5.2**	1.82	579	369
0.3090	0.0140	0.0428	0.0006	0.11	0.0533	0.0027	272	11	270	3.9	310	100	**270.0**	**3.9**	0.74	82	27
0.3500	0.0200	0.0431	0.0007	0.18	0.0599	0.0034	304	15	272.1	4.5	560	110	**272.1**	**4.5**	10.49	24	8
0.3260	0.0220	0.0432	0.0009	0.03	0.0543	0.0038	286	17	272.6	5.3	340	140	**272.6**	**5.3**	4.69	35	15
0.3134	0.0124	0.0435	0.0012	0.70	0.0523	0.0014	276.8	10.9	274.2	7.7	298.5	63.1	**274.2**	**7.7**	0.93	50	38
0.3230	0.0170	0.0436	0.0009	0.22	0.0533	0.0026	283	13	275.1	5.7	320	100	**275.1**	**5.7**	2.79	62	30
0.3287	0.0105	0.0455	0.0011	0.87	0.0525	0.0011	288.6	9.2	286.5	6.8	305.5	48.3	**286.5**	**6.8**	0.72	336	167
0.3611	0.0135	0.0456	0.0011	0.62	0.0574	0.0017	313.0	11.7	287.6	6.8	506.9	63.6	**287.6**	**6.8**	8.12	165	110
0.3279	0.0103	0.0457	0.0011	0.88	0.0520	0.0011	288.0	9.1	288.1	6.7	287.2	49.0	**288.1**	**6.7**	0.03	282	284
0.3337	0.0113	0.0458	0.0012	0.90	0.0529	0.0011	292.4	9.9	288.5	7.5	323.7	49.4	**288.5**	**7.5**	1.33	257	112
0.3268	0.0109	0.0458	0.0011	0.86	0.0518	0.0011	287.1	9.6	288.6	7.2	275.3	50.5	**288.6**	**7.2**	0.51	216	80
0.3420	0.0190	0.0461	0.0009	0.08	0.0534	0.0032	299	14	290.4	5.7	310	120	**290.4**	**5.7**	2.88	37	19
0.3410	0.0150	0.0461	0.0006	-0.12	0.0541	0.0026	300	11	290.5	3.9	360	100	**290.5**	**3.9**	3.17	39	5
0.3368	0.0117	0.0465	0.0013	0.95	0.0525	0.0011	294.8	10.3	293.2	8.1	307.2	48.3	**293.2**	**8.1**	0.53	426	187
0.3420	0.0112	0.0471	0.0012	0.91	0.0527	0.0011	298.7	9.7	296.6	7.3	315.0	48.5	**296.6**	**7.3**	0.70	232	114
0.3487	0.0112	0.0480	0.0012	0.90	0.0527	0.0011	303.7	9.8	302.0	7.3	317.2	48.6	**302.0**	**7.3**	0.58	289	185
0.3489	0.0119	0.0482	0.0012	0.81	0.0525	0.0012	303.9	10.4	303.4	7.5	307.7	53.1	**303.4**	**7.5**	0.16	158	288
0.3640	0.0122	0.0499	0.0012	0.86	0.0529	0.0012	315.2	10.5	313.8	7.9	325.4	50.4	**313.8**	**7.9**	0.44	239	80
0.3660	0.0340	0.0506	0.0013	0.03	0.0520	0.0050	312	26	318.3	8.2	220	180	**318.3**	**8.2**	2.02	10	14
0.4128	0.0144	0.0556	0.0014	0.80	0.0539	0.0013	350.9	12.2	348.8	8.9	364.8	53.3	**348.8**	**8.9**	0.60	84	52
0.4490	0.0200	0.0571	0.0011	0.14	0.0552	0.0024	375	14	357.9	6.6	430	100	**357.9**	**6.6**	4.56	92	29
0.4333	0.0149	0.0581	0.0015	0.91	0.0541	0.0012	365.5	12.5	364.1	9.6	374.8	49.4	**364.1**	**9.6**	0.40	117	28
0.4342	0.0143	0.0585	0.0014	0.83	0.0539	0.0012	366.2	12.0	366.3	9.0	365.2	49.2	**366.3**	**9.0**	0.04	90	138
0.4400	0.0141	0.0591	0.0014	0.89	0.0540	0.0011	370.2	11.9	370.0	8.9	371.5	47.7	**370.0**	**8.9**	0.05	238	219
0.4810	0.0250	0.0600	0.0010	-0.02	0.0578	0.0032	398	17	375.7	6	500	110	**375.7**	**6.0**	5.60	26	8
0.4600	0.0190	0.0600	0.0010	0.17	0.0562	0.0024	387	13	375.8	6.3	438	93	**375.8**	**6.3**	2.89	49	18
0.4850	0.0190	0.0603	0.0009	0.33	0.0577	0.0021	402	13	377.3	5.3	502	79	**377.3**	**5.3**	6.14	64	19
0.4563	0.0160	0.0608	0.0017	0.96	0.0544	0.0011	381.6	13.4	380.5	10.7	388.7	46.8	**380.5**	**10.7**	0.30	398	83
0.4620	0.0260	0.0615	0.0011	0.23	0.0548	0.0028	386	18	384.6	6.9	390	110	**384.6**	**6.9**	0.36	288	46
0.4900	0.0210	0.0625	0.0009	0.25	0.0565	0.0023	402	14	390.8	5.5	464	89	**390.8**	**5.5**	2.79	28	7
0.4290	0.0450	0.0626	0.0018	-0.04	0.0507	0.0056	360	31	391	11	200	210	**391.0**	**11.0**	8.61	104	42
0.4810	0.0260	0.0626	0.0013	0.23	0.0554	0.0028	401	18	391.2	7.6	410	110	**391.2**	**7.6**	2.44	140	81
0.4794	0.0151	0.0628	0.0015	0.91	0.0554	0.0012	397.7	12.5	392.8	9.2	426.4	46.4	**392.8**	**9.2**	1.24	334	126
0.4500	0.0290	0.0630	0.0015	0.04	0.0524	0.0034	377	21	394	9.2	280	130	**394.0**	**9.2**	4.51	126	38
0.4950	0.0310	0.0638	0.0010	-0.06	0.0566	0.0038	406	21	398.6	6.3	450	140	**398.6**	**6.3**	1.82	265	110
0.4860	0.0130	0.0643	0.0010	0.30	0.0535	0.0014	401.6	9.2	401.8	5.8	337	57	**401.8**	**5.8**	0.05	808	192
0.5200	0.0150	0.0647	0.0008	0.28	0.0566	0.0016	424	9.9	403.9	5	467	61	**403.9**	**5.0**	4.74	790	55
0.5240	0.0290	0.0651	0.0012	0.30	0.0561	0.0027	428	19	406.6	7	480	100	**406.6**	**7.0**	5.00	394	59
0.5049	0.0164	0.0655	0.0016	0.94	0.0559	0.0012	415.0	13.5	409.0	10.2	448.8	46.4	**409.0**	**10.2**	1.46	289	108
0.5230	0.0320	0.0671	0.0015	0.38	0.0582	0.0032	427	22	418.3	9.4	500	120	**418.3**	**9.4**	2.04	188	51
0.5230	0.0380	0.0675	0.0014	0.21	0.0553	0.0039	422	26	421.3	8.2	400	150	**421.3**	**8.2**	0.17	44	33
0.5280	0.0170	0.0680	0.0009	0.14	0.0564	0.0020	431	11	423.9	5.5	457	76	**423.9**	**5.5**	1.65	48	4
0.5380	0.0260	0.0684	0.0014	0.19	0.0565	0.0028	437	17	426.6	8.2	450	110	**426.6**	**8.2**	2.38	46	20
0.5420	0.0170	0.0688	0.0010	0.14	0.0570	0.0019	439	12	428.5	6	463	71	**428.5**	**6.0**	2.39	83	15
0.5810	0.0510	0.0700	0.0017	0.02	0.0594	0.0052	455	32	437.3	9.9	490	170	**437.3**	**9.9**	3.89	12	15
0.5630	0.0300	0.0701	0.0012	0.13	0.0572	0.0030	453	20	437.5	7.6	490	110	**437.5**	**7.6**	3.42	18	8
0.5540	0.0300	0.0709	0.0012	0.16	0.0555	0.0030	450	19	441.7	7.2	400	110	**441.7**	**7.2**	1.84	160	69
0.5860	0.0260	0.0719	0.0012	0.26	0.0596	0.0025	468	17	447.4	7.2	544	86	**447.4**	**7.2**	4.40	27	23
0.6110	0.0370	0.0738	0.0017	-0.03	0.0591	0.0039	489	22	459	10	520	140	**459.0**	**10.0**	6.13	132	72
0.6300	0.0350	0.0746	0.0013	0.01	0.0597	0.0035	493	21	463.6	7.9	580	120	**463.6**	**7.9**	5.96	23	18
0.5854	0.0198	0.0746	0.0020	0.95	0.0569	0.0012	467.9	15.8	463.8	12.3	488.0	46.1	**463.8**	**12.3**	0.88	259	123
0.6000	0.0340	0.0751	0.0015	0.10	0.0584	0.0034	473	22	466.8	9	490	120	**466.8**	**9.0**	1.31	10	6
0.5895	0.0187	0.0760	0.0018	0.92	0.0563	0.0012	470.6	14.9	472.2	11.3	462.6	46.1	**472.2**	**11.3**	0.35	213	105
0.5954	0.0192	0.0761	0.0019	0.94	0.0567	0.0012	474.3	15.3	472.8	11.7	481.4	45.9	**472.8**	**11.7**	0.31	277	215
0.6250	0.0270	0.0771	0.0014	0.23	0.0583	0.0026	491	17	479	8.2	544	97	**479.0**	**8.2**	2.44	29	17
0.5690	0.0350	0.0774	0.0014	0.11	0.0527	0.0033	457	22	480.7	8.5	320	130	**480.7**	**8.5**	5.19	179	200
0.6330	0.0330	0.0779	0.0014	-0.04	0.0587	0.0032	501	20	483.7	8.5	560	110	**483.7**	**8.5**	3.45	19	16
0.6520	0.0300	0.0780	0.0016	0.49	0.0604	0.0026	507	19	484	9.4	597	88	**484.0**	**9.4**	4.54	32	6
0.6130	0.0360	0.0781	0.0017	0.23	0.0558	0.0035	484	22	485	10	400	130	**485.0**	**10.0**	0.21	14	7
0.6500	0.1100	0.0787	0.0029	0.09	0.0618	0.0097	503	66	488	17	480	280	**488.0**	**17.0**	2.98	18	4
0.5850	0.0320	0.0787	0.0014	0.29	0.0530	0.0026	465	20	488	8.5	320	100	**488.0**	**8.5**	4.95	35	14
0.7390	0.0450	0.0789	0.0019	0.65	0.0670	0.0031	562	26	489	11	871	96	**489.0**	**11.0**	12.99	49	79
0.6210	0.0320	0.0794	0.0014	0.11	0.0564	0.0030	492	21	492.4	8.3	470	110	**492.4**	**8.3**	0.08	29	15
0.6470	0.0390	0.0798	0.0017	0.25	0.0581	0.0035	506	24	495	10	480	130	**495.0**	**10.0**	2.17	32	27
0.6130	0.0400	0.0799	0.0018	0.15	0.0543	0.0034	481	26	495	11	410	140	**495.0**	**11.0**	2.91	23	13
0.6249	0.0208	0.0799	0.0021	0.94	0.0567	0.0012	492.9	16.4	495.5	12.8	480.7	46.5	**495.5**	**12.8**	0.53	223	300
0.7140	0.0310	0.0800	0.0015	0.04	0.0631	0.0031	547	18	495.8	8.9	710	100	**495.8**	**8.9**	9.36	203	53
0.6470	0.0290	0.0802	0.0015	0.18	0.0579	0.0024	511	17	497.2	9	508	86	**497.2**	**9.0**	2.70	47	21

0.6350	0.0490	0.0809	0.0020	0.07	0.0584	0.0048	512	31	501	12	560	160	**501.0**	**12.0**	2.15	14	9
0.6600	0.0370	0.0811	0.0016	0.15	0.0594	0.0035	514	23	502.9	9.8	560	120	**502.9**	**9.8**	2.16	23	10
0.6578	0.0218	0.0818	0.0021	0.91	0.0583	0.0012	513.2	17.0	506.9	12.9	541.8	46.2	**506.9**	**12.9**	1.24	179	83
0.6830	0.0560	0.0818	0.0019	0.45	0.0602	0.0046	520	34	507	11	600	160	**507.0**	**11.0**	2.50	37	17
0.6710	0.0340	0.0829	0.0012	0.13	0.0594	0.0030	521	21	513.3	7.3	530	110	**513.3**	**7.3**	1.48	24	7
0.6643	0.0211	0.0830	0.0020	0.89	0.0581	0.0012	517.3	16.5	514.0	12.3	531.6	45.8	**514.0**	**12.3**	0.63	227	145
0.6980	0.0490	0.0834	0.0018	0.16	0.0607	0.0041	536	29	516	11	570	150	**516.0**	**11.0**	3.73	15	7
0.7460	0.0270	0.0842	0.0017	0.56	0.0637	0.0019	563	16	520.9	9.9	709	61	**520.9**	**9.9**	7.48	252	35
0.7120	0.0300	0.0842	0.0014	0.30	0.0601	0.0023	544	17	521	8.3	576	81	**521.0**	**8.3**	4.23	293	133
0.7228	0.0241	0.0854	0.0022	0.93	0.0614	0.0013	552.3	18.4	528.3	13.7	652.6	45.0	**528.3**	**13.7**	4.35	435	139
0.6863	0.0234	0.0857	0.0023	0.93	0.0581	0.0012	530.6	18.1	530.1	14.1	532.8	46.7	**530.1**	**14.1**	0.10	182	115
0.7160	0.0230	0.0863	0.0011	0.22	0.0588	0.0018	549	14	533.6	6.5	565	64	**533.6**	**6.5**	2.81	144	62
0.6944	0.0230	0.0863	0.0022	0.94	0.0584	0.0012	535.5	17.8	533.6	13.7	543.3	45.8	**533.6**	**13.7**	0.34	209	18
0.6903	0.0218	0.0864	0.0020	0.88	0.0580	0.0012	533.0	16.9	534.2	12.7	527.9	45.9	**534.2**	**12.7**	0.23	222	59
0.6909	0.0258	0.0864	0.0023	0.74	0.0580	0.0015	533.4	19.9	534.2	14.1	529.8	57.9	**534.2**	**14.1**	0.16	29	11
0.7200	0.0360	0.0862	0.0023	0.27	0.0586	0.0029	552	22	535	13	540	110	**535.0**	**13.0**	3.08	20	7
0.7500	0.0410	0.0874	0.0016	0.21	0.0618	0.0033	564	23	540.1	9.4	620	110	**540.1**	**9.4**	4.24	80	21
0.7400	0.0220	0.0885	0.0012	0.21	0.0602	0.0018	561	13	546.7	7.3	588	66	**546.7**	**7.3**	2.55	312	103
0.7187	0.0223	0.0885	0.0020	0.91	0.0589	0.0012	549.9	17.1	546.9	12.5	562.3	45.8	**546.9**	**12.5**	0.54	221	45
0.7130	0.0280	0.0893	0.0012	0.26	0.0582	0.0022	545	16	552.5	7.4	534	81	**552.5**	**7.4**	1.38	32	8
0.7297	0.0252	0.0900	0.0024	0.90	0.0588	0.0013	556.4	19.2	555.5	14.9	559.7	47.5	**555.5**	**14.9**	0.15	123	182
0.7870	0.0300	0.0908	0.0012	0.18	0.0630	0.0024	591	17	559.9	7	681	82	**559.9**	**7.0**	5.26	27	9
0.7520	0.0340	0.0919	0.0017	-0.03	0.0578	0.0026	568	20	566	10	530	110	**566.0**	**10.0**	0.35	25	14
0.8499	0.0291	0.0919	0.0025	0.96	0.0671	0.0014	624.6	21.4	566.8	15.4	839.9	43.2	**566.8**	**15.4**	9.26	173	72
0.8360	0.0490	0.0926	0.0024	0.47	0.0657	0.0035	614	27	571	14	750	110	**571.0**	**14.0**	7.00	32	11
0.7920	0.0240	0.0926	0.0012	0.37	0.0609	0.0016	592	13	571.6	7	621	58	**571.6**	**7.0**	3.45	48	36
0.7647	0.0246	0.0935	0.0022	0.87	0.0593	0.0013	576.7	18.6	576.2	13.7	578.9	47.3	**576.2**	**13.7**	0.09	157	328
0.7720	0.0300	0.0945	0.0012	0.20	0.0586	0.0022	579	17	582.1	7.2	535	85	**582.1**	**7.2**	0.54	394	125
0.8400	0.0180	0.0966	0.0012	0.42	0.0633	0.0013	620.1	9.8	594.2	7	713	42	**594.2**	**7.0**	4.18	91	29
0.8222	0.0265	0.0969	0.0024	0.95	0.0615	0.0013	609.3	19.6	596.2	14.7	658.2	44.3	**596.2**	**14.7**	2.15	240	128
0.8149	0.0265	0.0978	0.0024	0.92	0.0604	0.0013	605.2	19.7	601.5	14.8	619.0	45.8	**601.5**	**14.8**	0.61	130	66
0.8152	0.0316	0.0983	0.0030	0.88	0.0602	0.0015	605.4	23.5	604.4	18.1	609.0	52.9	**604.4**	**18.1**	0.16	36	20
0.8500	0.0190	0.0988	0.0012	0.14	0.0619	0.0015	626	10	607.3	7.1	675	51	**607.3**	**7.1**	2.99	43	8
0.8163	0.0271	0.0990	0.0025	0.93	0.0598	0.0013	606.0	20.1	608.5	15.6	596.3	45.3	**608.5**	**15.6**	0.43	223	237
0.8862	0.0302	0.1039	0.0028	0.96	0.0619	0.0013	644.3	21.9	637.2	17.3	669.4	44.1	**637.2**	**17.3**	1.11	384	50
0.8822	0.0282	0.1053	0.0026	0.98	0.0608	0.0012	642.2	20.5	645.4	15.8	630.9	44.0	**645.4**	**15.8**	0.50	569	68
0.9260	0.0270	0.1056	0.0016	0.44	0.0636	0.0015	665	14	647.2	9.6	720	53	**647.2**	**9.6**	2.68	28	2
0.9482	0.0301	0.1082	0.0026	0.91	0.0636	0.0013	677.2	21.5	662.3	15.8	727.0	44.6	**662.3**	**15.8**	2.20	115	68
1.1110	0.0300	0.1178	0.0019	0.55	0.0672	0.0015	756	15	717	11	832	45	**717.0**	**11.0**	5.16	540	66
1.1480	0.0550	0.1205	0.0029	-0.05	0.0678	0.0038	776	27	733	16	870	110	**733.0**	**16.0**	5.54	156	47
1.1709	0.0384	0.1233	0.0031	0.93	0.0689	0.0014	787.1	25.8	749.5	18.9	894.9	43.3	**749.5**	**18.9**	4.77	168	34
1.3500	0.2200	0.1318	0.0056	0.11	0.0740	0.0130	830	110	800	32	770	310	**770.0**	**310.0**	3.61	13	4
1.2207	0.0381	0.1346	0.0032	0.95	0.0658	0.0013	810.1	25.3	814.1	19.3	799.2	42.8	**799.2**	**42.8**	0.49	309	475
1.4490	0.0720	0.1512	0.0032	0.17	0.0674	0.0034	908	29	907	18	840	100	**840.0**	**100.0**	0.11	133	42
1.4900	0.1400	0.1512	0.0047	0.22	0.0704	0.0064	917	58	906	26	880	190	**880.0**	**190.0**	1.20	2	1
1.2552	0.0391	0.1328	0.0031	0.92	0.0686	0.0014	825.8	25.7	803.8	18.8	885.3	42.7	**885.3**	**42.7**	2.66	218	71
1.2615	0.0409	0.1333	0.0033	0.96	0.0686	0.0014	828.6	26.8	806.7	20.2	887.8	42.5	**887.8**	**42.5**	2.65	196	134
1.5200	0.1100	0.1545	0.0038	0.29	0.0707	0.0050	950	45	925	21	890	140	**890.0**	**140.0**	2.63	12	9
1.4839	0.0502	0.1510	0.0041	0.97	0.0713	0.0015	923.8	31.3	906.6	24.3	965.2	42.1	**965.2**	**42.1**	1.87	152	51
1.5170	0.0570	0.1508	0.0027	0.10	0.0715	0.0027	933	23	905	15	969	80	**969.0**	**80.0**	3.00	25	6
1.4970	0.0850	0.1499	0.0032	0.30	0.0718	0.0038	938	35	900	18	970	110	**970.0**	**110.0**	4.05	15	20
1.5979	0.0541	0.1617	0.0040	0.82	0.0717	0.0016	969.4	32.8	966.2	24.1	976.6	46.7	**976.6**	**46.7**	0.33	48	25
1.6157	0.0518	0.1633	0.0041	0.97	0.0718	0.0015	976.3	31.3	975.1	24.2	979.1	41.4	**979.1**	**41.4**	0.13	321	11
1.4180	0.0290	0.1404	0.0020	0.64	0.0721	0.0012	900	12	846	11	981	35	**981.0**	**35.0**	6.00	511	102
1.6270	0.0440	0.1637	0.0023	0.36	0.0722	0.0019	981	18	977	13	989	56	**989.0**	**56.0**	0.41	39	7
1.5340	0.0580	0.1499	0.0027	0.23	0.0733	0.0027	944	24	900	15	998	75	**998.0**	**75.0**	4.66	14	3
1.7665	0.0563	0.1762	0.0043	0.97	0.0727	0.0015	1033.2	32.9	1046.2	25.7	1005.9	41.4	**1005.9**	**41.4**	1.25	401	83
1.4940	0.0490	0.1458	0.0021	0.33	0.0727	0.0023	929	21	877	12	1017	66	**1017.0**	**66.0**	5.60	167	56
1.4640	0.0400	0.1451	0.0022	0.08	0.0727	0.0022	915	17	873	12	1019	58	**1019.0**	**58.0**	4.59	34	15
1.8153	0.0571	0.1797	0.0043	0.98	0.0733	0.0015	1051.0	33.1	1065.3	25.6	1021.3	41.1	**1021.3**	**41.1**	1.37	276	95
1.4810	0.0650	0.1501	0.0031	0.39	0.0733	0.0030	928	27	901	17	1027	74	**1027.0**	**74.0**	2.91	137	30
1.6450	0.0490	0.1618	0.0024	0.19	0.0737	0.0021	987	18	966	13	1033	59	**1033.0**	**59.0**	2.13	18	7
1.6979	0.0536	0.1671	0.0039	0.91	0.0737	0.0015	1007.7	31.8	996.1	23.5	1033.1	42.4	**1033.1**	**42.4**	1.15	105	145
1.7260	0.0420	0.1671	0.0027	0.32	0.0740	0.0017	1016	16	996	15	1035	48	**1035.0**	**48.0**	1.97	21	6
1.7619	0.0631	0.1732	0.0049	0.87	0.0738	0.0016	1031.6	36.9	1029.7	28.9	1035.5	45.0	**1035.5**	**45.0**	0.18	39	40
1.4420	0.0490	0.1388	0.0022	0.28	0.0744	0.0023	910	20	840	13	1042	67	**1042.0**	**67.0**	7.69	232	47
1.7206	0.0631	0.1682	0.0049	0.91	0.0742	0.0017	1016.2	37.3	1002.2	28.9	1046.6	45.7	**1046.6**	**45.7**	1.38	31	27
1.8900	0.0632	0.1846	0.0049	0.98	0.0743	0.0015	1077.6	36.0	1092.1	28.9	1048.4	41.1	**1048.4**	**41.1**	1.34	246	85
1.8636	0.0610	0.1817	0.0046	0.97	0.0744	0.0015	1068.3	34.9	1076.3	27.5	1052.0	41.0	**1052.0**	**41.0**	0.75	239	92
1.6508	0.0593	0.1609	0.0046	0.94	0.0744	0.0016	989.8	35.6	961.8	27.6	1052.6	43.8	**1052.6**	**43.8**	2.84	52	41
1.5540	0.0750	0.1481	0.0036	-0.01	0.0728	0.0042	952	30	889	20	1060	120	**1060.0**	**120.0**	6.62	28	27
1.6880	0.0500	0.1588	0.0022	0.19	0.0749	0.0021	999	19	950	12	1065	59	**1065.0**	**59.0**	4.90	119	47
1.7380	0.0510	0.1658	0.0024	0.26	0.0752	0.0022	1023	18	989	13	1075	60	**1075.0**	**60.0**	3.32	46	30
1.9161	0.0617	0.1845	0.0044	0.88	0.0753	0.0016	1086.7	35.0	1091.5	26.0	1077.1	43.4	**1077.1**	**43.4**	0.44	64	36
1.5480	0.0990	0.1422	0.0033	0.23	0.0773	0.0051	945	38	860	18	1080	130	**1080.0**	**130.0**	8.99	41	13

1.7980	0.0610	0.1678	0.0034	0.38	0.0765	0.0025	1052	21	1003	19	1092	65	**1092.0**	**65.0**	4.66	286	49
1.9143	0.0644	0.1820	0.0048	0.96	0.0763	0.0016	1086.1	36.5	1077.9	28.7	1102.6	41.1	**1102.6**	**41.1**	0.76	115	33
1.9600	0.0510	0.1845	0.0037	0.52	0.0762	0.0017	1101	18	1091	20	1109	42	**1109.0**	**42.0**	0.91	51	29
1.6820	0.0820	0.1596	0.0027	0.13	0.0775	0.0040	989	31	954	15	1110	100	**1110.0**	**100.0**	3.54	8	3
1.9791	0.0656	0.1863	0.0048	0.93	0.0770	0.0016	1108.4	36.8	1101.3	28.3	1122.4	41.8	**1122.4**	**41.8**	0.64	88	38
1.9998	0.0648	0.1877	0.0046	0.90	0.0773	0.0016	1115.4	36.2	1108.9	27.3	1128.2	42.0	**1128.2**	**42.0**	0.59	59	23
1.9470	0.0770	0.1762	0.0033	0.38	0.0781	0.0029	1095	26	1046	18	1149	73	**1149.0**	**73.0**	4.47	145	38
1.8300	0.0550	0.1664	0.0028	0.43	0.0793	0.0023	1059	19	992	16	1179	55	**1179.0**	**55.0**	6.33	26	8
2.0290	0.0770	0.1838	0.0027	0.14	0.0798	0.0029	1123	25	1087	15	1185	78	**1185.0**	**78.0**	3.21	33	11
2.1892	0.0684	0.1992	0.0047	0.96	0.0797	0.0016	1177.6	36.8	1171.0	27.6	1189.7	40.4	**1189.7**	**40.4**	0.56	174	108
1.8890	0.0320	0.1679	0.0019	0.52	0.0801	0.0011	1076	11	1000	11	1195	29	**1195.0**	**29.0**	7.06	779	274
1.9546	0.0767	0.1772	0.0050	0.75	0.0800	0.0022	1100.0	43.2	1051.7	29.6	1197.0	54.1	**1197.0**	**54.1**	4.40	31	12
2.0250	0.0650	0.1862	0.0028	0.24	0.0804	0.0028	1127	22	1100	15	1212	63	**1212.0**	**63.0**	2.40	41	12
2.1170	0.0480	0.1899	0.0027	0.44	0.0809	0.0018	1157	16	1121	15	1220	43	**1220.0**	**43.0**	3.11	54	12
2.2310	0.0430	0.1975	0.0026	0.55	0.0808	0.0012	1190	14	1161	14	1231	31	**1231.0**	**31.0**	2.44	69	39
2.1000	0.1100	0.1772	0.0036	0.22	0.0858	0.0044	1148	35	1051	20	1296	98	**1296.0**	**98.0**	8.45	11	11
2.4618	0.0859	0.2057	0.0052	0.84	0.0868	0.0021	1260.9	44.0	1205.9	30.2	1356.1	46.9	**1356.1**	**46.9**	4.36	63	24
2.4010	0.0720	0.1917	0.0038	0.41	0.0885	0.0026	1240	21	1130	21	1382	57	**1382.0**	**57.0**	8.87	238	72
3.4300	0.1300	0.2526	0.0047	0.27	0.0970	0.0032	1505	29	1451	24	1589	66	**1589.0**	**66.0**	3.59	23	32
2.8820	0.0980	0.2085	0.0045	0.38	0.1001	0.0035	1373	25	1220	24	1607	65	**1607.0**	**65.0**	11.14	147	40
4.6006	0.1520	0.3212	0.0084	0.99	0.1039	0.0021	1749.4	57.8	1795.6	46.9	1694.6	37.3	**1694.6**	**37.3**	2.64	221	219
4.1330	0.0990	0.2775	0.0062	0.44	0.1053	0.0025	1665	20	1578	31	1712	43	**1712.0**	**43.0**	5.23	273	317
4.1000	0.2600	0.2767	0.0085	0.05	0.1050	0.0070	1663	51	1580	42	1740	110	**1740.0**	**110.0**	4.99	12	3
5.0365	0.1622	0.3395	0.0084	0.97	0.1076	0.0022	1825.5	58.8	1884.3	46.8	1759.1	37.5	**1759.1**	**37.5**	3.22	261	378
4.7350	0.0840	0.3055	0.0042	0.34	0.1108	0.0021	1774	15	1718	21	1810	33	**1810.0**	**33.0**	3.16	62	33
5.3643	0.1752	0.3499	0.0090	0.98	0.1112	0.0023	1879.2	61.4	1934.1	49.5	1819.0	36.8	**1819.0**	**36.8**	2.92	163	155
4.3428	0.1357	0.2795	0.0066	0.97	0.1127	0.0023	1701.5	53.2	1588.8	37.5	1843.3	37.1	**1843.3**	**37.1**	6.62	388	213
4.7770	0.0930	0.3026	0.0039	0.31	0.1150	0.0024	1778	16	1704	19	1871	39	**1871.0**	**39.0**	4.16	15	14
4.7400	0.0750	0.2980	0.0036	0.37	0.1149	0.0019	1773	13	1681	18	1873	30	**1873.0**	**30.0**	5.19	55	16
8.9569	0.2781	0.5048	0.0119	0.99	0.1287	0.0026	2333.6	72.5	2634.4	62.1	2080.1	35.6	**2080.1**	**35.6**	12.89	222	141
5.8100	0.1200	0.3186	0.0050	0.13	0.1325	0.0037	1946	19	1782	24	2128	48	**2128.0**	**48.0**	8.43	25	41
6.0400	0.1700	0.3291	0.0072	0.78	0.1326	0.0020	1976	24	1831	35	2130	27	**2130.0**	**27.0**	7.34	19	6
5.9600	0.1400	0.3225	0.0045	0.39	0.1332	0.0029	1971	21	1801	22	2140	39	**2140.0**	**39.0**	8.63	134	35
12.2096	0.4059	0.5001	0.0131	0.98	0.1771	0.0036	2620.6	87.1	2614.2	68.6	2625.6	33.9	**2625.6**	**33.9**	0.24	44	21
16.7900	0.2800	0.5075	0.0061	0.58	0.2349	0.0028	2922	16	2645	26	3083	19	**3083.0**	**19.0**	9.48	162	42
23.0900	0.3400	0.5681	0.0084	0.63	0.2912	0.0043	3229	14	2904	33	3421	23	**3421.0**	**23.0**	10.07	14	5

HU-13 **Lat:** **-30.20867** **Lon:** **-68.36056**

Isotope Ratios							**Apparent Ages**								**Concentrations**	
$^{207}Pb/^{235}U$	± (2σ)	$^{206}Pb/^{238}U$	± (2σ)	Error Corr.	$^{207}Pb/^{206}Pb$	± (2σ)	$^{207}Pb/^{235}U$ (Ma)	± (Ma)	$^{206}Pb/^{238}U$ (Ma)	± (Ma)	$^{207}Pb/^{206}Pb$ (Ma)	± (Ma)	Plot Age (Ma)	± (Ma)	conc. (%)	U (ppm)	Th (ppm)
0.0290	0.0110	0.0033	0.0002	0.02	0.0890	0.0410	27	11	21.2	1.4	-	-	**20.0**	**1.4**	21.48	94	132
0.0130	0.0240	0.0035	0.0005	0.01	0.0720	0.0710	9	24	22.3	3.2	-	-	**21.6**	**3.2**	147.78	49	29
0.0123	0.0077	0.0036	0.0002	0.04	0.0260	0.0150	12.1	7.7	23	1.2	-	-	**23.6**	**1.2**	90.08	151	141
0.0203	0.0032	0.0036	0.0001	-0.05	0.0426	0.0070	20.5	3.2	23.03	0.67	-	-	**23.1**	**0.7**	12.34	1480	1320
0.1130	0.0470	0.0043	0.0005	0.14	0.1670	0.0820	93	44	27.4	3.2	-	-	**23.3**	**3.2**	70.54	153	96
0.0340	0.0250	0.0048	0.0005	0.16	0.0440	0.0410	30	24	30.7	3.2	-	-	**30.8**	**3.2**	2.33	66	57
0.0259	0.0080	0.0053	0.0003	0.16	0.0400	0.0120	25.1	7.9	33.9	1.6	-	-	**34.2**	**1.6**	35.06	158	87
0.0460	0.0310	0.0056	0.0006	-0.18	0.1800	0.1200	43	31	35.8	3.8	-	-	**29.8**	**3.8**	16.74	36	24
0.0440	0.0075	0.0056	0.0002	0.05	0.0600	0.0098	43.7	7.1	36.2	1.3	-	-	**35.6**	**1.3**	17.16	203	63
0.0719	0.0079	0.0104	0.0004	-0.02	0.0516	0.0060	69.8	7.5	67	2.3	-	-	**67.0**	**2.3**	4.01	188	197
0.3080	0.0160	0.0426	0.0008	0.08	0.0535	0.0030	271	13	268.9	5	300	110	**268.9**	**5.0**	0.77	409	421
0.3120	0.0350	0.0433	0.0011	-0.11	0.0522	0.0063	268	27	273	6.9	210	220	**273.0**	**6.9**	1.87	134	117
0.3160	0.0130	0.0435	0.0007	0.22	0.0518	0.0020	278	10	274.5	4.1	273	81	**274.5**	**4.1**	1.26	922	732
0.3440	0.0690	0.0470	0.0018	0.22	0.0519	0.0098	300	49	296	11	240	310	**296.0**	**11.0**	1.33	49	39
0.3230	0.0150	0.0473	0.0007	0.21	0.0508	0.0024	287	12	297.9	4.4	225	95	**297.9**	**4.4**	3.80	212	108
0.3070	0.0490	0.0472	0.0021	0.33	0.0466	0.0070	282	35	299	13	210	240	**299.0**	**13.0**	6.03	72	25
0.3570	0.0410	0.0498	0.0015	0.14	0.0514	0.0059	301	31	313.1	9.1	220	210	**313.1**	**9.1**	4.02	56	25
0.4670	0.0210	0.0618	0.0010	0.08	0.0545	0.0026	387	14	386.6	6.3	387	96	**386.6**	**6.3**	0.10	543	456
0.4840	0.0220	0.0629	0.0012	0.09	0.0545	0.0027	399	15	393.3	7.2	390	100	**393.3**	**7.2**	1.43	217	288
0.4930	0.0280	0.0644	0.0011	0.24	0.0553	0.0030	408	19	403.2	6.6	420	120	**403.2**	**6.6**	1.18	1374	708
0.5390	0.0180	0.0662	0.0009	0.18	0.0572	0.0020	438	12	413.2	5.5	480	75	**413.2**	**5.5**	5.66	236	176
0.6030	0.0340	0.0709	0.0016	-0.03	0.0562	0.0032	483	20	441.5	9.8	450	110	**441.5**	**9.8**	8.59	73	70
0.5980	0.0350	0.0729	0.0016	0.31	0.0607	0.0034	485	21	453.5	9.4	590	120	**453.5**	**9.4**	6.49	144	140
0.5490	0.0290	0.0731	0.0014	0.08	0.0540	0.0029	448	18	454.4	8.1	360	110	**454.4**	**8.1**	1.43	97	28
0.5930	0.0230	0.0732	0.0012	0.20	0.0560	0.0022	472	14	455.4	7.3	458	83	**455.4**	**7.3**	3.52	119	80
0.5790	0.0300	0.0732	0.0013	0.08	0.0567	0.0030	462	19	455.5	7.6	450	110	**455.5**	**7.6**	1.41	116	50
0.6010	0.0420	0.0738	0.0014	0.16	0.0573	0.0038	472	26	459.2	8.3	460	140	**459.2**	**8.3**	2.71	58	56
0.6010	0.0270	0.0747	0.0010	0.28	0.0587	0.0026	479	17	464.7	6	534	93	**464.7**	**6.0**	2.99	184	171
0.5630	0.0240	0.0748	0.0013	0.08	0.0548	0.0024	457	15	464.9	8	377	90	**464.9**	**8.0**	1.73	110	85
0.5540	0.0430	0.0750	0.0019	0.31	0.0527	0.0037	448	27	466	11	310	140	**466.0**	**11.0**	4.02	65	18
0.5700	0.0230	0.0750	0.0010	0.09	0.0551	0.0023	455	15	466.8	6.1	422	90	**466.8**	**6.1**	2.59	484	371
0.5640	0.0450	0.0758	0.0017	0.19	0.0526	0.0039	453	30	471	10	360	150	**471.0**	**10.0**	3.97	288	377

0.5990	0.0310	0.0768	0.0011	0.28	0.0558	0.0027	478	19	476.9	6.8	470	100	**476.9**	**6.8**	0.23	313	167
0.5900	0.0290	0.0771	0.0022	0.66	0.0568	0.0022	471	19	481	13	469	82	**481.0**	**13.0**	2.12	615	225
0.6000	0.0310	0.0778	0.0014	-0.07	0.0549	0.0031	472	20	483.1	8.3	360	110	**483.1**	**8.3**	2.35	108	64
0.6440	0.0800	0.0781	0.0027	0.05	0.0600	0.0077	485	53	484	16	420	250	**484.0**	**16.0**	0.21	20	18
0.6120	0.0260	0.0792	0.0018	0.06	0.0569	0.0027	487	17	491	10	483	98	**491.0**	**10.0**	0.82	282	710
0.5580	0.0830	0.0813	0.0029	0.20	0.0489	0.0072	451	55	503	17	210	280	**503.0**	**17.0**	11.53	34	24
0.6320	0.0460	0.0814	0.0015	0.28	0.0562	0.0042	496	28	504.1	9.2	450	150	**504.1**	**9.2**	1.63	55	13
0.6780	0.0440	0.0831	0.0019	0.04	0.0596	0.0044	521	27	514	11	530	150	**514.0**	**11.0**	1.34	55	37
0.6590	0.0360	0.0830	0.0013	0.07	0.0570	0.0031	513	22	515	7.7	470	110	**515.0**	**7.7**	0.39	91	39
0.6620	0.0490	0.0833	0.0021	0.05	0.0579	0.0044	520	28	516	12	510	140	**516.0**	**12.0**	0.77	41	28
0.7130	0.0350	0.0843	0.0016	0.07	0.0597	0.0031	547	20	521.6	9.4	610	100	**521.6**	**9.4**	4.64	133	53
0.6710	0.0180	0.0844	0.0010	0.27	0.0575	0.0014	523	11	522	5.9	507	57	**522.0**	**5.9**	0.19	809	69
0.6600	0.0280	0.0844	0.0012	0.14	0.0577	0.0024	518	17	522.5	7.2	482	90	**522.5**	**7.2**	0.87	459	222
0.6720	0.0260	0.0845	0.0013	0.01	0.0576	0.0024	526	16	522.9	7.5	493	89	**522.9**	**7.5**	0.59	152	8
0.7000	0.0640	0.0845	0.0021	0.04	0.0607	0.0056	539	40	523	13	560	190	**523.0**	**13.0**	2.97	163	204
0.6760	0.0270	0.0846	0.0015	0.02	0.0582	0.0025	521	17	523.6	9	507	92	**523.6**	**9.0**	0.50	355	282
0.7060	0.0390	0.0856	0.0019	0.23	0.0609	0.0031	540	24	530	11	620	120	**530.0**	**11.0**	1.85	266	247
0.6430	0.0310	0.0858	0.0017	0.19	0.0536	0.0026	504	19	530.7	9.8	340	100	**530.7**	**9.8**	5.30	91	77
0.6980	0.0610	0.0863	0.0022	0.13	0.0578	0.0050	532	38	533	13	490	180	**533.0**	**13.0**	0.19	44	17
0.7110	0.0290	0.0867	0.0011	0.14	0.0602	0.0024	543	17	536.2	6.6	594	82	**536.2**	**6.6**	1.25	160	114
0.7250	0.0570	0.0870	0.0025	0.25	0.0604	0.0044	561	34	538	15	600	150	**538.0**	**15.0**	4.10	353	19
0.8010	0.0340	0.0872	0.0015	0.14	0.0631	0.0028	599	19	539.1	9.2	662	94	**539.1**	**9.2**	10.00	104	93
0.7230	0.0340	0.0875	0.0014	0.11	0.0596	0.0029	554	19	541.2	8.3	590	100	**541.2**	**8.3**	2.31	92	55
0.7290	0.0390	0.0876	0.0015	0.08	0.0591	0.0032	554	24	541.4	9.1	520	110	**541.4**	**9.1**	2.27	91	50
0.7090	0.0240	0.0877	0.0012	0.28	0.0588	0.0019	546	14	541.6	6.9	554	66	**541.6**	**6.9**	0.81	217	32
0.7270	0.0190	0.0884	0.0009	0.26	0.0598	0.0015	554	11	546.2	5.4	594	56	**546.2**	**5.4**	1.41	1376	271
0.7360	0.0350	0.0884	0.0014	0.07	0.0579	0.0028	555	20	546.8	8.3	520	100	**546.8**	**8.3**	1.48	81	62
0.7120	0.0440	0.0894	0.0021	0.16	0.0571	0.0036	550	26	551	12	500	130	**551.0**	**12.0**	0.18	53	30
0.7550	0.0290	0.0899	0.0013	-0.05	0.0607	0.0025	568	17	554.7	7.4	597	88	**554.7**	**7.4**	2.34	432	363
0.7700	0.0530	0.0903	0.0023	0.09	0.0613	0.0045	571	30	557	14	580	150	**557.0**	**14.0**	2.45	105	187
0.8100	0.0630	0.0903	0.0022	0.10	0.0647	0.0053	594	35	557	13	700	170	**557.0**	**13.0**	6.23	106	55
0.7130	0.0170	0.0903	0.0010	0.35	0.0571	0.0014	546	10	557.1	5.8	473	54	**557.1**	**5.8**	2.03	1807	246
0.7710	0.0370	0.0907	0.0013	0.05	0.0603	0.0029	578	21	559.8	7.8	590	100	**559.8**	**7.8**	3.15	89	40
0.7360	0.0250	0.0910	0.0011	0.03	0.0584	0.0021	562	15	561.5	6.5	528	79	**561.5**	**6.5**	0.09	140	70
0.7560	0.0230	0.0915	0.0012	0.19	0.0590	0.0018	570	14	565.2	7	565	70	**565.2**	**7.0**	0.84	250	84
0.7250	0.0260	0.0925	0.0012	0.25	0.0571	0.0020	552	15	570.4	7.3	473	74	**570.4**	**7.3**	3.33	526	554
0.7860	0.0940	0.0928	0.0031	0.34	0.0619	0.0066	604	51	572	18	650	210	**572.0**	**18.0**	5.30	56	38
0.7700	0.0470	0.0931	0.0024	0.30	0.0594	0.0035	581	27	573	14	540	120	**573.0**	**14.0**	1.38	49	13
0.8500	0.0910	0.0975	0.0027	0.08	0.0627	0.0067	612	49	599	16	570	200	**599.0**	**16.0**	2.12	26	13
0.7950	0.0350	0.0990	0.0017	0.25	0.0583	0.0026	597	20	608.5	9.9	518	95	**608.5**	**9.9**	1.93	384	247
0.8400	0.0350	0.0994	0.0017	0.13	0.0624	0.0027	621	20	610.7	9.8	636	91	**610.7**	**9.8**	1.66	217	74
0.8110	0.0320	0.0997	0.0018	0.37	0.0591	0.0022	605	18	612	10	542	81	**612.0**	**10.0**	1.16	159	145
0.8240	0.0270	0.1003	0.0015	0.09	0.0591	0.0020	609	15	615.9	8.5	568	75	**615.9**	**8.5**	1.13	150	79
0.8670	0.0320	0.1007	0.0015	0.25	0.0627	0.0025	631	18	618.1	9	671	83	**618.1**	**9.0**	2.04	162	14
0.8260	0.0250	0.1009	0.0013	0.20	0.0583	0.0018	610	14	619.8	7.5	516	69	**619.8**	**7.5**	1.61	218	86
0.9150	0.0600	0.1009	0.0019	0.18	0.0655	0.0042	657	32	621	11	700	140	**621.0**	**11.0**	5.48	55	35
0.8540	0.0630	0.1034	0.0028	0.23	0.0601	0.0046	617	35	634	16	570	160	**634.0**	**16.0**	2.76	41	31
0.8990	0.0240	0.1034	0.0014	0.20	0.0631	0.0018	652	12	635	8.4	709	58	**635.0**	**8.4**	2.61	308	168
0.9170	0.0480	0.1050	0.0022	0.05	0.0642	0.0036	664	26	643	13	710	120	**643.0**	**13.0**	3.16	77	38
0.9490	0.0340	0.1096	0.0017	0.59	0.0630	0.0018	673	17	671	10	695	61	**671.0**	**10.0**	0.30	289	37
1.5900	0.1500	0.1781	0.0046	0.16	0.0644	0.0059	955	58	1056	25	680	190	**680.0**	**190.0**	10.58	28	8
1.2330	0.0710	0.1238	0.0025	0.18	0.0705	0.0038	806	32	754	15	920	130	**754.0**	**15.0**	6.45	74	32
1.6040	0.0580	0.1659	0.0026	-0.01	0.0697	0.0027	970	22	989	14	899	81	**899.0**	**81.0**	1.96	72	35
1.9160	0.0740	0.1865	0.0037	-0.05	0.0740	0.0032	1083	27	1101	20	1014	91	**1014.0**	**91.0**	1.66	47	16
1.8600	0.0460	0.1826	0.0029	0.11	0.0733	0.0020	1065	16	1081	16	1022	52	**1022.0**	**52.0**	1.50	553	218
1.8380	0.0480	0.1786	0.0026	0.54	0.0733	0.0016	1063	17	1059	14	1032	43	**1032.0**	**43.0**	0.38	161	61
1.6760	0.0430	0.1670	0.0024	0.42	0.0735	0.0018	1002	17	995	13	1034	50	**1034.0**	**50.0**	0.70	486	363
1.7570	0.0520	0.1693	0.0024	0.24	0.0750	0.0023	1025	19	1008	13	1058	63	**1058.0**	**63.0**	1.66	143	154
2.0560	0.0800	0.1967	0.0033	0.32	0.0753	0.0029	1135	27	1157	18	1065	72	**1065.0**	**72.0**	1.94	119	44
1.7770	0.0470	0.1725	0.0026	0.35	0.0756	0.0019	1035	17	1025	14	1066	52	**1066.0**	**52.0**	0.97	608	226
1.8290	0.0860	0.1746	0.0034	0.33	0.0760	0.0035	1051	32	1037	19	1075	89	**1075.0**	**89.0**	1.33	43	30
1.9070	0.0650	0.1821	0.0024	0.26	0.0752	0.0023	1081	22	1078	13	1080	62	**1080.0**	**62.0**	0.28	262	96
1.8630	0.0370	0.1772	0.0019	0.38	0.0754	0.0014	1069	13	1052	10	1085	38	**1085.0**	**38.0**	1.59	570	335
1.8480	0.0900	0.1757	0.0030	0.32	0.0760	0.0033	1058	31	1043	16	1094	98	**1094.0**	**98.0**	1.42	190	114
1.8590	0.0700	0.1730	0.0032	0.20	0.0780	0.0030	1067	25	1028	18	1102	79	**1102.0**	**79.0**	3.66	50	29
1.8980	0.0640	0.1800	0.0037	0.43	0.0769	0.0025	1080	21	1066	20	1104	61	**1104.0**	**61.0**	1.30	161	40
1.8670	0.0550	0.1779	0.0025	0.39	0.0767	0.0020	1072	20	1057	13	1113	54	**1113.0**	**54.0**	1.40	236	65
1.8090	0.0730	0.1741	0.0030	0.26	0.0750	0.0029	1053	26	1034	16	1113	74	**1113.0**	**74.0**	1.80	65	42
1.8290	0.0670	0.1689	0.0027	0.38	0.0769	0.0025	1056	24	1007	15	1123	71	**1123.0**	**71.0**	4.64	88	65
1.9780	0.0550	0.1821	0.0027	0.48	0.0774	0.0021	1106	19	1078	15	1124	57	**1124.0**	**57.0**	2.53	138	24
2.1420	0.0600	0.1999	0.0030	0.25	0.0783	0.0022	1162	19	1174	16	1128	56	**1128.0**	**56.0**	1.03	228	118
2.1070	0.0480	0.1941	0.0025	0.32	0.0785	0.0017	1151	15	1143	13	1147	43	**1147.0**	**43.0**	0.70	505	329
2.0210	0.0840	0.1882	0.0032	-0.07	0.0786	0.0035	1119	29	1111	17	1149	91	**1149.0**	**91.0**	0.71	79	42
1.8990	0.0640	0.1768	0.0029	0.18	0.0784	0.0026	1082	22	1049	16	1164	67	**1164.0**	**67.0**	3.05	569	850
1.9900	0.1100	0.1772	0.0044	0.20	0.0805	0.0048	1099	40	1051	24	1170	130	**1170.0**	**130.0**	4.37	25	12

2.0410	0.0510	0.1880	0.0028	-0.03	0.0795	0.0022	1130	18	1111	15	1186	58	**1186.0**	**58.0**	1.68	298	81
1.9410	0.0610	0.1784	0.0029	0.14	0.0799	0.0027	1090	21	1058	16	1195	65	**1195.0**	**65.0**	2.94	245	149
1.9200	0.1100	0.1683	0.0053	0.02	0.0791	0.0052	1087	40	1005	30	1200	140	**1200.0**	**140.0**	7.54	27	7
2.3480	0.0580	0.2068	0.0035	0.42	0.0821	0.0020	1224	18	1211	19	1241	50	**1241.0**	**50.0**	1.06	647	609
2.8190	0.0720	0.2339	0.0037	0.29	0.0881	0.0022	1362	18	1354	19	1370	47	**1370.0**	**47.0**	0.59	335	381
2.8800	0.1900	0.2302	0.0078	0.02	0.0885	0.0065	1374	48	1339	40	1370	130	**1370.0**	**130.0**	2.55	20	16
3.0590	0.0600	0.2503	0.0033	0.39	0.0886	0.0017	1423	15	1441	17	1392	37	**1392.0**	**37.0**	1.26	438	215
4.0500	0.1100	0.2950	0.0047	0.46	0.1004	0.0022	1647	22	1666	23	1631	40	**1631.0**	**40.0**	1.15	99	38
4.0400	0.1000	0.2892	0.0039	0.39	0.1018	0.0025	1641	21	1637	19	1652	45	**1652.0**	**45.0**	0.24	179	68
4.1610	0.0890	0.2658	0.0041	0.58	0.1108	0.0019	1664	18	1519	21	1807	31	**1807.0**	**31.0**	8.71	129	327
4.9100	0.1100	0.3103	0.0041	0.38	0.1141	0.0026	1801	19	1744	20	1852	42	**1852.0**	**42.0**	3.16	80	66
11.3800	0.1800	0.4680	0.0061	0.61	0.1753	0.0023	2553	15	2474	27	2607	22	**2607.0**	**22.0**	3.09	555	497
11.9400	0.3700	0.4680	0.0130	0.55	0.1829	0.0052	2596	29	2473	56	2682	45	**2682.0**	**45.0**	4.74	108	43
10.0600	0.1300	0.3915	0.0043	0.47	0.1851	0.0021	2439	12	2131	20	2698	19	**2698.0**	**19.0**	12.63	154	70
13.1800	0.2000	0.5016	0.0062	0.57	0.1883	0.0026	2690	14	2622	27	2731	23	**2731.0**	**23.0**	2.53	191	187
13.4200	0.1900	0.4704	0.0052	0.59	0.2068	0.0025	2711	13	2484	23	2881	19	**2881.0**	**19.0**	8.37	167	83
26.7200	0.6400	0.6370	0.0130	0.76	0.3015	0.0043	3373	24	3177	50	3478	22	**3478.0**	**22.0**	5.81	132	127

HU-14 **Lat:** **-30.21895** **Lon:** **-68.34872**

Isotope Ratios							**Apparent Ages**								**Concentrations**		
$^{207}Pb/^{235}U$	**± (2σ)**	**$^{206}Pb/^{238}U$**	**± (2σ)**	**Error Corr.**	**$^{207}Pb/^{206}Pb$**	**± (2σ)**	**$^{207}Pb/^{235}U$ (Ma)**	**± (Ma)**	**$^{206}Pb/^{238}U$ (Ma)**	**± (Ma)**	**$^{207}Pb/^{206}Pb$ (Ma)**	**± (Ma)**	**Plot Age (Ma)**	**± (Ma)**	**conc. (%)**	**U (ppm)**	**Th (ppm)**
0.0042	0.0080	0.0010	0.0001	-0.11	0.1620	0.0970	3.8	8.1	6.13	0.85	-	-	**5.2**	**0.9**	61.32	129	75
0.0073	0.0058	0.0010	0.0001	0.04	0.0490	0.0690	6.1	5.6	6.25	0.63	-	-	**6.2**	**0.6**	2.46	542	404
0.0060	0.0088	0.0010	0.0002	0.08	0.0900	0.1600	5.3	8.8	6.6	1.1	-	-	**6.2**	**1.1**	24.53	143	92
0.0051	0.0033	0.0011	0.0001	0.00	0.0340	0.0250	5	3.3	6.85	0.48	-	-	**7.0**	**0.5**	37.00	607	299
0.0128	0.0037	0.0011	0.0001	0.23	0.0840	0.0250	12.7	3.7	7.09	0.44	-	-	**6.8**	**0.4**	44.17	1078	679
0.0084	0.0022	0.0011	0.0001	0.19	0.0530	0.0140	8.5	2.2	7.12	0.33	-	-	**7.1**	**0.3**	16.24	1228	781
0.0069	0.0028	0.0012	0.0001	-0.12	0.0580	0.0210	6.9	2.9	7.53	0.47	-	-	**7.4**	**0.5**	9.13	689	330
-0.0300	0.0320	0.0013	0.0004	-0.13	0.0400	0.2200	-36	33	8.3	2.8	-	-	**8.4**	**2.8**	123.06	55	51
0.0520	0.0260	0.0014	0.0003	-0.10	0.1400	0.3800	42	23	8.9	1.7	-	-	**7.8**	**1.7**	78.81	60	37
0.0114	0.0024	0.0015	0.0001	0.08	0.0540	0.0110	11.5	2.4	9.85	0.53	-	-	**9.8**	**0.5**	14.35	1580	2070
0.0260	0.0270	0.0017	0.0004	0.15	-0.0500	0.2500	20	27	10.9	2.8	-	-	**12.2**	**2.8**	45.50	86	54
0.0110	0.0140	0.0018	0.0002	0.01	0.0500	0.1300	8	14	11.5	1.3	-	-	**11.4**	**1.3**	43.75	99	64
0.0113	0.0040	0.0018	0.0001	0.03	0.0500	0.0190	11.2	4	11.7	0.77	-	-	**11.7**	**0.8**	4.46	544	464
0.0170	0.0250	0.0019	0.0004	-0.08	-0.3300	0.4900	13	24	12.3	2.8	-	-	**18.1**	**2.8**	5.38	147	106
0.0037	0.0056	0.0020	0.0002	0.10	0.0120	0.0260	3.3	5.6	12.9	1	-	-	**13.4**	**1.0**	290.91	162	130
0.0128	0.0040	0.0021	0.0001	-0.16	0.0510	0.0150	12.8	4	13.51	0.79	-	-	**13.4**	**0.8**	5.55	330	571
0.0125	0.0047	0.0022	0.0001	0.01	0.0440	0.0160	12.4	4.7	14.25	0.78	-	-	**14.3**	**0.8**	14.92	322	163
0.0181	0.0063	0.0023	0.0001	0.21	0.0640	0.0200	17.8	6.3	15.01	0.89	-	-	**14.7**	**0.9**	15.67	263	137
0.0130	0.0150	0.0026	0.0003	-0.07	0.0400	0.1200	10	15	16.7	1.9	-	-	**16.8**	**1.9**	67.00	174	156
0.0190	0.0570	0.0031	0.0007	0.06	-0.1100	0.2200	19	62	19.8	4.7	-	-	**23.7**	**4.7**	4.21	30	15
0.0140	0.0250	0.0032	0.0004	-0.12	0.0680	0.0850	18	27	20.3	2.6	-	-	**19.7**	**2.6**	12.78	103	78
0.0154	0.0080	0.0032	0.0002	0.03	0.0380	0.0200	14.6	8	20.6	1.2	-	-	**20.8**	**1.2**	41.10	273	152
0.0180	0.0150	0.0033	0.0003	-0.04	0.0760	0.0590	15	15	21	2	-	-	**20.2**	**2.0**	40.00	140	169
0.0193	0.0039	0.0034	0.0001	0.20	0.0462	0.0092	19.3	3.9	21.59	0.74	-	-	**21.6**	**0.7**	11.87	854	366
0.0170	0.0100	0.0034	0.0002	0.02	0.0420	0.0250	16	10	21.6	1.4	-	-	**21.7**	**1.4**	35.00	256	285
0.0270	0.0120	0.0034	0.0003	-0.02	0.0760	0.0320	25	11	22.1	1.6	-	-	**21.3**	**1.6**	11.60	191	225
0.0250	0.0220	0.0036	0.0004	0.02	0.0690	0.0520	23	21	22.8	2.6	-	-	**22.2**	**2.6**	0.87	116	135
0.0170	0.0250	0.0037	0.0006	0.07	0.0280	0.0720	14	25	24	3.6	-	-	**24.6**	**3.6**	71.43	62	30
0.0254	0.0064	0.0038	0.0002	-0.01	0.0510	0.0130	24.9	6.3	24.4	1	-	-	**24.2**	**1.0**	2.01	290	164
0.0750	0.0310	0.0038	0.0005	0.08	0.1260	0.0850	68	30	24.6	3.1	-	-	**22.2**	**3.1**	63.82	55	82
0.1120	0.0430	0.0040	0.0005	0.54	0.2070	0.0760	85	35	25.5	3	-	-	**20.4**	**3.0**	70.00	111	86
0.0250	0.0230	0.0041	0.0005	-0.05	0.1150	0.0760	25	24	26.1	3.2	-	-	**23.9**	**3.2**	4.40	97	42
0.0290	0.0220	0.0061	0.0005	-0.09	0.0700	0.0420	23	22	39	3	-	-	**37.9**	**3.0**	69.57	102	78
0.0381	0.0061	0.0061	0.0003	-0.25	0.0494	0.0083	39.7	6.4	39.3	1.8	-	-	**39.1**	**1.8**	1.01	833	492
0.0720	0.0110	0.0098	0.0004	0.00	0.0567	0.0083	69	10	62.9	2.4	-	-	**62.9**	**2.4**	8.84	276	315
0.0713	0.0098	0.0109	0.0003	-0.06	0.0446	0.0064	70.5	9.4	69.9	2.2	-	-	**69.9**	**2.2**	0.85	208	160
0.0760	0.0180	0.0115	0.0005	0.04	0.0500	0.0110	80	16	73.4	3	-	-	**73.4**	**3.0**	8.25	155	98
0.2790	0.0580	0.0360	0.0017	0.19	0.0494	0.0099	236	45	228	11	210	330	**228.0**	**11.0**	3.39	86	38
0.2660	0.0190	0.0369	0.0007	0.03	0.0521	0.0038	239	15	233.2	4.6	260	140	**233.2**	**4.6**	2.43	294	216
0.2670	0.0470	0.0380	0.0018	-0.28	0.0540	0.0100	243	38	240	11	200	330	**240.0**	**11.0**	1.23	140	173
0.3040	0.0220	0.0386	0.0009	0.20	0.0571	0.0041	269	17	244.4	5.2	440	140	**244.4**	**5.2**	9.14	283	130
0.2820	0.0140	0.0399	0.0007	0.18	0.0514	0.0025	253	11	252.3	4.6	241	98	**252.3**	**4.6**	0.28	493	95
0.2660	0.0590	0.0402	0.0023	0.04	0.0500	0.0110	235	47	254	15	110	360	**254.0**	**15.0**	8.09	87	156
0.2860	0.0300	0.0402	0.0011	-0.11	0.0509	0.0054	252	24	254.3	6.9	210	200	**254.3**	**6.9**	0.91	232	193
0.2790	0.0110	0.0409	0.0006	0.04	0.0486	0.0020	250.8	8.9	258.1	3.5	124	81	**258.1**	**3.5**	2.91	266	236
0.3080	0.0120	0.0419	0.0008	0.35	0.0529	0.0019	272.3	9.3	264.7	5.1	307	77	**264.7**	**5.1**	2.79	879	139
0.3080	0.0160	0.0420	0.0008	0.19	0.0525	0.0026	271	12	265.2	5	310	100	**265.2**	**5.0**	2.14	526	353
0.3510	0.0770	0.0423	0.0022	-0.09	0.0600	0.0160	275	60	266	14	40	390	**266.0**	**14.0**	3.27	41	39
0.2790	0.0280	0.0425	0.0012	0.01	0.0469	0.0049	253	21	267.9	7.5	80	180	**267.9**	**7.5**	5.89	165	141
0.3000	0.0130	0.0425	0.0008	0.38	0.0502	0.0019	266	10	268.6	4.8	198	80	**268.6**	**4.8**	0.98	1032	571
0.3020	0.0180	0.0427	0.0007	-0.10	0.0501	0.0033	269	14	269.3	4.3	240	120	**269.3**	**4.3**	0.11	244	154
0.3060	0.0290	0.0433	0.0010	0.13	0.0512	0.0049	269	22	273.2	6	190	170	**273.2**	**6.0**	1.56	130	136

0.3100	0.0170	0.0433	0.0009	0.10	0.0513	0.0029	273	13	273.3	5.4	240	110	**273.3**	**5.4**	0.11	728	510
0.3010	0.0200	0.0443	0.0009	0.09	0.0497	0.0031	266	15	279.5	5.3	150	120	**279.5**	**5.3**	5.08	297	102
0.3180	0.0140	0.0444	0.0006	0.18	0.0513	0.0023	280	11	280.7	3.7	255	92	**280.7**	**3.7**	0.25	313	273
0.2890	0.0280	0.0453	0.0012	0.06	0.0466	0.0047	267	20	285.4	7.5	110	170	**285.4**	**7.5**	6.89	105	88
0.3500	0.0480	0.0453	0.0016	0.01	0.0560	0.0081	297	35	285.4	9.8	360	260	**285.4**	**9.8**	3.91	122	114
0.3410	0.0220	0.0458	0.0009	0.12	0.0546	0.0036	295	17	288.4	5.6	370	130	**288.4**	**5.6**	2.24	252	149
0.3410	0.0280	0.0465	0.0009	0.08	0.0534	0.0045	295	21	292.6	5.8	290	160	**292.6**	**5.8**	0.81	89	48
0.3200	0.0270	0.0481	0.0011	0.03	0.0494	0.0042	279	20	302.9	7	140	160	**302.9**	**7.0**	8.57	195	114
0.3810	0.0560	0.0487	0.0014	-0.19	0.0550	0.0084	333	40	306.7	8.9	310	280	**306.7**	**8.9**	7.90	116	55
0.3660	0.0300	0.0490	0.0011	0.02	0.0543	0.0046	311	23	308.3	6.8	290	160	**308.3**	**6.8**	0.87	105	74
0.3760	0.0160	0.0512	0.0008	0.11	0.0527	0.0024	323	12	321.8	5.1	296	94	**321.8**	**5.1**	0.37	567	143
0.3760	0.0120	0.0513	0.0008	0.08	0.0534	0.0018	324.3	9.1	322.6	4.9	337	69	**322.6**	**4.9**	0.52	492	102
0.3740	0.0200	0.0526	0.0009	0.00	0.0505	0.0028	320	15	330.1	5.3	220	110	**330.1**	**5.3**	3.16	337	52
0.4400	0.0240	0.0529	0.0007	0.17	0.0603	0.0032	369	17	332.4	4.3	580	110	**332.4**	**4.3**	9.92	324	201
0.3940	0.0530	0.0535	0.0019	0.20	0.0550	0.0073	346	43	336	11	370	260	**336.0**	**11.0**	2.89	84	132
0.4100	0.0140	0.0554	0.0008	0.34	0.0536	0.0017	347	10	347.5	4.8	343	70	**347.5**	**4.8**	0.14	877	412
0.4460	0.0250	0.0623	0.0012	0.07	0.0514	0.0030	373	18	389.6	7.5	240	110	**389.6**	**7.5**	4.45	279	133
0.5680	0.0290	0.0746	0.0012	0.11	0.0563	0.0029	461	19	463.9	6.9	420	110	**463.9**	**6.9**	0.63	295	194
0.5770	0.0300	0.0753	0.0013	0.05	0.0577	0.0033	458	20	467.6	7.6	440	120	**467.6**	**7.6**	2.10	232	164
0.5850	0.0130	0.0755	0.0009	0.40	0.0561	0.0012	467.6	8.5	469	5.6	449	46	**469.0**	**5.6**	0.30	1535	81
0.6250	0.0160	0.0769	0.0011	0.30	0.0588	0.0013	494	9.8	477.5	6.4	550	51	**477.5**	**6.4**	3.34	472	197
0.6510	0.0740	0.0811	0.0029	0.08	0.0594	0.0073	514	46	502	17	490	250	**502.0**	**17.0**	2.33	49	21
0.6800	0.0200	0.0840	0.0010	0.18	0.0583	0.0017	526	12	520.1	5.8	536	62	**520.1**	**5.8**	1.12	258	162
0.6520	0.0320	0.0863	0.0014	0.05	0.0539	0.0026	515	19	534.5	8.2	370	110	**534.5**	**8.2**	3.79	113	80
0.7540	0.0830	0.0877	0.0026	-0.06	0.0634	0.0074	564	47	542	15	600	230	**542.0**	**15.0**	3.90	58	33
0.6590	0.0560	0.0881	0.0025	0.14	0.0557	0.0050	512	35	545	15	330	170	**545.0**	**15.0**	6.45	152	47
0.7420	0.0280	0.0886	0.0013	0.08	0.0607	0.0024	562	16	547.2	7.8	575	86	**547.2**	**7.8**	2.63	296	133
0.7210	0.0150	0.0897	0.0010	0.25	0.0588	0.0012	551.4	9.1	553.7	5.7	561	43	**553.7**	**5.7**	0.42	1256	512
0.7270	0.0220	0.0909	0.0011	0.17	0.0576	0.0018	557	13	560.9	6.4	516	69	**560.9**	**6.4**	0.70	316	126
0.7590	0.0230	0.0912	0.0012	0.12	0.0602	0.0018	574	13	562.3	7	604	68	**562.3**	**7.0**	2.04	198	96
0.7190	0.0400	0.0922	0.0017	0.01	0.0564	0.0032	549	24	568	10	410	120	**568.0**	**10.0**	3.46	73	29
0.8550	0.0440	0.0929	0.0017	0.06	0.0642	0.0034	627	24	574	10	720	120	**574.0**	**10.0**	8.45	94	101
0.7130	0.0490	0.0936	0.0017	0.02	0.0559	0.0040	541	30	576	10	350	140	**576.0**	**10.0**	6.47	54	181
0.7740	0.0230	0.0944	0.0012	-0.09	0.0595	0.0020	584	13	581.4	7.1	568	73	**581.4**	**7.1**	0.45	256	297
0.7820	0.0200	0.0967	0.0015	0.29	0.0583	0.0015	588	12	594.8	8.8	534	57	**594.8**	**8.8**	1.16	1254	58
0.7860	0.0320	0.0976	0.0017	0.01	0.0578	0.0027	595	18	600	10	487	99	**600.0**	**10.0**	0.84	352	164
0.8210	0.0420	0.0979	0.0018	0.27	0.0597	0.0030	607	23	602	10	590	100	**602.0**	**10.0**	0.82	196	46
0.8170	0.0170	0.0999	0.0011	0.29	0.0585	0.0012	606.3	9.4	613.5	6.2	556	45	**613.5**	**6.2**	1.19	522	78
0.9940	0.0320	0.1086	0.0016	0.30	0.0660	0.0019	701	16	665.5	9.3	786	64	**665.5**	**9.3**	5.06	332	243
1.2860	0.0570	0.1389	0.0026	0.28	0.0655	0.0028	838	25	838	15	815	93	**815.0**	**93.0**	0.00	151	108
1.3560	0.0380	0.1382	0.0019	0.37	0.0692	0.0018	866	16	835	10	896	54	**896.0**	**54.0**	3.58	447	385
1.4330	0.0300	0.1459	0.0019	0.33	0.0700	0.0015	904	12	878	11	928	45	**928.0**	**45.0**	2.88	213	69
1.8100	0.1000	0.1777	0.0034	0.12	0.0739	0.0043	1041	38	1054	19	980	130	**980.0**	**130.0**	1.25	73	32
1.7140	0.0330	0.1704	0.0018	0.22	0.0722	0.0015	1013	13	1014	10	982	42	**982.0**	**42.0**	0.10	655	191
1.7910	0.0550	0.1729	0.0037	0.45	0.0744	0.0024	1043	19	1028	21	1043	66	**1043.0**	**66.0**	1.44	252	51
1.5760	0.0670	0.1508	0.0030	0.24	0.0741	0.0030	963	26	905	17	1044	89	**1044.0**	**89.0**	6.02	101	70
1.8450	0.0510	0.1763	0.0022	0.06	0.0745	0.0023	1059	18	1047	12	1048	61	**1048.0**	**61.0**	1.13	179	138
1.7530	0.0420	0.1673	0.0020	0.23	0.0752	0.0017	1031	16	997	11	1065	46	**1065.0**	**46.0**	3.30	221	71
1.8600	0.1400	0.1714	0.0043	0.19	0.0783	0.0061	1053	52	1021	23	1070	170	**1070.0**	**170.0**	3.04	30	9
1.8290	0.0730	0.1700	0.0029	0.33	0.0757	0.0031	1059	28	1012	16	1077	81	**1077.0**	**81.0**	4.44	219	61
1.8800	0.0880	0.1779	0.0033	0.25	0.0754	0.0034	1071	32	1057	19	1114	88	**1114.0**	**88.0**	1.31	119	57
1.7900	0.0560	0.1700	0.0029	0.31	0.0762	0.0022	1042	20	1011	16	1115	57	**1115.0**	**57.0**	2.98	290	187
1.9250	0.0320	0.1791	0.0019	0.37	0.0768	0.0012	1091	11	1062	10	1122	32	**1122.0**	**32.0**	2.66	327	84
1.8520	0.0690	0.1730	0.0028	0.26	0.0766	0.0029	1064	24	1028	15	1122	74	**1122.0**	**74.0**	3.38	145	87
1.8510	0.0450	0.1719	0.0019	0.12	0.0777	0.0020	1062	16	1022	11	1131	50	**1131.0**	**50.0**	3.77	219	78
1.9190	0.0630	0.1762	0.0030	0.18	0.0780	0.0026	1089	22	1046	17	1137	72	**1137.0**	**72.0**	3.95	187	83
1.9320	0.0840	0.1783	0.0031	0.34	0.0779	0.0032	1100	30	1057	17	1151	81	**1151.0**	**81.0**	3.91	113	38
2.0050	0.0640	0.1797	0.0031	0.14	0.0790	0.0028	1118	22	1067	16	1161	70	**1161.0**	**70.0**	4.56	184	59
2.1410	0.0740	0.1925	0.0030	0.26	0.0795	0.0027	1157	24	1136	16	1182	67	**1182.0**	**67.0**	1.82	146	71
2.1760	0.0480	0.1974	0.0020	0.30	0.0792	0.0018	1170	15	1161	11	1184	42	**1184.0**	**42.0**	0.77	140	55
2.3350	0.0450	0.2079	0.0024	0.31	0.0810	0.0016	1222	14	1219	13	1213	39	**1213.0**	**39.0**	0.25	174	156
2.5280	0.0880	0.2172	0.0042	0.34	0.0832	0.0028	1277	25	1266	22	1262	69	**1262.0**	**69.0**	0.86	51	12
2.1410	0.0460	0.1767	0.0035	0.32	0.0877	0.0023	1162	15	1051	20	1380	48	**1380.0**	**48.0**	9.55	326	185
2.8750	0.0530	0.2335	0.0031	0.40	0.0888	0.0015	1373	14	1352	16	1398	32	**1398.0**	**32.0**	1.53	583	272
3.2500	0.1300	0.2354	0.0033	0.39	0.0995	0.0037	1459	30	1365	18	1608	68	**1608.0**	**68.0**	6.44	182	90
3.9900	0.1600	0.2760	0.0070	0.65	0.1028	0.0029	1625	34	1570	36	1675	50	**1675.0**	**50.0**	3.38	205	89
5.2300	0.1700	0.3444	0.0067	0.44	0.1106	0.0032	1853	28	1916	31	1790	53	**1790.0**	**53.0**	3.40	146	86
6.2400	0.1700	0.3555	0.0062	0.43	0.1283	0.0031	2013	23	1959	30	2072	44	**2072.0**	**44.0**	2.68	119	73
13.3100	0.2500	0.5232	0.0080	0.56	0.1858	0.0028	2702	17	2710	34	2708	24	**2708.0**	**24.0**	0.30	159	39
13.3900	0.2400	0.5168	0.0088	0.64	0.1878	0.0029	2706	18	2683	37	2725	25	**2725.0**	**25.0**	0.85	164	89
13.1000	0.1400	0.4667	0.0051	0.57	0.2031	0.0021	2685.8	9.9	2468	23	2850	16	**2850.0**	**16.0**	8.11	186	164

HU-15 **Lat:** **-28.68006** **Lon:** **-68.30332**

Isotope Ratios **Apparent Ages** **Concentrations**

$^{207}Pb/^{235}U$	± (2σ)	$^{206}Pb/^{238}U$	± (2σ)	Error Corr.	$^{207}Pb/^{206}Pb$	± (2σ)	$^{207}Pb/^{235}U$ (Ma)	± (Ma)	$^{206}Pb/^{238}U$ (Ma)	± (Ma)	$^{207}Pb/^{206}Pb$ (Ma)	± (Ma)	Plot Age (Ma)	± (Ma)	conc. (%)	U (ppm)	Th (ppm)
0.0070	0.0010	0.0000	0.0000	0.04	0.0440	0.0100	7.1	3.7	6.7	0.4	-	-	**6.7**	**0.5**	0.94	96	100
0.0070	0.0010	0.0000	0.0000	0.19	0.0430	0.0060	6.9	1.4	7.2	0.2	-	-	**7.3**	**0.3**	1.05	197	147
0.0100	0.0010	0.0000	0.0000	-0.07	0.0490	0.0060	6.9	1.0	7.4	0.1	-	-	**7.4**	**0.2**	1.07	459	167
0.0120	0.0010	0.0000	0.0000	-0.07	0.0490	0.0020	9.6	1.2	9.1	0.3	-	-	**9.1**	**0.3**	0.95	211	56
0.0110	0.0020	0.0000	0.0000	-0.22	0.0460	0.0080	12.2	0.5	11.7	0.2	-	-	**11.6**	**0.3**	0.95	988	722
0.0130	0.0010	0.0000	0.0000	-0.16	0.0490	0.0050	11.5	1.8	11.7	0.3	-	-	**11.7**	**0.4**	1.01	172	126
0.0160	0.0020	0.0000	0.0000	0.29	0.0590	0.0070	12.7	1.2	12.0	0.4	-	-	**12.0**	**0.4**	0.95	330	185
0.0120	0.0010	0.0000	0.0000	0.38	0.0450	0.0040	15.6	1.6	12.5	0.2	-	-	**12.2**	**0.3**	0.80	472	660
0.0120	0.0090	0.0000	0.0000	-0.33	0.0390	0.0340	12.1	1.0	12.3	0.2	-	-	**12.3**	**0.4**	1.02	510	259
0.0130	0.0100	0.0000	0.0000	0.14	0.0590	0.0410	11.4	9.1	12.3	0.9	-	-	**12.4**	**1.1**	1.06	41	26
0.0140	0.0010	0.0000	0.0000	0.34	0.0500	0.0050	16.0	11.0	12.9	1.2	-	-	**12.7**	**1.4**	0.95	26	13
0.0130	0.0020	0.0000	0.0000	0.02	0.0470	0.0060	13.8	1.3	13.2	0.2	-	-	**13.1**	**0.4**	0.95	434	262
0.0140	0.0010	0.0000	0.0000	0.19	0.0480	0.0020	13.3	1.6	13.3	0.4	-	-	**13.3**	**0.5**	1.00	221	259
0.0140	0.0010	0.0000	0.0000	0.29	0.0470	0.0040	14.0	0.7	13.6	0.2	-	-	**13.5**	**0.3**	0.97	459	708
0.0140	0.0010	0.0000	0.0000	0.38	0.0470	0.0020	13.9	1.3	13.6	0.4	-	-	**13.6**	**0.5**	0.99	368	748
0.0150	0.0010	0.0000	0.0000	0.07	0.0480	0.0040	13.6	0.5	13.7	0.3	-	-	**13.7**	**0.4**	1.00	533	319
0.0150	0.0010	0.0000	0.0000	0.38	0.0480	0.0040	14.8	1.1	14.5	0.2	-	-	**14.5**	**0.4**	0.98	383	182
0.0180	0.0020	0.0000	0.0000	0.42	0.0510	0.0040	15.5	1.3	15.0	0.3	-	-	**15.0**	**0.4**	0.97	340	201
0.0170	0.0020	0.0000	0.0000	0.35	0.0500	0.0070	17.9	1.5	16.5	0.4	-	-	**16.4**	**0.5**	0.92	345	213
0.0170	0.0010	0.0000	0.0000	0.38	0.0460	0.0020	17.2	2.0	16.6	0.4	-	-	**16.5**	**0.5**	0.97	188	151
0.0170	0.0000	0.0000	0.0000	0.45	0.0460	0.0010	16.9	0.8	16.9	0.3	-	-	**16.9**	**0.4**	1.00	831	266
0.0170	0.0030	0.0000	0.0000	0.02	0.0470	0.0100	17.3	0.3	17.1	0.2	-	-	**17.1**	**0.4**	0.99	2855	1063
0.0170	0.0010	0.0000	0.0000	0.24	0.0470	0.0020	17.0	3.4	17.2	0.5	-	-	**17.1**	**0.6**	1.00	144	96
0.0180	0.0020	0.0000	0.0000	0.08	0.0480	0.0050	17.4	0.8	17.2	0.4	-	-	**17.2**	**0.5**	0.99	672	1574
0.0170	0.0020	0.0000	0.0000	0.14	0.0450	0.0050	18.2	1.5	18.0	0.3	-	-	**17.9**	**0.5**	0.99	225	155
0.0160	0.0040	0.0000	0.0000	0.16	0.0400	0.0100	17.5	2.1	18.2	0.4	-	-	**18.2**	**0.6**	1.04	180	70
0.0200	0.0020	0.0000	0.0000	-0.17	0.0460	0.0050	16.1	4.2	18.7	0.8	-	-	**18.9**	**1.0**	1.16	64	39
0.0210	0.0020	0.0000	0.0000	0.25	0.0470	0.0040	19.6	1.7	19.3	0.4	-	-	**19.3**	**0.6**	0.98	196	159
0.0220	0.0010	0.0000	0.0000	0.35	0.0480	0.0020	21.0	1.4	20.5	0.4	-	-	**20.5**	**0.6**	0.98	418	222
0.0240	0.0010	0.0000	0.0000	-0.01	0.0520	0.0040	22.3	0.7	21.7	0.5	-	-	**21.7**	**0.6**	0.97	737	236
0.0690	0.0160	0.0000	0.0000	0.21	0.1310	0.0320	24.1	1.6	21.9	0.5	-	-	**21.8**	**0.7**	0.93	188	136
0.0240	0.0010	0.0000	0.0000	0.45	0.0510	0.0030	67.0	15.0	24.5	1.7	-	-	**21.8**	**1.9**	0.36	23	10
0.0230	0.0010	0.0000	0.0000	0.25	0.0490	0.0030	23.7	1.3	21.9	0.5	-	-	**21.8**	**0.6**	0.92	330	218
0.0340	0.0110	0.0000	0.0000	0.30	0.0690	0.0210	22.8	1.2	22.0	0.5	-	-	**21.9**	**0.6**	0.96	368	116
0.0230	0.0010	0.0000	0.0000	0.34	0.0470	0.0020	34.0	10.0	22.6	1.0	-	-	**21.9**	**1.2**	0.67	35	28
0.0240	0.0010	0.0000	0.0000	0.06	0.0490	0.0020	22.9	1.2	22.1	0.4	-	-	**22.1**	**0.6**	0.96	450	153
0.0240	0.0030	0.0000	0.0000	-0.18	0.0510	0.0060	23.8	0.6	22.3	0.4	-	-	**22.2**	**0.6**	0.94	764	367
0.0220	0.0010	0.0000	0.0000	0.11	0.0460	0.0030	24.4	2.7	22.5	0.4	-	-	**22.4**	**0.6**	0.92	103	78
0.0240	0.0020	0.0000	0.0000	0.35	0.0490	0.0040	22.2	1.2	22.6	0.2	-	-	**22.6**	**0.5**	1.02	462	412
0.0250	0.0020	0.0000	0.0000	0.46	0.0500	0.0040	23.9	1.8	22.8	0.4	-	-	**22.7**	**0.6**	0.95	320	240
0.0310	0.0090	0.0000	0.0000	0.30	0.0620	0.0180	24.6	2.1	22.8	0.5	-	-	**22.7**	**0.7**	0.93	163	137
0.0280	0.0080	0.0000	0.0000	0.10	0.0570	0.0160	30.9	9.0	23.2	1.0	-	-	**22.7**	**1.2**	0.74	38	26
0.0240	0.0020	0.0000	0.0000	0.05	0.0500	0.0030	27.8	7.7	23.1	0.7	-	-	**22.7**	**1.0**	0.82	53	25
0.0220	0.0010	0.0000	0.0000	0.11	0.0450	0.0030	24.4	1.5	22.8	0.4	-	-	**22.7**	**0.6**	0.94	311	475
0.0530	0.0040	0.0000	0.0000	-0.02	0.0970	0.0080	22.2	1.4	22.7	0.4	-	-	**22.7**	**0.6**	1.02	329	305
0.0260	0.0020	0.0000	0.0000	0.34	0.0500	0.0040	52.7	3.4	24.5	0.6	-	-	**22.9**	**0.8**	0.46	97	74
0.0240	0.0010	0.0000	0.0000	0.70	0.0470	0.0020	25.5	2.2	23.4	0.5	-	-	**23.3**	**0.7**	0.91	225	79
0.0270	0.0080	0.0000	0.0000	0.13	0.0510	0.0130	23.9	0.9	24.2	0.2	-	-	**24.2**	**0.5**	1.01	726	525
0.0240	0.0020	0.0000	0.0000	-0.09	0.0460	0.0040	26.9	7.5	24.6	1.0	-	-	**24.5**	**1.2**	0.91	52	45
0.0260	0.0010	0.0000	0.0000	0.15	0.0480	0.0020	24.3	1.9	24.5	0.5	-	-	**24.5**	**0.7**	1.01	236	162
0.0260	0.0030	0.0000	0.0000	0.11	0.0480	0.0050	26.0	0.7	24.7	0.3	-	-	**24.6**	**0.6**	0.95	715	535
0.0470	0.0060	0.0000	0.0000	0.70	0.0830	0.0100	26.2	2.6	25.0	0.5	-	-	**24.9**	**0.7**	0.95	156	98
0.0270	0.0020	0.0000	0.0000	0.13	0.0490	0.0040	46.1	5.7	26.4	0.7	-	-	**25.1**	**0.9**	0.57	290	673
0.0320	0.0100	0.0000	0.0000	0.07	0.0580	0.0180	27.5	2.0	25.8	0.6	-	-	**25.7**	**0.8**	0.94	286	132
0.0390	0.0160	0.0000	0.0000	0.45	0.0610	0.0250	34.0	10.0	26.2	0.7	-	-	**25.8**	**1.1**	0.82	42	31
0.0440	0.0050	0.0100	0.0000	-0.01	0.0610	0.0070	39.0	15.0	30.0	1.5	-	-	**29.4**	**1.9**	0.77	26	17
0.0430	0.0120	0.0100	0.0000	-0.33	0.0600	0.0170	43.7	5.2	33.8	0.9	-	-	**33.2**	**1.2**	0.77	59	38
0.0420	0.0040	0.0100	0.0000	0.46	0.0550	0.0050	42.0	12.0	34.0	1.0	-	-	**33.4**	**1.4**	0.80	32	21
0.0330	0.0030	0.0100	0.0000	0.65	0.0450	0.0030	41.3	3.7	34.3	1.0	-	-	**34.0**	**1.2**	0.83	144	31
0.0340	0.0020	0.0100	0.0000	-0.13	0.0450	0.0030	33.2	2.6	34.6	0.8	-	-	**34.7**	**1.0**	1.04	259	324
0.0380	0.0010	0.0100	0.0000	0.54	0.0490	0.0020	33.9	2.1	34.9	0.9	-	-	**35.0**	**1.1**	1.03	243	87
0.0410	0.0020	0.0100	0.0000	0.15	0.0520	0.0030	37.5	1.2	35.3	0.8	-	-	**35.1**	**1.0**	0.94	317	96
0.0440	0.0030	0.0100	0.0000	0.30	0.0530	0.0030	40.8	1.7	37.0	0.8	-	-	**36.7**	**1.1**	0.91	205	105
0.0390	0.0020	0.0100	0.0000	0.42	0.0470	0.0030	43.8	2.9	38.3	0.7	-	-	**38.0**	**1.1**	0.87	109	57
0.0460	0.0040	0.0100	0.0000	0.61	0.0520	0.0040	38.5	2.2	38.0	0.6	-	-	**38.0**	**1.0**	0.99	237	151
0.0400	0.0090	0.0100	0.0000	0.24	0.0420	0.0100	45.7	3.5	42.9	0.9	-	-	**42.6**	**1.2**	0.94	122	122
0.0550	0.0050	0.0100	0.0000	0.47	0.0500	0.0040	39.4	9.1	44.9	1.4	-	-	**45.2**	**1.8**	1.13	39	37
0.0670	0.0040	0.0100	0.0000	0.34	0.0480	0.0030	54.5	5.0	50.7	2.0	-	-	**50.6**	**2.2**	0.93	84	46
0.0700	0.0040	0.0100	0.0000	0.18	0.0490	0.0030	65.8	3.2	65.8	1.8	-	-	**65.8**	**2.2**	1.00	147	77
0.0710	0.0040	0.0100	0.0000	0.74	0.0470	0.0020	68.2	3.3	66.3	1.5	-	-	**66.2**	**2.0**	0.97	189	164
0.1510	0.0070	0.0200	0.0000	0.72	0.0520	0.0020	69.3	3.5	70.2	2.0	-	-	**70.2**	**2.4**	1.01	257	76
0.2130	0.0100	0.0300	0.0000	0.86	0.0520	0.0020	143.0	5.5	134.8	3.9	-	-	**134.3**	**4.7**	0.94	202	127
0.2680	0.0080	0.0400	0.0000	0.55	0.0520	0.0010	196.0	7.2	187.4	4.6	-	-	**187.0**	**6.0**	0.96	211	59

0.2680	0.0080	0.0400	0.0000	0.65	0.0510	0.0010	240.8	4.7	237.1	2.9	265	40	**236.9**	**5.5**	0.98	212	104
0.2750	0.0080	0.0400	0.0000	0.88	0.0520	0.0010	241.1	4.4	240.2	3.6	233	44	240.3	6.0	1.00	191	183
0.2830	0.0140	0.0400	0.0000	0.57	0.0520	0.0020	246.7	4.9	241.6	5.0	292	27	241.3	7.0	0.98	434	448
0.2910	0.0070	0.0400	0.0000	0.83	0.0520	0.0010	253.1	9.9	248.8	4.6	284	83	248.5	6.8	0.98	89	76
0.4290	0.0340	0.0400	0.0000	0.46	0.0730	0.0050	259.3	3.5	256.6	3.0	289.0	25.0	**256.4**	**5.9**	0.99	652	623
0.2910	0.0090	0.0400	0.0000	0.60	0.0520	0.0010	259.5	5	258.8	3	267	42	**258.7**	**5.9**	1.00	284	227
0.0070	0.0040	0.0000	0.0000	-0.30	0.0520	0.0260	368.0	25.0	265.9	4.4	1010.0	140.0	**258.9**	**6.9**	0.73	72	81
0.2910	0.0070	0.0400	0.0000	0.87	0.0510	0.0010	259.2	2.7	259.5	2.6	254	14	**259.6**	**5.8**	1.00	964	413
0.2930	0.0070	0.0400	0.0000	0.86	0.0520	0.0010	260.9	2.8	260.9	3.2	286	17	**260.8**	**6.1**	1.00	1510	949
0.3010	0.0090	0.0400	0.0000	0.75	0.0530	0.0010	266.9	5.5	262.9	4.4	327	34	**262.4**	**6.8**	0.98	200	170
0.3000	0.0090	0.0400	0.0000	0.69	0.0520	0.0010	266.1	5.3	262.6	2.9	273	34	**262.5**	**6.0**	0.99	324	181
0.2950	0.0110	0.0400	0.0000	0.94	0.0520	0.0010	261.9	7.2	262.5	7.6	276	23	**262.6**	**9.1**	1.00	337	266
0.3000	0.0090	0.0400	0.0000	0.94	0.0520	0.0010	266	5.6	263.5	5.9	287	15	**263.3**	**7.9**	0.99	796	805
0.3030	0.0080	0.0400	0.0000	0.69	0.0520	0.0010	268.5	4	265.5	2.6	272	31	**265.4**	**5.9**	0.99	317	331
0.3010	0.0080	0.0400	0.0000	0.71	0.0520	0.0010	266.8	4.3	266.4	6	272	42	**266.3**	**8.0**	1.00	206	100
0.3020	0.0080	0.0400	0.0000	0.81	0.0520	0.0010	267.9	4.2	266.9	3.5	278	27	**266.9**	**6.4**	1.00	492	311
0.3060	0.0170	0.0400	0.0000	0.55	0.0520	0.0030	271	12	267	5.5	260	98	**266.9**	**7.7**	0.98	37	22
0.3020	0.0100	0.0400	0.0000	0.63	0.0520	0.0010	267.5	5.9	266.9	6.6	278	41	**267.1**	**8.7**	1.00	193	108
0.2940	0.0090	0.0400	0.0000	0.80	0.0510	0.0010	261.9	5.5	267.1	5.7	237	37	**267.3**	**7.8**	1.02	183	147
0.3000	0.0080	0.0400	0.0000	0.81	0.0510	0.0010	266.1	4.6	267.2	3.6	254	24	**267.4**	**6.4**	1.00	800	567
0.3020	0.0080	0.0400	0.0000	0.91	0.0520	0.0010	268.1	3.9	267.5	4.3	273	13	**267.4**	**6.8**	1.00	614	364
0.3040	0.0080	0.0400	0.0000	0.57	0.0530	0.0010	269.4	3.7	267.8	2	312	24	**267.4**	**5.7**	0.99	362	168
0.2990	0.0070	0.0400	0.0000	0.85	0.0510	0.0010	265.3	3.1	267.8	3.4	253	22	**267.9**	**6.3**	1.01	358	242
0.3040	0.0070	0.0400	0.0000	0.94	0.0520	0.0010	269.1	3.4	268.3	3.7	277	13	**268.2**	**6.5**	1.00	2173	1650
0.3100	0.0080	0.0400	0.0000	0.75	0.0520	0.0010	275.2	4.1	268.9	3.5	298	26	**268.7**	**6.4**	0.98	407	93
0.3060	0.0080	0.0400	0.0000	0.77	0.0520	0.0010	270.9	4.4	268.7	3.8	265	29	**268.8**	**6.6**	0.99	241	218
0.3010	0.0090	0.0400	0.0000	0.30	0.0510	0.0020	267.2	5.7	268.9	3	240	56	**269.1**	**6.1**	1.01	123	70
0.3110	0.0130	0.0400	0.0000	0.94	0.0530	0.0010	274.5	8.3	269.5	7.8	321	24	**269.2**	**9.7**	0.98	780	484
0.3110	0.0100	0.0400	0.0000	0.71	0.0520	0.0010	275.1	6	269.4	4.1	302	39	**269.2**	**6.7**	0.98	187	120
0.3110	0.0110	0.0400	0.0000	0.75	0.0520	0.0010	274.5	6.9	269.3	4.6	267	43	**269.3**	**7.1**	0.98	183	129
0.3080	0.0090	0.0400	0.0000	0.94	0.0520	0.0010	272.3	5.1	269.3	6.3	287	18	**269.4**	**8.2**	0.99	546	406
0.3000	0.0090	0.0400	0.0000	0.80	0.0510	0.0010	266.2	4.8	269.8	4.7	224	29	**270.2**	**7.1**	1.01	138	95
0.3100	0.0140	0.0400	0.0000	0.69	0.0530	0.0020	273.8	9.5	271.2	4.7	309	79	**270.8**	**7.1**	0.99	67	34
0.3020	0.0130	0.0400	0.0000	0.56	0.0520	0.0020	268.1	9.1	271.2	7.1	258	71	**271.5**	**9.2**	1.01	81	75
0.3160	0.0090	0.0400	0.0000	0.79	0.0520	0.0010	278.8	4.4	272.8	3.1	293	27	**272.6**	**6.2**	0.98	573	295
0.3070	0.0090	0.0400	0.0000	0.94	0.0520	0.0010	271.8	4.6	272.6	4.2	264	18	**272.6**	**6.9**	1.00	822	924
0.3170	0.0150	0.0400	0.0000	0.54	0.0540	0.0030	279	11	274.5	7.6	351	95	**273.7**	**9.2**	0.98	40	32
0.3080	0.0090	0.0400	0.0000	0.79	0.0520	0.0010	272.9	4.6	274.3	4.8	298	27	**274.2**	**7.3**	1.01	253	237
0.3050	0.0130	0.0400	0.0000	0.81	0.0520	0.0010	270.2	9.1	274.3	7.4	268	47	**274.5**	**9.2**	1.02	129	147
0.3070	0.0080	0.0400	0.0000	0.85	0.0510	0.0010	271.6	4.2	274.8	3.6	250	21	**275.1**	**6.5**	1.01	800	280
0.3130	0.0100	0.0400	0.0000	0.69	0.0520	0.0010	276.7	5.9	278.2	4.2	275	35	**278.2**	**6.9**	1.00	201	121
0.3110	0.0110	0.0400	0.0000	0.80	0.0520	0.0010	274.5	7	278.2	4.4	270	39	**278.3**	**7.1**	1.01	144	89
0.3170	0.0090	0.0400	0.0000	0.89	0.0520	0.0010	279.2	5	279	6.3	275	25	**278.8**	**8.3**	1.00	514	239
0.3200	0.0100	0.0400	0.0000	0.91	0.0520	0.0010	281.8	5.9	281	5.8	274	17	**281.1**	**8.0**	1.00	402	234
0.3170	0.0110	0.0400	0.0000	0.69	0.0520	0.0020	279.6	7.1	282.7	4.8	270	46	**282.9**	**7.4**	1.01	219	268
0.3260	0.0100	0.0500	0.0000	0.74	0.0520	0.0010	286.6	6.2	284.5	4.8	289	38	**284.4**	**7.4**	0.99	378	210
0.3250	0.0080	0.0500	0.0000	0.48	0.0520	0.0010	285.7	3.7	284.3	4.2	262	37	**284.5**	**7.1**	1.00	209	92
0.3290	0.0110	0.0500	0.0000	0.96	0.0520	0.0010	289	7.2	288.9	7.5	279	20	**288.8**	**9.4**	1.00	509	191
0.3250	0.0100	0.0500	0.0000	0.87	0.0520	0.0010	285.3	6.2	288.6	5	263	35	**288.8**	**7.6**	1.01	148	84
0.4720	0.0130	0.0600	0.0000	0.89	0.0550	0.0010	392.7	6	392	5.6	410	17	**391.8**	**9.6**	1.00	519	452
0.5020	0.0120	0.0700	0.0000	0.83	0.0550	0.0010	412.9	4	410.6	4.4	427	19	**410.3**	**9.2**	0.99	432	88
0.5760	0.0170	0.0700	0.0000	0.83	0.0570	0.0010	461.7	8.6	459.8	8.9	476	31	**459.3**	**12.8**	1.00	80	40
0.5820	0.0180	0.0700	0.0000	0.76	0.0570	0.0010	465.7	8.9	463.2	6.1	504	29	**462.6**	**10.9**	0.99	160	113
0.5830	0.0160	0.0800	0.0000	0.85	0.0560	0.0010	466.1	7.3	468.9	6.9	468	20	**469.3**	**11.4**	1.01	153	113
0.6040	0.0160	0.0800	0.0000	0.73	0.0570	0.0010	480	6.5	481.6	5.9	502	25	**481.3**	**11.2**	1.00	190	237
0.6180	0.0230	0.0800	0.0000	0.96	0.0560	0.0010	488	12	490	12	461	16	**490.6**	**15.5**	1.00	437	251
0.6530	0.0190	0.0800	0.0000	0.95	0.0570	0.0010	510.4	8.6	508	8.9	509	13	**508**	**13.5**	1.00	877	162
0.6930	0.0200	0.0900	0.0000	0.86	0.0580	0.0010	534.4	9.1	534.1	9	535	31	**534.2**	**13.9**	1.00	155	92
0.7090	0.0210	0.0900	0.0000	0.87	0.0590	0.0010	544.1	9.2	540.6	7.9	559	26	**540.4**	**13.2**	0.99	295	170
0.7330	0.0260	0.0900	0.0000	0.34	0.0600	0.0020	558	13	550.3	7.2	611	62	**548.9**	**13.0**	0.99	55	1
0.7580	0.0210	0.0900	0.0000	0.88	0.0600	0.0010	572.6	8.7	565.3	9	599	21	**565**	**14.3**	0.99	166	28
0.7550	0.0210	0.0900	0.0000	0.62	0.0600	0.0010	571.2	8.5	569	5.6	590	28	**568.6**	**12.5**	1.00	120	24
0.7500	0.0210	0.0900	0.0000	0.88	0.0590	0.0010	568	8.9	569.6	7.4	561	22	**569.9**	**13.6**	1.00	298	233
0.7840	0.0270	0.0900	0.0000	0.71	0.0600	0.0020	587	13	577.3	9.6	593	51	**577**	**14.9**	0.98	68	45
0.7920	0.0190	0.1000	0.0000	0.83	0.0600	0.0010	592.2	6.2	587.5	7.3	613	19	**586.8**	**13.6**	0.99	329	149
0.7940	0.0240	0.1000	0.0000	0.94	0.0600	0.0010	593	10	597	12	606	15	**596.7**	**17.2**	1.01	578	69
0.8270	0.0200	0.1000	0.0000	0.95	0.0600	0.0010	611.9	6.7	613	11	613	13	**613.8**	**16.5**	1.00	472	127
0.8680	0.0210	0.1000	0.0000	0.58	0.0610	0.0010	634.3	6.4	632.6	5.2	650	22	**632.2**	**13.5**	1.00	198	212
0.8500	0.0220	0.1000	0.0000	0.90	0.0600	0.0010	624.3	7.6	633	8.8	596	16	**634**	**15.3**	1.01	291	90
0.8630	0.0250	0.1000	0.0000	0.93	0.0610	0.0010	634	10	634	12	632	16	**634.4**	**17.6**	1.00	408	351
0.9340	0.0240	0.1100	0.0000	0.92	0.0620	0.0010	669.4	7.8	674.5	9.4	673	14	**674.5**	**16.3**	1.01	617	422
1.1380	0.0340	0.1300	0.0000	0.82	0.0650	0.0020	771	12	773	11	784	24	**772.7**	**19.1**	1.00	123	69
1.3840	0.0400	0.1500	0.0000	0.97	0.0690	0.0010	882	12	878	16	897	10	**877.2**	**24.0**	1.00	413	257
1.6980	0.0450	0.1600	0.0000	0.96	0.0760	0.0020	1008	11	974	15	1088	8.7	**968.8**	**24.8**	0.97	629	391
1.6790	0.0420	0.1700	0.0000	0.81	0.0730	0.0020	1000.4	9.9	1000	11	1018	21	**998.5**	**22.6**	1.00	136	71

1.6890	0.0400	0.1700	0.0000	0.68	0.0730	0.0020	1004.1	8.2	1005.5	8.3	1008	23	**1005.4**	**21.4**	1.00	168	85
1.7920	0.0600	0.1700	0.0100	0.71	0.0760	0.0020	1042	18	1025	20	1096	35	**1022.2**	**29.2**	0.98	36	18
1.7380	0.0550	0.1700	0.0100	0.94	0.0740	0.0020	1022	16	1025	25	1029	16	**1024.6**	**32.7**	1.00	103	43
1.7620	0.0500	0.1700	0.0000	0.92	0.0730	0.0020	1034	12	1036	17	1022	16	**1037**	**27.3**	1.00	166	72
1.8380	0.0470	0.1800	0.0000	0.85	0.0750	0.0020	1059	11	1059	14	1058	21	**1059.4**	**25.2**	1.00	98	39
1.8310	0.0470	0.1800	0.0000	0.95	0.0740	0.0020	1056	11	1062	15	1044	10	**1063**	**26.2**	1.01	400	301
1.8810	0.0440	0.1800	0.0000	0.75	0.0760	0.0020	1074.3	8	1068	7.3	1102	15	**1066.3**	**22.1**	0.99	155	48
1.9290	0.0500	0.1800	0.0000	0.82	0.0770	0.0020	1091	11	1072	16	1109	17	**1069.9**	**27.0**	0.98	52	18
1.9460	0.0600	0.1800	0.0000	0.96	0.0780	0.0020	1096	16	1076	16	1153	16	**1072.1**	**26.7**	0.98	340	158
1.8970	0.0540	0.1800	0.0000	0.72	0.0760	0.0020	1080	13	1074	17	1084	25	**1072.9**	**27.8**	0.99	37	10
1.9180	0.0550	0.1800	0.0100	0.94	0.0760	0.0020	1087	14	1078	20	1096	12	**1077**	**29.8**	0.99	415	135
1.9210	0.0490	0.1800	0.0000	0.62	0.0760	0.0020	1088	11	1079	12	1086	29	**1078**	**24.2**	0.99	57	32
1.9190	0.0770	0.1800	0.0100	0.87	0.0770	0.0020	1086	23	1084	23	1121	39	**1082.3**	**32.0**	1.00	23	4
1.9480	0.0590	0.1800	0.0100	0.84	0.0770	0.0020	1097	15	1093	21	1108	22	**1092.3**	**30.9**	1.00	72	41
1.9920	0.0480	0.1900	0.0000	0.90	0.0760	0.0020	1112.5	9	1110	11	1105	11	**1109.7**	**24.5**	1.00	157	140
2.0750	0.0510	0.1900	0.0000	0.86	0.0780	0.0020	1143	11	1126	17	1153	21	**1125.2**	**28.2**	0.99	130	88
2.0270	0.0610	0.1900	0.0000	0.82	0.0770	0.0020	1124	15	1128	17	1119	25	**1128.9**	**28.3**	1.00	63	28
2.0550	0.0530	0.1900	0.0000	0.89	0.0780	0.0020	1134	11	1143	15	1140	17	**1143.1**	**27.1**	1.01	151	83
2.0730	0.0590	0.2000	0.0100	0.90	0.0770	0.0020	1139	14	1151	17	1132	18	**1152.2**	**29.0**	1.01	156	35
2.1460	0.0740	0.2000	0.0100	0.94	0.0790	0.0020	1163	19	1161	22	1165	21	**1161.6**	**32.6**	1.00	74	37
2.1400	0.0540	0.2000	0.0000	0.85	0.0780	0.0020	1162	11	1173	15	1148	15	**1174.6**	**27.9**	1.01	297	148
2.2960	0.0730	0.2100	0.0100	0.91	0.0800	0.0020	1210	18	1218	22	1205	19	**1218.9**	**33.3**	1.01	117	42
2.3340	0.0910	0.2100	0.0100	0.98	0.0800	0.0020	1221	24	1233	38	1201	9.9	**1235.7**	**47.5**	1.01	304	117
2.9370	0.1120	0.2300	0.0100	0.78	0.0920	0.0030	1390	24	1354	20	1460	40	**1345.4**	**34.5**	0.97	187	64
3.2710	0.1030	0.2600	0.0100	0.95	0.0920	0.0020	1473	19	1480	31	1471	14	**1471.3**	**40.4**	1.00	94	48
4.0800	0.1370	0.2900	0.0100	0.98	0.1010	0.0020	1648	23	1651	41	1651	8.5	**1650.8**	**38.0**	1.00	553	89
4.1850	0.0960	0.2900	0.0100	0.93	0.1060	0.0020	1670.9	9.3	1626	16	1724	7.5	**1724.3**	**37.5**	0.97	285	64
5.7800	0.1740	0.3600	0.0100	0.96	0.1170	0.0020	1947	20	1991	41	1905	8	**1905.2**	**36.8**	1.02	112	606
12.8400	0.3730	0.5000	0.0100	0.97	0.1860	0.0040	2667	20	2600	38	2709	7.9	**2708.9**	**33.9**	0.97	63	44
13.4900	0.3240	0.5200	0.0100	0.97	0.1880	0.0040	2718	11	2703	30	2723	5	**2723.2**	**33.3**	1.00	200	173
18.5900	0.4590	0.6000	0.0100	0.98	0.2220	0.0040	3020	14	3047	36	2996	5.1	**2995.8**	**32.6**	1.01	199	72